

CALCULUS

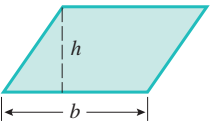
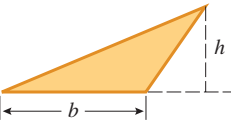
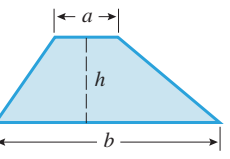
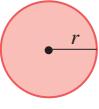
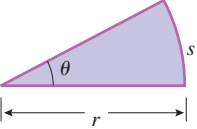
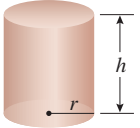
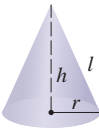
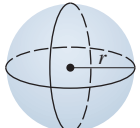
ANTON BIVENS DAVIS

10TH EDITION



GEOMETRY FORMULAS

A = area, S = lateral surface area, V = volume, h = height, B = area of base, r = radius, l = slant height, C = circumference, s = arc length

Parallelogram	Triangle	Trapezoid	Circle	Sector
 $A = bh$	 $A = \frac{1}{2}bh$	 $A = \frac{1}{2}(a + b)h$	 $A = \pi r^2, C = 2\pi r$	 $A = \frac{1}{2}r^2\theta, s = r\theta$ $(\theta \text{ in radians})$
Right Circular Cylinder	Right Circular Cone	Any Cylinder or Prism with Parallel Bases		Sphere
 $V = \pi r^2 h, S = 2\pi r h$	 $V = \frac{1}{3}\pi r^2 h, S = \pi r l$	 $V = Bh$		 $V = \frac{4}{3}\pi r^3, S = 4\pi r^2$

ALGEBRA FORMULAS

THE QUADRATIC FORMULA	THE BINOMIAL FORMULA
The solutions of the quadratic equation $ax^2 + bx + c = 0$ are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	$(x + y)^n = x^n + nx^{n-1}y + \frac{n(n-1)}{1 \cdot 2}x^{n-2}y^2 + \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3}x^{n-3}y^3 + \cdots + nxy^{n-1} + y^n$ $(x - y)^n = x^n - nx^{n-1}y + \frac{n(n-1)}{1 \cdot 2}x^{n-2}y^2 - \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3}x^{n-3}y^3 + \cdots \pm nxy^{n-1} \mp y^n$

TABLE OF INTEGRALS

BASIC FUNCTIONS

1. $\int u^n du = \frac{u^{n+1}}{n+1} + C$
2. $\int \frac{du}{u} = \ln |u| + C$
3. $\int e^u du = e^u + C$
4. $\int \sin u du = -\cos u + C$
5. $\int \cos u du = \sin u + C$
6. $\int \tan u du = \ln |\sec u| + C$
7. $\int \sin^{-1} u du = u \sin^{-1} u + \sqrt{1 - u^2} + C$
8. $\int \cos^{-1} u du = u \cos^{-1} u - \sqrt{1 - u^2} + C$
9. $\int \tan^{-1} u du = u \tan^{-1} u - \ln \sqrt{1 + u^2} + C$
10. $\int a^u du = \frac{a^u}{\ln a} + C$
11. $\int \ln u du = u \ln u - u + C$
12. $\int \cot u du = \ln |\sin u| + C$
13. $\int \sec u du = \ln |\sec u + \tan u| + C$
 $= \ln \left| \tan \left(\frac{1}{4}\pi + \frac{1}{2}u \right) \right| + C$
14. $\int \csc u du = \ln |\csc u - \cot u| + C$
 $= \ln \left| \tan \frac{1}{2}u \right| + C$
15. $\int \cot^{-1} u du = u \cot^{-1} u + \ln \sqrt{1 + u^2} + C$
16. $\int \sec^{-1} u du = u \sec^{-1} u - \ln |u + \sqrt{u^2 - 1}| + C$
17. $\int \csc^{-1} u du = u \csc^{-1} u + \ln |u + \sqrt{u^2 - 1}| + C$

RECIPROCAL OF BASIC FUNCTIONS

$$18. \int \frac{1}{1 \pm \sin u} du = \tan u \mp \sec u + C$$

$$19. \int \frac{1}{1 \pm \cos u} du = -\cot u \pm \csc u + C$$

$$20. \int \frac{1}{1 \pm \tan u} du = \frac{1}{2}(u \pm \ln |\cos u \pm \sin u|) + C$$

$$21. \int \frac{1}{\sin u \cos u} du = \ln |\tan u| + C$$

$$22. \int \frac{1}{1 \pm \cot u} du = \frac{1}{2}(u \mp \ln |\sin u \pm \cos u|) + C$$

$$23. \int \frac{1}{1 \pm \sec u} du = u + \cot u \mp \csc u + C$$

$$24. \int \frac{1}{1 \pm \csc u} du = u - \tan u \pm \sec u + C$$

$$25. \int \frac{1}{1 \pm e^u} du = u - \ln(1 \pm e^u) + C$$

POWERS OF TRIGONOMETRIC FUNCTIONS

$$26. \int \sin^2 u du = \frac{1}{2}u - \frac{1}{4}\sin 2u + C$$

$$27. \int \cos^2 u du = \frac{1}{2}u + \frac{1}{4}\sin 2u + C$$

$$28. \int \tan^2 u du = \tan u - u + C$$

$$29. \int \sin^n u du = -\frac{1}{n}\sin^{n-1} u \cos u + \frac{n-1}{n} \int \sin^{n-2} u du$$

$$30. \int \cos^n u du = \frac{1}{n}\cos^{n-1} u \sin u + \frac{n-1}{n} \int \cos^{n-2} u du$$

$$31. \int \tan^n u du = \frac{1}{n-1}\tan^{n-1} u - \int \tan^{n-2} u du$$

$$32. \int \cot^2 u du = -\cot u - u + C$$

$$33. \int \sec^2 u du = \tan u + C$$

$$34. \int \csc^2 u du = -\cot u + C$$

$$35. \int \cot^n u du = -\frac{1}{n-1}\cot^{n-1} u - \int \cot^{n-2} u du$$

$$36. \int \sec^n u du = \frac{1}{n-1}\sec^{n-2} u \tan u + \frac{n-2}{n-1} \int \sec^{n-2} u du$$

$$37. \int \csc^n u du = -\frac{1}{n-1}\csc^{n-2} u \cot u + \frac{n-2}{n-1} \int \csc^{n-2} u du$$

PRODUCTS OF TRIGONOMETRIC FUNCTIONS

$$38. \int \sin mu \sin nu du = -\frac{\sin(m+n)u}{2(m+n)} + \frac{\sin(m-n)u}{2(m-n)} + C$$

$$39. \int \cos mu \cos nu du = \frac{\sin(m+n)u}{2(m+n)} + \frac{\sin(m-n)u}{2(m-n)} + C$$

$$40. \int \sin mu \cos nu du = -\frac{\cos(m+n)u}{2(m+n)} - \frac{\cos(m-n)u}{2(m-n)} + C$$

$$41. \int \sin^m u \cos^n u du = -\frac{\sin^{m-1} u \cos^{n+1} u}{m+n} + \frac{m-1}{m+n} \int \sin^{m-2} u \cos^n u du$$
$$= \frac{\sin^{m+1} u \cos^{n-1} u}{m+n} + \frac{n-1}{m+n} \int \sin^m u \cos^{n-2} u du$$

PRODUCTS OF TRIGONOMETRIC AND EXPONENTIAL FUNCTIONS

$$42. \int e^{au} \sin bu du = \frac{e^{au}}{a^2 + b^2}(a \sin bu - b \cos bu) + C$$

$$43. \int e^{au} \cos bu du = \frac{e^{au}}{a^2 + b^2}(a \cos bu + b \sin bu) + C$$

POWERS OF u MULTIPLYING OR DIVIDING BASIC FUNCTIONS

$$44. \int u \sin u du = \sin u - u \cos u + C$$

$$45. \int u \cos u du = \cos u + u \sin u + C$$

$$46. \int u^2 \sin u du = 2u \sin u + (2 - u^2) \cos u + C$$

$$47. \int u^2 \cos u du = 2u \cos u + (u^2 - 2) \sin u + C$$

$$48. \int u^n \sin u du = -u^n \cos u + n \int u^{n-1} \cos u du$$

$$49. \int u^n \cos u du = u^n \sin u - n \int u^{n-1} \sin u du$$

$$50. \int u^n \ln u du = \frac{u^{n+1}}{(n+1)^2}[(n+1) \ln u - 1] + C$$

$$51. \int u e^u du = e^u(u - 1) + C$$

$$52. \int u^n e^u du = u^n e^u - n \int u^{n-1} e^u du$$

$$53. \int u^n a^u du = \frac{u^n a^u}{\ln a} - \frac{n}{\ln a} \int u^{n-1} a^u du + C$$

$$54. \int \frac{e^u}{u^n} du = -\frac{e^u}{(n-1)u^{n-1}} + \frac{1}{n-1} \int \frac{e^u}{u^{n-1}} du$$

$$55. \int \frac{a^u}{u^n} du = -\frac{a^u}{(n-1)u^{n-1}} + \frac{\ln a}{n-1} \int \frac{a^u}{u^{n-1}} du$$

$$56. \int \frac{du}{u \ln u} = \ln |\ln u| + C$$

POLYNOMIALS MULTIPLYING BASIC FUNCTIONS

$$57. \int p(u) e^{au} du = \frac{1}{a} p(u) e^{au} - \frac{1}{a^2} p'(u) e^{au} + \frac{1}{a^3} p''(u) e^{au} - \dots \quad [\text{signs alternate: } + - + - \dots]$$

$$58. \int p(u) \sin au du = -\frac{1}{a} p(u) \cos au + \frac{1}{a^2} p'(u) \sin au + \frac{1}{a^3} p''(u) \cos au - \dots \quad [\text{signs alternate in pairs after first term: } + + - - + + - - \dots]$$

$$59. \int p(u) \cos au du = \frac{1}{a} p(u) \sin au + \frac{1}{a^2} p'(u) \cos au - \frac{1}{a^3} p''(u) \sin au - \dots \quad [\text{signs alternate in pairs: } + + - - + + - - \dots]$$

Calculus provides a way of viewing and analyzing the physical world. As with all mathematics courses, calculus involves equations and formulas. However, if you successfully learn to use all the formulas and solve all of the problems in the text but do not master the underlying *ideas*, you will have missed the most important part of calculus. If you master these ideas, you will have a widely applicable tool that goes far beyond textbook exercises.

Before starting your studies, you may find it helpful to leaf through this text to get a general feeling for its different parts:

- The opening page of each chapter gives you an overview of what that chapter is about, and the opening page of each section within a chapter gives you an overview of what that section is about. To help you locate specific information, sections are subdivided into topics that are marked with a box like this ■.
- Each section ends with a set of exercises. The answers to most odd-numbered exercises appear in the back of the book. If you find that your answer to an exercise does not match that in the back of the book, do not assume immediately that yours is incorrect—there may be more than one way to express the answer. For example, if your answer is $\sqrt{2}/2$ and the text answer is $1/\sqrt{2}$, then both are correct since your answer can be obtained by “rationalizing” the text answer. In general, if your answer does not match that in the text, then your best first step is to look for an algebraic manipulation or a trigonometric identity that might help you determine if the two answers are equivalent. If the answer is in the form of a decimal approximation, then your answer might differ from that in the text because of a difference in the number of decimal places used in the computations.
- The section exercises include regular exercises and four special categories: *Quick Check*, *Focus on Concepts*, *True/False*, and *Writing*.
 - The *Quick Check* exercises are intended to give you quick feedback on whether you understand the key ideas in the section; they involve relatively little computation, and have answers provided at the end of the exercise set.
 - The *Focus on Concepts* exercises, as their name suggests, key in on the main ideas in the section.
 - *True/False* exercises focus on key ideas in a different way. You must decide whether the statement is true in *all possible circumstances*, in which case you would declare it to be “true,” or whether there are some circumstances in which it is not true, in which case you would declare it to be “false.” In each such exercise you are asked to “Explain your answer.” You might do this by noting a theorem in the text that shows the statement to be true or by finding a particular example in which the statement is not true.
- *Writing* exercises are intended to test your ability to explain mathematical ideas in words rather than relying solely on numbers and symbols. All exercises requiring writing should be answered in complete, correctly punctuated logical sentences—not with fragmented phrases and formulas.
- Each chapter ends with two additional sets of exercises: *Chapter Review Exercises*, which, as the name suggests, is a select set of exercises that provide a review of the main concepts and techniques in the chapter, and *Making Connections*, in which exercises require you to draw on and combine various ideas developed throughout the chapter.
- Your instructor may choose to incorporate technology in your calculus course. Exercises whose solution involves the use of some kind of technology are tagged with icons to alert you and your instructor. Those exercises tagged with the icon  require graphing technology—either a graphing calculator or a computer program that can graph equations. Those exercises tagged with the icon  require a computer algebra system (CAS) such as *Mathematica*, *Maple*, or available on some graphing calculators.
- At the end of the text you will find a set of four appendices covering various topics such as a detailed review of trigonometry and graphing techniques using technology. Inside the front and back covers of the text you will find endpapers that contain useful formulas.
- The ideas in this text were created by real people with interesting personalities and backgrounds. Pictures and biographical sketches of many of these people appear throughout the book.
- Notes in the margin are intended to clarify or comment on important points in the text.

A Word of Encouragement

As you work your way through this text you will find some ideas that you understand immediately, some that you don’t understand until you have read them several times, and others that you do not seem to understand, even after several readings. Do not become discouraged—some ideas are intrinsically difficult and take time to “percolate.” You may well find that a hard idea becomes clear later when you least expect it.

Web Sites for this Text

www.antontextbooks.com
www.wiley.com/go/global/anton



David Henderson/Getty Images

10th
EDITION

CALCULUS

- **HOWARD ANTON** *Drexel University*
- **IRL BIVENS** *Davidson College*
- **STEPHEN DAVIS** *Davidson College*



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About HOWARD ANTON

Howard Anton obtained his B.A. from Lehigh University, his M.A. from the University of Illinois, and his Ph.D. from the Polytechnic University of Brooklyn, all in mathematics. In the early 1960s he worked for Burroughs Corporation and Avco Corporation at Cape Canaveral, Florida, where he was involved with the manned space program. In 1968 he joined the Mathematics Department at Drexel University, where he taught full time until 1983. Since that time he has been an Emeritus Professor at Drexel and has devoted the majority of his time to textbook writing and activities for mathematical associations. Dr. Anton was president of the EPADEL section of the Mathematical Association of America (MAA), served on the Board of Governors of that organization, and guided the creation of the student chapters of the MAA. He has published numerous research papers in functional analysis, approximation theory, and topology, as well as pedagogical papers. He is best known for his textbooks in mathematics, which are among the most widely used in the world. There are currently more than one hundred versions of his books, including translations into Spanish, Arabic, Portuguese, Italian, Indonesian, French, Japanese, Chinese, Hebrew, and German. His textbook in linear algebra has won both the Textbook Excellence Award and the McGuffey Award from the Textbook Author's Association. For relaxation, Dr. Anton enjoys traveling and photography.

About IRL BIVENS

Irl C. Bivens, recipient of the George Polya Award and the Merten M. Hasse Prize for Expository Writing in Mathematics, received his A.B. from Pfeiffer College and his Ph.D. from the University of North Carolina at Chapel Hill, both in mathematics. Since 1982, he has taught at Davidson College, where he currently holds the position of professor of mathematics. A typical academic year sees him teaching courses in calculus, topology, and geometry. Dr. Bivens also enjoys mathematical history, and his annual History of Mathematics seminar is a perennial favorite with Davidson mathematics majors. He has published numerous articles on undergraduate mathematics, as well as research papers in his specialty, differential geometry. He has served on the editorial boards of the MAA Problem Book series, the MAA Dolciani Mathematical Expositions series and *The College Mathematics Journal*. When he is not pursuing mathematics, Professor Bivens enjoys reading, juggling, swimming, and walking.

About STEPHEN DAVIS

Stephen L. Davis received his B.A. from Lindenwood College and his Ph.D. from Rutgers University in mathematics. Having previously taught at Rutgers University and Ohio State University, Dr. Davis came to Davidson College in 1981, where he is currently a professor of mathematics. He regularly teaches calculus, linear algebra, abstract algebra, and computer science. A sabbatical in 1995–1996 took him to Swarthmore College as a visiting associate professor. Professor Davis has published numerous articles on calculus reform and testing, as well as research papers on finite group theory, his specialty. Professor Davis has held several offices in the Southeastern section of the MAA, including chair and secretary-treasurer and has served on the MAA Board of Governors. He is currently a faculty consultant for the Educational Testing Service for the grading of the Advanced Placement Calculus Exam, webmaster for the North Carolina Association of Advanced Placement Mathematics Teachers, and is actively involved in nurturing mathematically talented high school students through leadership in the Charlotte Mathematics Club. For relaxation, he plays basketball, juggles, and travels. Professor Davis and his wife Elisabeth have three children, Laura, Anne, and James, all former calculus students.

To
my wife Pat and my children: Brian, David, and Lauren

In Memory of
my mother Shirley
my father Benjamin
my thesis advisor and inspiration, George Bachman
my benefactor in my time of need, Stephen Girard (1750–1831)
—HA

To
my son Robert
—IB

To
my wife Elisabeth
my children: Laura, Anne, and James
—SD



PREFACE

This tenth edition of *Calculus* maintains those aspects of previous editions that have led to the series' success—we continue to strive for student comprehension without sacrificing mathematical accuracy, and the exercise sets are carefully constructed to avoid unhappy surprises that can derail a calculus class.

All of the changes to the tenth edition were carefully reviewed by outstanding teachers comprised of both users and nonusers of the previous edition. The charge of this committee was to ensure that all changes did not alter those aspects of the text that attracted users of the ninth edition and at the same time provide freshness to the new edition that would attract new users.



NEW TO THIS EDITION

- Exercise sets have been modified to correspond more closely to questions in *WileyPLUS*. In addition, more *WileyPLUS* questions now correspond to specific exercises in the text.
- New applied exercises have been added to the book and existing applied exercises have been updated.
- Where appropriate, additional skill/practice exercises were added.



OTHER FEATURES

Flexibility This edition has a built-in flexibility that is designed to serve a broad spectrum of calculus philosophies—from traditional to “reform.” Technology can be emphasized or not, and the order of many topics can be permuted freely to accommodate each instructor’s specific needs.

Rigor The challenge of writing a good calculus book is to strike the right balance between rigor and clarity. Our goal is to present precise mathematics to the fullest extent possible in an introductory treatment. Where clarity and rigor conflict, we choose clarity; however, we believe it to be important that the student understand the difference between a careful proof and an informal argument, so we have informed the reader when the arguments being presented are informal or motivational. Theory involving ϵ - δ arguments appears in separate sections so that they can be covered or not, as preferred by the instructor.

Rule of Four The “rule of four” refers to presenting concepts from the verbal, algebraic, visual, and numerical points of view. In keeping with current pedagogical philosophy, we used this approach whenever appropriate.

Visualization This edition makes extensive use of modern computer graphics to clarify concepts and to develop the student’s ability to visualize mathematical objects, particularly those in 3-space. For those students who are working with graphing technology, there are

many exercises that are designed to develop the student's ability to generate and analyze mathematical curves and surfaces.

Quick Check Exercises Each exercise set begins with approximately five exercises (answers included) that are designed to provide students with an immediate assessment of whether they have mastered key ideas from the section. They require a minimum of computation and are answered by filling in the blanks.

Focus on Concepts Exercises Each exercise set contains a clearly identified group of problems that focus on the main ideas of the section.

Technology Exercises Most sections include exercises that are designed to be solved using either a graphing calculator or a computer algebra system such as *Mathematica*, *Maple*, or the open source program *Sage*. These exercises are marked with an icon for easy identification.

Applicability of Calculus One of the primary goals of this text is to link calculus to the real world and the student's own experience. This theme is carried through in the examples and exercises.

Career Preparation This text is written at a mathematical level that will prepare students for a wide variety of careers that require a sound mathematics background, including engineering, the various sciences, and business.

Trigonometry Review Deficiencies in trigonometry plague many students, so we have included a substantial trigonometry review in Appendix B.

Appendix on Polynomial Equations Because many calculus students are weak in solving polynomial equations, we have included an appendix (Appendix C) that reviews the Factor Theorem, the Remainder Theorem, and procedures for finding rational roots.

Principles of Integral Evaluation The traditional Techniques of Integration is entitled "Principles of Integral Evaluation" to reflect its more modern approach to the material. The chapter emphasizes general methods and the role of technology rather than specific tricks for evaluating complicated or obscure integrals.

Historical Notes The biographies and historical notes have been a hallmark of this text from its first edition and have been maintained. All of the biographical materials have been distilled from standard sources with the goal of capturing and bringing to life for the student the personalities of history's greatest mathematicians.

Margin Notes and Warnings These appear in the margins throughout the text to clarify or expand on the text exposition or to alert the reader to some pitfall.



SUPPLEMENTS

The **Student Solutions Manual**, which is printed in two volumes, provides detailed solutions to the odd-numbered exercises in the text. The structure of the step-by-step solutions matches those of the worked examples in the textbook. The Student Solutions Manual is also provided in digital format to students in *WileyPLUS*.

Volume I (Single-Variable Calculus ISBN: 978-1-118-17382-4

Volume II (Multivariable Calculus ISBN: 978-1-118-17383-1

The **Student Study Guide** is available for download from the book companion Web site at www.wiley.com/college/anton or at www.howardanton.com and to users of *WileyPLUS*.

The **Instructor's Solutions Manual** contains detailed solutions to all of the exercises in the text. The Instructor's Solutions Manual is also available in PDF format on the password-protected Instructor Companion Site at www.wiley.com/college/anton or at www.howardanton.com and in *WileyPLUS*.

ISBN: 978-1-118-17380-0

The **Instructor's Manual** suggests time allocations and teaching plans for each section in the text. Most of the teaching plans contain a bulleted list of key points to emphasize. The discussion of each section concludes with a sample homework assignment. The Instructor's Manual is available in PDF format on the password-protected Instructor Companion Site at www.wiley.com/college/anton or at www.howardanton.com and in *WileyPLUS*.

The **Web Projects (Expanding the Calculus Horizon)** referenced in the text can also be downloaded from the companion Web sites and from *WileyPLUS*.

Instructors can also access the following materials from the book companion site or *WileyPLUS*:

- **Interactive Illustrations** can be used in the classroom or computer lab to present and explore key ideas graphically and dynamically. They are especially useful for display of three-dimensional graphs in multivariable calculus.
- The **Computerized Test Bank** features more than 4000 questions—mostly algorithmically generated—that allow for varied questions and numerical inputs.
- The **Printable Test Bank** features questions and answers for every section of the text.
- **PowerPoint lecture slides** cover the major concepts and themes of each section of the book. Personal-Response System questions (“Clicker Questions”) appear at the end of each PowerPoint presentation and provide an easy way to gauge classroom understanding.
- **Additional calculus content** covers analytic geometry in calculus, mathematical modeling with differential equations and parametric equations, as well as an introduction to linear algebra.

WileyPLUS

WileyPLUS, Wiley's digital-learning environment, is loaded with all of the supplements listed on the previous page, and also features the following:

- **Homework management tools**, which easily allow you to assign and grade algorithmic questions, as well as gauge student comprehension.
- **Algorithmic** questions with randomized numeric values and an answer-entry palette for symbolic notation are provided online through *WileyPLUS*. Students can click on “help” buttons for hints, link to the relevant section of the text, show their work or query their instructor using a white board, or see a step-by-step solution (depending on instructor-selecting settings).
- **Interactive Illustrations** can be used in the classroom or computer lab, or for student practice.
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F ABSOLUTE VALUE

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ITERATION AND DYNAMICAL SYSTEMS

RAILROAD DESIGN

ROBOTICS

THE ROOTS OF CALCULUS

Today's exciting applications of calculus have roots that can be traced to the work of the Greek mathematician Archimedes, but the actual discovery of the fundamental principles of calculus was made independently by Isaac Newton (English) and Gottfried Leibniz (German) in the late seventeenth century. The work of Newton and Leibniz was motivated by four major classes of scientific and mathematical problems of the time:

- Find the tangent line to a general curve at a given point.
- Find the area of a general region, the length of a general curve, and the volume of a general solid.
- Find the maximum or minimum value of a quantity—for example, the maximum and minimum distances of a planet from the Sun, or the maximum range attainable for a projectile by varying its angle of fire.
- Given a formula for the distance traveled by a body in any specified amount of time, find the velocity and acceleration of the body at any instant. Conversely, given a formula that

specifies the acceleration of velocity at any instant, find the distance traveled by the body in a specified period of time.

Newton and Leibniz found a fundamental relationship between the problem of finding a tangent line to a curve and the problem of determining the area of a region. Their realization of this connection is considered to be the “discovery of calculus.” Though Newton saw how these two problems are related ten years before Leibniz did, Leibniz published his work twenty years before Newton. This situation led to a stormy debate over who was the rightful discoverer of calculus. The debate engulfed Europe for half a century, with the scientists of the European continent supporting Leibniz and those from England supporting Newton. The conflict was extremely unfortunate because Newton's inferior notation badly hampered scientific development in England, and the Continent in turn lost the benefit of Newton's discoveries in astronomy and physics for nearly fifty years. In spite of it all, Newton and Leibniz were sincere admirers of each other's work.



[Image: Public domain image from <http://commons.wikimedia.org/wiki/File:Hw-newton.jpg>. Image provided courtesy of the University of Texas Libraries, The University of Texas at Austin.]

ISAAC NEWTON (1642–1727)

Newton was born in the village of Woolsthorpe, England. His father died before he was born and his mother raised him on the family farm. As a youth he showed little evidence of his later brilliance, except for an unusual talent with mechanical devices—he apparently built a working water clock and a toy flour mill powered by a mouse. In 1661 he entered Trinity College in Cambridge with a deficiency in geometry. Fortunately, Newton caught the eye of Isaac Barrow, a gifted mathematician and teacher. Under Barrow's guidance Newton immersed himself in mathematics and science, but he graduated without any special distinction. Because the bubonic plague was spreading rapidly through London, Newton returned to his home in Woolsthorpe and stayed there during the years of 1665 and 1666. In those two momentous years the entire framework of modern science was miraculously created in Newton's mind. He discovered calculus, recognized the underlying principles of planetary motion and gravity, and determined that “white” sunlight was composed of all colors, red to violet. For whatever reasons he kept his discoveries to himself. In 1667 he returned to Cambridge to obtain his Master's degree and upon graduation became a teacher at Trinity. Then in 1669 Newton succeeded his teacher, Isaac Barrow, to the Lucasian chair of mathematics at Trinity, one of the most honored chairs of mathematics in the world.

Thereafter, brilliant discoveries flowed from Newton steadily. He formulated the law of gravitation and used it to explain the motion of the moon, the planets, and the tides; he formulated basic theories of light, thermodynamics, and hydrodynamics; and he devised and constructed the first modern reflecting telescope. Throughout his life Newton was hesitant to publish his major discoveries, revealing them only to a select

circle of friends, perhaps because of a fear of criticism or controversy. In 1687, only after intense coaxing by the astronomer, Edmond Halley (discoverer of Halley's comet), did Newton publish his masterpiece, *Philosophiae Naturalis Principia Mathematica* (The Mathematical Principles of Natural Philosophy). This work is generally considered to be the most important and influential scientific book ever written. In it Newton explained the workings of the solar system and formulated the basic laws of motion, which to this day are fundamental in engineering and physics. However, not even the pleas of his friends could convince Newton to publish his discovery of calculus. Only after Leibniz published his results did Newton relent and publish his own work on calculus.

After twenty-five years as a professor, Newton suffered depression and a nervous breakdown. He gave up research in 1695 to accept a position as warden and later master of the London mint. During the twenty-five years that he worked at the mint, he did virtually no scientific or mathematical work. He was knighted in 1705 and on his death was buried in Westminster Abbey with all the honors his country could bestow. It is interesting to note that Newton was a learned theologian who viewed the primary value of his work to be its support of the existence of God. Throughout his life he worked passionately to date biblical events by relating them to astronomical phenomena. He was so consumed with this passion that he spent years searching the Book of Daniel for clues to the end of the world and the geography of hell.

Newton described his brilliant accomplishments as follows: "I seem to have been only like a boy playing on the seashore and diverting myself in now and then finding a smoother pebble or prettier shell than ordinary, whilst the great ocean of truth lay all undiscovered before me."



[Image: Public domain image from http://commons.wikimedia.org/wiki/File:Gottfried_Wilhelm_von_Leibniz.jpg]

GOTTFRIED WILHELM LEIBNIZ (1646–1716)

This gifted genius was one of the last people to have mastered most major fields of knowledge—an impossible accomplishment in our own era of specialization. He was an expert in law, religion, philosophy, literature, politics, geology, metaphysics, alchemy, history, and mathematics.

Leibniz was born in Leipzig, Germany. His father, a professor of moral philosophy at the University of Leipzig, died when Leibniz was six years old. The precocious boy then gained access to his father's library and began reading voraciously on a wide range of subjects, a habit that he maintained throughout his life. At age fifteen he entered the University of Leipzig as a law student and by the age of twenty received a doctorate from the University of Altdorf. Subsequently, Leibniz followed a career in law and international politics, serving as counsel to kings and princes. During his numerous foreign missions, Leibniz came in contact with outstanding mathematicians and scientists who stimulated his interest in mathematics—most notably, the physicist Christian Huygens. In mathematics Leibniz was self-taught, learning the subject by reading papers and journals. As a result of this fragmented mathematical education, Leibniz often rediscovered the results of others, and this helped to fuel the debate over the discovery of calculus.

Leibniz never married. He was moderate in his habits, quick-tempered but easily appeased, and charitable in his judgment of other people's work. In spite of his great achievements, Leibniz never received the honors showered on Newton, and he spent his final years as a lonely embittered man. At his funeral there was one mourner, his secretary. An eyewitness stated, "He was buried more like a robber than what he really was—an ornament of his country."

0

BEFORE CALCULUS



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The development of calculus in the seventeenth and eighteenth centuries was motivated by the need to understand physical phenomena such as the tides, the phases of the moon, the nature of light, and gravity.

One of the important themes in calculus is the analysis of relationships between physical or mathematical quantities. Such relationships can be described in terms of graphs, formulas, numerical data, or words. In this chapter we will develop the concept of a “function,” which is the basic idea that underlies almost all mathematical and physical relationships, regardless of the form in which they are expressed. We will study properties of some of the most basic functions that occur in calculus.

0.1 FUNCTIONS

In this section we will define and develop the concept of a “function,” which is the basic mathematical object that scientists and mathematicians use to describe relationships between variable quantities. Functions play a central role in calculus and its applications.

■ DEFINITION OF A FUNCTION

Many scientific laws and engineering principles describe how one quantity depends on another. This idea was formalized in 1673 by Gottfried Wilhelm Leibniz (see p. xx) who coined the term *function* to indicate the dependence of one quantity on another, as described in the following definition.

0.1.1 DEFINITION If a variable y depends on a variable x in such a way that each value of x determines exactly one value of y , then we say that y is a *function of x* .

Four common methods for representing functions are:

- Numerically by tables
- Geometrically by graphs
- Algebraically by formulas
- Verbally

Table 0.1.1

INDIANAPOLIS 500
QUALIFYING SPEEDS

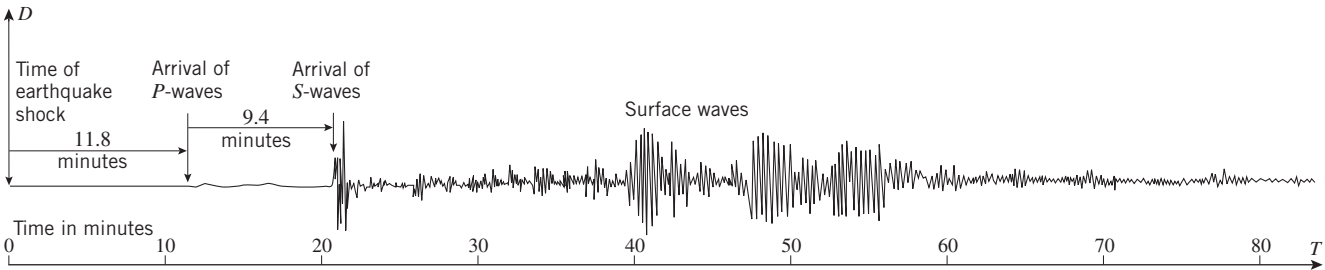
YEAR t	SPEED S (mi/h)
1994	228.011
1995	231.604
1996	233.100
1997	218.263
1998	223.503
1999	225.179
2000	223.471
2001	226.037
2002	231.342
2003	231.725
2004	222.024
2005	227.598
2006	228.985
2007	225.817
2008	226.366
2009	224.864
2010	227.970
2011	227.472

The method of representation often depends on how the function arises. For example:

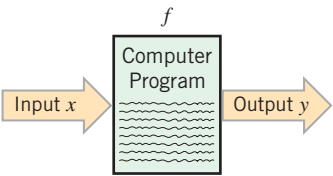
- Table 0.1.1 shows the top qualifying speed S for the Indianapolis 500 auto race as a function of the year t . There is exactly one value of S for each value of t .
- Figure 0.1.1 is a graphical record of an earthquake recorded on a seismograph. The graph describes the deflection D of the seismograph needle as a function of the time T elapsed since the wave left the earthquake’s epicenter. There is exactly one value of D for each value of T .
- Some of the most familiar functions arise from formulas; for example, the formula $C = 2\pi r$ expresses the circumference C of a circle as a function of its radius r . There is exactly one value of C for each value of r .
- Sometimes functions are described in words. For example, Isaac Newton’s Law of Universal Gravitation is often stated as follows: The gravitational force of attraction between two bodies in the Universe is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. This is the verbal description of the formula

$$F = G \frac{m_1 m_2}{r^2}$$

in which F is the force of attraction, m_1 and m_2 are the masses, r is the distance between them, and G is a constant. If the masses are constant, then the verbal description defines F as a function of r . There is exactly one value of F for each value of r .



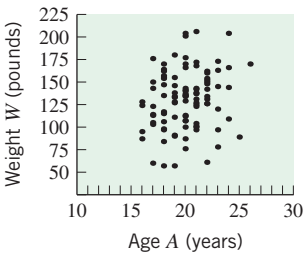
▲ Figure 0.1.1



▲ Figure 0.1.2

In the mid-eighteenth century the Swiss mathematician Leonhard Euler (pronounced “oiler”) conceived the idea of denoting functions by letters of the alphabet, thereby making it possible to refer to functions without stating specific formulas, graphs, or tables. To understand Euler’s idea, think of a function as a computer program that takes an *input* x , operates on it in some way, and produces exactly one *output* y . The computer program is an object in its own right, so we can give it a name, say f . Thus, the function f (the computer program) associates a unique output y with each input x (Figure 0.1.2). This suggests the following definition.

0.1.2 DEFINITION A **function** f is a rule that associates a unique output with each input. If the input is denoted by x , then the output is denoted by $f(x)$ (read “ f of x ”).



▲ Figure 0.1.3

In this definition the term *unique* means “exactly one.” Thus, a function cannot assign two different outputs to the same input. For example, Figure 0.1.3 shows a plot of weight versus age for a random sample of 100 college students. This plot does *not* describe W as a function of A because there are some values of A with more than one corresponding

value of W . This is to be expected, since two people with the same age can have different weights.

■ INDEPENDENT AND DEPENDENT VARIABLES

For a given input x , the output of a function f is called the **value** of f at x or the **image** of x under f . Sometimes we will want to denote the output by a single letter, say y , and write

$$y = f(x)$$

This equation expresses y as a function of x ; the variable x is called the **independent variable** (or **argument**) of f , and the variable y is called the **dependent variable** of f . This terminology is intended to suggest that x is free to vary, but that once x has a specific value a corresponding value of y is determined. For now we will only consider functions in which the independent and dependent variables are real numbers, in which case we say that f is a **real-valued function of a real variable**. Later, we will consider other kinds of functions.

Table 0.1.2

x	0	1	2	3
y	3	4	-1	6

► **Example 1** Table 0.1.2 describes a functional relationship $y = f(x)$ for which

$f(0) = 3$	f associates $y = 3$ with $x = 0$.
$f(1) = 4$	f associates $y = 4$ with $x = 1$.
$f(2) = -1$	f associates $y = -1$ with $x = 2$.
$f(3) = 6$	f associates $y = 6$ with $x = 3$. ◀

► **Example 2** The equation

$$y = 3x^2 - 4x + 2$$

has the form $y = f(x)$ in which the function f is given by the formula

$$f(x) = 3x^2 - 4x + 2$$



Leonhard Euler (1707–1783) Euler was probably the most prolific mathematician who ever lived. It has been said that “Euler wrote mathematics as effortlessly as most men breathe.” He was born in Basel, Switzerland, and was the son of a Protestant minister who had himself studied mathematics. Euler’s genius developed early. He attended the University of Basel, where by age 16 he obtained both a Bachelor of Arts degree and a Master’s degree in philosophy. While at Basel, Euler had the good fortune to be tutored one day a week in mathematics by a distinguished mathematician, Johann Bernoulli. At the urging of his father, Euler then began to study theology. The lure of mathematics was too great, however, and by age 18 Euler had begun to do mathematical research. Nevertheless, the influence of his father and his theological studies remained, and throughout his life Euler was a deeply religious, unaffected person. At various times Euler taught at St. Petersburg Academy of Sciences (in Russia), the University of Basel, and the Berlin Academy of Sciences. Euler’s energy and capacity for work were virtually boundless. His collected works form more than 100 quarto-sized volumes and it is believed that much of his work has been lost. What is particularly

astonishing is that Euler was blind for the last 17 years of his life, and this was one of his most productive periods! Euler’s flawless memory was phenomenal. Early in his life he memorized the entire *Aeneid* by Virgil, and at age 70 he could not only recite the entire work but could also state the first and last sentence on each page of the book from which he memorized the work. His ability to solve problems in his head was beyond belief. He worked out in his head major problems of lunar motion that baffled Isaac Newton and once did a complicated calculation in his head to settle an argument between two students whose computations differed in the fiftieth decimal place.

Following the development of calculus by Leibniz and Newton, results in mathematics developed rapidly in a disorganized way. Euler’s genius gave coherence to the mathematical landscape. He was the first mathematician to bring the full power of calculus to bear on problems from physics. He made major contributions to virtually every branch of mathematics as well as to the theory of optics, planetary motion, electricity, magnetism, and general mechanics.

[Image: http://commons.wikimedia.org/wiki/File:Leonhard_Euler_by_Handmann_.png]

For each input x , the corresponding output y is obtained by substituting x in this formula. For example,

$$f(0) = 3(0)^2 - 4(0) + 2 = 2$$

f associates $y = 2$ with $x = 0$.

$$f(-1.7) = 3(-1.7)^2 - 4(-1.7) + 2 = 17.47$$

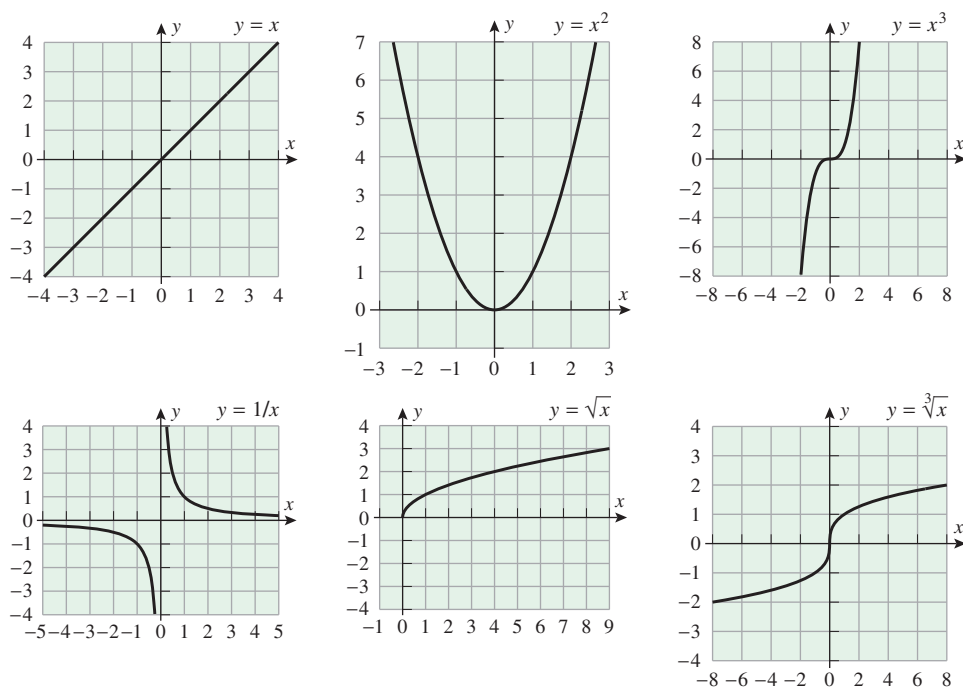
f associates $y = 17.47$ with $x = -1.7$.

$$f(\sqrt{2}) = 3(\sqrt{2})^2 - 4\sqrt{2} + 2 = 8 - 4\sqrt{2}$$

f associates $y = 8 - 4\sqrt{2}$ with $x = \sqrt{2}$. ◀

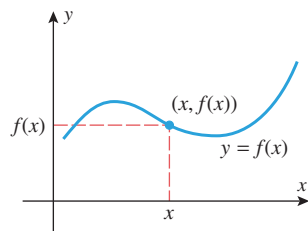
GRAPHS OF FUNCTIONS

Figure 0.1.4 shows only portions of the graphs. Where appropriate, and unless indicated otherwise, it is understood that graphs shown in this text extend indefinitely beyond the boundaries of the displayed figure.



Since $\sqrt{x}$ is imaginary for negative values of x , there are no points on the graph of $y = \sqrt{x}$ in the region where $x < 0$.

▲ Figure 0.1.4



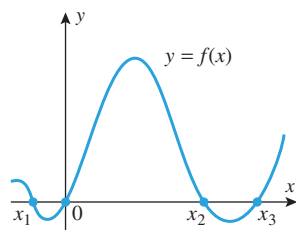
▲ Figure 0.1.5 The y -coordinate of a point on the graph of $y = f(x)$ is the value of f at the corresponding x -coordinate.

Graphs can provide valuable visual information about a function. For example, since the graph of a function f in the xy -plane is the graph of the equation $y = f(x)$, the points on the graph of f are of the form $(x, f(x))$; that is, the y -coordinate of a point on the graph of f is the value of f at the corresponding x -coordinate (Figure 0.1.5). The values of x for which $f(x) = 0$ are the x -coordinates of the points where the graph of f intersects the x -axis (Figure 0.1.6). These values are called the **zeros** of f , the **roots** of $f(x) = 0$, or the **x -intercepts** of the graph of $y = f(x)$.

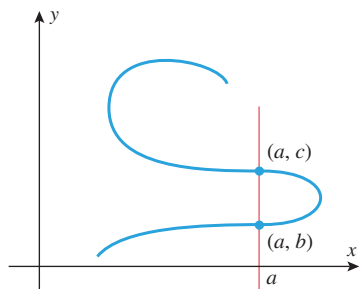
THE VERTICAL LINE TEST

Not every curve in the xy -plane is the graph of a function. For example, consider the curve in Figure 0.1.7, which is cut at two distinct points, (a, b) and (a, c) , by a vertical line. This curve cannot be the graph of $y = f(x)$ for any function f ; otherwise, we would have

$$f(a) = b \quad \text{and} \quad f(a) = c$$

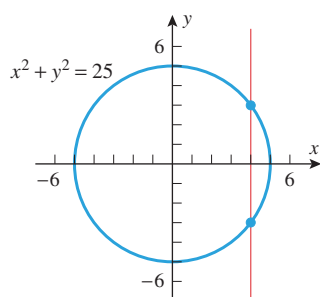


▲ **Figure 0.1.6** f has zeros at x_1 , 0 , x_2 , and x_3 .



▲ **Figure 0.1.7** This curve cannot be the graph of a function.

Symbols such as $+x$ and $-x$ are deceptive, since it is tempting to conclude that $+x$ is positive and $-x$ is negative. However, this need not be so, since x itself can be positive or negative. For example, if x is negative, say $x = -3$, then $-x = 3$ is positive and $+x = -3$ is negative.



▲ **Figure 0.1.8**

WARNING

To denote the negative square root you must write $-\sqrt{x}$. For example, the positive square root of 9 is $\sqrt{9} = 3$, whereas the negative square root of 9 is $-\sqrt{9} = -3$. (Do not make the mistake of writing $\sqrt{9} = \pm 3$.)

which is impossible, since f cannot assign two different values to a . Thus, there is no function f whose graph is the given curve. This illustrates the following general result, which we will call the **vertical line test**.

0.1.3 THE VERTICAL LINE TEST A curve in the xy -plane is the graph of some function f if and only if no vertical line intersects the curve more than once.

► **Example 3** The graph of the equation

$$x^2 + y^2 = 25$$

is a circle of radius 5 centered at the origin and hence there are vertical lines that cut the graph more than once (Figure 0.1.8). Thus this equation does not define y as a function of x . ◀

THE ABSOLUTE VALUE FUNCTION

Recall that the **absolute value** or **magnitude** of a real number x is defined by

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

The effect of taking the absolute value of a number is to strip away the minus sign if the number is negative and to leave the number unchanged if it is nonnegative. Thus,

$$|5| = 5, \quad \left| -\frac{4}{7} \right| = \frac{4}{7}, \quad |0| = 0$$

A more detailed discussion of the properties of absolute value is given in Web Appendix F. However, for convenience we provide the following summary of its algebraic properties.

0.1.4 PROPERTIES OF ABSOLUTE VALUE If a and b are real numbers, then

- | | |
|------------------------------------|--|
| (a) $ -a = a $ | A number and its negative have the same absolute value. |
| (b) $ ab = a b $ | The absolute value of a product is the product of the absolute values. |
| (c) $ a/b = a / b $, $b \neq 0$ | The absolute value of a ratio is the ratio of the absolute values. |
| (d) $ a + b \leq a + b $ | The triangle inequality |

The graph of the function $f(x) = |x|$ can be obtained by graphing the two parts of the equation

$$y = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

separately. Combining the two parts produces the V-shaped graph in Figure 0.1.9.

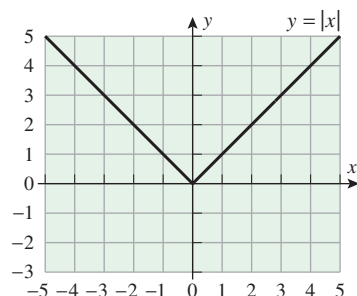
Absolute values have important relationships to square roots. To see why this is so, recall from algebra that every positive real number x has two square roots, one positive and one negative. By definition, the symbol $\sqrt{x}$ denotes the *positive* square root of x .

Care must be exercised in simplifying expressions of the form $\sqrt{x^2}$, since it is *not* always true that $\sqrt{x^2} = x$. This equation is correct if x is nonnegative, but it is false if x is negative. For example, if $x = -4$, then

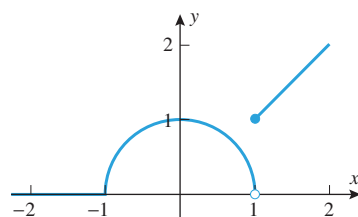
$$\sqrt{x^2} = \sqrt{(-4)^2} = \sqrt{16} = 4 \neq x$$

TECHNOLOGY MASTERY

Verify (1) by using a graphing utility to show that the equations $y = \sqrt{x^2}$ and $y = |x|$ have the same graph.



▲ Figure 0.1.9



▲ Figure 0.1.10

REMARK

In Figure 0.1.10 the solid dot and open circle at the breakpoint $x = 1$ serve to emphasize that the point on the graph lies on the ray and not the semicircle. There is no ambiguity at the breakpoint $x = -1$ because the two parts of the graph join together continuously there.



© Brian Horisk/Alamy

The wind chill index measures the sensation of coldness that we feel from the combined effect of temperature and wind speed.

A statement that is correct for all real values of x is

$$\sqrt{x^2} = |x| \quad (1)$$

PIECEWISE-DEFINED FUNCTIONS

The absolute value function $f(x) = |x|$ is an example of a function that is defined *piecewise* in the sense that the formula for f changes, depending on the value of x .

► **Example 4** Sketch the graph of the function defined piecewise by the formula

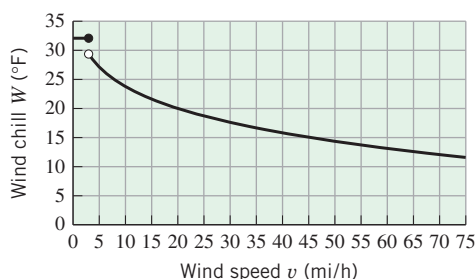
$$f(x) = \begin{cases} 0, & x \leq -1 \\ \sqrt{1 - x^2}, & -1 < x < 1 \\ x, & x \geq 1 \end{cases}$$

Solution. The formula for f changes at the points $x = -1$ and $x = 1$. (We call these the *breakpoints* for the formula.) A good procedure for graphing functions defined piecewise is to graph the function separately over the open intervals determined by the breakpoints, and then graph f at the breakpoints themselves. For the function f in this example the graph is the horizontal ray $y = 0$ on the interval $(-\infty, -1]$, it is the semicircle $y = \sqrt{1 - x^2}$ on the interval $(-1, 1)$, and it is the ray $y = x$ on the interval $[1, +\infty)$. The formula for f specifies that the equation $y = 0$ applies at the breakpoint -1 [so $y = f(-1) = 0$], and it specifies that the equation $y = x$ applies at the breakpoint 1 [so $y = f(1) = 1$]. The graph of f is shown in Figure 0.1.10. ◀

► **Example 5** Increasing the speed at which air moves over a person's skin increases the rate of moisture evaporation and makes the person feel cooler. (This is why we fan ourselves in hot weather.) The *wind chill index* is the temperature at a wind speed of 4 mi/h that would produce the same sensation on exposed skin as the current temperature and wind speed combination. An empirical formula (i.e., a formula based on experimental data) for the wind chill index W at 32°F for a wind speed of v mi/h is

$$W = \begin{cases} 32, & 0 \leq v \leq 3 \\ 55.628 - 22.07v^{0.16}, & 3 < v \end{cases}$$

A computer-generated graph of $W(v)$ is shown in Figure 0.1.11. ◀



► **Figure 0.1.11** Wind chill versus wind speed at 32°F

■ DOMAIN AND RANGE

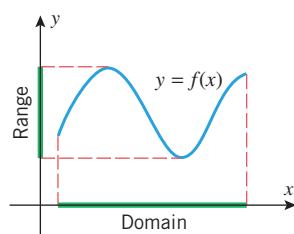
If x and y are related by the equation $y = f(x)$, then the set of all allowable inputs (x -values) is called the **domain** of f , and the set of outputs (y -values) that result when x varies over the domain is called the **range** of f . For example, if f is the function defined by the table in Example 1, then the domain is the set $\{0, 1, 2, 3\}$ and the range is the set $\{-1, 3, 4, 6\}$.

Sometimes physical or geometric considerations impose restrictions on the allowable inputs of a function. For example, if y denotes the area of a square of side x , then these variables are related by the equation $y = x^2$. Although this equation produces a unique value of y for every real number x , the fact that lengths must be nonnegative imposes the requirement that $x \geq 0$.

When a function is defined by a mathematical formula, the formula itself may impose restrictions on the allowable inputs. For example, if $y = 1/x$, then $x = 0$ is not an allowable input since division by zero is undefined, and if $y = \sqrt{x}$, then negative values of x are not allowable inputs because they produce imaginary values for y and we have agreed to consider only real-valued functions of a real variable. In general, we make the following definition.

One might argue that a physical square cannot have a side of length zero. However, it is often convenient mathematically to allow zero lengths, and we will do so throughout this text where appropriate.

0.1.5 DEFINITION If a real-valued function of a real variable is defined by a formula, and if no domain is stated explicitly, then it is to be understood that the domain consists of all real numbers for which the formula yields a real value. This is called the **natural domain** of the function.



▲ **Figure 0.1.12** The projection of $y = f(x)$ on the x -axis is the set of allowable x -values for f , and the projection on the y -axis is the set of corresponding y -values.

The domain and range of a function f can be pictured by projecting the graph of $y = f(x)$ onto the coordinate axes as shown in Figure 0.1.12.

► **Example 6** Find the natural domain of

- (a) $f(x) = x^3$ (b) $f(x) = 1/[(x-1)(x-3)]$
 (c) $f(x) = \tan x$ (d) $f(x) = \sqrt{x^2 - 5x + 6}$

Solution (a). The function f has real values for all real x , so its natural domain is the interval $(-\infty, +\infty)$.

Solution (b). The function f has real values for all real x , except $x = 1$ and $x = 3$, where divisions by zero occur. Thus, the natural domain is

$$\{x : x \neq 1 \text{ and } x \neq 3\} = (-\infty, 1) \cup (1, 3) \cup (3, +\infty)$$

Solution (c). Since $f(x) = \tan x = \sin x / \cos x$, the function f has real values except where $\cos x = 0$, and this occurs when x is an odd integer multiple of $\pi/2$. Thus, the natural domain consists of all real numbers except

$$x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots$$

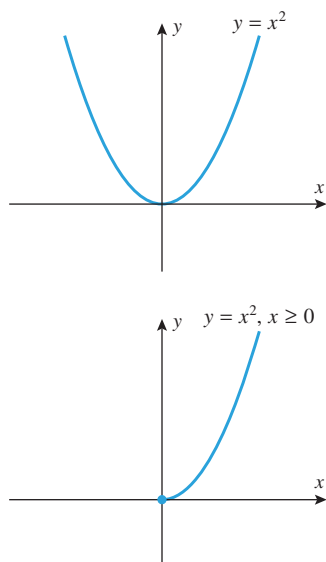
Solution (d). The function f has real values, except when the expression inside the radical is negative. Thus the natural domain consists of all real numbers x such that

$$x^2 - 5x + 6 = (x-3)(x-2) \geq 0$$

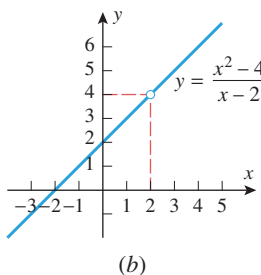
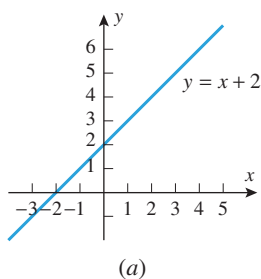
This inequality is satisfied if $x \leq 2$ or $x \geq 3$ (verify), so the natural domain of f is

$$(-\infty, 2] \cup [3, +\infty) \quad \blacktriangleleft$$

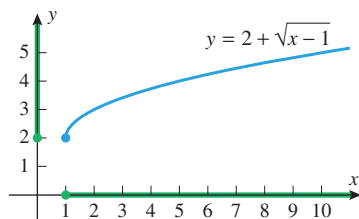
For a review of trigonometry see Appendix B.



▲ Figure 0.1.13



▲ Figure 0.1.14



▲ Figure 0.1.15

In some cases we will state the domain explicitly when defining a function. For example, if $f(x) = x^2$ is the area of a square of side x , then we can write

$$f(x) = x^2, \quad x \geq 0$$

to indicate that we take the domain of f to be the set of nonnegative real numbers (Figure 0.1.13).

THE EFFECT OF ALGEBRAIC OPERATIONS ON THE DOMAIN

Algebraic expressions are frequently simplified by canceling common factors in the numerator and denominator. However, care must be exercised when simplifying formulas for functions in this way, since this process can alter the domain.

► **Example 7** The natural domain of the function

$$f(x) = \frac{x^2 - 4}{x - 2} \quad (2)$$

consists of all real x except $x = 2$. However, if we factor the numerator and then cancel the common factor in the numerator and denominator, we obtain

$$f(x) = \frac{(x - 2)(x + 2)}{x - 2} = x + 2 \quad (3)$$

Since the right side of (3) has a value of $f(2) = 4$ and $f(2)$ was undefined in (2), the algebraic simplification has changed the function. Geometrically, the graph of (3) is the line in Figure 0.1.14a, whereas the graph of (2) is the same line but with a hole at $x = 2$, since the function is undefined there (Figure 0.1.14b). In short, the geometric effect of the algebraic cancellation is to eliminate the hole in the original graph. ◀

Sometimes alterations to the domain of a function that result from algebraic simplification are irrelevant to the problem at hand and can be ignored. However, if the domain must be preserved, then one must impose the restrictions on the simplified function explicitly. For example, if we wanted to preserve the domain of the function in Example 7, then we would have to express the simplified form of the function as

$$f(x) = x + 2, \quad x \neq 2$$

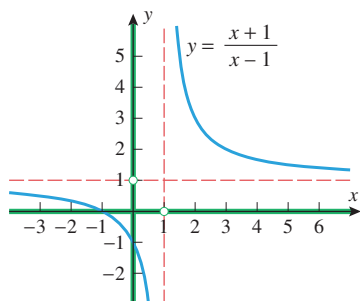
► **Example 8** Find the domain and range of

$$(a) f(x) = 2 + \sqrt{x - 1} \quad (b) f(x) = (x + 1)/(x - 1)$$

Solution (a). Since no domain is stated explicitly, the domain of f is its natural domain, $[1, +\infty)$. As x varies over the interval $[1, +\infty)$, the value of $\sqrt{x - 1}$ varies over the interval $[0, +\infty)$, so the value of $f(x) = 2 + \sqrt{x - 1}$ varies over the interval $[2, +\infty)$, which is the range of f . The domain and range are highlighted in green on the x - and y -axes in Figure 0.1.15.

Solution (b). The given function f is defined for all real x , except $x = 1$, so the natural domain of f is

$$\{x : x \neq 1\} = (-\infty, 1) \cup (1, +\infty)$$



▲ Figure 0.1.16

To determine the range it will be convenient to introduce a dependent variable

$$y = \frac{x+1}{x-1} \quad (4)$$

Although the set of possible y -values is not immediately evident from this equation, the graph of (4), which is shown in Figure 0.1.16, suggests that the range of f consists of all y , except $y = 1$. To see that this is so, we solve (4) for x in terms of y :

$$(x-1)y = x+1$$

$$xy - y = x + 1$$

$$xy - x = y + 1$$

$$x(y-1) = y+1$$

$$x = \frac{y+1}{y-1}$$

It is now evident from the right side of this equation that $y = 1$ is not in the range; otherwise we would have a division by zero. No other values of y are excluded by this equation, so the range of the function f is $\{y : y \neq 1\} = (-\infty, 1) \cup (1, +\infty)$, which agrees with the result obtained graphically. ◀

■ DOMAIN AND RANGE IN APPLIED PROBLEMS

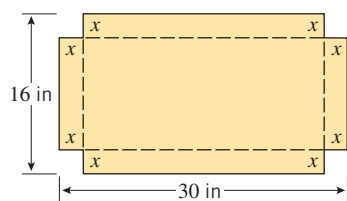
In applications, physical considerations often impose restrictions on the domain and range of a function.

► **Example 9** An open box is to be made from a 16-inch by 30-inch piece of cardboard by cutting out squares of equal size from the four corners and bending up the sides (Figure 0.1.17a).

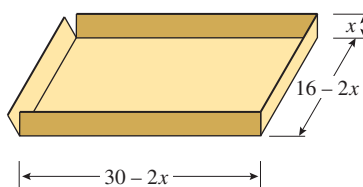
- Let V be the volume of the box that results when the squares have sides of length x . Find a formula for V as a function of x .
- Find the domain of V .
- Use the graph of V given in Figure 0.1.17c to estimate the range of V .
- Describe in words what the graph tells you about the volume.

Solution (a). As shown in Figure 0.1.17b, the resulting box has dimensions $16 - 2x$ by $30 - 2x$ by x , so the volume $V(x)$ is given by

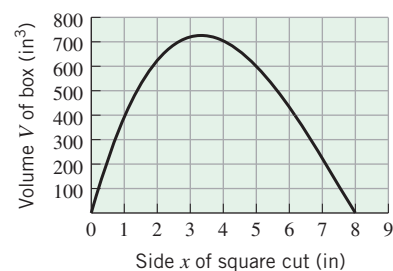
$$V(x) = (16 - 2x)(30 - 2x)x = 480x - 92x^2 + 4x^3$$



(a)



(b)



(c)

▲ Figure 0.1.17

Solution (b). The domain is the set of x -values and the range is the set of V -values. Because x is a length, it must be nonnegative, and because we cannot cut out squares whose sides are more than 8 in long (why?), the x -values in the domain must satisfy

$$0 \leq x \leq 8$$

Solution (c). From the graph of V versus x in Figure 0.1.17c we estimate that the V -values in the range satisfy

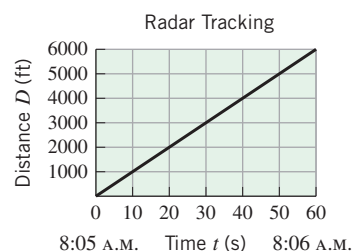
$$0 \leq V \leq 725$$

Note that this is an approximation. Later we will show how to find the range exactly.

Solution (d). The graph tells us that the box of maximum volume occurs for a value of x that is between 3 and 4 and that the maximum volume is approximately 725 in^3 . The graph also shows that the volume decreases toward zero as x gets closer to 0 or 8, which should make sense to you intuitively. ◀

In applications involving time, formulas for functions are often expressed in terms of a variable t whose starting value is taken to be $t = 0$.

► **Example 10** At 8:05 A.M. a car is clocked at 100 ft/s by a radar detector that is positioned at the edge of a straight highway. Assuming that the car maintains a constant speed between 8:05 A.M. and 8:06 A.M., find a function $D(t)$ that expresses the distance traveled by the car during that time interval as a function of the time t .



▲ Figure 0.1.18

Solution. It would be clumsy to use the actual clock time for the variable t , so let us agree to use the *elapsed* time in seconds, starting with $t = 0$ at 8:05 A.M. and ending with $t = 60$ at 8:06 A.M. At each instant, the distance traveled (in ft) is equal to the speed of the car (in ft/s) multiplied by the elapsed time (in s). Thus,

$$D(t) = 100t, \quad 0 \leq t \leq 60$$

The graph of D versus t is shown in Figure 0.1.18. ◀

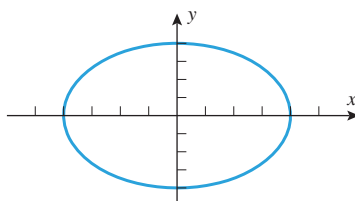
ISSUES OF SCALE AND UNITS

In geometric problems where you want to preserve the “true” shape of a graph, you must use units of equal length on both axes. For example, if you graph a circle in a coordinate system in which 1 unit in the y -direction is smaller than 1 unit in the x -direction, then the circle will be squashed vertically into an elliptical shape (Figure 0.1.19).

However, sometimes it is inconvenient or impossible to display a graph using units of equal length. For example, consider the equation

$$y = x^2$$

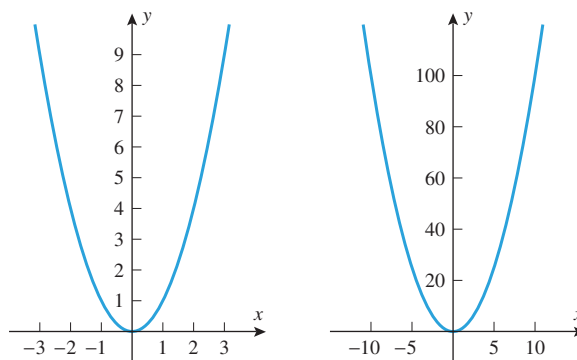
If we want to show the portion of the graph over the interval $-3 \leq x \leq 3$, then there is no problem using units of equal length, since y only varies from 0 to 9 over that interval. However, if we want to show the portion of the graph over the interval $-10 \leq x \leq 10$, then there is a problem keeping the units equal in length, since the value of y varies between 0 and 100. In this case the only reasonable way to show all of the graph that occurs over the interval $-10 \leq x \leq 10$ is to compress the unit of length along the y -axis, as illustrated in Figure 0.1.20.



The circle is squashed because 1 unit on the y -axis has a smaller length than 1 unit on the x -axis.

▲ Figure 0.1.19

In applications where the variables on the two axes have unrelated units (say, centimeters on the y -axis and seconds on the x -axis), then nothing is gained by requiring the units to have equal lengths; choose the lengths to make the graph as clear as possible.



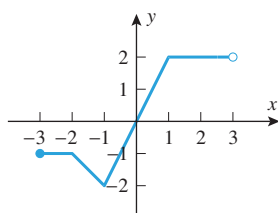
► Figure 0.1.20

✓ QUICK CHECK EXERCISES 0.1 (See page 15 for answers.)

- Let $f(x) = \sqrt{x+1} + 4$.
 - The natural domain of f is _____.
 - $f(3) =$ _____
 - $f(t^2 - 1) =$ _____
 - $f(x) = 7$ if $x =$ _____
 - The range of f is _____.
- Line segments in an xy -plane form “letters” as depicted.



- If the y -axis is parallel to the letter I, which of the letters represent the graph of $y = f(x)$ for some function f ?
 - If the y -axis is perpendicular to the letter I, which of the letters represent the graph of $y = f(x)$ for some function f ?
- The accompanying figure shows the complete graph of $y = f(x)$.
 - The domain of f is _____.
 - The range of f is _____.
 - $f(-3) =$ _____
 - $f\left(\frac{1}{2}\right) =$ _____
 - The solutions to $f(x) = -\frac{3}{2}$ are $x =$ _____ and $x =$ _____.



◀ Figure Ex-3

- The accompanying table gives a 5-day forecast of high and low temperatures in degrees Fahrenheit ($^{\circ}\text{F}$).
 - Suppose that x and y denote the respective high and low temperature predictions for each of the 5 days. Is y a function of x ? If so, give the domain and range of this function.
 - Suppose that x and y denote the respective low and high temperature predictions for each of the 5 days. Is y a function of x ? If so, give the domain and range of this function.

	MON	TUE	WED	THURS	FRI
HIGH	75	71	65	70	73
LOW	52	56	48	50	52

▲ Table Ex-4

- Let l , w , and A denote the length, width, and area of a rectangle, respectively, and suppose that the width of the rectangle is half the length.
 - If l is expressed as a function of w , then $l =$ _____.
 - If A is expressed as a function of l , then $A =$ _____.
 - If w is expressed as a function of A , then $w =$ _____.

EXERCISE SET 0.1



Graphing Utility

- Use the accompanying graph to answer the following questions, making reasonable approximations where needed.
 - For what values of x is $y = 1$?
 - For what values of x is $y = 3$?
 - For what values of y is $x = 3$?
 - For what values of x is $y \leq 0$?
 - What are the maximum and minimum values of y and for what values of x do they occur?

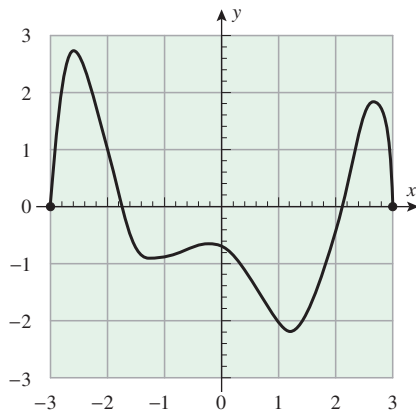


Figure Ex-1

- Use the accompanying table to answer the questions posed in Exercise 1.

x	-2	-1	0	2	3	4	5	6
y	5	1	-2	7	-1	1	0	9

Table Ex-2

- In each part of the accompanying figure, determine whether the graph defines y as a function of x .

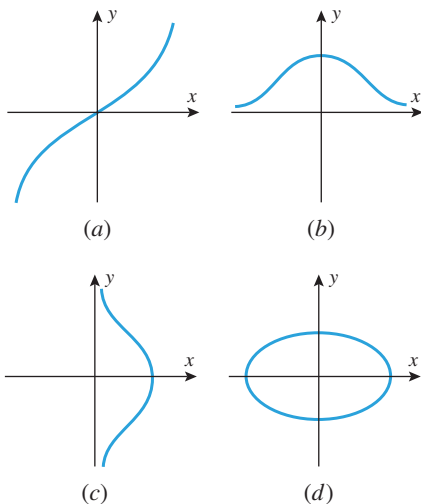


Figure Ex-3

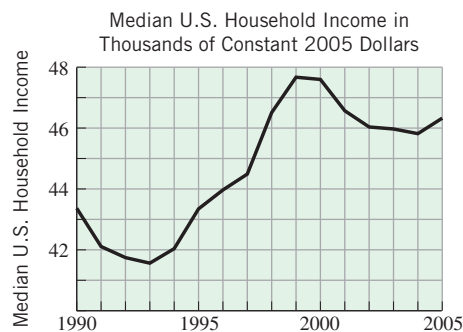
- In each part, compare the natural domains of f and g .

(a) $f(x) = \frac{x^2 + x}{x + 1}$; $g(x) = x$

(b) $f(x) = \frac{x\sqrt{x} + \sqrt{x}}{x + 1}$; $g(x) = \sqrt{x}$

FOCUS ON CONCEPTS

- The accompanying graph shows the median income in U.S. households (adjusted for inflation) between 1990 and 2005. Use the graph to answer the following questions, making reasonable approximations where needed.
 - When was the median income at its maximum value, and what was the median income when that occurred?
 - When was the median income at its minimum value, and what was the median income when that occurred?
 - The median income was declining during the 2-year period between 2000 and 2002. Was it declining more rapidly during the first year or the second year of that period? Explain your reasoning.



Source: U.S. Census Bureau, August 2006.

Figure Ex-5

- Use the median income graph in Exercise 5 to answer the following questions, making reasonable approximations where needed.
 - What was the average yearly growth of median income between 1993 and 1999?
 - The median income was increasing during the 6-year period between 1993 and 1999. Was it increasing more rapidly during the first 3 years or the last 3 years of that period? Explain your reasoning.
 - Consider the statement: "After years of decline, median income this year was finally higher than that of last year." In what years would this statement have been correct?

7. Find $f(0)$, $f(2)$, $f(-2)$, $f(3)$, $f(\sqrt{2})$, and $f(3t)$.

$$(a) f(x) = 3x^2 - 2 \quad (b) f(x) = \begin{cases} \frac{1}{x}, & x > 3 \\ 2x, & x \leq 3 \end{cases}$$

8. Find $g(3)$, $g(-1)$, $g(\pi)$, $g(-1.1)$, and $g(t^2 - 1)$.

$$(a) g(x) = \frac{x+1}{x-1} \quad (b) g(x) = \begin{cases} \sqrt{x+1}, & x \geq 1 \\ 3, & x < 1 \end{cases}$$

 **9–10** Find the natural domain and determine the range of each function. If you have a graphing utility, use it to confirm that your result is consistent with the graph produced by your graphing utility. [Note: Set your graphing utility in radian mode when graphing trigonometric functions.] ■

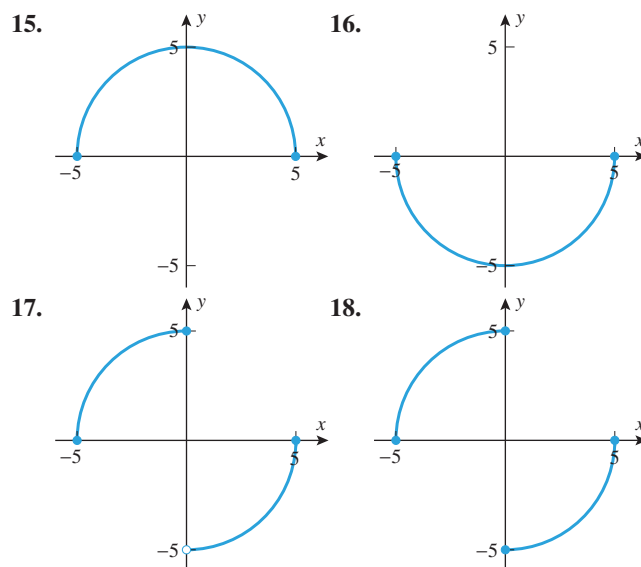
9. (a) $f(x) = \frac{1}{x-3}$ (b) $F(x) = \frac{x}{|x|}$
 (c) $g(x) = \sqrt{x^2 - 3}$ (d) $G(x) = \sqrt{x^2 - 2x + 5}$
 (e) $h(x) = \frac{1}{1 - \sin x}$ (f) $H(x) = \sqrt{\frac{x^2 - 4}{x - 2}}$
10. (a) $f(x) = \sqrt{3 - x}$ (b) $F(x) = \sqrt{4 - x^2}$
 (c) $g(x) = 3 + \sqrt{x}$ (d) $G(x) = x^3 + 2$
 (e) $h(x) = 3 \sin x$ (f) $H(x) = (\sin \sqrt{x})^{-2}$

FOCUS ON CONCEPTS

11. (a) If you had a device that could record the Earth's population continuously, would you expect the graph of population versus time to be a continuous (unbroken) curve? Explain what might cause breaks in the curve.
 (b) Suppose that a hospital patient receives an injection of an antibiotic every 8 hours and that between injections the concentration C of the antibiotic in the bloodstream decreases as the antibiotic is absorbed by the tissues. What might the graph of C versus the elapsed time t look like?
12. (a) If you had a device that could record the temperature of a room continuously over a 24-hour period, would you expect the graph of temperature versus time to be a continuous (unbroken) curve? Explain your reasoning.
 (b) If you had a computer that could track the number of boxes of cereal on the shelf of a market continuously over a 1-week period, would you expect the graph of the number of boxes on the shelf versus time to be a continuous (unbroken) curve? Explain your reasoning.
13. A boat is bobbing up and down on some gentle waves. Suddenly it gets hit by a large wave and sinks. Sketch a rough graph of the height of the boat above the ocean floor as a function of time.

14. A cup of hot coffee sits on a table. You pour in some cool milk and let it sit for an hour. Sketch a rough graph of the temperature of the coffee as a function of time.

15–18 As seen in Example 3, the equation $x^2 + y^2 = 25$ does not define y as a function of x . Each graph in these exercises is a portion of the circle $x^2 + y^2 = 25$. In each case, determine whether the graph defines y as a function of x , and if so, give a formula for y in terms of x . ■



19–22 True-False Determine whether the statement is true or false. Explain your answer. ■

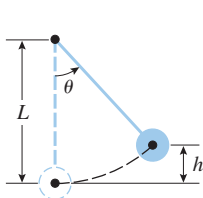
19. A curve that crosses the x -axis at two different points cannot be the graph of a function.
20. The natural domain of a real-valued function defined by a formula consists of all those real numbers for which the formula yields a real value.
21. The range of the absolute value function is all positive real numbers.
22. If $g(x) = 1/\sqrt{f(x)}$, then the domain of g consists of all those real numbers x for which $f(x) \neq 0$.
23. Use the equation $y = x^2 - 6x + 8$ to answer the following questions.
 (a) For what values of x is $y = 0$?
 (b) For what values of x is $y = -10$?
 (c) For what values of x is $y \geq 0$?
 (d) Does y have a minimum value? A maximum value? If so, find them.
24. Use the equation $y = 1 + \sqrt{x}$ to answer the following questions.
 (a) For what values of x is $y = 4$?
 (b) For what values of x is $y = 0$?
 (c) For what values of x is $y \geq 6$?

(cont.)

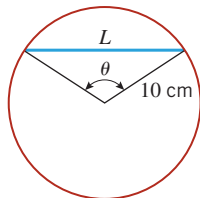
(d) Does y have a minimum value? A maximum value? If so, find them.

25. As shown in the accompanying figure, a pendulum of constant length L makes an angle θ with its vertical position. Express the height h as a function of the angle θ .

26. Express the length L of a chord of a circle with radius 10 cm as a function of the central angle θ (see the accompanying figure).



▲ Figure Ex-25



▲ Figure Ex-26

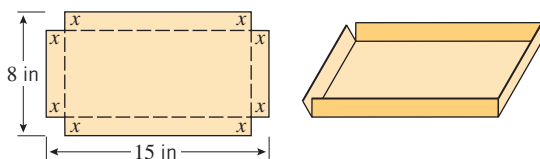
- 27–28 Express the function in piecewise form without using absolute values. [Suggestion: It may help to generate the graph of the function.] ■

27. (a) $f(x) = |x| + 3x + 1$ (b) $g(x) = |x| + |x - 1|$

28. (a) $f(x) = 3 + |2x - 5|$ (b) $g(x) = 3|x - 2| - |x + 1|$

29. As shown in the accompanying figure, an open box is to be constructed from a rectangular sheet of metal, 8 in by 15 in, by cutting out squares with sides of length x from each corner and bending up the sides.

- (a) Express the volume V as a function of x .
 (b) Find the domain of V .
 (c) Plot the graph of the function V obtained in part (a) and estimate the range of this function.
 (d) In words, describe how the volume V varies with x , and discuss how one might construct boxes of maximum volume.



▲ Figure Ex-29

30. Repeat Exercise 29 assuming the box is constructed in the same fashion from a 6-inch-square sheet of metal.

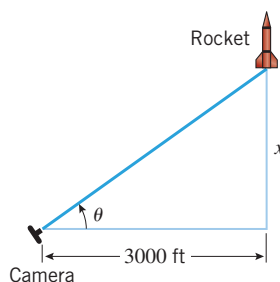
31. A construction company has adjoined a 1000 ft² rectangular enclosure to its office building. Three sides of the enclosure are fenced in. The side of the building adjacent to the enclosure is 100 ft long and a portion of this side is used as the fourth side of the enclosure. Let x and y be the dimensions of the enclosure, where x is measured parallel to the building, and let L be the length of fencing required for those dimensions.

- (a) Find a formula for L in terms of x and y .
 (b) Find a formula that expresses L as a function of x alone.
 (c) What is the domain of the function in part (b)?

- (d) Plot the function in part (b) and estimate the dimensions of the enclosure that minimize the amount of fencing required.

32. As shown in the accompanying figure, a camera is mounted at a point 3000 ft from the base of a rocket launching pad. The rocket rises vertically when launched, and the camera's elevation angle is continually adjusted to follow the bottom of the rocket.

- (a) Express the height x as a function of the elevation angle θ .
 (b) Find the domain of the function in part (a).
 (c) Plot the graph of the function in part (a) and use it to estimate the height of the rocket when the elevation angle is $\pi/4 \approx 0.7854$ radian. Compare this estimate to the exact height.



◀ Figure Ex-32

33. A soup company wants to manufacture a can in the shape of a right circular cylinder that will hold 500 cm³ of liquid. The material for the top and bottom costs 0.02 cent/cm², and the material for the sides costs 0.01 cent/cm².

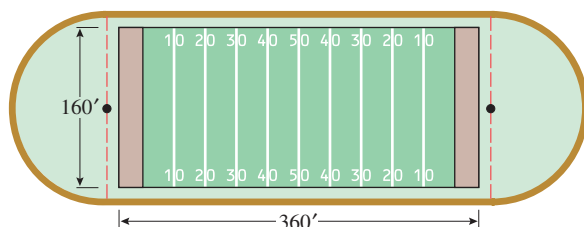
- (a) Estimate the radius r and the height h of the can that costs the least to manufacture. [Suggestion: Express the cost C in terms of r .]
 (b) Suppose that the tops and bottoms of radius r are punched out from square sheets with sides of length $2r$ and the scraps are waste. If you allow for the cost of the waste, would you expect the can of least cost to be taller or shorter than the one in part (a)? Explain.
 (c) Estimate the radius, height, and cost of the can in part (b), and determine whether your conjecture was correct.

34. The designer of a sports facility wants to put a quarter-mile (1320 ft) running track around a football field, oriented as in the accompanying figure on the next page. The football field is 360 ft long (including the end zones) and 160 ft wide. The track consists of two straightaways and two semicircles, with the straightaways extending at least the length of the football field.

- (a) Show that it is possible to construct a quarter-mile track around the football field. [Suggestion: Find the shortest track that can be constructed around the field.]
 (b) Let L be the length of a straightaway (in feet), and let x be the distance (in feet) between a sideline of the football field and a straightaway. Make a graph of L versus x .

(cont.)

- (c) Use the graph to estimate the value of x that produces the shortest straightaways, and then find this value of x exactly.
- (d) Use the graph to estimate the length of the longest possible straightaways, and then find that length exactly.



▲ Figure Ex-34

- 35–36** (i) Explain why the function f has one or more holes in its graph, and state the x -values at which those holes occur. (ii) Find a function g whose graph is identical to that of f , but without the holes. ■

35. $f(x) = \frac{(x+2)(x^2-1)}{(x+2)(x-1)}$ 36. $f(x) = \frac{x^2+|x|}{|x|}$

37. In 2001 the National Weather Service introduced a new wind chill temperature (WCT) index. For a given outside temper-

ature T and wind speed v , the wind chill temperature index is the equivalent temperature that exposed skin would feel with a wind speed of v mi/h. Based on a more accurate model of cooling due to wind, the new formula is

$$\text{WCT} = \begin{cases} T, & 0 \leq v \leq 3 \\ 35.74 + 0.6215T - 35.75v^{0.16} + 0.4275Tv^{0.16}, & 3 < v \end{cases}$$

where T is the temperature in $^{\circ}\text{F}$, v is the wind speed in mi/h, and WCT is the equivalent temperature in $^{\circ}\text{F}$. Find the WCT to the nearest degree if $T = 25^{\circ}\text{F}$ and

- (a) $v = 3$ mi/h (b) $v = 15$ mi/h (c) $v = 46$ mi/h.

Source: Adapted from UMAP Module 658, *Windchill*, W. Bosch and L. Cobb, COMAP, Arlington, MA.

38–40 Use the formula for the wind chill temperature index described in Exercise 37. ■

38. Find the air temperature to the nearest degree if the WCT is reported as -60°F with a wind speed of 48 mi/h.
39. Find the air temperature to the nearest degree if the WCT is reported as -10°F with a wind speed of 48 mi/h.
40. Find the wind speed to the nearest mile per hour if the WCT is reported as 5°F with an air temperature of 20°F .

✓ QUICK CHECK ANSWERS 0.1

1. (a) $[-1, +\infty)$ (b) 6 (c) $|t| + 4$ (d) 8 (e) $[4, +\infty)$ 2. (a) M (b) I 3. (a) $[-3, 3)$ (b) $[-2, 2]$ (c) -1 (d) 1 (e) $-\frac{3}{4}$; $-\frac{3}{2}$ 4. (a) yes; domain: $\{65, 70, 71, 73, 75\}$; range: $\{48, 50, 52, 56\}$ (b) no 5. (a) $l = 2w$ (b) $A = l^2/2$ (c) $w = \sqrt{A/2}$

0.2 NEW FUNCTIONS FROM OLD

Just as numbers can be added, subtracted, multiplied, and divided to produce other numbers, so functions can be added, subtracted, multiplied, and divided to produce other functions. In this section we will discuss these operations and some others that have no analogs in ordinary arithmetic.

■ ARITHMETIC OPERATIONS ON FUNCTIONS

Two functions, f and g , can be added, subtracted, multiplied, and divided in a natural way to form new functions $f + g$, $f - g$, fg , and f/g . For example, $f + g$ is defined by the formula

$$(f + g)(x) = f(x) + g(x) \quad (1)$$

which states that for each input the value of $f + g$ is obtained by adding the values of f and g . Equation (1) provides a formula for $f + g$ but does not say anything about the domain of $f + g$. However, for the right side of this equation to be defined, x must lie in the domains of both f and g , so we define the domain of $f + g$ to be the intersection of these two domains. More generally, we make the following definition.

If f is a constant function, that is, $f(x) = c$ for all x , then the product of f and g is cg , so multiplying a function by a constant is a special case of multiplying two functions.

0.2.1 DEFINITION Given functions f and g , we define

$$(f + g)(x) = f(x) + g(x)$$

$$(f - g)(x) = f(x) - g(x)$$

$$(fg)(x) = f(x)g(x)$$

$$(f/g)(x) = f(x)/g(x)$$

For the functions $f + g$, $f - g$, and fg we define the domain to be the intersection of the domains of f and g , and for the function f/g we define the domain to be the intersection of the domains of f and g but with the points where $g(x) = 0$ excluded (to avoid division by zero).

► **Example 1** Let

$$f(x) = 1 + \sqrt{x - 2} \quad \text{and} \quad g(x) = x - 3$$

Find the domains and formulas for the functions $f + g$, $f - g$, fg , f/g , and $7f$.

Solution. First, we will find the formulas and then the domains. The formulas are

$$(f + g)(x) = f(x) + g(x) = (1 + \sqrt{x - 2}) + (x - 3) = x - 2 + \sqrt{x - 2} \quad (2)$$

$$(f - g)(x) = f(x) - g(x) = (1 + \sqrt{x - 2}) - (x - 3) = 4 - x + \sqrt{x - 2} \quad (3)$$

$$(fg)(x) = f(x)g(x) = (1 + \sqrt{x - 2})(x - 3) \quad (4)$$

$$(f/g)(x) = f(x)/g(x) = \frac{1 + \sqrt{x - 2}}{x - 3} \quad (5)$$

$$(7f)(x) = 7f(x) = 7 + 7\sqrt{x - 2} \quad (6)$$

The domains of f and g are $[2, +\infty)$ and $(-\infty, +\infty)$, respectively (their natural domains). Thus, it follows from Definition 0.2.1 that the domains of $f + g$, $f - g$, and fg are the intersection of these two domains, namely,

$$[2, +\infty) \cap (-\infty, +\infty) = [2, +\infty) \quad (7)$$

Moreover, since $g(x) = 0$ if $x = 3$, the domain of f/g is (7) with $x = 3$ removed, namely,

$$[2, 3) \cup (3, +\infty)$$

Finally, the domain of $7f$ is the same as the domain of f . ◀

We saw in the last example that the domains of the functions $f + g$, $f - g$, fg , and f/g were the natural domains resulting from the formulas obtained for these functions. The following example shows that this will not always be the case.

► **Example 2** Show that if $f(x) = \sqrt{x}$, $g(x) = \sqrt{x}$, and $h(x) = x$, then the domain of fg is not the same as the natural domain of h .

Solution. The natural domain of $h(x) = x$ is $(-\infty, +\infty)$. Note that

$$(fg)(x) = \sqrt{x}\sqrt{x} = x = h(x)$$

on the domain of fg . The domains of both f and g are $[0, +\infty)$, so the domain of fg is

$$[0, +\infty) \cap [0, +\infty) = [0, +\infty)$$

by Definition 0.2.1. Since the domains of fg and h are different, it would be misleading to write $(fg)(x) = x$ without including the restriction that this formula holds only for $x \geq 0$. ◀

COMPOSITION OF FUNCTIONS

We now consider an operation on functions, called *composition*, which has no direct analog in ordinary arithmetic. Informally stated, the operation of composition is performed by substituting some function for the independent variable of another function. For example, suppose that

$$f(x) = x^2 \quad \text{and} \quad g(x) = x + 1$$

If we substitute $g(x)$ for x in the formula for f , we obtain a new function

$$f(g(x)) = (g(x))^2 = (x + 1)^2$$

which we denote by $f \circ g$. Thus,

$$(f \circ g)(x) = f(g(x)) = (g(x))^2 = (x + 1)^2$$

In general, we make the following definition.

Although the domain of $f \circ g$ may seem complicated at first glance, it makes sense intuitively: To compute $f(g(x))$ one needs x in the domain of g to compute $g(x)$, and one needs $g(x)$ in the domain of f to compute $f(g(x))$.

0.2.2 DEFINITION Given functions f and g , the *composition* of f with g , denoted by $f \circ g$, is the function defined by

$$(f \circ g)(x) = f(g(x))$$

The domain of $f \circ g$ is defined to consist of all x in the domain of g for which $g(x)$ is in the domain of f .

► **Example 3** Let $f(x) = x^2 + 3$ and $g(x) = \sqrt{x}$. Find

$$(a) (f \circ g)(x) \quad (b) (g \circ f)(x)$$

Solution (a). The formula for $f(g(x))$ is

$$f(g(x)) = [g(x)]^2 + 3 = (\sqrt{x})^2 + 3 = x + 3$$

Since the domain of g is $[0, +\infty)$ and the domain of f is $(-\infty, +\infty)$, the domain of $f \circ g$ consists of all x in $[0, +\infty)$ such that $g(x) = \sqrt{x}$ lies in $(-\infty, +\infty)$; thus, the domain of $f \circ g$ is $[0, +\infty)$. Therefore,

$$(f \circ g)(x) = x + 3, \quad x \geq 0$$

Solution (b). The formula for $g(f(x))$ is

$$g(f(x)) = \sqrt{f(x)} = \sqrt{x^2 + 3}$$

Since the domain of f is $(-\infty, +\infty)$ and the domain of g is $[0, +\infty)$, the domain of $g \circ f$ consists of all x in $(-\infty, +\infty)$ such that $f(x) = x^2 + 3$ lies in $[0, +\infty)$. Thus, the domain of $g \circ f$ is $(-\infty, +\infty)$. Therefore,

$$(g \circ f)(x) = \sqrt{x^2 + 3}$$

There is no need to indicate that the domain is $(-\infty, +\infty)$, since this is the natural domain of $\sqrt{x^2 + 3}$. ◀

Note that the functions $f \circ g$ and $g \circ f$ in Example 3 are not the same. Thus, the order in which functions are composed can (and usually will) make a difference in the end result.

Compositions can also be defined for three or more functions; for example, $(f \circ g \circ h)(x)$ is computed as

$$(f \circ g \circ h)(x) = f(g(h(x)))$$

In other words, first find $h(x)$, then find $g(h(x))$, and then find $f(g(h(x)))$.

► **Example 4** Find $(f \circ g \circ h)(x)$ if

$$f(x) = \sqrt{x}, \quad g(x) = 1/x, \quad h(x) = x^3$$

Solution.

$$(f \circ g \circ h)(x) = f(g(h(x))) = f(g(x^3)) = f(1/x^3) = \sqrt{1/x^3} = 1/x^{3/2} \quad \blacktriangleleft$$

■ EXPRESSING A FUNCTION AS A COMPOSITION

Many problems in mathematics are solved by “decomposing” functions into compositions of simpler functions. For example, consider the function h given by

$$h(x) = (x + 1)^2$$

To evaluate $h(x)$ for a given value of x , we would first compute $x + 1$ and then square the result. These two operations are performed by the functions

$$g(x) = x + 1 \quad \text{and} \quad f(x) = x^2$$

We can express h in terms of f and g by writing

$$h(x) = (x + 1)^2 = [g(x)]^2 = f(g(x))$$

so we have succeeded in expressing h as the composition $h = f \circ g$.

The thought process in this example suggests a general procedure for decomposing a function h into a composition $h = f \circ g$:

- Think about how you would evaluate $h(x)$ for a specific value of x , trying to break the evaluation into two steps performed in succession.
- The first operation in the evaluation will determine a function g and the second a function f .
- The formula for h can then be written as $h(x) = f(g(x))$.

For descriptive purposes, we will refer to g as the “inside function” and f as the “outside function” in the expression $f(g(x))$. The inside function performs the first operation and the outside function performs the second.

► **Example 5** Express $\sin(x^3)$ as a composition of two functions.

Solution. To evaluate $\sin(x^3)$, we would first compute x^3 and then take the sine, so $g(x) = x^3$ is the inside function and $f(x) = \sin x$ the outside function. Therefore,

$$\sin(x^3) = f(g(x)) \quad \boxed{g(x) = x^3 \text{ and } f(x) = \sin x} \quad \blacktriangleleft$$

Table 0.2.1 gives some more examples of decomposing functions into compositions.

Table 0.2.1
COMPOSING FUNCTIONS

FUNCTION	$g(x)$ INSIDE	$f(x)$ OUTSIDE	COMPOSITION
$(x^2 + 1)^{10}$	$x^2 + 1$	x^{10}	$(x^2 + 1)^{10} = f(g(x))$
$\sin^3 x$	$\sin x$	x^3	$\sin^3 x = f(g(x))$
$\tan(x^5)$	x^5	$\tan x$	$\tan(x^5) = f(g(x))$
$\sqrt{4 - 3x}$	$4 - 3x$	$\sqrt{x}$	$\sqrt{4 - 3x} = f(g(x))$
$8 + \sqrt{x}$	$\sqrt{x}$	$8 + x$	$8 + \sqrt{x} = f(g(x))$
$\frac{1}{x+1}$	$x+1$	$\frac{1}{x}$	$\frac{1}{x+1} = f(g(x))$

REMARK There is always more than one way to express a function as a composition. For example, here are two ways to express $(x^2 + 1)^{10}$ as a composition that differ from that in Table 0.2.1:

$$(x^2 + 1)^{10} = [(x^2 + 1)^2]^5 = f(g(x))$$

$$g(x) = (x^2 + 1)^2 \text{ and } f(x) = x^5$$

$$(x^2 + 1)^{10} = [(x^2 + 1)^3]^{10/3} = f(g(x))$$

$$g(x) = (x^2 + 1)^3 \text{ and } f(x) = x^{10/3}$$

NEW FUNCTIONS FROM OLD

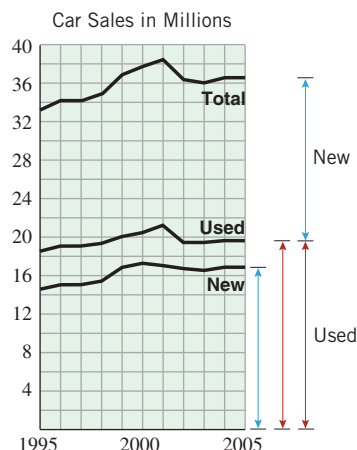
The remainder of this section will be devoted to considering the geometric effect of performing basic operations on functions. This will enable us to use known graphs of functions to visualize or sketch graphs of related functions. For example, Figure 0.2.1 shows the graphs of yearly new car sales $N(t)$ and used car sales $U(t)$ over a certain time period. Those graphs can be used to construct the graph of the total car sales

$$T(t) = N(t) + U(t)$$

by adding the values of $N(t)$ and $U(t)$ for each value of t . In general, the graph of $y = f(x) + g(x)$ can be constructed from the graphs of $y = f(x)$ and $y = g(x)$ by adding corresponding y -values for each x .

► **Example 6** Referring to Figure 0.1.4 for the graphs of $y = \sqrt{x}$ and $y = 1/x$, make a sketch that shows the general shape of the graph of $y = \sqrt{x} + 1/x$ for $x \geq 0$.

Solution. To add the corresponding y -values of $y = \sqrt{x}$ and $y = 1/x$ graphically, just imagine them to be “stacked” on top of one another. This yields the sketch in Figure 0.2.2.



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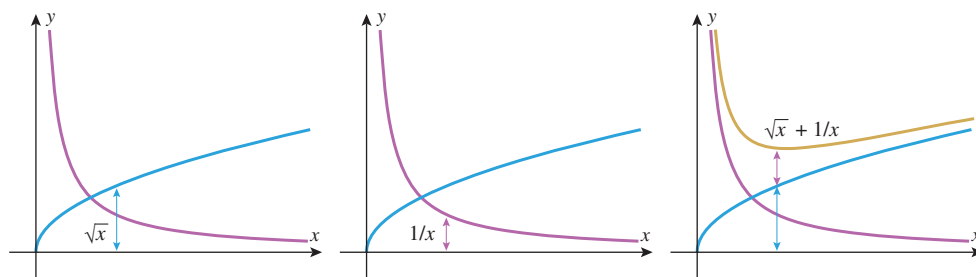
▲ Figure 0.2.1

Use the technique in Example 6 to sketch the graph of the function

$$\sqrt{x} + \frac{1}{x}$$

► Figure 0.2.2

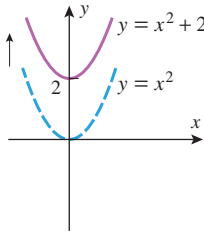
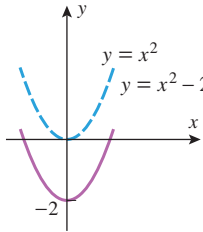
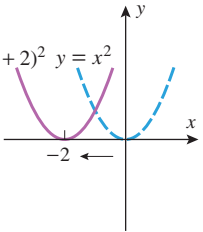
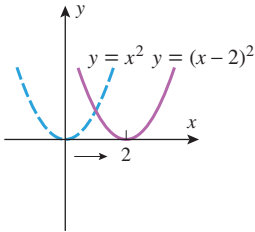
Add the y -coordinates of $\sqrt{x}$ and $1/x$ to obtain the y -coordinate of $\sqrt{x} + 1/x$.

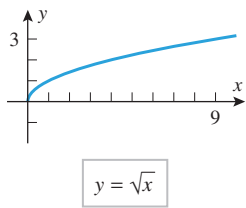


TRANSLATIONS

Table 0.2.2 illustrates the geometric effect on the graph of $y = f(x)$ of adding or subtracting a *positive* constant c to f or to its independent variable x . For example, the first result in the table illustrates that adding a positive constant c to a function f adds c to each y -coordinate of its graph, thereby shifting the graph of f up by c units. Similarly, subtracting c from f shifts the graph down by c units. On the other hand, if a positive constant c is added to x , then the value of $y = f(x + c)$ at $x - c$ is $f(x)$; and since the point $x - c$ is c units to the left of x on the x -axis, the graph of $y = f(x + c)$ must be the graph of $y = f(x)$ shifted left by c units. Similarly, subtracting c from x shifts the graph of $y = f(x)$ right by c units.

Table 0.2.2
TRANSLATION PRINCIPLES

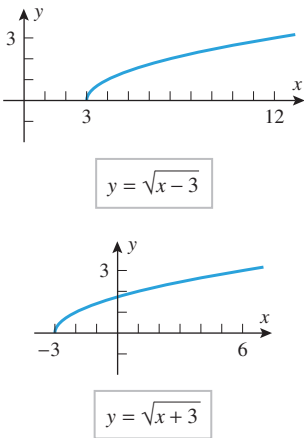
OPERATION ON $y = f(x)$	Add a positive constant c to $f(x)$	Subtract a positive constant c from $f(x)$	Add a positive constant c to x	Subtract a positive constant c from x
NEW EQUATION	$y = f(x) + c$	$y = f(x) - c$	$y = f(x + c)$	$y = f(x - c)$
GEOMETRIC EFFECT	Translates the graph of $y = f(x)$ up c units	Translates the graph of $y = f(x)$ down c units	Translates the graph of $y = f(x)$ left c units	Translates the graph of $y = f(x)$ right c units
EXAMPLE				



Before proceeding to the next examples, it will be helpful to review the graphs in Figures 0.1.4 and 0.1.9.

► **Example 7** Sketch the graph of
(a) $y = \sqrt{x - 3}$ (b) $y = \sqrt{x + 3}$

Solution. Using the translation principles given in Table 0.2.2, the graph of the equation $y = \sqrt{x - 3}$ can be obtained by translating the graph of $y = \sqrt{x}$ right 3 units. The graph of $y = \sqrt{x + 3}$ can be obtained by translating the graph of $y = \sqrt{x}$ left 3 units (Figure 0.2.3). ◀

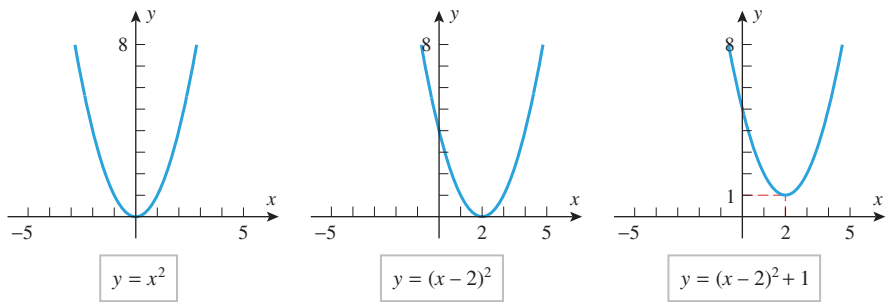


▲ Figure 0.2.3

► **Example 8** Sketch the graph of $y = x^2 - 4x + 5$.

Solution. Completing the square on the first two terms yields
$$y = (x^2 - 4x + 4) - 4 + 5 = (x - 2)^2 + 1$$

(see Web Appendix H for a review of this technique). In this form we see that the graph can be obtained by translating the graph of $y = x^2$ right 2 units because of the $x - 2$, and up 1 unit because of the $+1$ (Figure 0.2.4). ◀

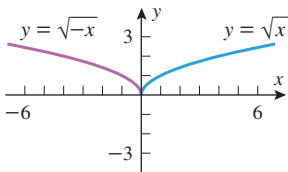
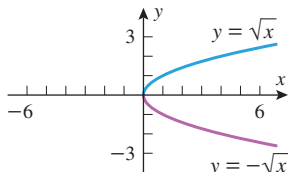


► Figure 0.2.4

REFLECTIONS

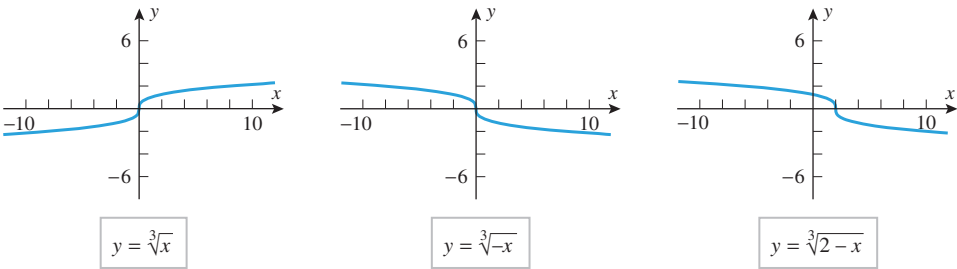
The graph of $y = f(-x)$ is the reflection of the graph of $y = f(x)$ about the y -axis because the point (x, y) on the graph of $f(x)$ is replaced by $(-x, y)$. Similarly, the graph of $y = -f(x)$ is the reflection of the graph of $y = f(x)$ about the x -axis because the point (x, y) on the graph of $f(x)$ is replaced by $(x, -y)$ [the equation $y = -f(x)$ is equivalent to $-y = f(x)$]. This is summarized in Table 0.2.3.

Table 0.2.3
REFLECTION PRINCIPLES

OPERATION ON $y = f(x)$	Replace x by $-x$	Multiply $f(x)$ by -1
NEW EQUATION	$y = f(-x)$	$y = -f(x)$
GEOMETRIC EFFECT	Reflects the graph of $y = f(x)$ about the y -axis	Reflects the graph of $y = f(x)$ about the x -axis
EXAMPLE		

► **Example 9** Sketch the graph of $y = \sqrt[3]{2-x}$.

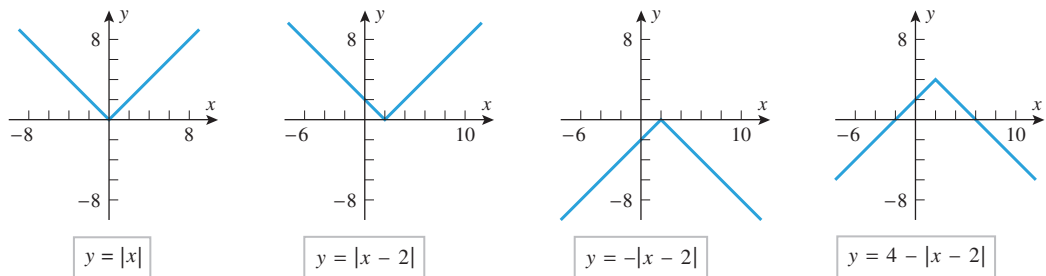
Solution. Using the translation and reflection principles in Tables 0.2.2 and 0.2.3, we can obtain the graph by a reflection followed by a translation as follows: First reflect the graph of $y = \sqrt[3]{x}$ about the y -axis to obtain the graph of $y = \sqrt[3]{-x}$, then translate this graph right 2 units to obtain the graph of the equation $y = \sqrt[3]{-(x-2)} = \sqrt[3]{2-x}$ (Figure 0.2.5).



► Figure 0.2.5

► **Example 10** Sketch the graph of $y = 4 - |x - 2|$.

Solution. The graph can be obtained by a reflection and two translations: First translate the graph of $y = |x|$ right 2 units to obtain the graph of $y = |x - 2|$; then reflect this graph about the x -axis to obtain the graph of $y = -|x - 2|$; and then translate this graph up 4 units to obtain the graph of the equation $y = -|x - 2| + 4 = 4 - |x - 2|$ (Figure 0.2.6).



▲ Figure 0.2.6

■ **STRETCHES AND COMPRESSIONS**

Describe the geometric effect of multiplying a function f by a *negative* constant in terms of reflection and stretching or compressing. What is the geometric effect of multiplying the independent variable of a function f by a *negative* constant?

Multiplying $f(x)$ by a *positive* constant c has the geometric effect of stretching the graph of $y = f(x)$ in the y -direction by a factor of c if $c > 1$ and compressing it in the y -direction by a factor of $1/c$ if $0 < c < 1$. For example, multiplying $f(x)$ by 2 doubles each y -coordinate, thereby stretching the graph vertically by a factor of 2, and multiplying by $\frac{1}{2}$ cuts each y -coordinate in half, thereby compressing the graph vertically by a factor of 2. Similarly, multiplying x by a *positive* constant c has the geometric effect of compressing the graph of $y = f(x)$ by a factor of c in the x -direction if $c > 1$ and stretching it by a factor of $1/c$ if $0 < c < 1$. [If this seems backwards to you, then think of it this way: The value of $2x$ changes twice as fast as x , so a point moving along the x -axis from the origin will only have to move half as far for $y = f(2x)$ to have the same value as $y = f(x)$, thereby creating a horizontal compression of the graph.] All of this is summarized in Table 0.2.4.

Table 0.2.4
STRETCHING AND COMPRESSION PRINCIPLES

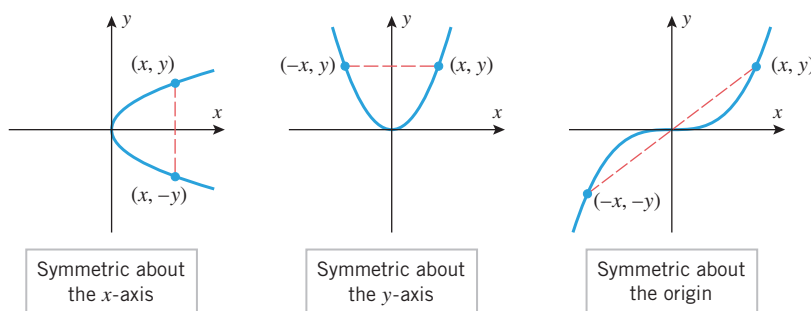
OPERATION ON $y = f(x)$	Multiply $f(x)$ by c ($c > 1$)	Multiply $f(x)$ by c ($0 < c < 1$)	Multiply x by c ($c > 1$)	Multiply x by c ($0 < c < 1$)
NEW EQUATION	$y = cf(x)$	$y = cf(x)$	$y = f(cx)$	$y = f(cx)$
GEOMETRIC EFFECT	Stretches the graph of $y = f(x)$ vertically by a factor of c	Compresses the graph of $y = f(x)$ vertically by a factor of $1/c$	Compresses the graph of $y = f(x)$ horizontally by a factor of c	Stretches the graph of $y = f(x)$ horizontally by a factor of $1/c$
EXAMPLE				

SYMMETRY

Figure 0.2.7 illustrates three types of symmetries: *symmetry about the x-axis*, *symmetry about the y-axis*, and *symmetry about the origin*. As illustrated in the figure, a curve is symmetric about the x-axis if for each point (x, y) on the graph the point $(x, -y)$ is also on the graph, and it is symmetric about the y-axis if for each point (x, y) on the graph the point $(-x, y)$ is also on the graph. A curve is symmetric about the origin if for each point (x, y) on the graph, the point $(-x, -y)$ is also on the graph. (Equivalently, a graph is symmetric about the origin if rotating the graph 180° about the origin leaves it unchanged.) This suggests the following symmetry tests.

Explain why the graph of a nonzero function cannot be symmetric about the x-axis.

► Figure 0.2.7

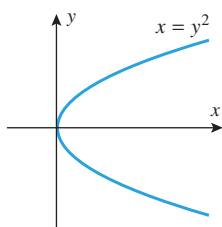


0.2.3 THEOREM (Symmetry Tests)

- (a) A plane curve is symmetric about the y-axis if and only if replacing x by $-x$ in its equation produces an equivalent equation.
- (b) A plane curve is symmetric about the x-axis if and only if replacing y by $-y$ in its equation produces an equivalent equation.
- (c) A plane curve is symmetric about the origin if and only if replacing both x by $-x$ and y by $-y$ in its equation produces an equivalent equation.

► **Example 11** Use Theorem 0.2.3 to identify symmetries in the graph of $x = y^2$.

Solution. Replacing y by $-y$ yields $x = (-y)^2$, which simplifies to the original equation $x = y^2$. Thus, the graph is symmetric about the x-axis. The graph is not symmetric about the y-axis because replacing x by $-x$ yields $-x = y^2$, which is not equivalent to the original equation $x = y^2$. Similarly, the graph is not symmetric about the origin because replacing x by $-x$ and y by $-y$ yields $-x = (-y)^2$, which simplifies to $-x = y^2$, and this is again not equivalent to the original equation. These results are consistent with the graph of $x = y^2$ shown in Figure 0.2.8. ◀



▲ Figure 0.2.8

EVEN AND ODD FUNCTIONS

A function f is said to be an *even function* if

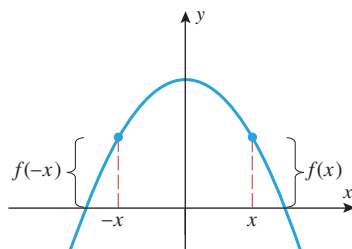
$$f(-x) = f(x) \quad (8)$$

and is said to be an *odd function* if

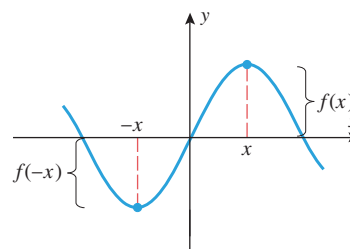
$$f(-x) = -f(x) \quad (9)$$

Geometrically, the graphs of even functions are symmetric about the y-axis because replacing x by $-x$ in the equation $y = f(x)$ yields $y = f(-x)$, which is equivalent to the original

equation $y = f(x)$ by (8) (see Figure 0.2.9). Similarly, it follows from (9) that graphs of odd functions are symmetric about the origin (see Figure 0.2.10). Some examples of even functions are x^2 , x^4 , x^6 , and $\cos x$; and some examples of odd functions are x^3 , x^5 , x^7 , and $\sin x$.



▲ **Figure 0.2.9** This is the graph of an even function since $f(-x) = f(x)$.



▲ **Figure 0.2.10** This is the graph of an odd function since $f(-x) = -f(x)$.

✓ QUICK CHECK EXERCISES 0.2 (See page 27 for answers.)

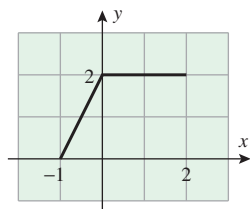
- Let $f(x) = 3\sqrt{x} - 2$ and $g(x) = |x|$. In each part, give the formula for the function and state the corresponding domain.
 - $f + g$: _____ Domain: _____
 - $f - g$: _____ Domain: _____
 - fg : _____ Domain: _____
 - f/g : _____ Domain: _____
- Let $f(x) = 2 - x^2$ and $g(x) = \sqrt{x}$. In each part, give the formula for the composition and state the corresponding domain.
 - $f \circ g$: _____ Domain: _____
 - $g \circ f$: _____ Domain: _____
- The graph of $y = 1 + (x - 2)^2$ may be obtained by shifting the graph of $y = x^2$ _____ (left/right) by _____ unit(s) and then shifting this new graph _____ (up/down) by _____ unit(s).
- Let

$$f(x) = \begin{cases} |x + 1|, & -2 \leq x \leq 0 \\ |x - 1|, & 0 < x \leq 2 \end{cases}$$
 - The letter of the alphabet that most resembles the graph of f is _____.
 - Is f an even function?

EXERCISE SET 0.2 Graphing Utility

FOCUS ON CONCEPTS

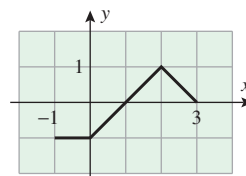
- The graph of a function f is shown in the accompanying figure. Sketch the graphs of the following equations.
 - $y = f(x) - 1$
 - $y = f(x - 1)$
 - $y = \frac{1}{2}f(x)$
 - $y = f(-\frac{1}{2}x)$



◀ **Figure Ex-1**

- Use the graph in Exercise 1 to sketch the graphs of the following equations.
 - $y = -f(-x)$
 - $y = f(2 - x)$
 - $y = 1 - f(2 - x)$
 - $y = \frac{1}{2}f(2x)$

- The graph of a function f is shown in the accompanying figure. Sketch the graphs of the following equations.
 - $y = f(x + 1)$
 - $y = f(2x)$
 - $y = |f(x)|$
 - $y = 1 - |f(x)|$



◀ **Figure Ex-3**

- Use the graph in Exercise 3 to sketch the graph of the equation $y = f(|x|)$.

 **5–24** Sketch the graph of the equation by translating, reflecting, compressing, and stretching the graph of $y = x^2$, $y = \sqrt{x}$, $y = 1/x$, $y = |x|$, or $y = \sqrt[3]{x}$ appropriately. Then use a graphing utility to confirm that your sketch is correct. ■

5. $y = -2(x+1)^2 - 3$ 6. $y = \frac{1}{2}(x-3)^2 + 2$
 7. $y = x^2 + 6x$ 8. $y = \frac{1}{2}(x^2 - 2x + 3)$
 9. $y = 3 - \sqrt{x+1}$ 10. $y = 1 + \sqrt{x-4}$
 11. $y = \frac{1}{2}\sqrt{x} + 1$ 12. $y = -\sqrt{3x}$
 13. $y = \frac{1}{x-3}$ 14. $y = \frac{1}{1-x}$
 15. $y = 2 - \frac{1}{x+1}$ 16. $y = \frac{x-1}{x}$
 17. $y = |x+2| - 2$ 18. $y = 1 - |x-3|$
 19. $y = |2x-1| + 1$ 20. $y = \sqrt{x^2 - 4x + 4}$
 21. $y = 1 - 2\sqrt[3]{x}$ 22. $y = \sqrt[3]{x-2} - 3$
 23. $y = 2 + \sqrt[3]{x+1}$ 24. $y + \sqrt[3]{x-2} = 0$

25. (a) Sketch the graph of $y = x + |x|$ by adding the corresponding y -coordinates on the graphs of $y = x$ and $y = |x|$.
 (b) Express the equation $y = x + |x|$ in piecewise form with no absolute values, and confirm that the graph you obtained in part (a) is consistent with this equation.
26. Sketch the graph of $y = x + (1/x)$ by adding corresponding y -coordinates on the graphs of $y = x$ and $y = 1/x$. Use a graphing utility to confirm that your sketch is correct.

27–28 Find formulas for $f + g$, $f - g$, fg , and f/g , and state the domains of the functions. ■

27. $f(x) = 2\sqrt{x-1}$, $g(x) = \sqrt{x-1}$

28. $f(x) = \frac{x}{1+x^2}$, $g(x) = \frac{1}{x}$

29. Let $f(x) = \sqrt{x}$ and $g(x) = x^3 + 1$. Find
 (a) $f(g(2))$ (b) $g(f(4))$ (c) $f(f(16))$
 (d) $g(g(0))$ (e) $f(2+h)$ (f) $g(3+h)$.

30. Let $g(x) = \sqrt{x}$. Find
 (a) $g(5s+2)$ (b) $g(\sqrt{x}+2)$ (c) $3g(5x)$
 (d) $\frac{1}{g(x)}$ (e) $g(g(x))$ (f) $(g(x))^2 - g(x^2)$
 (g) $g(1/\sqrt{x})$ (h) $g((x-1)^2)$ (i) $g(x+h)$.

31–34 Find formulas for $f \circ g$ and $g \circ f$, and state the domains of the compositions. ■

31. $f(x) = x^2$, $g(x) = \sqrt{1-x}$

32. $f(x) = \sqrt{x-3}$, $g(x) = \sqrt{x^2+3}$

33. $f(x) = \frac{1+x}{1-x}$, $g(x) = \frac{x}{1-x}$

34. $f(x) = \frac{x}{1+x^2}$, $g(x) = \frac{1}{x}$

35–36 Find a formula for $f \circ g \circ h$. ■

35. $f(x) = x^2 + 1$, $g(x) = \frac{1}{x}$, $h(x) = x^3$

36. $f(x) = \frac{1}{1+x}$, $g(x) = \sqrt[3]{x}$, $h(x) = \frac{1}{x^3}$

37–42 Express f as a composition of two functions; that is, find g and h such that $f = g \circ h$. [Note: Each exercise has more than one solution.] ■

37. (a) $f(x) = \sqrt{x+2}$

(b) $f(x) = |x^2 - 3x + 5|$

38. (a) $f(x) = x^2 + 1$

(b) $f(x) = \frac{1}{x-3}$

39. (a) $f(x) = \sin^2 x$

(b) $f(x) = \frac{3}{5 + \cos x}$

40. (a) $f(x) = 3 \sin(x^2)$

(b) $f(x) = 3 \sin^2 x + 4 \sin x$

41. (a) $f(x) = (1 + \sin(x^2))^3$

(b) $f(x) = \sqrt{1 - \sqrt[3]{x}}$

42. (a) $f(x) = \frac{1}{1-x^2}$

(b) $f(x) = |5 + 2x|$

43–46 True-False Determine whether the statement is true or false. Explain your answer. ■

43. The domain of $f + g$ is the intersection of the domains of f and g .
 44. The domain of $f \circ g$ consists of all values of x in the domain of g for which $g(x) \neq 0$.
 45. The graph of an even function is symmetric about the y -axis.
 46. The graph of $y = f(x+2) + 3$ is obtained by translating the graph of $y = f(x)$ right 2 units and up 3 units.

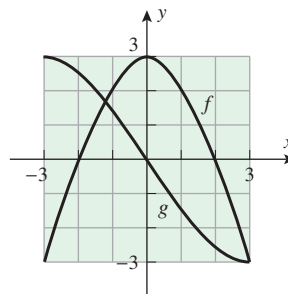
FOCUS ON CONCEPTS

47. Use the data in the accompanying table to make a plot of $y = f(g(x))$.

x	-3	-2	-1	0	1	2	3
$f(x)$	-4	-3	-2	-1	0	1	2
$g(x)$	-1	0	1	2	3	-2	-3

▲ Table Ex-47

48. Find the domain of $g \circ f$ for the functions f and g in Exercise 47.
 49. Sketch the graph of $y = f(g(x))$ for the functions graphed in the accompanying figure.



◀ Figure Ex-49

50. Sketch the graph of $y = g(f(x))$ for the functions graphed in Exercise 49.
 51. Use the graphs of f and g in Exercise 49 to estimate the solutions of the equations $f(g(x)) = 0$ and $g(f(x)) = 0$.
 52. Use the table given in Exercise 47 to solve the equations $f(g(x)) = 0$ and $g(f(x)) = 0$.

53–56 Find

$$\frac{f(x+h) - f(x)}{h} \quad \text{and} \quad \frac{f(w) - f(x)}{w-x}$$

Simplify as much as possible. ■

53. $f(x) = 3x^2 - 5$

54. $f(x) = x^2 + 6x$

55. $f(x) = 1/x$

56. $f(x) = 1/x^2$

57. Classify the functions whose values are given in the accompanying table as even, odd, or neither.

x	-3	-2	-1	0	1	2	3
$f(x)$	5	3	2	3	1	-3	5
$g(x)$	4	1	-2	0	2	-1	-4
$h(x)$	2	-5	8	-2	8	-5	2

▲ Table Ex-57

58. Complete the accompanying table so that the graph of $y = f(x)$ is symmetric about

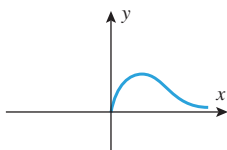
- (a) the
- y
- axis (b) the origin.

x	-3	-2	-1	0	1	2	3
$f(x)$	1		-1	0		-5	

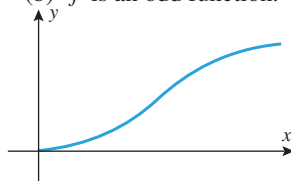
▲ Table Ex-58

59. The accompanying figure shows a portion of a graph. Complete the graph so that the entire graph is symmetric about

- (a) the
- x
- axis (b) the
- y
- axis (c) the origin.



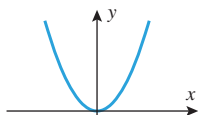
▲ Figure Ex-59



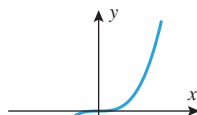
▲ Figure Ex-60

61–62 Classify the functions graphed in the accompanying figures as even, odd, or neither. ■

61.



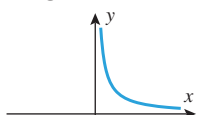
(a)



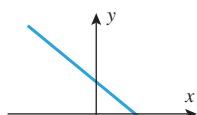
(b)

▲ Figure Ex-61

62.



(a)



(b)

▲ Figure Ex-62

63. In each part, classify the function as even, odd, or neither.

(a) $f(x) = x^2$

(b) $f(x) = x^3$

(c) $f(x) = |x|$

(d) $f(x) = x + 1$

(e) $f(x) = \frac{x^5 - x}{1 + x^2}$

(f) $f(x) = 2$

64. Suppose that the function f has domain all real numbers. Determine whether each function can be classified as even or odd. Explain.

(a) $g(x) = \frac{f(x) + f(-x)}{2}$

(b) $h(x) = \frac{f(x) - f(-x)}{2}$

65. Suppose that the function f has domain all real numbers. Show that f can be written as the sum of an even function and an odd function. [Hint: See Exercise 64.]66–67 Use Theorem 0.2.3 to determine whether the graph has symmetries about the x -axis, the y -axis, or the origin. ■

66. (a) $x = 5y^2 + 9$

(b) $x^2 - 2y^2 = 3$

(c) $xy = 5$

67. (a) $x^4 = 2y^3 + y$

(b) $y = \frac{x}{3 + x^2}$

(c) $y^2 = |x| - 5$

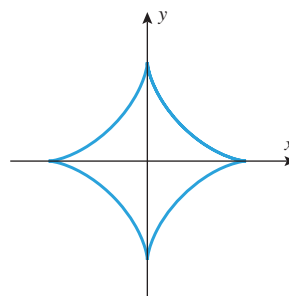
68–69 (i) Use a graphing utility to graph the equation in the first quadrant. [Note: To do this you will have to solve the equation for y in terms of x .] (ii) Use symmetry to make a hand-drawn sketch of the entire graph. (iii) Confirm your work by generating the graph of the equation in the remaining three quadrants. ■

68. $9x^2 + 4y^2 = 36$

69. $4x^2 + 16y^2 = 16$

70. The graph of the equation $x^{2/3} + y^{2/3} = 1$, which is shown in the accompanying figure, is called a **four-cusped hypocycloid**.(a) Use Theorem 0.2.3 to confirm that this graph is symmetric about the x -axis, the y -axis, and the origin.(b) Find a function f whose graph in the first quadrant coincides with the four-cusped hypocycloid, and use a graphing utility to confirm your work.

(c) Repeat part (b) for the remaining three quadrants.



Four-cusped hypocycloid

◀ Figure Ex-70

71. The equation $y = |f(x)|$ can be written as

$$y = \begin{cases} f(x), & f(x) \geq 0 \\ -f(x), & f(x) < 0 \end{cases}$$

which shows that the graph of $y = |f(x)|$ can be obtained from the graph of $y = f(x)$ by retaining the portion that lies

on or above the x -axis and reflecting about the x -axis the portion that lies below the x -axis. Use this method to obtain the graph of $y = |2x - 3|$ from the graph of $y = 2x - 3$.

72–73 Use the method described in Exercise 71. ■

72. Sketch the graph of $y = |1 - x^2|$.

73. Sketch the graph of

(a) $f(x) = |\cos x|$ (b) $f(x) = \cos x + |\cos x|$.

74. The **greatest integer function**, $\lfloor x \rfloor$, is defined to be the greatest integer that is less than or equal to x . For example, $\lfloor 2.7 \rfloor = 2$, $\lfloor -2.3 \rfloor = -3$, and $\lfloor 4 \rfloor = 4$. In each part, sketch the graph of $y = f(x)$.

(a) $f(x) = \lfloor x \rfloor$ (b) $f(x) = \lfloor x^2 \rfloor$
 (c) $f(x) = \lfloor x \rfloor^2$ (d) $f(x) = \lfloor \sin x \rfloor$

75. Is it ever true that $f \circ g = g \circ f$ if f and g are nonconstant functions? If not, prove it; if so, give some examples for which it is true.

✓ QUICK CHECK ANSWERS 0.2

1. (a) $(f + g)(x) = 3\sqrt{x} - 2 + x$; $x \geq 0$ (b) $(f - g)(x) = 3\sqrt{x} - 2 - x$; $x \geq 0$ (c) $(fg)(x) = 3x^{3/2} - 2x$; $x \geq 0$
 (d) $(f/g)(x) = \frac{3\sqrt{x} - 2}{x}$; $x > 0$ 2. (a) $(f \circ g)(x) = 2 - x$; $x \geq 0$ (b) $(g \circ f)(x) = \sqrt{2 - x^2}$; $-\sqrt{2} \leq x \leq \sqrt{2}$
 3. right; 2; up; 1 4. (a) W (b) yes

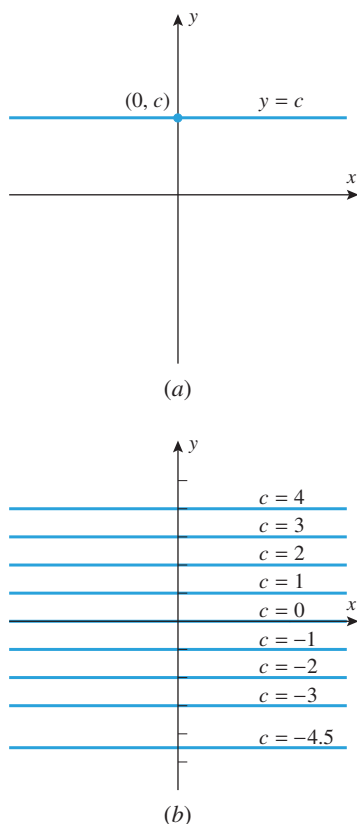
0.3 FAMILIES OF FUNCTIONS

Functions are often grouped into families according to the form of their defining formulas or other common characteristics. In this section we will discuss some of the most basic families of functions.

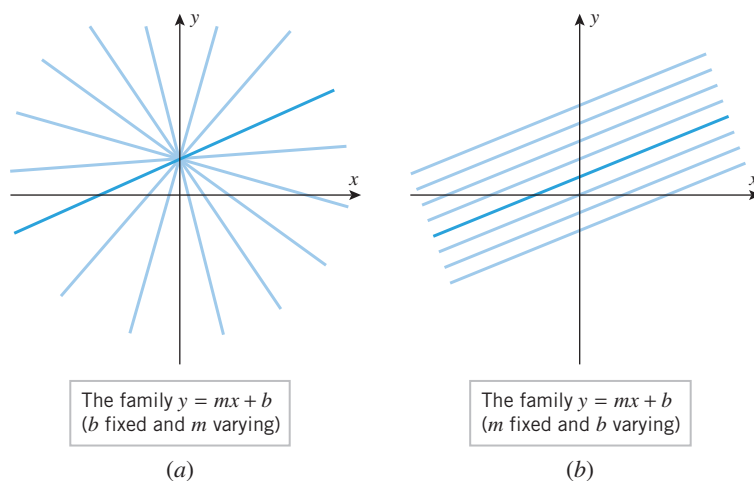
■ FAMILIES OF CURVES

The graph of a constant function $f(x) = c$ is the graph of the equation $y = c$, which is the horizontal line shown in Figure 0.3.1a. If we vary c , then we obtain a set or **family** of horizontal lines such as those in Figure 0.3.1b.

Constants that are varied to produce families of curves are called **parameters**. For example, recall that an equation of the form $y = mx + b$ represents a line of slope m and y -intercept b . If we keep b fixed and treat m as a parameter, then we obtain a family of lines whose members all have y -intercept b (Figure 0.3.2a), and if we keep m fixed and treat b as a parameter, we obtain a family of parallel lines whose members all have slope m (Figure 0.3.2b).



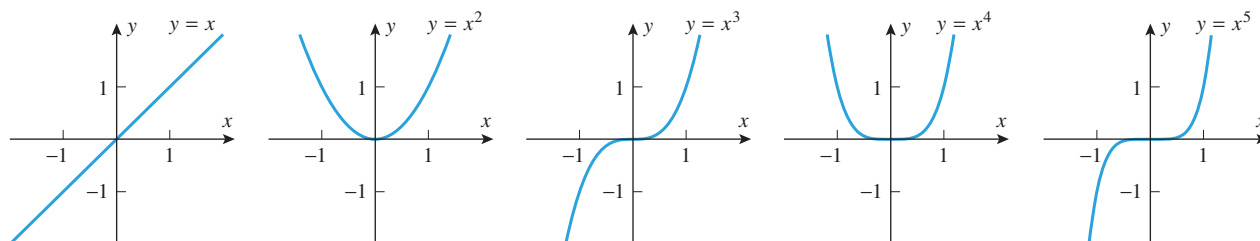
▲ Figure 0.3.1



► Figure 0.3.2

POWER FUNCTIONS; THE FAMILY $y = x^n$

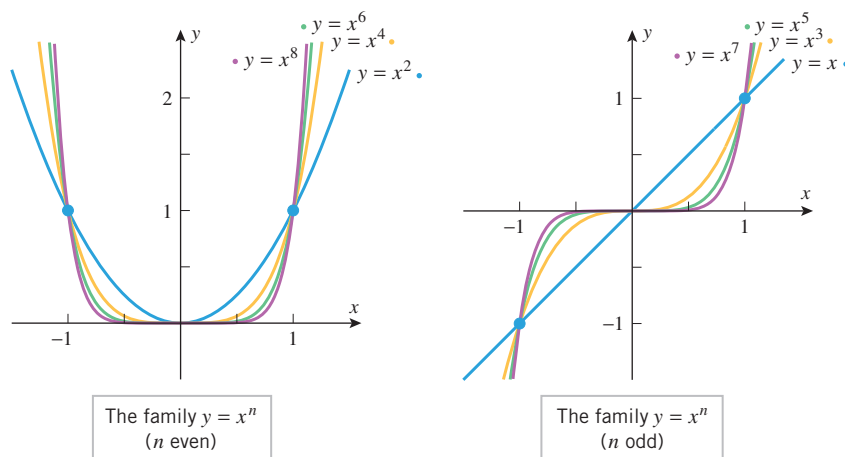
A function of the form $f(x) = x^p$, where p is constant, is called a **power function**. For the moment, let us consider the case where p is a positive integer, say $p = n$. The graphs of the curves $y = x^n$ for $n = 1, 2, 3, 4$, and 5 are shown in Figure 0.3.3. The first graph is the line with slope 1 that passes through the origin, and the second is a parabola that opens up and has its vertex at the origin (see Web Appendix H).



▲ Figure 0.3.3

For $n \geq 2$ the shape of the curve $y = x^n$ depends on whether n is even or odd (Figure 0.3.4):

- For even values of n , the functions $f(x) = x^n$ are even, so their graphs are symmetric about the y -axis. The graphs all have the general shape of the graph of $y = x^2$, and each graph passes through the points $(-1, 1)$, $(0, 0)$, and $(1, 1)$. As n increases, the graphs become flatter over the interval $-1 < x < 1$ and steeper over the intervals $x > 1$ and $x < -1$.
- For odd values of n , the functions $f(x) = x^n$ are odd, so their graphs are symmetric about the origin. The graphs all have the general shape of the curve $y = x^3$, and each graph passes through the points $(-1, -1)$, $(0, 0)$, and $(1, 1)$. As n increases, the graphs become flatter over the interval $-1 < x < 1$ and steeper over the intervals $x > 1$ and $x < -1$.



► Figure 0.3.4

REMARK

The flattening and steepening effects can be understood by considering what happens when a number x is raised to higher and higher powers: If $-1 < x < 1$, then the absolute value of x^n *decreases* as n increases, thereby causing the graphs to become flatter on this interval as n increases (try raising $\frac{1}{2}$ or $-\frac{1}{2}$ to higher and higher powers). On the other hand, if $x > 1$ or $x < -1$, then the absolute value of x^n *increases* as n increases, thereby causing the graphs to become steeper on these intervals as n increases (try raising 2 or -2 to higher and higher powers).

THE FAMILY $y = x^{-n}$

If p is a negative integer, say $p = -n$, then the power functions $f(x) = x^p$ have the form $f(x) = x^{-n} = 1/x^n$. Figure 0.3.5 shows the graphs of $y = 1/x$ and $y = 1/x^2$. The graph of $y = 1/x$ is called an **equilateral hyperbola** (for reasons to be discussed later).

As illustrated in Figure 0.3.5, the shape of the curve $y = 1/x^n$ depends on whether n is even or odd:

- For even values of n , the functions $f(x) = 1/x^n$ are even, so their graphs are symmetric about the y -axis. The graphs all have the general shape of the curve $y = 1/x^2$, and each graph passes through the points $(-1, 1)$ and $(1, 1)$. As n increases, the graphs become steeper over the intervals $-1 < x < 0$ and $0 < x < 1$ and become flatter over the intervals $x > 1$ and $x < -1$.
- For odd values of n , the functions $f(x) = 1/x^n$ are odd, so their graphs are symmetric about the origin. The graphs all have the general shape of the curve $y = 1/x$, and each graph passes through the points $(1, 1)$ and $(-1, -1)$. As n increases, the graphs become steeper over the intervals $-1 < x < 0$ and $0 < x < 1$ and become flatter over the intervals $x > 1$ and $x < -1$.
- For both even and odd values of n the graph $y = 1/x^n$ has a break at the origin (called a **discontinuity**), which occurs because division by zero is undefined.

By considering the value of $1/x^n$ for a fixed x as n increases, explain why the graphs become flatter or steeper as described here for increasing values of n .

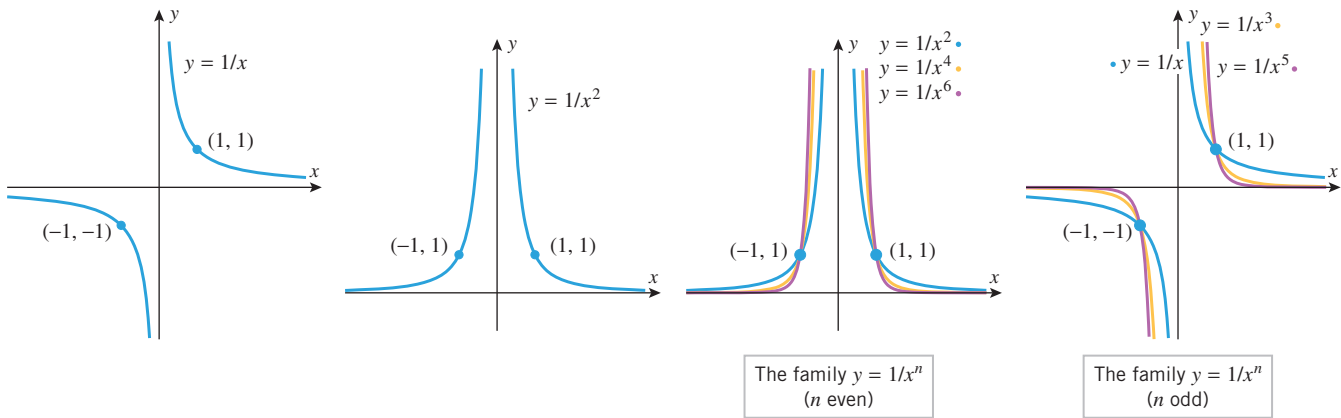


Figure 0.3.5

INVERSE PROPORTIONS

Recall that a variable y is said to be **inversely proportional to a variable x** if there is a positive constant k , called the **constant of proportionality**, such that

$$y = \frac{k}{x} \tag{1}$$

Since k is assumed to be positive, the graph of (1) has the same shape as $y = 1/x$ but is compressed or stretched in the y -direction. Also, it should be evident from (1) that doubling x multiplies y by $\frac{1}{2}$, tripling x multiplies y by $\frac{1}{3}$, and so forth.

Equation (1) can be expressed as $xy = k$, which tells us that the product of inversely proportional variables is a positive constant. This is a useful form for identifying inverse proportionality in experimental data.

► **Example 1** Table 0.3.1 shows some experimental data.

Table 0.3.1							
x	0.8	1	2.5	4	6.25	10	
y	6.25	5	2	1.25	0.8	0.5	

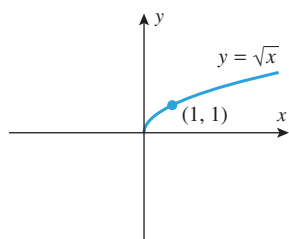
- (a) Explain why the data suggest that y is inversely proportional to x .
- (b) Express y as a function of x .
- (c) Graph your function and the data together for $x > 0$.

Solution. For every data point we have $xy = 5$, so y is inversely proportional to x and $y = 5/x$. The graph of this equation with the data points is shown in Figure 0.3.6. ◀

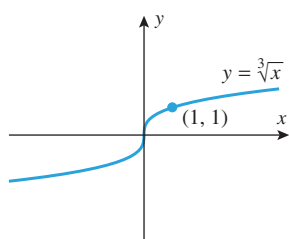
Inverse proportions arise in various laws of physics. For example, **Boyle's law** in physics states that *if a fixed amount of an ideal gas is held at a constant temperature, then the product of the pressure P exerted by the gas and the volume V that it occupies is constant*; that is,

$$PV = k$$

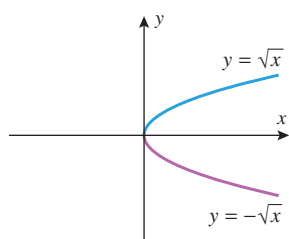
This implies that the variables P and V are inversely proportional to one another. Figure 0.3.7 shows a typical graph of volume versus pressure under the conditions of Boyle's law. Note how doubling the pressure corresponds to halving the volume, as expected.



(a)

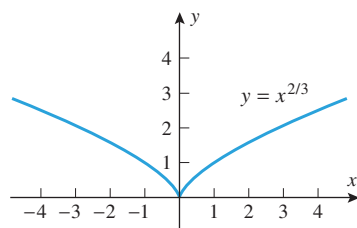


(b)

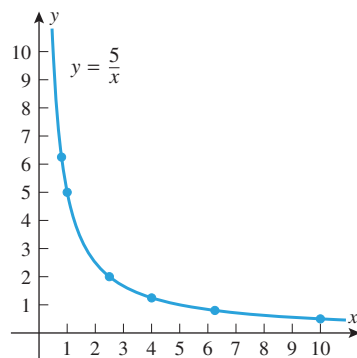


(c)

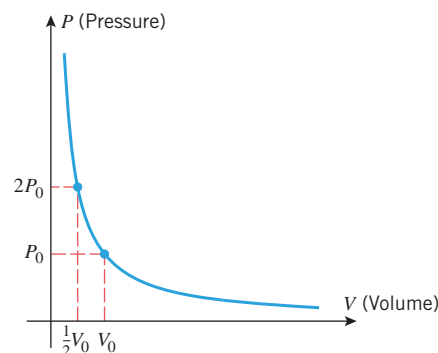
▲ Figure 0.3.8



▲ Figure 0.3.9



▲ Figure 0.3.6



▲ Figure 0.3.7 Doubling pressure corresponds to halving volume

POWER FUNCTIONS WITH NONINTEGER EXPONENTS

If $p = 1/n$, where n is a positive integer, then the power functions $f(x) = x^p$ have the form

$$f(x) = x^{1/n} = \sqrt[n]{x}$$

In particular, if $n = 2$, then $f(x) = \sqrt{x}$, and if $n = 3$, then $f(x) = \sqrt[3]{x}$. The graphs of these functions are shown in parts (a) and (b) of Figure 0.3.8.

Since every real number has a real cube root, the domain of the function $f(x) = \sqrt[3]{x}$ is $(-\infty, +\infty)$, and hence the graph of $y = \sqrt[3]{x}$ extends over the entire x -axis. In contrast, the graph of $y = \sqrt{x}$ extends only over the interval $[0, +\infty)$ because $\sqrt{x}$ is imaginary for negative x . As illustrated in Figure 0.3.8c, the graphs of $y = \sqrt{x}$ and $y = -\sqrt{x}$ form the upper and lower halves of the parabola $x = y^2$. In general, the graph of $y = \sqrt[n]{x}$ extends over the entire x -axis if n is odd, but extends only over the interval $[0, +\infty)$ if n is even.

Power functions can have other fractional exponents. Some examples are

$$f(x) = x^{2/3}, \quad f(x) = \sqrt[5]{x^3}, \quad f(x) = x^{-7/8} \quad (2)$$

The graph of $f(x) = x^{2/3}$ is shown in Figure 0.3.9. We will discuss expressions involving irrational exponents later.

TECHNOLOGY MASTERY

Graphing utilities sometimes omit portions of the graph of a function involving fractional exponents (or radicals). If $f(x) = x^{p/q}$, where p/q is a positive fraction in *lowest terms*, then you can circumvent this problem as follows:

- If p is even and q is odd, then graph $g(x) = |x|^{p/q}$ instead of $f(x)$.
- If p is odd and q is odd, then graph $g(x) = (|x|/x)|x|^{p/q}$ instead of $f(x)$.

Use a graphing utility to generate graphs of $f(x) = \sqrt[5]{x^3}$ and $f(x) = x^{-7/8}$ that show all of their significant features.

POLYNOMIALS

A **polynomial in x** is a function that is expressible as a sum of finitely many terms of the form cx^n , where c is a constant and n is a nonnegative integer. Some examples of polynomials are

$$2x + 1, \quad 3x^2 + 5x - \sqrt{2}, \quad x^3, \quad 4 (= 4x^0), \quad 5x^7 - x^4 + 3$$

The function $(x^2 - 4)^3$ is also a polynomial because it can be expanded by the binomial formula (see the inside front cover) and expressed as a sum of terms of the form cx^n :

$$(x^2 - 4)^3 = (x^2)^3 - 3(x^2)^2(4) + 3(x^2)(4^2) - (4^3) = x^6 - 12x^4 + 48x^2 - 64 \quad (3)$$

A general polynomial can be written in either of the following forms, depending on whether one wants the powers of x in ascending or descending order:

$$c_0 + c_1x + c_2x^2 + \cdots + c_nx^n$$

$$c_nx^n + c_{n-1}x^{n-1} + \cdots + c_1x + c_0$$

The constants $c_0, c_1, \dots, c_n$ are called the **coefficients** of the polynomial. When a polynomial is expressed in one of these forms, the highest power of x that occurs with a nonzero coefficient is called the **degree** of the polynomial. Nonzero constant polynomials are considered to have degree 0, since we can write $c = cx^0$. Polynomials of degree 1, 2, 3, 4, and 5 are described as **linear**, **quadratic**, **cubic**, **quartic**, and **quintic**, respectively. For example,

$$3 + 5x$$

Has degree 1 (linear)

$$x^2 - 3x + 1$$

Has degree 2 (quadratic)

$$2x^3 - 7$$

Has degree 3 (cubic)

$$8x^4 - 9x^3 + 5x - 3$$

Has degree 4 (quartic)

$$\sqrt{3} + x^3 + x^5$$

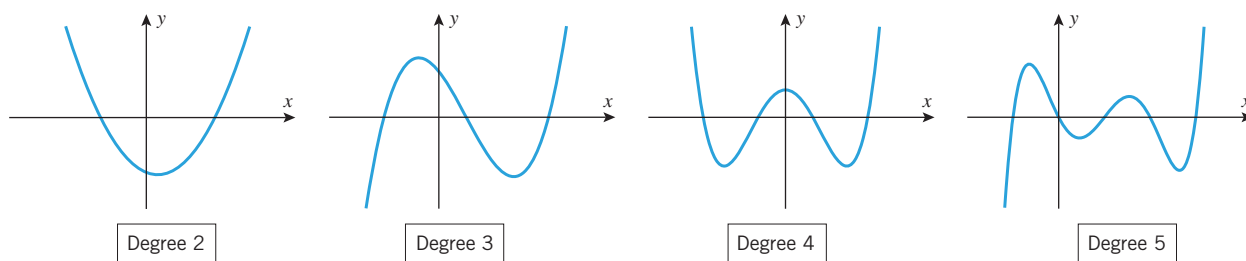
Has degree 5 (quintic)

$$(x^2 - 4)^3$$

Has degree 6 [see (3)]

The constant 0 is a polynomial called the **zero polynomial**. In this text we will take the degree of the zero polynomial to be undefined. Other texts may use different conventions for the degree of the zero polynomial.

The natural domain of a polynomial in x is $(-\infty, +\infty)$, since the only operations involved are multiplication and addition; the range depends on the particular polynomial. We already know that the graphs of polynomials of degree 0 and 1 are lines and that the graphs of polynomials of degree 2 are parabolas. Figure 0.3.10 shows the graphs of some typical polynomials of higher degree. Later, we will discuss polynomial graphs in detail, but for now it suffices to observe that graphs of polynomials are very well behaved in the sense that they have no discontinuities or sharp corners. As illustrated in Figure 0.3.10, the graphs of polynomials wander up and down for awhile in a roller-coaster fashion, but eventually that behavior stops and the graphs steadily rise or fall indefinitely as one travels along the curve in either the positive or negative direction. We will see later that the number of peaks and valleys is less than the degree of the polynomial.



▲ Figure 0.3.10

RATIONAL FUNCTIONS

A function that can be expressed as a ratio of two polynomials is called a **rational function**. If $P(x)$ and $Q(x)$ are polynomials, then the domain of the rational function

$$f(x) = \frac{P(x)}{Q(x)}$$

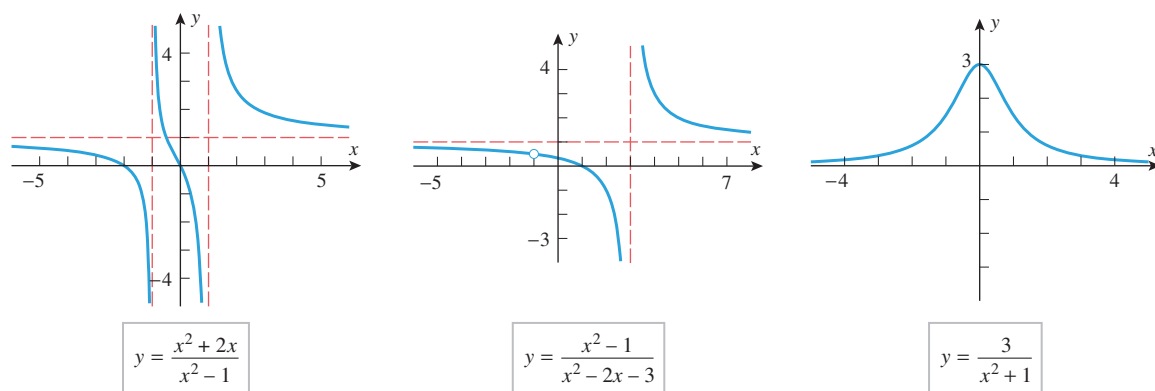
consists of all values of x such that $Q(x) \neq 0$. For example, the domain of the rational function

$$f(x) = \frac{x^2 + 2x}{x^2 - 1}$$

consists of all values of x , except $x = 1$ and $x = -1$. Its graph is shown in Figure 0.3.11 along with the graphs of two other typical rational functions.

The graphs of rational functions with nonconstant denominators differ from the graphs of polynomials in some essential ways:

- Unlike polynomials whose graphs are continuous (unbroken) curves, the graphs of rational functions have discontinuities at the points where the denominator is zero.
- Unlike polynomials, rational functions may have numbers at which they are not defined. Near such points, many rational functions have graphs that closely approximate a vertical line, called a **vertical asymptote**. These are represented by the dashed vertical lines in Figure 0.3.11.
- Unlike the graphs of nonconstant polynomials, which eventually rise or fall indefinitely, the graphs of many rational functions eventually get closer and closer to some horizontal line, called a **horizontal asymptote**, as one traverses the curve in either the positive or negative direction. The horizontal asymptotes are represented by the dashed horizontal lines in the first two parts of Figure 0.3.11. In the third part of the figure the x -axis is a horizontal asymptote.



▲ Figure 0.3.11

■ ALGEBRAIC FUNCTIONS

Functions that can be constructed from polynomials by applying finitely many algebraic operations (addition, subtraction, multiplication, division, and root extraction) are called **algebraic functions**. Some examples are

$$f(x) = \sqrt{x^2 - 4}, \quad f(x) = 3\sqrt[3]{x}(2 + x), \quad f(x) = x^{2/3}(x + 2)^2$$

As illustrated in Figure 0.3.12, the graphs of algebraic functions vary widely, so it is difficult to make general statements about them. Later in this text we will develop general calculus methods for analyzing such functions.

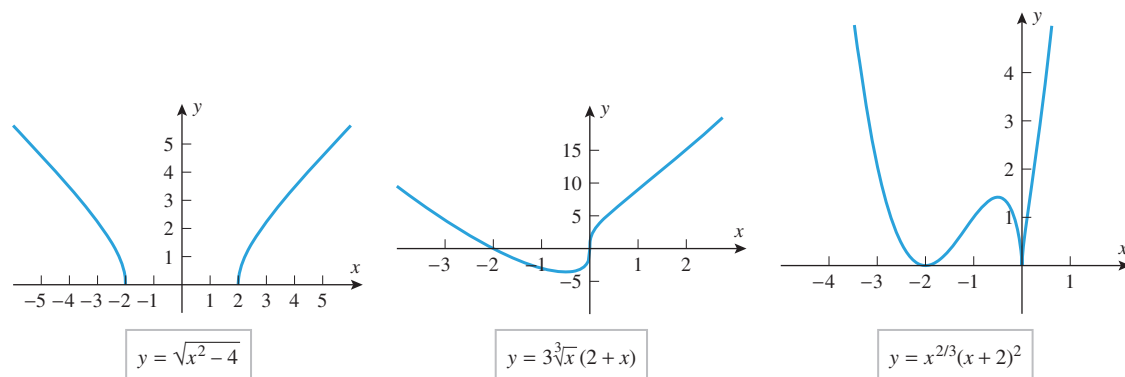
■ THE FAMILIES $y = A \sin Bx$ AND $y = A \cos Bx$

Many important applications lead to trigonometric functions of the form

$$f(x) = A \sin(Bx - C) \quad \text{and} \quad g(x) = A \cos(Bx - C) \quad (4)$$

where A , B , and C are nonzero constants. The graphs of such functions can be obtained by stretching, compressing, translating, and reflecting the graphs of $y = \sin x$ and $y = \cos x$

In this text we will assume that the independent variable of a trigonometric function is in radians unless otherwise stated. A review of trigonometric functions can be found in Appendix B.

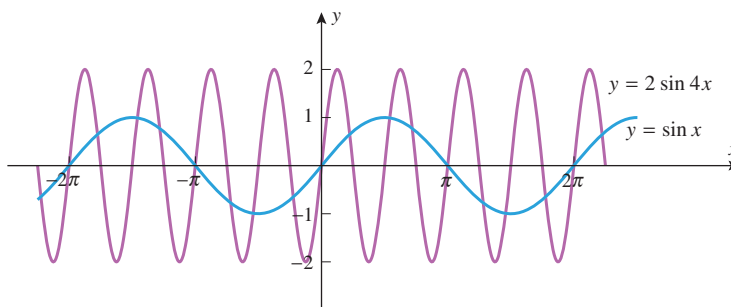


▲ Figure 0.3.12

appropriately. To see why this is so, let us start with the case where $C = 0$ and consider how the graphs of the equations

$$y = A \sin Bx \quad \text{and} \quad y = A \cos Bx$$

relate to the graphs of $y = \sin x$ and $y = \cos x$. If A and B are positive, then the effect of the constant A is to stretch or compress the graphs of $y = \sin x$ and $y = \cos x$ vertically and the effect of the constant B is to compress or stretch the graphs of $\sin x$ and $\cos x$ horizontally. For example, the graph of $y = 2 \sin 4x$ can be obtained by stretching the graph of $y = \sin x$ vertically by a factor of 2 and compressing it horizontally by a factor of 4. (Recall from Section 0.2 that the multiplier of x *stretches* when it is less than 1 and *compresses* when it is greater than 1.) Thus, as shown in Figure 0.3.13, the graph of $y = 2 \sin 4x$ varies between -2 and 2 , and repeats every $2\pi/4 = \pi/2$ units.



► Figure 0.3.13

In general, if A and B are positive numbers, then the graphs of

$$y = A \sin Bx \quad \text{and} \quad y = A \cos Bx$$

oscillate between $-A$ and A and repeat every $2\pi/B$ units, so we say that these functions have **amplitude** A and **period** $2\pi/B$. In addition, we define the **frequency** of these functions to be the reciprocal of the period, that is, the frequency is $B/(2\pi)$. If A or B is negative, then these constants cause reflections of the graphs about the axes as well as compressing or stretching them; and in this case the amplitude, period, and frequency are given by

$$\text{amplitude} = |A|, \quad \text{period} = \frac{2\pi}{|B|}, \quad \text{frequency} = \frac{|B|}{2\pi}$$

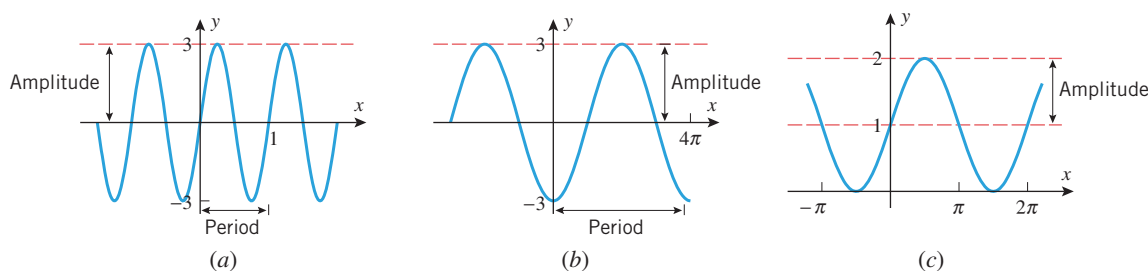
► **Example 2** Make sketches of the following graphs that show the period and amplitude.

- (a) $y = 3 \sin 2\pi x$ (b) $y = -3 \cos 0.5x$ (c) $y = 1 + \sin x$

Solution (a). The equation is of the form $y = A \sin Bx$ with $A = 3$ and $B = 2\pi$, so the graph has the shape of a sine function, but it has an amplitude of $A = 3$ and a period of $2\pi/B = 2\pi/(2\pi) = 1$ (Figure 0.3.14a).

Solution (b). The equation is of the form $y = A \cos Bx$ with $A = -3$ and $B = 0.5$, so the graph has the shape of a cosine curve that has been reflected about the x -axis (because $A = -3$ is negative), but with amplitude $|A| = 3$ and period $2\pi/B = 2\pi/0.5 = 4\pi$ (Figure 0.3.14b).

Solution (c). The graph has the shape of a sine curve that has been translated up 1 unit (Figure 0.3.14c). ◀



▲ Figure 0.3.14

■ THE FAMILIES $y = A \sin(Bx - C)$ AND $y = A \cos(Bx - C)$

To investigate the graphs of the more general families

$$y = A \sin(Bx - C) \quad \text{and} \quad y = A \cos(Bx - C)$$

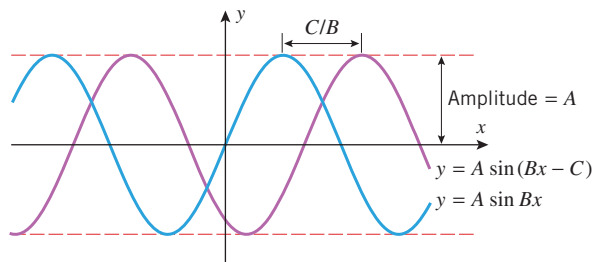
it will be helpful to rewrite these equations as

$$y = A \sin \left[B \left(x - \frac{C}{B} \right) \right] \quad \text{and} \quad y = A \cos \left[B \left(x - \frac{C}{B} \right) \right]$$

In this form we see that the graphs of these equations can be obtained by translating the graphs of $y = A \sin Bx$ and $y = A \cos Bx$ to the left or right, depending on the sign of C/B . For example, if $C/B > 0$, then the graph of

$$y = A \sin[B(x - C/B)] = A \sin(Bx - C)$$

can be obtained by translating the graph of $y = A \sin Bx$ to the right by C/B units (Figure 0.3.15). If $C/B < 0$, the graph of $y = A \sin(Bx - C)$ is obtained by translating the graph of $y = A \sin Bx$ to the left by $|C/B|$ units.



► Figure 0.3.15

► **Example 3** Find the amplitude and period of

$$y = 3 \cos \left(2x + \frac{\pi}{2} \right)$$

and determine how the graph of $y = 3 \cos 2x$ should be translated to produce the graph of this equation. Confirm your results by graphing the equation on a calculator or computer.

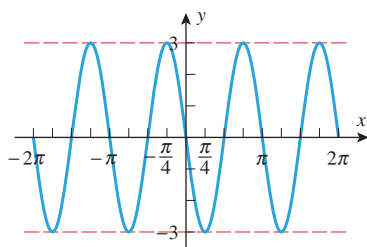
Solution. The equation can be rewritten as

$$y = 3 \cos \left[2x - \left(-\frac{\pi}{2} \right) \right] = 3 \cos \left[2 \left(x - \left(-\frac{\pi}{4} \right) \right) \right]$$

which is of the form

$$y = A \cos \left[B \left(x - \frac{C}{B} \right) \right]$$

with $A = 3$, $B = 2$, and $C/B = -\pi/4$. It follows that the amplitude is $A = 3$, the period is $2\pi/B = \pi$, and the graph is obtained by translating the graph of $y = 3 \cos 2x$ left by $|C/B| = \pi/4$ units (Figure 0.3.16). ◀



▲ Figure 0.3.16

✓ QUICK CHECK EXERCISES 0.3 (See page 38 for answers.)

- Consider the family of functions $y = x^n$, where n is an integer. The graphs of $y = x^n$ are symmetric with respect to the y -axis if n is _____. These graphs are symmetric with respect to the origin if n is _____. The y -axis is a vertical asymptote for these graphs if n is _____.
- What is the natural domain of a polynomial?
- Consider the family of functions $y = x^{1/n}$, where n is a nonzero integer. Find the natural domain of these functions if n is
 - positive and even
 - positive and odd
 - negative and even
 - negative and odd.
- Classify each equation as a polynomial, rational, algebraic, or not an algebraic function.
 - $y = \sqrt{x} + 2$
 - $y = \sqrt{3}x^4 - x + 1$
 - $y = 5x^3 + \cos 4x$
 - $y = \frac{x^2 + 5}{2x - 7}$
 - $y = 3x^2 + 4x^{-2}$
- The graph of $y = A \sin Bx$ has amplitude _____ and is periodic with period _____.

EXERCISE SET 0.3 Graphing Utility

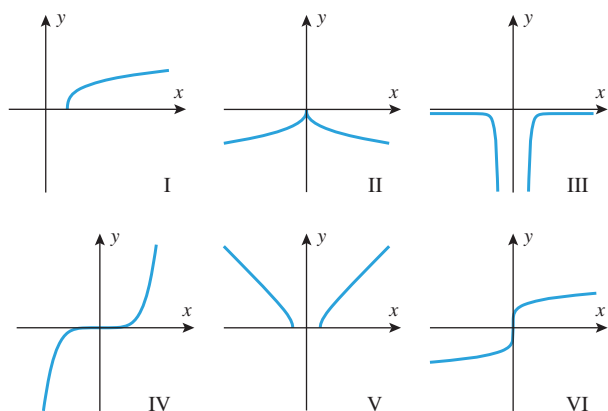
- Find an equation for the family of lines whose members have slope $m = 3$.
 - Find an equation for the member of the family that passes through $(-1, 3)$.
 - Sketch some members of the family, and label them with their equations. Include the line in part (b).
- Find an equation for the family of lines whose members are perpendicular to those in Exercise 1.
- Find an equation for the family of lines with y -intercept $b = 2$.
 - Find an equation for the member of the family whose angle of inclination is 135° .
 - Sketch some members of the family, and label them with their equations. Include the line in part (b).
- Find an equation for
 - the family of lines that pass through the origin
 - the family of lines with x -intercept $a = 1$
 - the family of lines that pass through the point $(1, -2)$
 - the family of lines parallel to $2x + 4y = 1$.
- Find an equation for the family of lines tangent to the circle with center at the origin and radius 3.
- Find an equation for the family of lines that pass through the intersection of $5x - 3y + 11 = 0$ and $2x - 9y + 7 = 0$.
- The U.S. Internal Revenue Service uses a 10-year linear depreciation schedule to determine the value of various business items. This means that an item is assumed to have a value of zero at the end of the tenth year and that at intermediate times the value is a linear function of the elapsed time. Sketch some typical depreciation lines, and explain the practical significance of the y -intercepts.
- Find all lines through $(6, -1)$ for which the product of the x - and y -intercepts is 3.

FOCUS ON CONCEPTS

9–10 State a geometric property common to all lines in the family, and sketch five of the lines. ■

- The family $y = -x + b$
 - The family $y = mx - 1$
 - The family $y = m(x + 4) + 2$
 - The family $x - ky = 1$

10. (a) The family $y = b$
 (b) The family $Ax + 2y + 1 = 0$
 (c) The family $2x + By + 1 = 0$
 (d) The family $y - 1 = m(x + 1)$
11. In each part, match the equation with one of the accompanying graphs.
- (a) $y = \sqrt[5]{x}$ (b) $y = 2x^5$
 (c) $y = -1/x^8$ (d) $y = \sqrt{x^2 - 1}$
 (e) $y = \sqrt[4]{x - 2}$ (f) $y = -\sqrt[5]{x^2}$



▲ Figure Ex-11

12. The accompanying table gives approximate values of three functions: one of the form kx^2 , one of the form kx^{-3} , and one of the form $kx^{3/2}$. Identify which is which, and estimate k in each case.

x	0.25	0.37	2.1	4.0	5.8	6.2	7.9	9.3
$f(x)$	640	197	1.08	0.156	0.0513	0.0420	0.0203	0.0124
$g(x)$	0.0312	0.0684	2.20	8.00	16.8	19.2	31.2	43.2
$h(x)$	0.250	0.450	6.09	16.0	27.9	30.9	44.4	56.7

▲ Table Ex-12

- 13–14 Sketch the graph of the equation for $n = 1, 3$, and 5 in one coordinate system and for $n = 2, 4$, and 6 in another coordinate system. If you have a graphing utility, use it to check your work. ■

13. (a) $y = -x^n$ (b) $y = 2x^{-n}$ (c) $y = (x - 1)^{1/n}$
 14. (a) $y = 2x^n$ (b) $y = -x^{-n}$
 (c) $y = -3(x + 2)^{1/n}$
15. (a) Sketch the graph of $y = ax^2$ for $a = \pm 1, \pm 2$, and ± 3 in a single coordinate system.
 (b) Sketch the graph of $y = x^2 + b$ for $b = \pm 1, \pm 2$, and ± 3 in a single coordinate system.
 (c) Sketch some typical members of the family of curves $y = ax^2 + b$.
16. (a) Sketch the graph of $y = a\sqrt{x}$ for $a = \pm 1, \pm 2$, and ± 3 in a single coordinate system.

- (b) Sketch the graph of $y = \sqrt{x} + b$ for $b = \pm 1, \pm 2$, and ± 3 in a single coordinate system.
 (c) Sketch some typical members of the family of curves $y = a\sqrt{x} + b$.

17–18 Sketch the graph of the equation by making appropriate transformations to the graph of a basic power function. If you have a graphing utility, use it to check your work. ■

17. (a) $y = 2(x + 1)^2$ (b) $y = -3(x - 2)^3$
 (c) $y = \frac{-3}{(x + 1)^2}$ (d) $y = \frac{1}{(x - 3)^5}$
18. (a) $y = 1 - \sqrt{x + 2}$ (b) $y = 1 - \sqrt[3]{x + 2}$
 (c) $y = \frac{5}{(1 - x)^3}$ (d) $y = \frac{2}{(4 + x)^4}$
19. Sketch the graph of $y = x^2 + 2x$ by completing the square and making appropriate transformations to the graph of $y = x^2$.
20. (a) Use the graph of $y = \sqrt{x}$ to help sketch the graph of $y = \sqrt{|x|}$.
 (b) Use the graph of $y = \sqrt[3]{x}$ to help sketch the graph of $y = \sqrt[3]{|x|}$.

21. As discussed in this section, Boyle's law states that at a constant temperature the pressure P exerted by a gas is related to the volume V by the equation $PV = k$.

- (a) Find the appropriate units for the constant k if pressure (which is force per unit area) is in newtons per square meter (N/m^2) and volume is in cubic meters (m^3).
 (b) Find k if the gas exerts a pressure of $20,000 \text{ N/m}^2$ when the volume is 1 liter (0.001 m^3).
 (c) Make a table that shows the pressures for volumes of 0.25, 0.5, 1.0, 1.5, and 2.0 liters.
 (d) Make a graph of P versus V .

22. A manufacturer of cardboard drink containers wants to construct a closed rectangular container that has a square base and will hold $\frac{1}{10}$ liter (100 cm^3). Estimate the dimensions of the container that will require the least amount of material for its manufacture.

23–24 A variable y is said to be *inversely proportional to the square of a variable* x if y is related to x by an equation of the form $y = k/x^2$, where k is a nonzero constant, called the *constant of proportionality*. This terminology is used in these exercises. ■

23. According to *Coulomb's law*, the force F of attraction between positive and negative point charges is inversely proportional to the square of the distance x between them.
 (a) Assuming that the force of attraction between two point charges is 0.0005 newton when the distance between them is 0.3 meter, find the constant of proportionality (with proper units).
 (b) Find the force of attraction between the point charges when they are 3 meters apart.
 (c) Make a graph of force versus distance for the two charges.

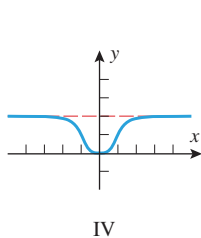
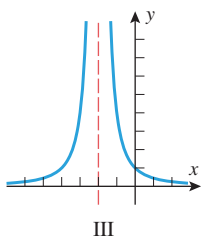
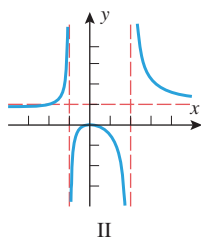
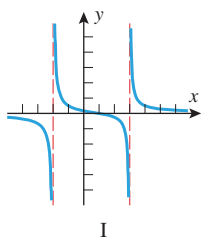
(cont.)

- (d) What happens to the force as the particles get closer and closer together? What happens as they get farther and farther apart?
24. It follows from Newton's Law of Universal Gravitation that the weight W of an object (relative to the Earth) is inversely proportional to the square of the distance x between the object and the center of the Earth, that is, $W = C/x^2$.
- (a) Assuming that a weather satellite weighs 2000 pounds on the surface of the Earth and that the Earth is a sphere of radius 4000 miles, find the constant C .
- (b) Find the weight of the satellite when it is 1000 miles above the surface of the Earth.
- (c) Make a graph of the satellite's weight versus its distance from the center of the Earth.
- (d) Is there any distance from the center of the Earth at which the weight of the satellite is zero? Explain your reasoning.
- 25–28 True–False** Determine whether the statement is true or false. Explain your answer. ■
25. Each curve in the family $y = 2x + b$ is parallel to the line $y = 2x$.
26. Each curve in the family $y = x^2 + bx + c$ is a translation of the graph of $y = x^2$.
27. If a curve passes through the point $(2, 6)$ and y is inversely proportional to x , then the constant of proportionality is 3.
28. Curves in the family $y = -5 \sin(A\pi x)$ have amplitude 5 and period $2/|A|$.

FOCUS ON CONCEPTS

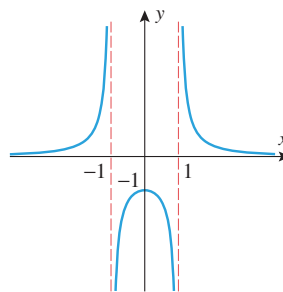
29. In each part, match the equation with one of the accompanying graphs, and give the equations for the horizontal and vertical asymptotes.

(a) $y = \frac{x^2}{x^2 - x - 2}$ (b) $y = \frac{x - 1}{x^2 - x - 6}$
 (c) $y = \frac{2x^4}{x^4 + 1}$ (d) $y = \frac{4}{(x + 2)^2}$



▲ Figure Ex-29

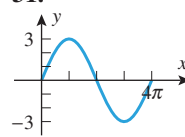
30. Find an equation of the form $y = k/(x^2 + bx + c)$ whose graph is a reasonable match to that in the accompanying figure. If you have a graphing utility, use it to check your work.



◀ Figure Ex-30

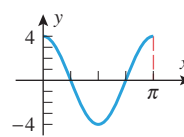
- 31–32** Find an equation of the form $y = D + A \sin Bx$ or $y = D + A \cos Bx$ for each graph. ■

31.



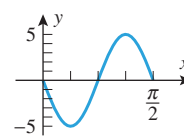
Not drawn to scale

(a)



Not drawn to scale

(b)

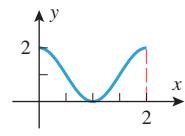


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(c)

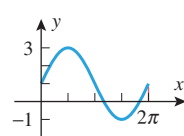
▲ Figure Ex-31

32.



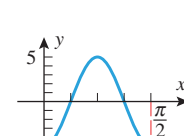
Not drawn to scale

(a)



Not drawn to scale

(b)

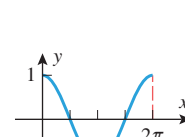


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(c)

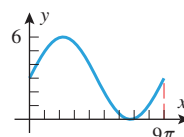
▲ Figure Ex-32

33. In each part, find an equation for the graph that has the form $y = y_0 + A \sin(Bx - C)$.



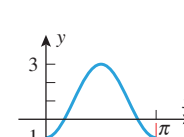
Not drawn to scale

(a)



Not drawn to scale

(b)



Not drawn to scale

(c)

▲ Figure Ex-33

34. In the United States, a standard electrical outlet supplies sinusoidal electrical current with a maximum voltage of $V = 120\sqrt{2}$ volts (V) at a frequency of 60 hertz (Hz). Write an equation that expresses V as a function of the time t , assuming that $V = 0$ if $t = 0$. [Note: 1 Hz = 1 cycle per second.]

 **35–36** Find the amplitude and period, and sketch at least two periods of the graph by hand. If you have a graphing utility, use it to check your work. ■

35. (a) $y = 3 \sin 4x$ (b) $y = -2 \cos \pi x$
 (c) $y = 2 + \cos\left(\frac{x}{2}\right)$

36. (a) $y = -1 - 4 \sin 2x$ (b) $y = \frac{1}{2} \cos(3x - \pi)$
 (c) $y = -4 \sin\left(\frac{x}{3} + 2\pi\right)$

 **37.** Equations of the form

$$x = A_1 \sin \omega t + A_2 \cos \omega t$$

arise in the study of vibrations and other periodic motion. Express the equation

$$x = \sqrt{2} \sin 2\pi t + \sqrt{6} \cos 2\pi t$$

in the form $x = A \sin(\omega t + \theta)$, and use a graphing utility to confirm that both equations have the same graph.

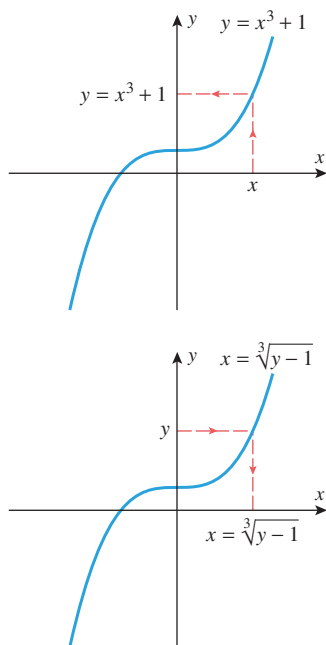
 **38.** Determine the number of solutions of $x = 2 \sin x$, and use a graphing or calculating utility to estimate them.

✓ QUICK CHECK ANSWERS 0.3

1. even; odd; negative 2. $(-\infty, +\infty)$ 3. (a) $[0, +\infty)$ (b) $(-\infty, +\infty)$ (c) $(0, +\infty)$ (d) $(-\infty, 0) \cup (0, +\infty)$ 4. (a) algebraic (b) polynomial (c) not algebraic (d) rational (e) rational 5. $|A|$; $2\pi/|B|$

0.4 INVERSE FUNCTIONS

*In everyday language the term “inversion” conveys the idea of a reversal. For example, in meteorology a temperature inversion is a reversal in the usual temperature properties of air layers, and in music a melodic inversion reverses an ascending interval to the corresponding descending interval. In mathematics the term **inverse** is used to describe functions that reverse one another in the sense that each undoes the effect of the other. In this section we discuss this fundamental mathematical idea.*



▲ Figure 0.4.1

■ INVERSE FUNCTIONS

The idea of solving an equation $y = f(x)$ for x as a function of y , say $x = g(y)$, is one of the most important ideas in mathematics. Sometimes, solving an equation is a simple process; for example, using basic algebra the equation

$$y = x^3 + 1 \quad \boxed{y = f(x)}$$

can be solved for x as a function of y :

$$x = \sqrt[3]{y - 1} \quad \boxed{x = g(y)}$$

The first equation is better for computing y if x is known, and the second is better for computing x if y is known (Figure 0.4.1).

Our primary interest in this section is to identify relationships that may exist between the functions f and g when an equation $y = f(x)$ is expressed as $x = g(y)$, or conversely. For example, consider the functions $f(x) = x^3 + 1$ and $g(y) = \sqrt[3]{y - 1}$ discussed above. When these functions are composed in either order, they cancel out the effect of one another in the sense that

$$\begin{aligned} g(f(x)) &= \sqrt[3]{f(x) - 1} = \sqrt[3]{(x^3 + 1) - 1} = x \\ f(g(y)) &= [g(y)]^3 + 1 = (\sqrt[3]{y - 1})^3 + 1 = y \end{aligned} \quad (1)$$

Pairs of functions with these two properties are so important that there is special terminology for them.

0.4.1 DEFINITION If the functions f and g satisfy the two conditions

$$g(f(x)) = x \text{ for every } x \text{ in the domain of } f$$

$$f(g(y)) = y \text{ for every } y \text{ in the domain of } g$$

then we say that f is an *inverse of* g and g is an *inverse of* f or that f and g are *inverse functions*.

WARNING

If f is a function, then the -1 in the symbol f^{-1} always denotes an inverse and *never* an exponent. That is,

$$f^{-1}(x) \text{ never means } \frac{1}{f(x)}$$

It can be shown (Exercise 35) that if a function f has an inverse, then that inverse is unique. Thus, if a function f has an inverse, then we are entitled to talk about “the” inverse of f , in which case we denote it by the symbol f^{-1} .

► **Example 1** The computations in (1) show that $g(y) = \sqrt[3]{y-1}$ is the inverse of $f(x) = x^3 + 1$. Thus, we can express g in inverse notation as

$$f^{-1}(y) = \sqrt[3]{y-1}$$

and we can express the equations in Definition 0.4.1 as

$$\begin{aligned} f^{-1}(f(x)) &= x && \text{for every } x \text{ in the domain of } f \\ f(f^{-1}(y)) &= y && \text{for every } y \text{ in the domain of } f^{-1} \end{aligned} \quad (2)$$

We will call these the *cancellation equations* for f and f^{-1} . ◀

CHANGING THE INDEPENDENT VARIABLE

The formulas in (2) use x as the independent variable for f and y as the independent variable for f^{-1} . Although it is often convenient to use different independent variables for f and f^{-1} , there will be occasions on which it is desirable to use the same independent variable for both. For example, if we want to graph the functions f and f^{-1} together in the same xy -coordinate system, then we would want to use x as the independent variable and y as the dependent variable for both functions. Thus, to graph the functions $f(x) = x^3 + 1$ and $f^{-1}(y) = \sqrt[3]{y-1}$ of Example 1 in the same xy -coordinate system, we would change the independent variable y to x , use y as the dependent variable for both functions, and graph the equations

$$y = x^3 + 1 \quad \text{and} \quad y = \sqrt[3]{x-1}$$

We will talk more about graphs of inverse functions later in this section, but for reference we give the following reformulation of the cancellation equations in (2) using x as the independent variable for both f and f^{-1} :

$$\begin{aligned} f^{-1}(f(x)) &= x && \text{for every } x \text{ in the domain of } f \\ f(f^{-1}(x)) &= x && \text{for every } x \text{ in the domain of } f^{-1} \end{aligned} \quad (3)$$

► **Example 2** Confirm each of the following.

(a) The inverse of $f(x) = 2x$ is $f^{-1}(x) = \frac{1}{2}x$.

(b) The inverse of $f(x) = x^3$ is $f^{-1}(x) = x^{1/3}$.

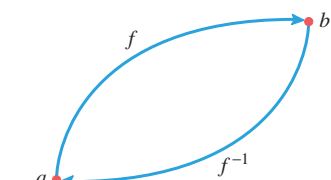
Solution (a).

$$f^{-1}(f(x)) = f^{-1}(2x) = \frac{1}{2}(2x) = x$$

$$f(f^{-1}(x)) = f\left(\frac{1}{2}x\right) = 2\left(\frac{1}{2}x\right) = x$$

The results in Example 2 should make sense to you intuitively, since the operations of multiplying by 2 and multiplying by $\frac{1}{2}$ in either order cancel the effect of one another, as do the operations of cubing and taking a cube root.

In general, if a function f has an inverse and $f(a) = b$, then the procedure in Example 3 shows that $a = f^{-1}(b)$; that is, f^{-1} maps each output of f back into the corresponding input (Figure 0.4.2).



▲ **Figure 0.4.2** If f maps a to b , then f^{-1} maps b back to a .

Solution (b).

$$f^{-1}(f(x)) = f^{-1}(x^3) = (x^3)^{1/3} = x$$

$$f(f^{-1}(x)) = f(x^{1/3}) = (x^{1/3})^3 = x \quad \blacktriangleleft$$

► **Example 3** Given that the function f has an inverse and that $f(3) = 5$, find $f^{-1}(5)$.

Solution. Apply f^{-1} to both sides of the equation $f(3) = 5$ to obtain

$$f^{-1}(f(3)) = f^{-1}(5)$$

and now apply the first equation in (3) to conclude that $f^{-1}(5) = 3$. ◀

■ DOMAIN AND RANGE OF INVERSE FUNCTIONS

The equations in (3) imply the following relationships between the domains and ranges of f and f^{-1} :

$$\begin{aligned} \text{domain of } f^{-1} &= \text{range of } f \\ \text{range of } f^{-1} &= \text{domain of } f \end{aligned} \quad (4)$$

One way to show that two sets are the same is to show that each is a subset of the other. Thus we can establish the first equality in (4) by showing that the domain of f^{-1} is a subset of the range of f and that the range of f is a subset of the domain of f^{-1} . We do this as follows: The first equation in (3) implies that f^{-1} is defined at $f(x)$ for all values of x in the domain of f , and this implies that the range of f is a subset of the domain of f^{-1} . Conversely, if x is in the domain of f^{-1} , then the second equation in (3) implies that x is in the range of f because it is the image of $f^{-1}(x)$. Thus, the domain of f^{-1} is a subset of the range of f . We leave the proof of the second equation in (4) as an exercise.

■ A METHOD FOR FINDING INVERSE FUNCTIONS

At the beginning of this section we observed that solving $y = f(x) = x^3 + 1$ for x as a function of y produces $x = f^{-1}(y) = \sqrt[3]{y-1}$. The following theorem shows that this is not accidental.

0.4.2 THEOREM If an equation $y = f(x)$ can be solved for x as a function of y , say $x = g(y)$, then f has an inverse and that inverse is $g(y) = f^{-1}(y)$.

PROOF Substituting $y = f(x)$ into $x = g(y)$ yields $x = g(f(x))$, which confirms the first equation in Definition 0.4.1, and substituting $x = g(y)$ into $y = f(x)$ yields $y = f(g(y))$, which confirms the second equation in Definition 0.4.1. ■

Theorem 0.4.2 provides us with the following procedure for finding the inverse of a function.

A Procedure for Finding the Inverse of a Function f

Step 1. Write down the equation $y = f(x)$.

Step 2. If possible, solve this equation for x as a function of y .

Step 3. The resulting equation will be $x = f^{-1}(y)$, which provides a formula for f^{-1} with y as the independent variable.

Step 4. If y is acceptable as the independent variable for the inverse function, then you are done, but if you want to have x as the independent variable, then you need to interchange x and y in the equation $x = f^{-1}(y)$ to obtain $y = f^{-1}(x)$.

An alternative way to obtain a formula for $f^{-1}(x)$ with x as the independent variable is to reverse the roles of x and y at the outset and solve the equation $x = f(y)$ for y as a function of x .

► **Example 4** Find a formula for the inverse of $f(x) = \sqrt{3x - 2}$ with x as the independent variable, and state the domain of f^{-1} .

Solution. Following the procedure stated above, we first write

$$y = \sqrt{3x - 2}$$

Then we solve this equation for x as a function of y :

$$\begin{aligned} y^2 &= 3x - 2 \\ x &= \frac{1}{3}(y^2 + 2) \end{aligned}$$

which tells us that

$$f^{-1}(y) = \frac{1}{3}(y^2 + 2) \quad (5)$$

Since we want x to be the independent variable, we reverse x and y in (5) to produce the formula

$$f^{-1}(x) = \frac{1}{3}(x^2 + 2) \quad (6)$$

We know from (4) that the domain of f^{-1} is the range of f . In general, this need not be the same as the natural domain of the formula for f^{-1} . Indeed, in this example the natural domain of (6) is $(-\infty, +\infty)$, whereas the range of $f(x) = \sqrt{3x - 2}$ is $[0, +\infty)$. Thus, if we want to make the domain of f^{-1} clear, we must express it explicitly by rewriting (6) as

$$f^{-1}(x) = \frac{1}{3}(x^2 + 2), \quad x \geq 0 \quad \blacktriangleleft$$

■ EXISTENCE OF INVERSE FUNCTIONS

The procedure we gave above for finding the inverse of a function f was based on solving the equation $y = f(x)$ for x as a function of y . This procedure can fail for two reasons—the function f may not have an inverse, or it may have an inverse but the equation $y = f(x)$ cannot be solved explicitly for x as a function of y . Thus, it is important to establish conditions that ensure the existence of an inverse, even if it cannot be found explicitly.

If a function f has an inverse, then it must assign distinct outputs to distinct inputs. For example, the function $f(x) = x^2$ cannot have an inverse because it assigns the same value to $x = 2$ and $x = -2$, namely,

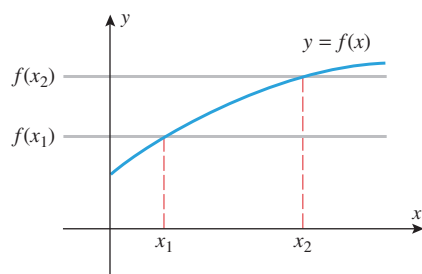
$$f(2) = f(-2) = 4$$

Thus, if $f(x) = x^2$ were to have an inverse, then the equation $f(2) = 4$ would imply that $f^{-1}(4) = 2$, and the equation $f(-2) = 4$ would imply that $f^{-1}(4) = -2$. But this is impossible because $f^{-1}(4)$ cannot have two different values. Another way to see that $f(x) = x^2$ has no inverse is to attempt to find the inverse by solving the equation $y = x^2$ for x as a function of y . We run into trouble immediately because the resulting equation $x = \pm\sqrt{y}$ does not express x as a *single* function of y .

A function that assigns distinct outputs to distinct inputs is said to be **one-to-one** or **invertible**, so we know from the preceding discussion that if a function f has an inverse, then it must be one-to-one. The converse is also true, thereby establishing the following theorem.

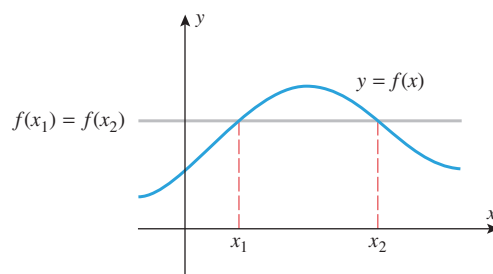
0.4.3 THEOREM A function has an inverse if and only if it is one-to-one.

Stated algebraically, a function f is one-to-one if and only if $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$; stated geometrically, a function f is one-to-one if and only if the graph of $y = f(x)$ is cut at most once by any horizontal line (Figure 0.4.3). The latter statement together with Theorem 0.4.3 provides the following geometric test for determining whether a function has an inverse.



One-to-one, since $f(x_1) \neq f(x_2)$
if $x_1 \neq x_2$

► Figure 0.4.3

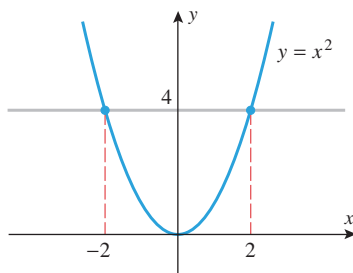


Not one-to-one, since
 $f(x_1) = f(x_2)$ and $x_1 \neq x_2$

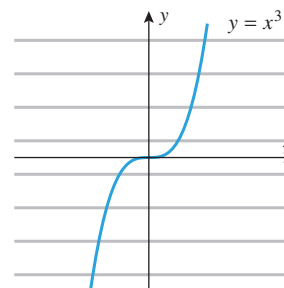
0.4.4 THEOREM (The Horizontal Line Test) A function has an inverse function if and only if its graph is cut at most once by any horizontal line.

► **Example 5** Use the horizontal line test to show that $f(x) = x^2$ has no inverse but that $f(x) = x^3$ does.

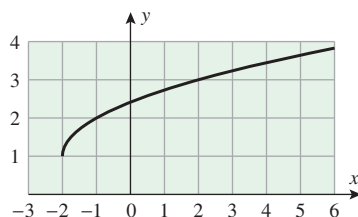
Solution. Figure 0.4.4 shows a horizontal line that cuts the graph of $y = x^2$ more than once, so $f(x) = x^2$ is not invertible. Figure 0.4.5 shows that the graph of $y = x^3$ is cut at most once by any horizontal line, so $f(x) = x^3$ is invertible. [Recall from Example 2 that the inverse of $f(x) = x^3$ is $f^{-1}(x) = x^{1/3}$.] ◀



▲ Figure 0.4.4



▲ Figure 0.4.5



▲ Figure 0.4.6

► **Example 6** Explain why the function f that is graphed in Figure 0.4.6 has an inverse, and find $f^{-1}(3)$.

Solution. The function f has an inverse since its graph passes the horizontal line test. To evaluate $f^{-1}(3)$, we view $f^{-1}(3)$ as that number x for which $f(x) = 3$. From the graph we see that $f(2) = 3$, so $f^{-1}(3) = 2$. ◀

INCREASING OR DECREASING FUNCTIONS ARE INVERTIBLE

A function whose graph is always rising as it is traversed from left to right is said to be an **increasing function**, and a function whose graph is always falling as it is traversed from left to right is said to be a **decreasing function**. If x_1 and x_2 are points in the domain of a function f , then f is increasing if

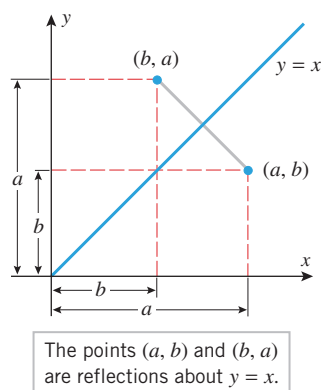
$$f(x_1) < f(x_2) \quad \text{whenever } x_1 < x_2$$

The function $f(x) = x^3$ in Figure 0.4.5 is an example of an increasing function. Give an example of a decreasing function and compute its inverse.

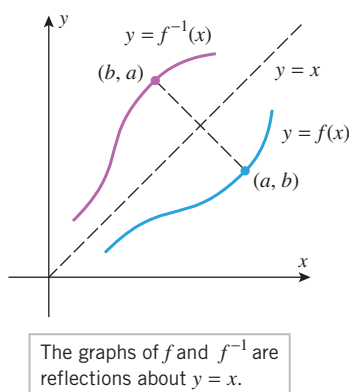
and f is decreasing if

$$f(x_1) > f(x_2) \quad \text{whenever } x_1 < x_2$$

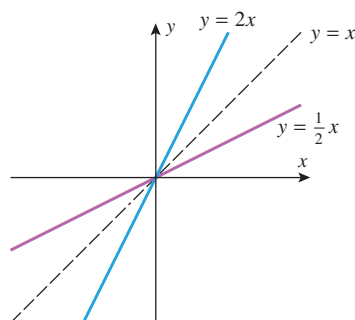
(Figure 0.4.7). It is evident geometrically that increasing and decreasing functions pass the horizontal line test and hence are invertible.



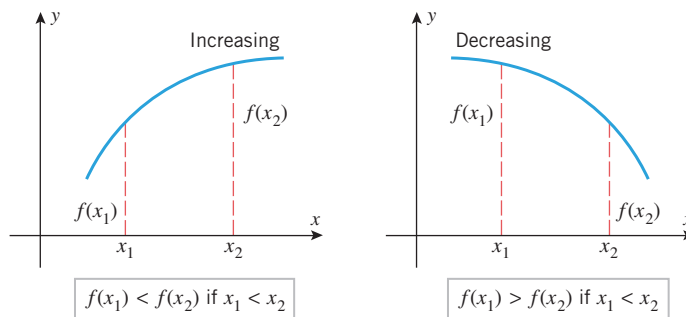
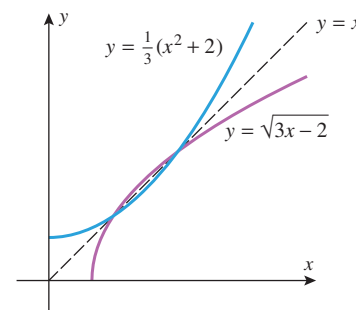
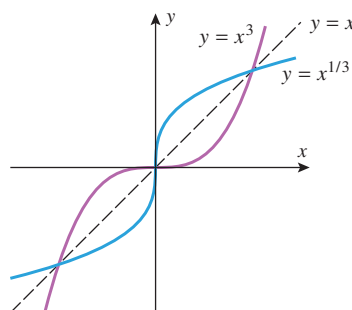
▲ Figure 0.4.8



▲ Figure 0.4.9



▲ Figure 0.4.10



► Figure 0.4.7

■ GRAPHS OF INVERSE FUNCTIONS

Our next objective is to explore the relationship between the graphs of f and f^{-1} . For this purpose, it will be desirable to use x as the independent variable for both functions so we can compare the graphs of $y = f(x)$ and $y = f^{-1}(x)$.

If (a, b) is a point on the graph $y = f(x)$, then $b = f(a)$. This is equivalent to the statement that $a = f^{-1}(b)$, which means that (b, a) is a point on the graph of $y = f^{-1}(x)$. In short, reversing the coordinates of a point on the graph of f produces a point on the graph of f^{-1} . Similarly, reversing the coordinates of a point on the graph of f^{-1} produces a point on the graph of f (verify). However, the geometric effect of reversing the coordinates of a point is to reflect that point about the line $y = x$ (Figure 0.4.8), and hence the graphs of $y = f(x)$ and $y = f^{-1}(x)$ are reflections of one another about this line (Figure 0.4.9). In summary, we have the following result.

0.4.5 THEOREM If f has an inverse, then the graphs of $y = f(x)$ and $y = f^{-1}(x)$ are reflections of one another about the line $y = x$; that is, each graph is the mirror image of the other with respect to that line.

► **Example 7** Figure 0.4.10 shows the graphs of the inverse functions discussed in Examples 2 and 4. ◀

■ RESTRICTING DOMAINS FOR INVERTIBILITY

If a function g is obtained from a function f by placing restrictions on the domain of f , then g is called a **restriction** of f . Thus, for example, the function

$$g(x) = x^3, \quad x \geq 0$$

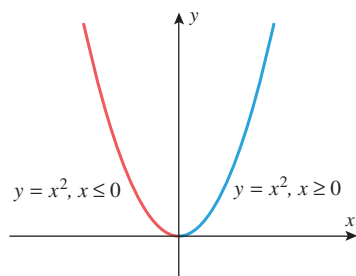
is a restriction of the function $f(x) = x^3$. More precisely, it is called the restriction of x^3 to the interval $[0, +\infty)$.

Sometimes it is possible to create an invertible function from a function that is not invertible by restricting the domain appropriately. For example, we showed earlier that $f(x) = x^2$ is not invertible. However, consider the restricted functions

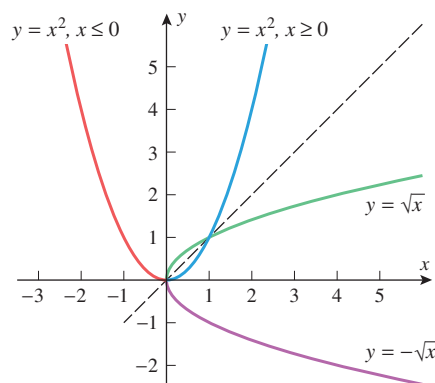
$$f_1(x) = x^2, \quad x \geq 0 \quad \text{and} \quad f_2(x) = x^2, \quad x \leq 0$$

the union of whose graphs is the complete graph of $f(x) = x^2$ (Figure 0.4.11). These restricted functions are each one-to-one (hence invertible), since their graphs pass the horizontal line test. As illustrated in Figure 0.4.12, their inverses are

$$f_1^{-1}(x) = \sqrt{x} \quad \text{and} \quad f_2^{-1}(x) = -\sqrt{x}$$



▲ Figure 0.4.11



▲ Figure 0.4.12

✓ QUICK CHECK EXERCISES 0.4 (See page 46 for answers.)

- In each part, determine whether the function f is one-to-one.
 - $f(t)$ is the number of people in line at a movie theater at time t .
 - $f(x)$ is the measured high temperature (rounded to the nearest $^{\circ}\text{F}$) in a city on the x th day of the year.
 - $f(v)$ is the weight of v cubic inches of lead.
- A student enters a number on a calculator, doubles it, adds 8 to the result, divides the sum by 2, subtracts 3 from the quotient, and then cubes the difference. If the resulting number is x , then _____ was the student's original number.
- If $(3, -2)$ is a point on the graph of an odd invertible function f , then _____ and _____ are points on the graph of f^{-1} .

EXERCISE SET 0.4 Graphing Utility

- In (a)–(d), determine whether f and g are inverse functions.
 - $f(x) = 4x$, $g(x) = \frac{1}{4}x$
 - $f(x) = 3x + 1$, $g(x) = 3x - 1$
 - $f(x) = \sqrt[3]{x-2}$, $g(x) = x^3 + 2$
 - $f(x) = x^4$, $g(x) = \sqrt[4]{x}$
-  Check your answers to Exercise 1 with a graphing utility by determining whether the graphs of f and g are reflections of one another about the line $y = x$.
- In each part, use the horizontal line test to determine whether the function f is one-to-one.
 - $f(x) = 3x + 2$
 - $f(x) = \sqrt{x-1}$
 - $f(x) = |x|$
 - $f(x) = x^3$
 - $f(x) = x^2 - 2x + 2$
 - $f(x) = \sin x$

4. In each part, generate the graph of the function f with a graphing utility, and determine whether f is one-to-one.
- (a) $f(x) = x^3 - 3x + 2$ (b) $f(x) = x^3 - 3x^2 + 3x - 1$

FOCUS ON CONCEPTS

5. In each part, determine whether the function f defined by the table is one-to-one.

(a)

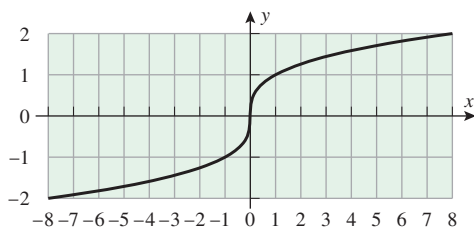
x	1	2	3	4	5	6
$f(x)$	-2	-1	0	1	2	3

(b)

x	1	2	3	4	5	6
$f(x)$	4	-7	6	-3	1	4

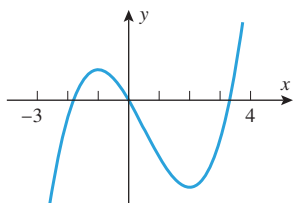
6. A face of a broken clock lies in the xy -plane with the center of the clock at the origin and 3:00 in the direction of the positive x -axis. When the clock broke, the tip of the hour hand stopped on the graph of $y = f(x)$, where f is a function that satisfies $f(0) = 0$.

- (a) Are there any times of the day that cannot appear in such a configuration? Explain.
- (b) How does your answer to part (a) change if f must be an invertible function?
- (c) How do your answers to parts (a) and (b) change if it was the tip of the minute hand that stopped on the graph of f ?
7. (a) The accompanying figure shows the graph of a function f over its domain $-8 \leq x \leq 8$. Explain why f has an inverse, and use the graph to find $f^{-1}(2)$, $f^{-1}(-1)$, and $f^{-1}(0)$.
- (b) Find the domain and range of f^{-1} .
- (c) Sketch the graph of f^{-1} .



▲ Figure Ex-7

8. (a) Explain why the function f graphed in the accompanying figure has no inverse function on its domain $-3 \leq x \leq 4$.
- (b) Subdivide the domain into three adjacent intervals on each of which the function f has an inverse.



◀ Figure Ex-8

- 9–16 Find a formula for $f^{-1}(x)$. ■

9. $f(x) = 7x - 6$ 10. $f(x) = \frac{x+1}{x-1}$
11. $f(x) = 3x^3 - 5$ 12. $f(x) = \sqrt[5]{4x+2}$
13. $f(x) = 3/x^2, \quad x < 0$ 14. $f(x) = 5/(x^2 + 1), \quad x \geq 0$
15. $f(x) = \begin{cases} 5/2 - x, & x < 2 \\ 1/x, & x \geq 2 \end{cases}$
16. $f(x) = \begin{cases} 2x, & x \leq 0 \\ x^2, & x > 0 \end{cases}$

- 17–20 Find a formula for $f^{-1}(x)$, and state the domain of the function f^{-1} . ■

17. $f(x) = (x+2)^4, \quad x \geq 0$
18. $f(x) = \sqrt{x+3}$ 19. $f(x) = -\sqrt{3-2x}$
20. $f(x) = x - 5x^2, \quad x \geq 1$

- 21–24 True-False Determine whether the statement is true or false. Explain your answer. ■

21. If f is an invertible function such that $f(2) = 2$, then $f^{-1}(2) = \frac{1}{2}$.
22. If f and g are inverse functions, then f and g have the same domain.
23. A one-to-one function is invertible.
24. If the graph of a function f is symmetric about the line $y = x$, then f is invertible.
25. Let $f(x) = ax^2 + bx + c, a > 0$. Find f^{-1} if the domain of f is restricted to
- (a) $x \geq -b/(2a)$ (b) $x \leq -b/(2a)$.

FOCUS ON CONCEPTS

26. The formula $F = \frac{9}{5}C + 32$, where $C \geq -273.15$ expresses the Fahrenheit temperature F as a function of the Celsius temperature C .
- (a) Find a formula for the inverse function.
- (b) In words, what does the inverse function tell you?
- (c) Find the domain and range of the inverse function.
27. (a) One meter is about 6.214×10^{-4} miles. Find a formula $y = f(x)$ that expresses a length y in meters as a function of the same length x in miles.
- (b) Find a formula for the inverse of f .
- (c) Describe what the formula $x = f^{-1}(y)$ tells you in practical terms.
28. Let $f(x) = x^2, x > 1$, and $g(x) = \sqrt{x}$.
- (a) Show that $f(g(x)) = x, x > 1$, and $g(f(x)) = x, x > 1$.
- (b) Show that f and g are *not* inverses by showing that the graphs of $y = f(x)$ and $y = g(x)$ are not reflections of one another about $y = x$.
- (c) Do parts (a) and (b) contradict one another? Explain.

29. (a) Show that $f(x) = (3 - x)/(1 - x)$ is its own inverse.
 (b) What does the result in part (a) tell you about the graph of f ?
 30. Sketch the graph of a function that is one-to-one on $(-\infty, +\infty)$, yet not increasing on $(-\infty, +\infty)$ and not decreasing on $(-\infty, +\infty)$.

31. Let $f(x) = 2x^3 + 5x + 3$. Find x if $f^{-1}(x) = 1$.

32. Let $f(x) = \frac{x^3}{x^2 + 1}$. Find x if $f^{-1}(x) = 2$.

33. Prove that if $a^2 + bc \neq 0$, then the graph of

$$f(x) = \frac{ax + b}{cx - a}$$

is symmetric about the line $y = x$.

34. (a) Prove: If f and g are one-to-one, then so is the composition $f \circ g$.

(b) Prove: If f and g are one-to-one, then

$$(f \circ g)^{-1} = g^{-1} \circ f^{-1}$$

35. Prove: A one-to-one function f cannot have two different inverses.

✓ QUICK CHECK ANSWERS 0.4

1. (a) not one-to-one (b) not one-to-one (c) one-to-one 2. $\sqrt[3]{x} - 1$ 3. $(-2, 3); (2, -3)$

CHAPTER 0 REVIEW EXERCISES

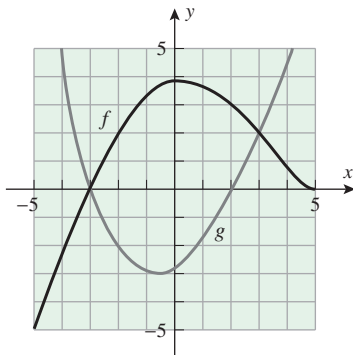


1. Sketch the graph of the function

$$f(x) = \begin{cases} -1, & x \leq -5 \\ \sqrt{25 - x^2}, & -5 < x < 5 \\ x - 5, & x \geq 5 \end{cases}$$

2. Use the graphs of the functions f and g in the accompanying figure to solve the following problems.

- (a) Find the values of $f(-2)$ and $g(3)$.
 (b) For what values of x is $f(x) = g(x)$?
 (c) For what values of x is $f(x) < 2$?
 (d) What are the domain and range of f ?
 (e) What are the domain and range of g ?
 (f) Find the zeros of f and g .



◀ Figure Ex-2

3. A glass filled with water that has a temperature of 40°F is placed in a room in which the temperature is a constant 70°F . Sketch a rough graph that reasonably describes the temperature of the water in the glass as a function of the elapsed time.
 4. You want to paint the top of a circular table. Find a formula that expresses the amount of paint required as a function

of the radius, and discuss all of the assumptions you have made in finding the formula.

5. A rectangular storage container with an open top and a square base has a volume of 8 cubic meters. Material for the base costs \$5 per square meter and material for the sides \$2 per square meter.

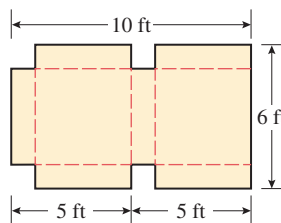
- (a) Find a formula that expresses the total cost of materials as a function of the length of a side of the base.
 (b) What is the domain of the cost function obtained in part (a)?

6. A ball of radius 3 inches is coated uniformly with plastic.

- (a) Express the volume of the plastic as a function of its thickness.
 (b) What is the domain of the volume function obtained in part (a)?

7. A box with a closed top is to be made from a 6 ft by 10 ft piece of cardboard by cutting out four squares of equal size (see the accompanying figure), folding along the dashed lines, and tucking the two extra flaps inside.

- (a) Find a formula that expresses the volume of the box as a function of the length of the sides of the cut-out squares.
 (b) Find an inequality that specifies the domain of the function in part (a).
 (c) Use the graph of the volume function to estimate the dimensions of the box of largest volume.



◀ Figure Ex-7

8. Let C denote the graph of $y = 1/x$, $x > 0$.
- Express the distance between the point $P(1, 0)$ and a point Q on C as a function of the x -coordinate of Q .
 - What is the domain of the distance function obtained in part (a)?
 - Use the graph of the distance function obtained in part (a) to estimate the point Q on C that is closest to the point P .
9. Sketch the graph of the equation $x^2 - 4y^2 = 0$.
10. Generate the graph of $f(x) = x^4 - 24x^3 - 25x^2$ in two different viewing windows, each of which illustrates a different property of f . Identify each viewing window and a characteristic of the graph of f that is illustrated well in the window.
11. Complete the following table.

x	-4	-3	-2	-1	0	1	2	3	4
$f(x)$	0	-1	2	1	3	-2	-3	4	-4
$g(x)$	3	2	1	-3	-1	-4	4	-2	0
$(f \circ g)(x)$									
$(g \circ f)(x)$									

▲ Table Ex-11

12. Let $f(x) = -x^2$ and $g(x) = 1/\sqrt{x}$. Find formulas for $f \circ g$ and $g \circ f$ and state the domain of each composition.
13. Given that $f(x) = x^2 + 1$ and $g(x) = 3x + 2$, find all values of x such that $f(g(x)) = g(f(x))$.
14. Let $f(x) = (2x - 1)/(x + 1)$ and $g(x) = 1/(x - 1)$.
- Find $f(g(x))$.
 - Is the natural domain of the function $h(x) = (3 - x)/x$ the same as the domain of $f \circ g$? Explain.
15. Given that

$$f(x) = \frac{x}{x-1}, \quad g(x) = \frac{1}{x}, \quad h(x) = x^2 - 1$$

find a formula for $f \circ g \circ h$ and state the domain of this composition.

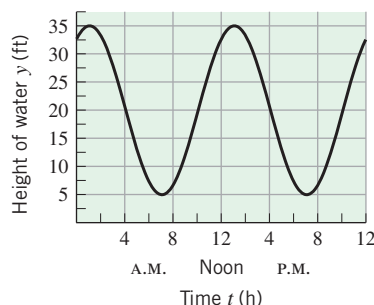
16. Given that $f(x) = 2x + 1$ and $h(x) = 2x^2 + 4x + 1$, find a function g such that $f(g(x)) = h(x)$.
17. In each part, classify the function as even, odd, or neither.
- $x^2 \sin x$ (b) $\sin^2 x$ (c) $x + x^2$ (d) $\sin x \tan x$
18. (a) Write an equation for the graph that is obtained by reflecting the graph of $y = |x - 1|$ about the y -axis, then stretching that graph vertically by a factor of 2, then translating that graph down 3 units, and then reflecting that graph about the x -axis.
- (b) Sketch the original graph and the final graph.
19. In each part, describe the family of curves.
- $(x - a)^2 + (y - a^2)^2 = 1$
 - $y = a + (x - 2a)^2$

20. Find an equation for a parabola that passes through the points $(2, 0)$, $(8, 18)$, and $(-8, 18)$.
21. Suppose that the expected low temperature in Anchorage, Alaska (in $^{\circ}\text{F}$), is modeled by the equation

$$T = 50 \sin \frac{2\pi}{365}(t - 101) + 25$$

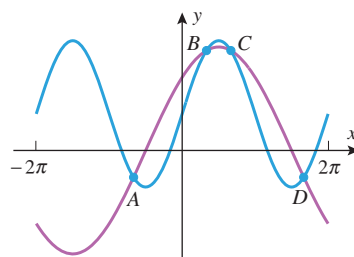
where t is in days and $t = 0$ corresponds to January 1.

- Sketch the graph of T versus t for $0 \leq t \leq 365$.
 - Use the model to predict when the coldest day of the year will occur.
 - Based on this model, how many days during the year would you expect the temperature to be below 0°F ?
22. The accompanying figure shows a model for the tide variation in an inlet to San Francisco Bay during a 24-hour period. Find an equation of the form $y = y_0 + y_1 \sin(at + b)$ for the model, assuming that $t = 0$ corresponds to midnight.



◀ Figure Ex-22

23. The accompanying figure shows the graphs of the equations $y = 1 + 2 \sin x$ and $y = 2 \sin(x/2) + 2 \cos(x/2)$ for $-2\pi \leq x \leq 2\pi$. Without the aid of a calculator, label each curve by its equation, and find the coordinates of the points A , B , C , and D . Explain your reasoning.



◀ Figure Ex-23

24. The electrical resistance R in ohms (Ω) for a pure metal wire is related to its temperature T in $^{\circ}\text{C}$ by the formula

$$R = R_0(1 + kT)$$

in which R_0 and k are positive constants.

- Make a hand-drawn sketch of the graph of R versus T , and explain the geometric significance of R_0 and k for your graph.
- In theory, the resistance R of a pure metal wire drops to zero when the temperature reaches absolute zero ($T = -273^{\circ}\text{C}$). What information does this give you about k ?

(cont.)

1



Joe McBride/Stone/Getty Images

LIMITS AND CONTINUITY

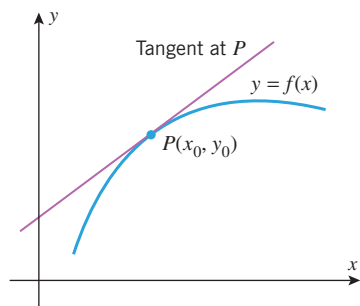
Air resistance prevents the velocity of a skydiver from increasing indefinitely. The velocity approaches a limit, called the “terminal velocity.”

The development of calculus in the seventeenth century by Newton and Leibniz provided scientists with their first real understanding of what is meant by an “instantaneous rate of change” such as velocity and acceleration. Once the idea was understood conceptually, efficient computational methods followed, and science took a quantum leap forward. The fundamental building block on which rates of change rest is the concept of a “limit,” an idea that is so important that all other calculus concepts are now based on it.

In this chapter we will develop the concept of a limit in stages, proceeding from an informal, intuitive notion to a precise mathematical definition. We will also develop theorems and procedures for calculating limits, and we will conclude the chapter by using the limits to study “continuous” curves.

1.1 LIMITS (AN INTUITIVE APPROACH)

The concept of a “limit” is the fundamental building block on which all calculus concepts are based. In this section we will study limits informally, with the goal of developing an intuitive feel for the basic ideas. In the next three sections we will focus on computational methods and precise definitions.



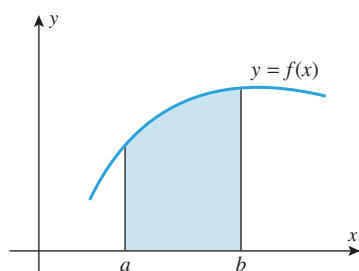
▲ Figure 1.1.1

Many of the ideas of calculus originated with the following two geometric problems:

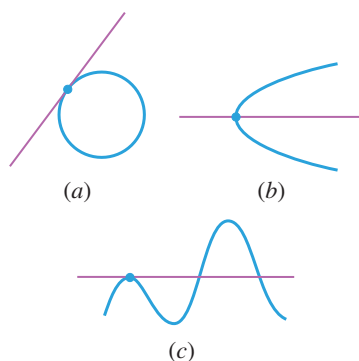
THE TANGENT LINE PROBLEM Given a function f and a point $P(x_0, y_0)$ on its graph, find an equation of the line that is tangent to the graph at P (Figure 1.1.1).

THE AREA PROBLEM Given a function f , find the area between the graph of f and an interval $[a, b]$ on the x -axis (Figure 1.1.2).

Traditionally, that portion of calculus arising from the tangent line problem is called *differential calculus* and that arising from the area problem is called *integral calculus*. However, we will see later that the tangent line and area problems are so closely related that the distinction between differential and integral calculus is somewhat artificial.



▲ Figure 1.1.2

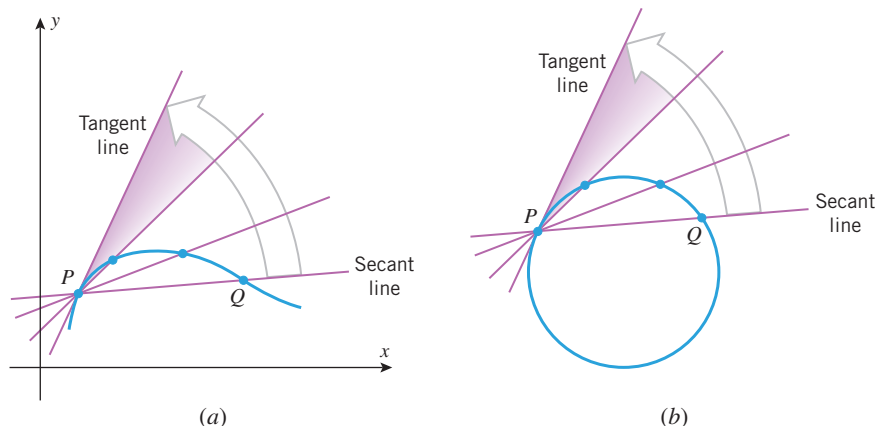


▲ Figure 1.1.3

TANGENT LINES AND LIMITS

In plane geometry, a line is called *tangent* to a circle if it meets the circle at precisely one point (Figure 1.1.3a). Although this definition is adequate for circles, it is not appropriate for more general curves. For example, in Figure 1.1.3b, the line meets the curve exactly once but is obviously not what we would regard to be a tangent line; and in Figure 1.1.3c, the line appears to be tangent to the curve, yet it intersects the curve more than once.

To obtain a definition of a tangent line that applies to curves other than circles, we must view tangent lines another way. For this purpose, suppose that we are interested in the tangent line at a point P on a curve in the xy -plane and that Q is any point that lies on the curve and is different from P . The line through P and Q is called a *secant line* for the curve at P . Intuition suggests that if we move the point Q along the curve toward P , then the secant line will rotate toward a *limiting position*. The line in this limiting position is what we will consider to be the *tangent line* at P (Figure 1.1.4a). As suggested by Figure 1.1.4b, this new concept of a tangent line coincides with the traditional concept when applied to circles.



► Figure 1.1.4

► **Example 1** Find an equation for the tangent line to the parabola $y = x^2$ at the point $P(1, 1)$.

Solution. If we can find the slope $m_{\tan}$ of the tangent line at P , then we can use the point P and the point-slope formula for a line (Web Appendix G) to write the equation of the tangent line as

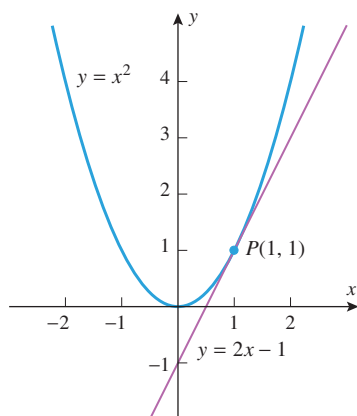
$$y - 1 = m_{\tan}(x - 1) \quad (1)$$

To find the slope $m_{\tan}$, consider the secant line through P and a point $Q(x, x^2)$ on the parabola that is distinct from P . The slope $m_{\sec}$ of this secant line is

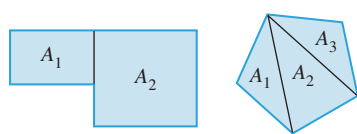
$$m_{\sec} = \frac{x^2 - 1}{x - 1} \quad (2)$$

Why are we requiring that P and Q be distinct?

Figure 1.1.4a suggests that if we now let Q move along the parabola, getting closer and closer to P , then the limiting position of the secant line through P and Q will coincide with that of the tangent line at P . This in turn suggests that the value of $m_{\sec}$ will get closer and closer to the value of $m_{\tan}$ as P moves toward Q along the curve. However, to say that $Q(x, x^2)$ gets closer and closer to $P(1, 1)$ is algebraically equivalent to saying that x gets closer and closer to 1. Thus, the problem of finding $m_{\tan}$ reduces to finding the “limiting value” of $m_{\sec}$ in Formula (2) as x gets closer and closer to 1 (but with $x \neq 1$ to ensure that P and Q remain distinct).



▲ Figure 1.1.5



▲ Figure 1.1.6

We can rewrite (2) as

$$m_{\text{sec}} = \frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{(x - 1)} = x + 1$$

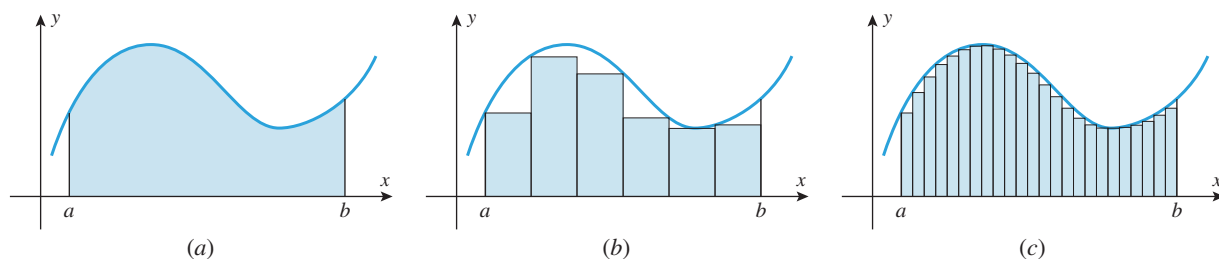
where the cancellation of the factor $(x - 1)$ is allowed because $x \neq 1$. It is now evident that m_{sec} gets closer and closer to 2 as x gets closer and closer to 1. Thus, $m_{\text{tan}} = 2$ and (1) implies that the equation of the tangent line is

$$y - 1 = 2(x - 1) \quad \text{or equivalently} \quad y = 2x - 1$$

Figure 1.1.5 shows the graph of $y = x^2$ and this tangent line. ◀

■ AREAS AND LIMITS

Just as the general notion of a tangent line leads to the concept of *limit*, so does the general notion of area. For plane regions with straight-line boundaries, areas can often be calculated by subdividing the region into rectangles or triangles and adding the areas of the constituent parts (Figure 1.1.6). However, for regions with curved boundaries, such as that in Figure 1.1.7a, a more general approach is needed. One such approach is to begin by approximating the area of the region by inscribing a number of rectangles of equal width under the curve and adding the areas of these rectangles (Figure 1.1.7b). Intuition suggests that if we repeat that approximation process using more and more rectangles, then the rectangles will tend to fill in the gaps under the curve, and the approximations will get closer and closer to the exact area under the curve (Figure 1.1.7c). This suggests that we can define the area under the curve to be the limiting value of these approximations. This idea will be considered in detail later, but the point to note here is that once again the concept of a limit comes into play.



▲ Figure 1.1.7

■ DECIMALS AND LIMITS

Limits also arise in the familiar context of decimals. For example, the decimal expansion of the fraction $\frac{1}{3}$ is

$$\frac{1}{3} = 0.3333 \dots \quad (3)$$

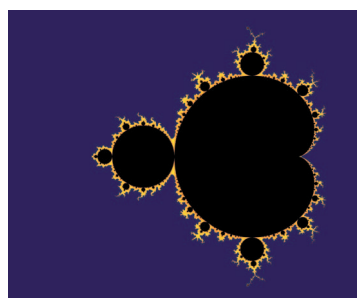
in which the dots indicate that the digit 3 repeats indefinitely. Although you may not have thought about decimals in this way, we can write (3) as

$$\frac{1}{3} = 0.3333 \dots = 0.3 + 0.03 + 0.003 + 0.0003 + 0.00003 + \dots \quad (4)$$

which is a sum with “infinitely many” terms. As we will discuss in more detail later, we interpret (4) to mean that the succession of finite sums

$$0.3, \quad 0.3 + 0.03, \quad 0.3 + 0.03 + 0.003, \quad 0.3 + 0.03 + 0.003 + 0.0003, \dots$$

gets closer and closer to a limiting value of $\frac{1}{3}$ as more and more terms are included. Thus, limits even occur in the familiar context of decimal representations of real numbers.



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This figure shows a region called the **Mandelbrot Set**. It illustrates how complicated a region in the plane can be and why the notion of area requires careful definition.

LIMITS

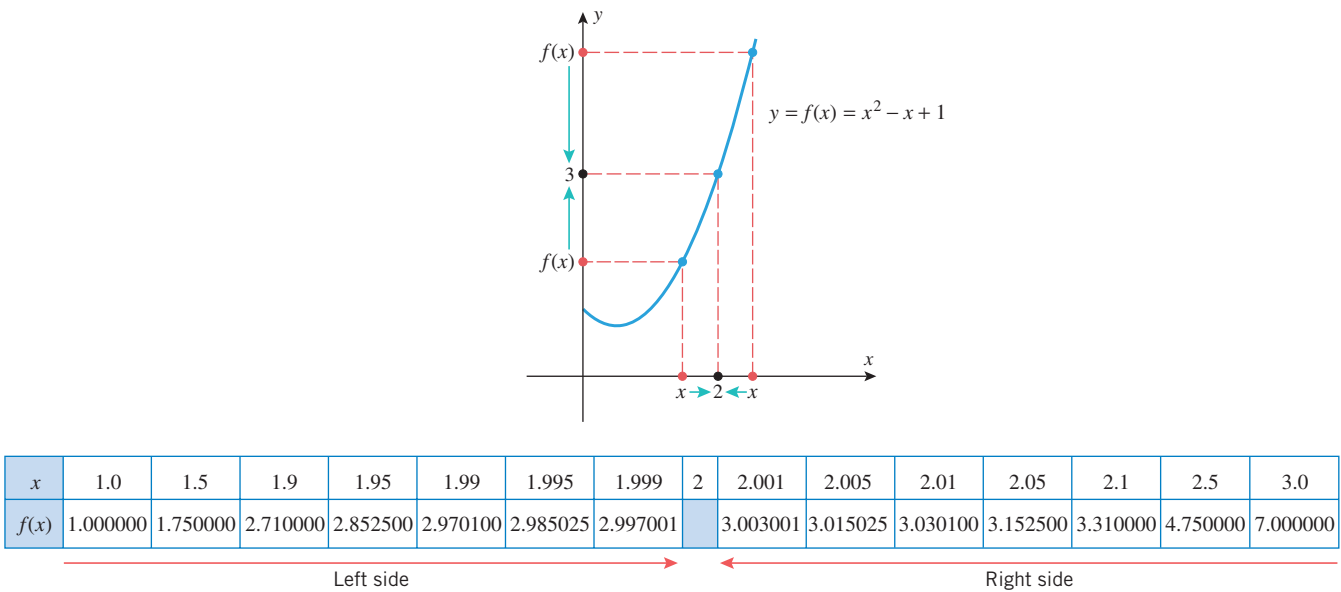
Now that we have seen how limits arise in various ways, let us focus on the limit concept itself.

The most basic use of limits is to describe how a function behaves as the independent variable approaches a given value. For example, let us examine the behavior of the function

$$f(x) = x^2 - x + 1$$

for x -values closer and closer to 2. It is evident from the graph and table in Figure 1.1.8 that the values of $f(x)$ get closer and closer to 3 as values of x are selected closer and closer to 2 on either the left or the right side of 2. We describe this by saying that the “limit of $x^2 - x + 1$ is 3 as x approaches 2 from either side,” and we write

$$\lim_{x \rightarrow 2} (x^2 - x + 1) = 3 \tag{5}$$



▲ Figure 1.1.8

This leads us to the following general idea.

1.1.1 LIMITS (AN INFORMAL VIEW) If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (but not equal to a), then we write

$$\lim_{x \rightarrow a} f(x) = L \tag{6}$$

which is read “the limit of $f(x)$ as x approaches a is L ” or “ $f(x)$ approaches L as x approaches a .” The expression in (6) can also be written as

$$f(x) \rightarrow L \quad \text{as} \quad x \rightarrow a \tag{7}$$

Since x is required to be different from a in (6), the value of f at a , or even whether f is defined at a , has no bearing on the limit L . The limit describes the behavior of f close to a but not at a .

► **Example 2** Use numerical evidence to make a conjecture about the value of

$$\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1} \quad (8)$$

Solution. Although the function

$$f(x) = \frac{x-1}{\sqrt{x}-1} \quad (9)$$

is undefined at $x = 1$, this has no bearing on the limit. Table 1.1.1 shows sample x -values approaching 1 from the left side and from the right side. In both cases the corresponding values of $f(x)$, calculated to six decimal places, appear to get closer and closer to 2, and hence we conjecture that

$$\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1} = 2$$

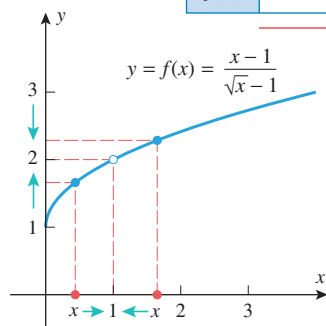
This is consistent with the graph of f shown in Figure 1.1.9. In the next section we will show how to obtain this result algebraically. ◀

TECHNOLOGY MASTERY

Use a graphing utility to generate the graph of the equation $y = f(x)$ for the function in (9). Find a window containing $x = 1$ in which all values of $f(x)$ are within 0.5 of $y = 2$ and one in which all values of $f(x)$ are within 0.1 of $y = 2$.

Table 1.1.1

x	0.99	0.999	0.9999	0.99999		1.00001	1.0001	1.001	1.01
$f(x)$	1.994987	1.999500	1.999950	1.999995		2.000005	2.000050	2.000500	2.004988



▲ Figure 1.1.9

Use numerical evidence to determine whether the limit in (11) changes if x is measured in degrees.

► **Example 3** Use numerical evidence to make a conjecture about the value of

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} \quad (10)$$

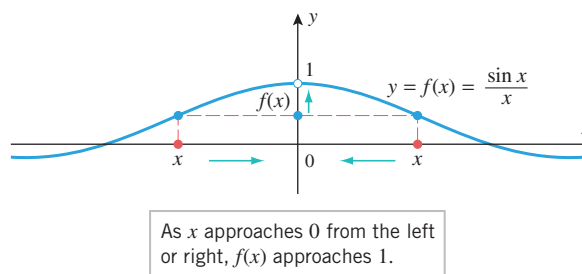
Solution. With the help of a calculating utility set in radian mode, we obtain Table 1.1.2. The data in the table suggest that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad (11)$$

The result is consistent with the graph of $f(x) = (\sin x)/x$ shown in Figure 1.1.10. Later in this chapter we will give a geometric argument to prove that our conjecture is correct. ◀

Table 1.1.2

x (RADIANs)	$y = \frac{\sin x}{x}$
± 1.0	0.84147
± 0.9	0.87036
± 0.8	0.89670
± 0.7	0.92031
± 0.6	0.94107
± 0.5	0.95885
± 0.4	0.97355
± 0.3	0.98507
± 0.2	0.99335
± 0.1	0.99833
± 0.01	0.99998



► Figure 1.1.10

SAMPLING PITFALLS

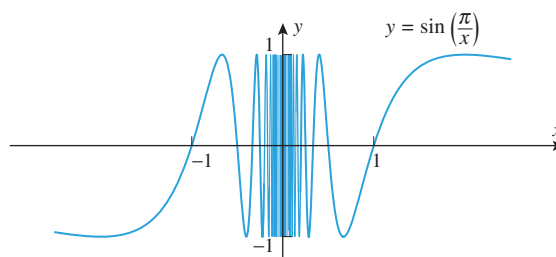
Numerical evidence can sometimes lead to incorrect conclusions about limits because of roundoff error or because the sample values chosen do not reveal the true limiting behavior. For example, one might *incorrectly* conclude from Table 1.1.3 that

$$\lim_{x \rightarrow 0} \sin\left(\frac{\pi}{x}\right) = 0$$

The fact that this is not correct is evidenced by the graph of f in Figure 1.1.11. The graph reveals that the values of f oscillate between -1 and 1 with increasing rapidity as $x \rightarrow 0$ and hence do not approach a limit. The data in the table deceived us because the x -values selected all happened to be x -intercepts for $f(x)$. This points out the need for having alternative methods for corroborating limits conjectured from numerical evidence.

Table 1.1.3

x	$\frac{\pi}{x}$	$f(x) = \sin\left(\frac{\pi}{x}\right)$
$x = \pm 1$	$\pm\pi$	$\sin(\pm\pi) = 0$
$x = \pm 0.1$	$\pm 10\pi$	$\sin(\pm 10\pi) = 0$
$x = \pm 0.01$	$\pm 100\pi$	$\sin(\pm 100\pi) = 0$
$x = \pm 0.001$	$\pm 1000\pi$	$\sin(\pm 1000\pi) = 0$
$x = \pm 0.0001$	$\pm 10,000\pi$	$\sin(\pm 10,000\pi) = 0$
$\vdots$	$\vdots$	$\vdots$



▲ Figure 1.1.11

ONE-SIDED LIMITS

The limit in (6) is called a **two-sided limit** because it requires the values of $f(x)$ to get closer and closer to L as values of x are taken from *either* side of $x = a$. However, some functions exhibit different behaviors on the two sides of an x -value a , in which case it is necessary to distinguish whether values of x near a are on the left side or on the right side of a for purposes of investigating limiting behavior. For example, consider the function

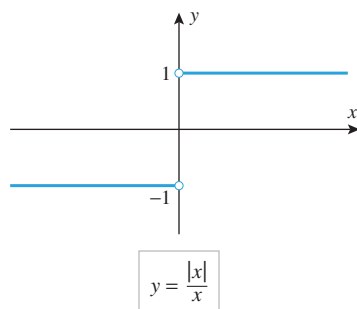
$$f(x) = \frac{|x|}{x} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases} \quad (12)$$

which is graphed in Figure 1.1.12. As x approaches 0 from the *right*, the values of $f(x)$ approach a limit of 1 [in fact, the values of $f(x)$ are exactly 1 for all such x], and similarly, as x approaches 0 from the *left*, the values of $f(x)$ approach a limit of -1 . We denote these limits by writing

$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1 \quad \text{and} \quad \lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1 \quad (13)$$

With this notation, the superscript “+” indicates a limit from the right and the superscript “−” indicates a limit from the left.

This leads to the general idea of a **one-sided limit**.



▲ Figure 1.1.12

As with two-sided limits, the one-sided limits in (14) and (15) can also be written as

$$f(x) \rightarrow L \quad \text{as} \quad x \rightarrow a^+$$

and

$$f(x) \rightarrow L \quad \text{as} \quad x \rightarrow a^-$$

respectively.

1.1.2 ONE-SIDED LIMITS (AN INFORMAL VIEW) If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (but greater than a), then we write

$$\lim_{x \rightarrow a^+} f(x) = L \quad (14)$$

and if the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (but less than a), then we write

$$\lim_{x \rightarrow a^-} f(x) = L \quad (15)$$

Expression (14) is read “the limit of $f(x)$ as x approaches a from the right is L ” or “ $f(x)$ approaches L as x approaches a from the right.” Similarly, expression (15) is read “the limit of $f(x)$ as x approaches a from the left is L ” or “ $f(x)$ approaches L as x approaches a from the left.”

THE RELATIONSHIP BETWEEN ONE-SIDED LIMITS AND TWO-SIDED LIMITS

In general, there is no guarantee that a function f will have a two-sided limit at a given point a ; that is, the values of $f(x)$ may not get closer and closer to any *single* real number L as $x \rightarrow a$. In this case we say that

$$\lim_{x \rightarrow a} f(x) \text{ does not exist}$$

Similarly, the values of $f(x)$ may not get closer and closer to a single real number L as $x \rightarrow a^+$ or as $x \rightarrow a^-$. In these cases we say that

$$\lim_{x \rightarrow a^+} f(x) \text{ does not exist}$$

or that

$$\lim_{x \rightarrow a^-} f(x) \text{ does not exist}$$

In order for the two-sided limit of a function $f(x)$ to exist at a point a , the values of $f(x)$ must approach some real number L as x approaches a , and this number must be the same regardless of whether x approaches a from the left or the right. This suggests the following result, which we state without formal proof.

1.1.3 THE RELATIONSHIP BETWEEN ONE-SIDED AND TWO-SIDED LIMITS The two-sided limit of a function $f(x)$ exists at a if and only if both of the one-sided limits exist at a and have the same value; that is,

$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x)$$

► **Example 4** Explain why

$$\lim_{x \rightarrow 0} \frac{|x|}{x}$$

does not exist.

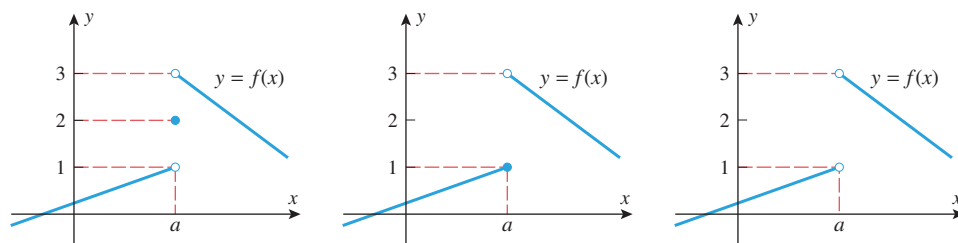
Solution. As x approaches 0, the values of $f(x) = |x|/x$ approach -1 from the left and approach 1 from the right [see (13)]. Thus, the one-sided limits at 0 are not the same. ◀

► **Example 5** For the functions in Figure 1.1.13, find the one-sided and two-sided limits at $x = a$ if they exist.

Solution. The functions in all three figures have the same one-sided limits as $x \rightarrow a$, since the functions are identical, except at $x = a$. These limits are

$$\lim_{x \rightarrow a^+} f(x) = 3 \quad \text{and} \quad \lim_{x \rightarrow a^-} f(x) = 1$$

In all three cases the two-sided limit does not exist as $x \rightarrow a$ because the one-sided limits are not equal. ◀



► Figure 1.1.13

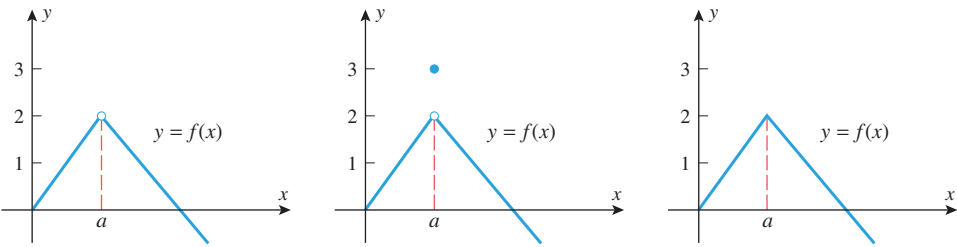
► **Example 6** For the functions in Figure 1.1.14, find the one-sided and two-sided limits at $x = a$ if they exist.

Solution. As in the preceding example, the value of f at $x = a$ has no bearing on the limits as $x \rightarrow a$, so in all three cases we have

$$\lim_{x \rightarrow a^+} f(x) = 2 \quad \text{and} \quad \lim_{x \rightarrow a^-} f(x) = 2$$

Since the one-sided limits are equal, the two-sided limit exists and

$$\lim_{x \rightarrow a} f(x) = 2 \quad \blacktriangleleft$$



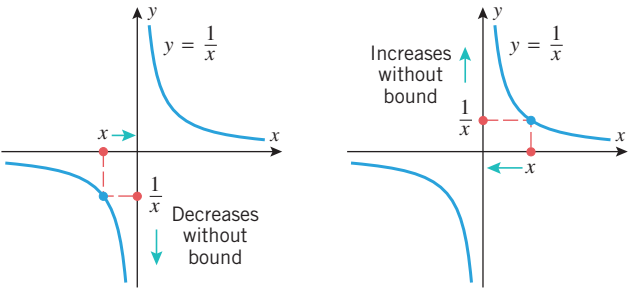
▲ Figure 1.1.14

■ **INFINITE LIMITS**

Sometimes one-sided or two-sided limits fail to exist because the values of the function increase or decrease without bound. For example, consider the behavior of $f(x) = 1/x$ for values of x near 0. It is evident from the table and graph in Figure 1.1.15 that as x -values are taken closer and closer to 0 from the right, the values of $f(x) = 1/x$ are positive and increase without bound; and as x -values are taken closer and closer to 0 from the left, the values of $f(x) = 1/x$ are negative and decrease without bound. We describe these limiting behaviors by writing

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty \quad \text{and} \quad \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$$

The symbols $+\infty$ and $-\infty$ here are *not* real numbers; they simply describe particular ways in which the limits fail to exist. Do not make the mistake of manipulating these symbols using rules of algebra. For example, it is *incorrect* to write $(+\infty) - (+\infty) = 0$.



x	-1	-0.1	-0.01	-0.001	-0.0001	0	0.0001	0.001	0.01	0.1	1
$\frac{1}{x}$	-1	-10	-100	-1000	-10,000		10,000	1000	100	10	1

Left side Right side

▲ Figure 1.1.15

1.1.4 INFINITE LIMITS (AN INFORMAL VIEW) The expressions

$$\lim_{x \rightarrow a^-} f(x) = +\infty \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) = +\infty$$

denote that $f(x)$ increases without bound as x approaches a from the left and from the right, respectively. If both are true, then we write

$$\lim_{x \rightarrow a} f(x) = +\infty$$

Similarly, the expressions

$$\lim_{x \rightarrow a^-} f(x) = -\infty \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) = -\infty$$

denote that $f(x)$ decreases without bound as x approaches a from the left and from the right, respectively. If both are true, then we write

$$\lim_{x \rightarrow a} f(x) = -\infty$$

► **Example 7** For the functions in Figure 1.1.16, describe the limits at $x = a$ in appropriate limit notation.

Solution (a). In Figure 1.1.16a, the function increases without bound as x approaches a from the right and decreases without bound as x approaches a from the left. Thus,

$$\lim_{x \rightarrow a^+} \frac{1}{x-a} = +\infty \quad \text{and} \quad \lim_{x \rightarrow a^-} \frac{1}{x-a} = -\infty$$

Solution (b). In Figure 1.1.16b, the function increases without bound as x approaches a from both the left and right. Thus,

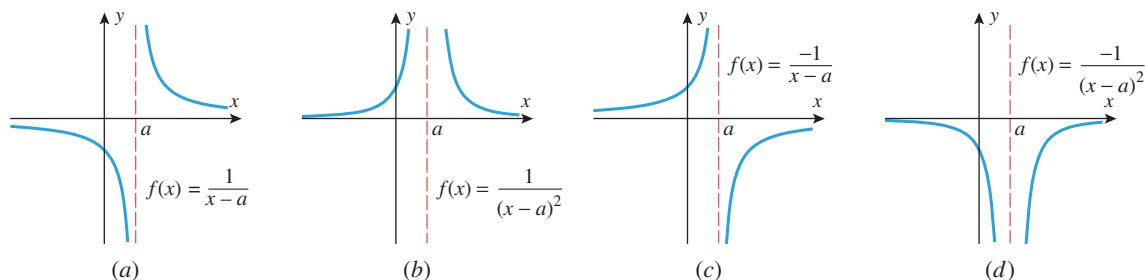
$$\lim_{x \rightarrow a} \frac{1}{(x-a)^2} = \lim_{x \rightarrow a^+} \frac{1}{(x-a)^2} = \lim_{x \rightarrow a^-} \frac{1}{(x-a)^2} = +\infty$$

Solution (c). In Figure 1.1.16c, the function decreases without bound as x approaches a from the right and increases without bound as x approaches a from the left. Thus,

$$\lim_{x \rightarrow a^+} \frac{-1}{x-a} = -\infty \quad \text{and} \quad \lim_{x \rightarrow a^-} \frac{-1}{x-a} = +\infty$$

Solution (d). In Figure 1.1.16d, the function decreases without bound as x approaches a from both the left and right. Thus,

$$\lim_{x \rightarrow a} \frac{-1}{(x-a)^2} = \lim_{x \rightarrow a^+} \frac{-1}{(x-a)^2} = \lim_{x \rightarrow a^-} \frac{-1}{(x-a)^2} = -\infty \quad \blacktriangleleft$$



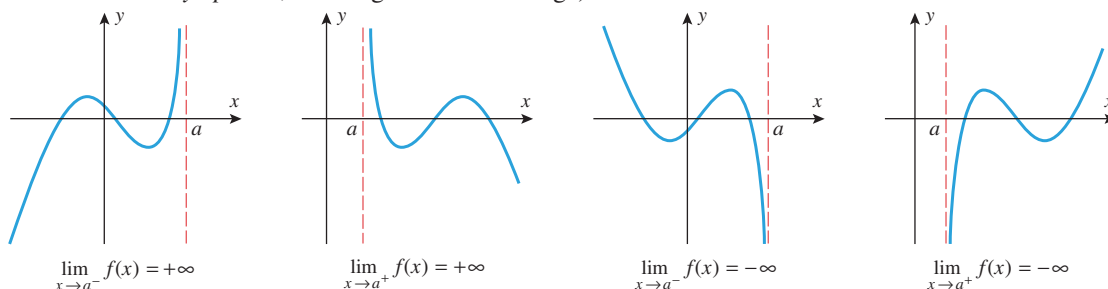
▲ Figure 1.1.16

VERTICAL ASYMPTOTES

Figure 1.1.17 illustrates geometrically what happens when any of the following situations occur:

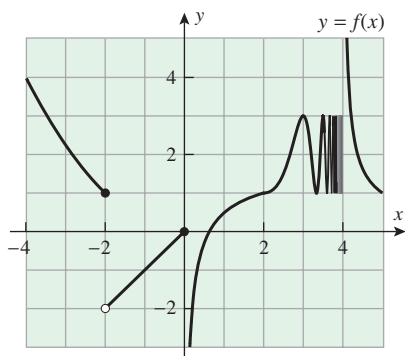
$$\lim_{x \rightarrow a^-} f(x) = +\infty, \quad \lim_{x \rightarrow a^+} f(x) = +\infty, \quad \lim_{x \rightarrow a^-} f(x) = -\infty, \quad \lim_{x \rightarrow a^+} f(x) = -\infty$$

In each case the graph of $y = f(x)$ either rises or falls without bound, squeezing closer and closer to the vertical line $x = a$ as x approaches a from the side indicated in the limit. The line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ (from the Greek word *asymptotos*, meaning “nonintersecting”).



▲ Figure 1.1.17

In general, the graph of a single function can display a wide variety of limits.



▲ Figure 1.1.18

► **Example 8** For the function f graphed in Figure 1.1.18, find

- (a) $\lim_{x \rightarrow -2^-} f(x)$ (b) $\lim_{x \rightarrow -2^+} f(x)$ (c) $\lim_{x \rightarrow 0^-} f(x)$ (d) $\lim_{x \rightarrow 0^+} f(x)$
 (e) $\lim_{x \rightarrow 4^-} f(x)$ (f) $\lim_{x \rightarrow 4^+} f(x)$ (g) the vertical asymptotes of the graph of f .

Solution (a) and (b).

$$\lim_{x \rightarrow -2^-} f(x) = 1 = f(-2) \quad \text{and} \quad \lim_{x \rightarrow -2^+} f(x) = -2$$

Solution (c) and (d).

$$\lim_{x \rightarrow 0^-} f(x) = 0 = f(0) \quad \text{and} \quad \lim_{x \rightarrow 0^+} f(x) = -\infty$$

Solution (e) and (f).

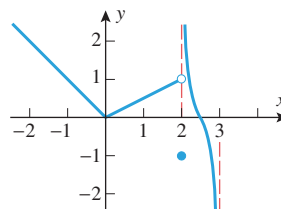
$$\lim_{x \rightarrow 4^-} f(x) \text{ does not exist due to oscillation} \quad \text{and} \quad \lim_{x \rightarrow 4^+} f(x) = +\infty$$

Solution (g). The y -axis and the line $x = 4$ are vertical asymptotes for the graph of f . ◀

QUICK CHECK EXERCISES 1.1 (See page 61 for answers.)

- We write $\lim_{x \rightarrow a} f(x) = L$ provided the values of _____ can be made as close to _____ as desired, by taking values of _____ sufficiently close to _____ but not _____.
- We write $\lim_{x \rightarrow a^-} f(x) = +\infty$ provided _____ increases without bound, as _____ approaches _____ from the left.
- State what must be true about $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ in order for it to be the case that $\lim_{x \rightarrow a} f(x) = L$.
- Use the accompanying graph of $y = f(x)$ ($-\infty < x < 3$) to determine the limits.
 (a) $\lim_{x \rightarrow 0} f(x) =$ _____

- (b) $\lim_{x \rightarrow 2^-} f(x) =$ _____
 (c) $\lim_{x \rightarrow 2^+} f(x) =$ _____
 (d) $\lim_{x \rightarrow 3^-} f(x) =$ _____



◀ Figure Ex-4

- The slope of the secant line through $P(2, 4)$ and $Q(x, x^2)$ on the parabola $y = x^2$ is $m_{\text{sec}} = x + 2$. It follows that the slope of the tangent line to this parabola at the point P is _____.

EXERCISE SET 1.1



Graphing Utility



CAS

1–10 In these exercises, make reasonable assumptions about the graph of the indicated function outside of the region depicted. ■

1. For the function g graphed in the accompanying figure, find

(a) $\lim_{x \rightarrow 0^-} g(x)$ (b) $\lim_{x \rightarrow 0^+} g(x)$
 (c) $\lim_{x \rightarrow 0} g(x)$ (d) $g(0)$.

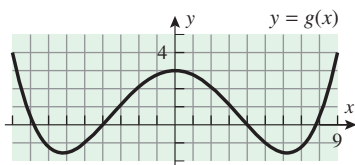


Figure Ex-1

2. For the function G graphed in the accompanying figure, find

(a) $\lim_{x \rightarrow 0^-} G(x)$ (b) $\lim_{x \rightarrow 0^+} G(x)$
 (c) $\lim_{x \rightarrow 0} G(x)$ (d) $G(0)$.

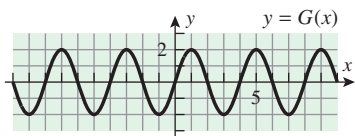


Figure Ex-2

3. For the function f graphed in the accompanying figure, find

(a) $\lim_{x \rightarrow 3^-} f(x)$ (b) $\lim_{x \rightarrow 3^+} f(x)$
 (c) $\lim_{x \rightarrow 3} f(x)$ (d) $f(3)$.

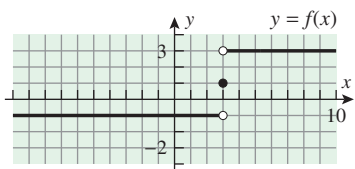


Figure Ex-3

4. For the function f graphed in the accompanying figure, find

(a) $\lim_{x \rightarrow 2^-} f(x)$ (b) $\lim_{x \rightarrow 2^+} f(x)$
 (c) $\lim_{x \rightarrow 2} f(x)$ (d) $f(2)$.

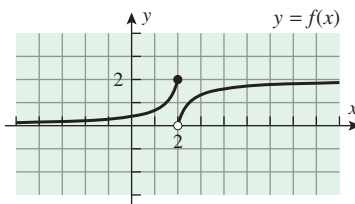


Figure Ex-4

5. For the function F graphed in the accompanying figure, find

(a) $\lim_{x \rightarrow -2^-} F(x)$ (b) $\lim_{x \rightarrow -2^+} F(x)$
 (c) $\lim_{x \rightarrow -2} F(x)$ (d) $F(-2)$.

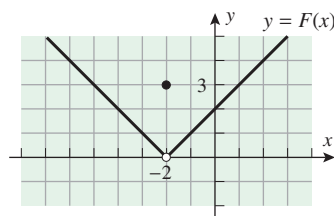


Figure Ex-5

6. For the function G graphed in the accompanying figure, find

(a) $\lim_{x \rightarrow 0^-} G(x)$ (b) $\lim_{x \rightarrow 0^+} G(x)$
 (c) $\lim_{x \rightarrow 0} G(x)$ (d) $G(0)$.

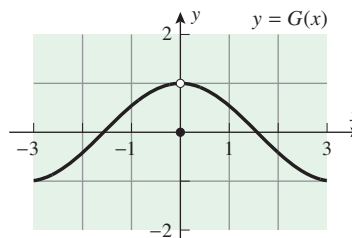


Figure Ex-6

7. For the function f graphed in the accompanying figure, find

(a) $\lim_{x \rightarrow 3^-} f(x)$ (b) $\lim_{x \rightarrow 3^+} f(x)$
 (c) $\lim_{x \rightarrow 3} f(x)$ (d) $f(3)$.

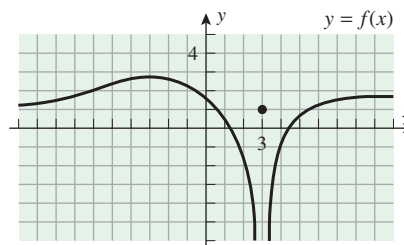


Figure Ex-7

8. For the function ϕ graphed in the accompanying figure, find

(a) $\lim_{x \rightarrow 4^-} \phi(x)$ (b) $\lim_{x \rightarrow 4^+} \phi(x)$
 (c) $\lim_{x \rightarrow 4} \phi(x)$ (d) $\phi(4)$.

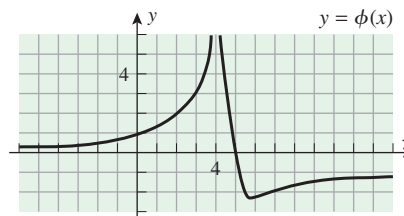


Figure Ex-8

9. For the function f graphed in the accompanying figure on the next page, find

(a) $\lim_{x \rightarrow -2} f(x)$ (b) $\lim_{x \rightarrow 0^-} f(x)$
 (c) $\lim_{x \rightarrow 0^+} f(x)$ (d) $\lim_{x \rightarrow 2^-} f(x)$
 (e) $\lim_{x \rightarrow 2^+} f(x)$
 (f) the vertical asymptotes of the graph of f .

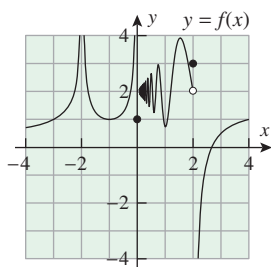


Figure Ex-9

10. For the function f graphed in the accompanying figure, find

- (a) $\lim_{x \rightarrow -2^-} f(x)$ (b) $\lim_{x \rightarrow -2^+} f(x)$ (c) $\lim_{x \rightarrow 0^-} f(x)$
 (d) $\lim_{x \rightarrow 0^+} f(x)$ (e) $\lim_{x \rightarrow 2^-} f(x)$ (f) $\lim_{x \rightarrow 2^+} f(x)$
 (g) the vertical asymptotes of the graph of f .

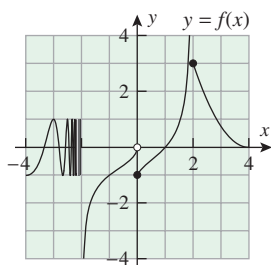


Figure Ex-10

11–12 (i) Complete the table and make a guess about the limit indicated. (ii) Confirm your conclusions about the limit by graphing a function over an appropriate interval. [Note: For the trigonometric functions, be sure to put your calculating and graphing utilities in radian mode.] ■

11. $f(x) = \frac{\sin 2x}{x}$; $\lim_{x \rightarrow 0} f(x)$

x	-0.1	-0.01	-0.001	0.001	0.01	0.1
$f(x)$						

Table Ex-11

12. $f(x) = \frac{\cos x - 1}{x^2}$; $\lim_{x \rightarrow 0} f(x)$

x	-0.5	-0.05	-0.005	0.005	0.05	0.5
$f(x)$						

Table Ex-12

13–16 (i) Make a guess at the limit (if it exists) by evaluating the function at the specified x -values. (ii) Confirm your conclusions about the limit by graphing the function over an appropriate interval. (iii) If you have a CAS, then use it to find the limit. [Note: For the trigonometric functions, be sure to put your calculating and graphing utilities in radian mode.] ■

13. (a) $\lim_{x \rightarrow 1} \frac{x-1}{x^3-1}$; $x = 2, 1.5, 1.1, 1.01, 1.001, 0, 0.5, 0.9, 0.99, 0.999$

(b) $\lim_{x \rightarrow 1^+} \frac{x+1}{x^3-1}$; $x = 2, 1.5, 1.1, 1.01, 1.001, 1.0001$

(c) $\lim_{x \rightarrow 1^-} \frac{x+1}{x^3-1}$; $x = 0, 0.5, 0.9, 0.99, 0.999, 0.9999$

14. (a) $\lim_{x \rightarrow 0} \frac{\sqrt{x+1}-1}{x}$; $x = \pm 0.25, \pm 0.1, \pm 0.001, \pm 0.0001$

(b) $\lim_{x \rightarrow 0^+} \frac{\sqrt{x+1}+1}{x}$; $x = 0.25, 0.1, 0.001, 0.0001$

(c) $\lim_{x \rightarrow 0^-} \frac{\sqrt{x+1}+1}{x}$; $x = -0.25, -0.1, -0.001, -0.0001$

15. (a) $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$; $x = \pm 0.25, \pm 0.1, \pm 0.001, \pm 0.0001$

(b) $\lim_{x \rightarrow -1} \frac{x \cos x}{x+1}$; $x = 0, -0.5, -0.9, -0.99, -0.999, -1.5, -1.1, -1.01, -1.001$

16. (a) $\lim_{x \rightarrow -1} \frac{\tan(x+1)}{x+1}$; $x = 0, -0.5, -0.9, -0.99, -0.999, -1.5, -1.1, -1.01, -1.001$

(b) $\lim_{x \rightarrow 0} \frac{\sin(5x)}{\sin(2x)}$; $x = \pm 0.25, \pm 0.1, \pm 0.001, \pm 0.0001$

17–20 True–False Determine whether the statement is true or false. Explain your answer. ■

17. If $f(a) = L$, then $\lim_{x \rightarrow a} f(x) = L$.

18. If $\lim_{x \rightarrow a} f(x)$ exists, then so do $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$.

19. If $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ exist, then so does $\lim_{x \rightarrow a} f(x)$.

20. If $\lim_{x \rightarrow a^+} f(x) = +\infty$, then $f(a)$ is undefined.

21–26 Sketch a possible graph for a function f with the specified properties. (Many different solutions are possible.) ■

21. (i) the domain of f is $[-1, 1]$

(ii) $f(-1) = f(0) = f(1) = 0$

(iii) $\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 1^-} f(x) = 1$

22. (i) the domain of f is $[-2, 1]$

(ii) $f(-2) = f(0) = f(1) = 0$

(iii) $\lim_{x \rightarrow -2^+} f(x) = 2$, $\lim_{x \rightarrow 0} f(x) = 0$, and $\lim_{x \rightarrow 1^-} f(x) = 1$

23. (i) the domain of f is $(-\infty, 0]$

(ii) $f(-2) = f(0) = 1$

(iii) $\lim_{x \rightarrow -2} f(x) = +\infty$

24. (i) the domain of f is $(0, +\infty)$

(ii) $f(1) = 0$

(iii) the y -axis is a vertical asymptote for the graph of f

(iv) $f(x) < 0$ if $0 < x < 1$

25. (i) $f(-3) = f(0) = f(2) = 0$
 (ii) $\lim_{x \rightarrow -2^-} f(x) = +\infty$ and $\lim_{x \rightarrow -2^+} f(x) = -\infty$
 (iii) $\lim_{x \rightarrow 1} f(x) = +\infty$
26. (i) $f(-1) = 0, f(0) = 1, f(1) = 0$
 (ii) $\lim_{x \rightarrow -1^-} f(x) = 0$ and $\lim_{x \rightarrow -1^+} f(x) = +\infty$
 (iii) $\lim_{x \rightarrow 1^-} f(x) = 1$ and $\lim_{x \rightarrow 1^+} f(x) = +\infty$

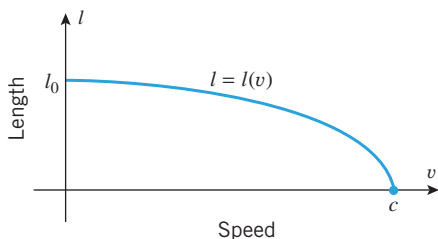
27–30 Modify the argument of Example 1 to find the equation of the tangent line to the specified graph at the point given. ■

27. the graph of $y = x^2$ at $(-1, 1)$
 28. the graph of $y = x^2$ at $(0, 0)$
 29. the graph of $y = x^4$ at $(1, 1)$
 30. the graph of $y = x^4$ at $(-1, 1)$

FOCUS ON CONCEPTS

31. In the special theory of relativity the length l of a narrow rod moving longitudinally is a function $l = l(v)$ of the rod's speed v . The accompanying figure, in which c denotes the speed of light, displays some of the qualitative features of this function.

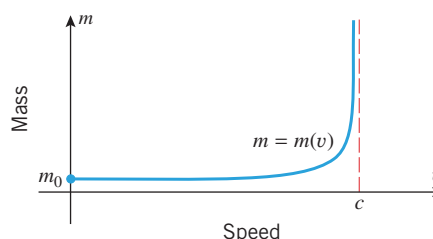
- (a) What is the physical interpretation of l_0 ?
 (b) What is $\lim_{v \rightarrow c^-} l(v)$? What is the physical significance of this limit?



▲ Figure Ex-31

32. In the special theory of relativity the mass m of a moving object is a function $m = m(v)$ of the object's speed v . The accompanying figure, in which c denotes the speed of light, displays some of the qualitative features of this function.
- (a) What is the physical interpretation of m_0 ?

- (b) What is $\lim_{v \rightarrow c^-} m(v)$? What is the physical significance of this limit?



▲ Figure Ex-32

33. Let

$$f(x) = (1 + x^2)^{1.1/x^2}$$

- (a) Graph f in the window

$$[-1, 1] \times [2.5, 3.5]$$

and use the calculator's trace feature to make a conjecture about the limit of $f(x)$ as $x \rightarrow 0$.

- (b) Graph f in the window

$$[-0.001, 0.001] \times [2.5, 3.5]$$

and use the calculator's trace feature to make a conjecture about the limit of $f(x)$ as $x \rightarrow 0$.

- (c) Graph f in the window

$$[-0.000001, 0.000001] \times [2.5, 3.5]$$

and use the calculator's trace feature to make a conjecture about the limit of $f(x)$ as $x \rightarrow 0$.

- (d) Later we will be able to show that

$$\lim_{x \rightarrow 0} (1 + x^2)^{1.1/x^2} \approx 3.00416602$$

What flaw do your graphs reveal about using numerical evidence (as revealed by the graphs you obtained) to make conjectures about limits?

34. **Writing** Two students are discussing the limit of $\sqrt{x}$ as x approaches 0. One student maintains that the limit is 0, while the other claims that the limit does not exist. Write a short paragraph that discusses the pros and cons of each student's position.

35. **Writing** Given a function f and a real number a , explain informally why

$$\lim_{x \rightarrow 0} f(x + a) = \lim_{x \rightarrow a} f(x)$$

(Here "equality" means that either both limits exist and are equal or that both limits fail to exist.)

QUICK CHECK ANSWERS 1.1

1. $f(x)$; L ; x ; a ; a 2. $f(x)$; x ; a 3. Both one-sided limits must exist and equal L . 4. (a) 0 (b) 1 (c) $+\infty$ (d) $-\infty$ 5. 4

1.2 COMPUTING LIMITS

In this section we will discuss techniques for computing limits of many functions. We base these results on the informal development of the limit concept discussed in the preceding section. A more formal derivation of these results is possible after Section 1.4.

■ SOME BASIC LIMITS

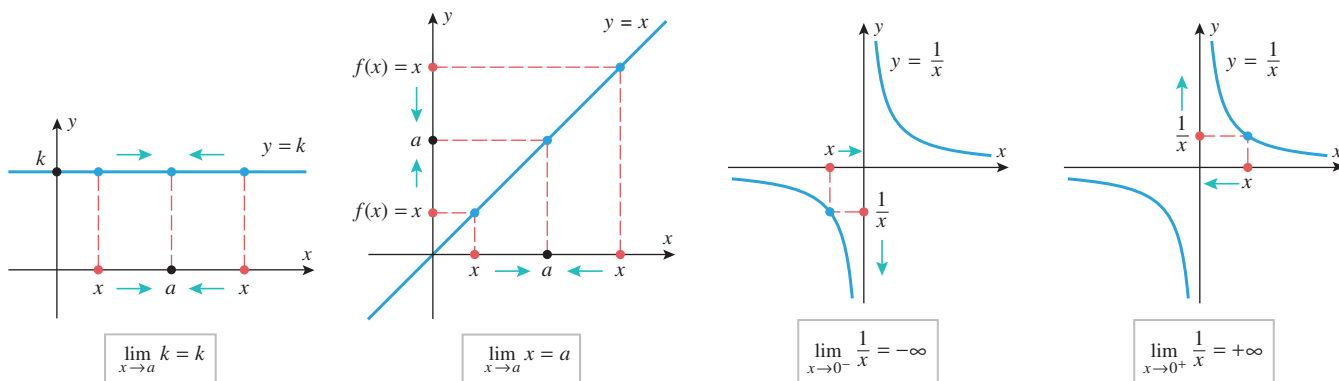
Our strategy for finding limits algebraically has two parts:

- First we will obtain the limits of some simple functions.
- Then we will develop a repertoire of theorems that will enable us to use the limits of those simple functions as building blocks for finding limits of more complicated functions.

We start with the following basic results, which are illustrated in Figure 1.2.1.

1.2.1 THEOREM Let a and k be real numbers.

$$(a) \lim_{x \rightarrow a} k = k \quad (b) \lim_{x \rightarrow a} x = a \quad (c) \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty \quad (d) \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$$



▲ Figure 1.2.1

The following examples explain these results further.

► **Example 1** If $f(x) = k$ is a constant function, then the values of $f(x)$ remain fixed at k as x varies, which explains why $f(x) \rightarrow k$ as $x \rightarrow a$ for all values of a . For example,

$$\lim_{x \rightarrow -25} 3 = 3, \quad \lim_{x \rightarrow 0} 3 = 3, \quad \lim_{x \rightarrow \pi} 3 = 3 \quad \blacktriangleleft$$

► **Example 2** If $f(x) = x$, then as $x \rightarrow a$ it must also be true that $f(x) \rightarrow a$. For example,

$$\lim_{x \rightarrow 0} x = 0, \quad \lim_{x \rightarrow -2} x = -2, \quad \lim_{x \rightarrow \pi} x = \pi \quad \blacktriangleleft$$

Do not confuse the algebraic size of a number with its closeness to zero. For positive numbers, the smaller the number the closer it is to zero, but for negative numbers, the larger the number the closer it is to zero. For example, -2 is larger than -4 , but it is closer to zero.

► **Example 3** You should know from your experience with fractions that for a fixed nonzero numerator, the closer the denominator is to zero, the larger the absolute value of the fraction. This fact and the data in Table 1.2.1 suggest why $1/x \rightarrow +\infty$ as $x \rightarrow 0^+$ and why $1/x \rightarrow -\infty$ as $x \rightarrow 0^-$. ◀

Table 1.2.1

	VALUES						CONCLUSION
x	-1	-0.1	-0.01	-0.001	-0.0001	...	As $x \rightarrow 0^-$ the value of $1/x$ decreases without bound.
$1/x$	-1	-10	-100	-1000	-10,000	...	
x	1	0.1	0.01	0.001	0.0001	...	As $x \rightarrow 0^+$ the value of $1/x$ increases without bound.
$1/x$	1	10	100	1000	10,000	...	

The following theorem, parts of which are proved in Appendix D, will be our basic tool for finding limits algebraically.

1.2.2 THEOREM Let a be a real number, and suppose that

$$\lim_{x \rightarrow a} f(x) = L_1 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = L_2$$

That is, the limits exist and have values L_1 and L_2 , respectively. Then:

- (a) $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = L_1 + L_2$
- (b) $\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) = L_1 - L_2$
- (c) $\lim_{x \rightarrow a} [f(x)g(x)] = \left(\lim_{x \rightarrow a} f(x) \right) \left(\lim_{x \rightarrow a} g(x) \right) = L_1 L_2$
- (d) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L_1}{L_2}$, provided $L_2 \neq 0$
- (e) $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} = \sqrt[n]{L_1}$, provided $L_1 > 0$ if n is even.

Moreover, these statements are also true for the one-sided limits as $x \rightarrow a^-$ or as $x \rightarrow a^+$.

Theorem 1.2.2(e) remains valid for n even and $L_1 = 0$, provided $f(x)$ is nonnegative for x near a with $x \neq a$.

This theorem can be stated informally as follows:

- (a) The limit of a sum is the sum of the limits.
- (b) The limit of a difference is the difference of the limits.
- (c) The limit of a product is the product of the limits.
- (d) The limit of a quotient is the quotient of the limits, provided the limit of the denominator is not zero.
- (e) The limit of an n th root is the n th root of the limit.

For the special case of part (c) in which $f(x) = k$ is a constant function, we have

$$\lim_{x \rightarrow a} (kg(x)) = \lim_{x \rightarrow a} k \cdot \lim_{x \rightarrow a} g(x) = k \lim_{x \rightarrow a} g(x) \quad (1)$$

and similarly for one-sided limits. This result can be rephrased as follows:

A constant factor can be moved through a limit symbol.

Although parts (a) and (c) of Theorem 1.2.2 are stated for two functions, the results hold for any finite number of functions. Moreover, the various parts of the theorem can be used in combination to reformulate expressions involving limits.

► **Example 4**

$$\lim_{x \rightarrow a} [f(x) - g(x) + 2h(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) + 2 \lim_{x \rightarrow a} h(x)$$

$$\lim_{x \rightarrow a} [f(x)g(x)h(x)] = \left(\lim_{x \rightarrow a} f(x) \right) \left(\lim_{x \rightarrow a} g(x) \right) \left(\lim_{x \rightarrow a} h(x) \right)$$

$$\lim_{x \rightarrow a} [f(x)]^3 = \left(\lim_{x \rightarrow a} f(x) \right)^3$$

$$\lim_{x \rightarrow a} [f(x)]^n = \left(\lim_{x \rightarrow a} f(x) \right)^n$$

$$\lim_{x \rightarrow a} x^n = \left(\lim_{x \rightarrow a} x \right)^n = a^n$$

Take $g(x) = h(x) = f(x)$ in the last equation.

The extension of Theorem 1.2.2(c) in which there are n factors, each of which is $f(x)$

Apply the previous result with $f(x) = x$. ◀

■ **LIMITS OF POLYNOMIALS AND RATIONAL FUNCTIONS AS $x \rightarrow a$**

► **Example 5** Find $\lim_{x \rightarrow 5} (x^2 - 4x + 3)$.

Solution.

$$\lim_{x \rightarrow 5} (x^2 - 4x + 3) = \lim_{x \rightarrow 5} x^2 - \lim_{x \rightarrow 5} 4x + \lim_{x \rightarrow 5} 3$$

$$= \lim_{x \rightarrow 5} x^2 - 4 \lim_{x \rightarrow 5} x + \lim_{x \rightarrow 5} 3$$

$$= 5^2 - 4(5) + 3$$

$$= 8 \quad \blacktriangleleft$$

Theorem 1.2.2(a), (b)

A constant can be moved through a limit symbol.

The last part of Example 4

Observe that in Example 5 the limit of the polynomial $p(x) = x^2 - 4x + 3$ as $x \rightarrow 5$ turned out to be the same as $p(5)$. This is not an accident. The next result shows that, in general, the limit of a polynomial $p(x)$ as $x \rightarrow a$ is the same as the value of the polynomial at a . Knowing this fact allows us to reduce the computation of limits of polynomials to simply evaluating the polynomial at the appropriate point.

1.2.3 THEOREM For any polynomial

$$p(x) = c_0 + c_1x + \cdots + c_nx^n$$

and any real number a ,

$$\lim_{x \rightarrow a} p(x) = c_0 + c_1a + \cdots + c_na^n = p(a)$$

PROOF

$$\begin{aligned}
\lim_{x \rightarrow a} p(x) &= \lim_{x \rightarrow a} (c_0 + c_1x + \cdots + c_nx^n) \\
&= \lim_{x \rightarrow a} c_0 + \lim_{x \rightarrow a} c_1x + \cdots + \lim_{x \rightarrow a} c_nx^n \\
&= \lim_{x \rightarrow a} c_0 + c_1 \lim_{x \rightarrow a} x + \cdots + c_n \lim_{x \rightarrow a} x^n \\
&= c_0 + c_1a + \cdots + c_na^n = p(a) \quad \blacksquare
\end{aligned}$$

► **Example 6** Find $\lim_{x \rightarrow 1} (x^7 - 2x^5 + 1)^{35}$.

Solution. The function involved is a polynomial (why?), so the limit can be obtained by evaluating this polynomial at $x = 1$. This yields

$$\lim_{x \rightarrow 1} (x^7 - 2x^5 + 1)^{35} = 0 \quad \blacktriangleleft$$

Recall that a rational function is a ratio of two polynomials. The following example illustrates how Theorems 1.2.2(d) and 1.2.3 can sometimes be used in combination to compute limits of rational functions.

► **Example 7** Find $\lim_{x \rightarrow 2} \frac{5x^3 + 4}{x - 3}$.

Solution.

$$\begin{aligned}
\lim_{x \rightarrow 2} \frac{5x^3 + 4}{x - 3} &= \frac{\lim_{x \rightarrow 2} (5x^3 + 4)}{\lim_{x \rightarrow 2} (x - 3)} && \text{Theorem 1.2.2(d)} \\
&= \frac{5 \cdot 2^3 + 4}{2 - 3} = -44 && \text{Theorem 1.2.3} \quad \blacktriangleleft
\end{aligned}$$

The method used in the last example will not work for rational functions in which the limit of the denominator is zero because Theorem 1.2.2(d) is not applicable. There are two cases of this type to be considered—the case where the limit of the denominator is zero and the limit of the numerator is not, and the case where the limits of the numerator and denominator are both zero. If the limit of the denominator is zero but the limit of the numerator is not, then one can prove that the limit of the rational function does not exist and that one of the following situations occurs:

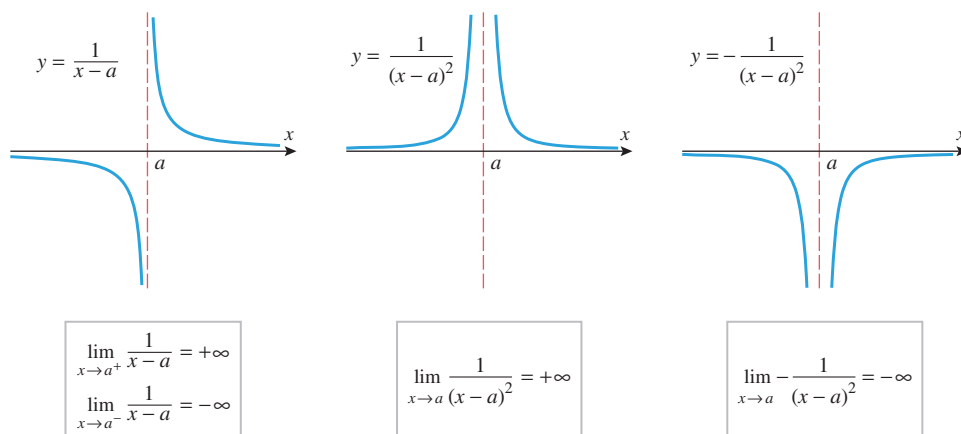
- The limit may be $-\infty$ from one side and $+\infty$ from the other.
- The limit may be $+\infty$.
- The limit may be $-\infty$.

Figure 1.2.2 illustrates these three possibilities graphically for rational functions of the form $1/(x - a)$, $1/(x - a)^2$, and $-1/(x - a)^2$.

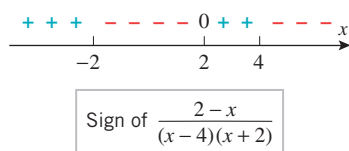
► **Example 8** Find

$$\begin{array}{lll}
\text{(a)} \quad \lim_{x \rightarrow 4^+} \frac{2 - x}{(x - 4)(x + 2)} & \text{(b)} \quad \lim_{x \rightarrow 4^-} \frac{2 - x}{(x - 4)(x + 2)} & \text{(c)} \quad \lim_{x \rightarrow 4} \frac{2 - x}{(x - 4)(x + 2)}
\end{array}$$

Solution. In all three parts the limit of the numerator is -2 , and the limit of the denominator is 0, so the limit of the ratio does not exist. To be more specific than this, we need



▲ Figure 1.2.2



▲ Figure 1.2.3

to analyze the sign of the ratio. The sign of the ratio, which is given in Figure 1.2.3, is determined by the signs of $2 - x$, $x - 4$, and $x + 2$. (The method of test points, discussed in Web Appendix E, provides a way of finding the sign of the ratio here.) It follows from this figure that as x approaches 4 from the right, the ratio is always negative; and as x approaches 4 from the left, the ratio is eventually positive. Thus,

$$\lim_{x \rightarrow 4^+} \frac{2-x}{(x-4)(x+2)} = -\infty \quad \text{and} \quad \lim_{x \rightarrow 4^-} \frac{2-x}{(x-4)(x+2)} = +\infty$$

Because the one-sided limits have opposite signs, all we can say about the two-sided limit is that it does not exist. ◀

In the case where $p(x)/q(x)$ is a rational function for which $p(a) = 0$ and $q(a) = 0$, the numerator and denominator must have one or more common factors of $x - a$. In this case the limit of $p(x)/q(x)$ as $x \rightarrow a$ can be found by canceling all common factors of $x - a$ and using one of the methods already considered to find the limit of the simplified function. Here is an example.

In Example 9(a), the simplified function $x - 3$ is defined at $x = 3$, but the original function is not. However, this has no effect on the limit as x approaches 3 since the two functions are identical if $x \neq 3$ (Exercise 50).

► **Example 9** Find

$$(a) \lim_{x \rightarrow 3} \frac{x^2 - 6x + 9}{x - 3} \quad (b) \lim_{x \rightarrow -4} \frac{2x + 8}{x^2 + x - 12} \quad (c) \lim_{x \rightarrow 5} \frac{x^2 - 3x - 10}{x^2 - 10x + 25}$$

Solution (a). The numerator and the denominator both have a zero at $x = 3$, so there is a common factor of $x - 3$. Then

$$\lim_{x \rightarrow 3} \frac{x^2 - 6x + 9}{x - 3} = \lim_{x \rightarrow 3} \frac{(x - 3)^2}{x - 3} = \lim_{x \rightarrow 3} (x - 3) = 0$$

Solution (b). The numerator and the denominator both have a zero at $x = -4$, so there is a common factor of $x - (-4) = x + 4$. Then

$$\lim_{x \rightarrow -4} \frac{2x + 8}{x^2 + x - 12} = \lim_{x \rightarrow -4} \frac{2(x + 4)}{(x + 4)(x - 3)} = \lim_{x \rightarrow -4} \frac{2}{x - 3} = -\frac{2}{7}$$

Solution (c). The numerator and the denominator both have a zero at $x = 5$, so there is a common factor of $x - 5$. Then

$$\lim_{x \rightarrow 5} \frac{x^2 - 3x - 10}{x^2 - 10x + 25} = \lim_{x \rightarrow 5} \frac{(x - 5)(x + 2)}{(x - 5)(x - 5)} = \lim_{x \rightarrow 5} \frac{x + 2}{x - 5}$$

However,

$$\lim_{x \rightarrow 5} (x + 2) = 7 \neq 0 \quad \text{and} \quad \lim_{x \rightarrow 5} (x - 5) = 0$$

so

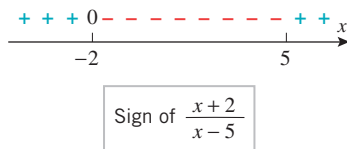
$$\lim_{x \rightarrow 5} \frac{x^2 - 3x - 10}{x^2 - 10x + 25} = \lim_{x \rightarrow 5} \frac{x + 2}{x - 5}$$

does not exist. More precisely, the sign analysis in Figure 1.2.4 implies that

$$\lim_{x \rightarrow 5^+} \frac{x^2 - 3x - 10}{x^2 - 10x + 25} = \lim_{x \rightarrow 5^+} \frac{x + 2}{x - 5} = +\infty$$

and

$$\lim_{x \rightarrow 5^-} \frac{x^2 - 3x - 10}{x^2 - 10x + 25} = \lim_{x \rightarrow 5^-} \frac{x + 2}{x - 5} = -\infty \quad \blacktriangleleft$$



▲ Figure 1.2.4

Discuss the logical errors in the following statement: An indeterminate form of type $0/0$ must have a limit of zero because zero divided by anything is zero.

A quotient $f(x)/g(x)$ in which the numerator and denominator both have a limit of zero as $x \rightarrow a$ is called an **indeterminate form of type $0/0$** . The problem with such limits is that it is difficult to tell by inspection whether the limit exists, and, if so, its value. Informally stated, this is because there are two conflicting influences at work. The value of $f(x)/g(x)$ would tend to zero as $f(x)$ approached zero if $g(x)$ were to remain at some fixed nonzero value, whereas the value of this ratio would tend to increase or decrease without bound as $g(x)$ approached zero if $f(x)$ were to remain at some fixed nonzero value. But with both $f(x)$ and $g(x)$ approaching zero, the behavior of the ratio depends on precisely how these conflicting tendencies offset one another for the particular f and g .

Sometimes, limits of indeterminate forms of type $0/0$ can be found by algebraic simplification, as in the last example, but frequently this will not work and other methods must be used. We will study such methods in later sections.

The following theorem summarizes our observations about limits of rational functions.

1.2.4 THEOREM Let

$$f(x) = \frac{p(x)}{q(x)}$$

be a rational function, and let a be any real number.

- (a) If $q(a) \neq 0$, then $\lim_{x \rightarrow a} f(x) = f(a)$.
- (b) If $q(a) = 0$ but $p(a) \neq 0$, then $\lim_{x \rightarrow a} f(x)$ does not exist.

■ LIMITS INVOLVING RADICALS

► **Example 10** Find $\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1}$.

Solution. In Example 2 of Section 1.1 we used numerical evidence to conjecture that this limit is 2. Here we will confirm this algebraically. Since this limit is an indeterminate form of type $0/0$, we will need to devise some strategy for making the limit (if it exists) evident. One such strategy is to rationalize the denominator of the function. This yields

$$\frac{x - 1}{\sqrt{x} - 1} = \frac{(x - 1)(\sqrt{x} + 1)}{(\sqrt{x} - 1)(\sqrt{x} + 1)} = \frac{(x - 1)(\sqrt{x} + 1)}{x - 1} = \sqrt{x} + 1 \quad (x \neq 1)$$

Confirm the limit in Example 10 by factoring the numerator.

Therefore,

$$\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1} = \lim_{x \rightarrow 1} (\sqrt{x}+1) = 2 \quad \blacktriangleleft$$

■ LIMITS OF PIECEWISE-DEFINED FUNCTIONS

For functions that are defined piecewise, a two-sided limit at a point where the formula changes is best obtained by first finding the one-sided limits at that point.

► **Example 11** Let

$$f(x) = \begin{cases} 1/(x+2), & x < -2 \\ x^2 - 5, & -2 < x \leq 3 \\ \sqrt{x+13}, & x > 3 \end{cases}$$

Find

$$(a) \lim_{x \rightarrow -2} f(x) \quad (b) \lim_{x \rightarrow 0} f(x) \quad (c) \lim_{x \rightarrow 3} f(x)$$

Solution (a). We will determine the stated two-sided limit by first considering the corresponding one-sided limits. For each one-sided limit, we must use that part of the formula that is applicable on the interval over which x varies. For example, as x approaches -2 from the left, the applicable part of the formula is

$$f(x) = \frac{1}{x+2}$$

and as x approaches -2 from the right, the applicable part of the formula near -2 is

$$f(x) = x^2 - 5$$

Thus,

$$\lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^-} \frac{1}{x+2} = -\infty$$

$$\lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} (x^2 - 5) = (-2)^2 - 5 = -1$$

from which it follows that $\lim_{x \rightarrow -2} f(x)$ does not exist.

Solution (b). The applicable part of the formula is $f(x) = x^2 - 5$ on both sides of 0, so there is no need to consider one-sided limits here. We see directly that

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (x^2 - 5) = 0^2 - 5 = -5$$

Solution (c). Using the applicable parts of the formula for $f(x)$, we obtain

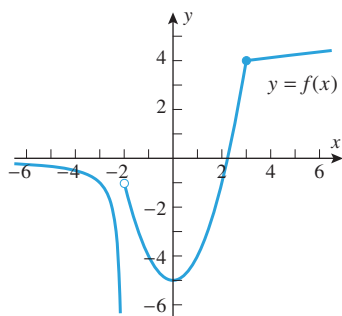
$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (x^2 - 5) = 3^2 - 5 = 4$$

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} \sqrt{x+13} = \sqrt{\lim_{x \rightarrow 3^+} (x+13)} = \sqrt{3+13} = 4$$

Since the one-sided limits are equal, we have

$$\lim_{x \rightarrow 3} f(x) = 4$$

We note that the limit calculations in parts (a), (b), and (c) are consistent with the graph of f shown in Figure 1.2.5. $\blacktriangleleft$



▲ Figure 1.2.5

QUICK CHECK EXERCISES 1.2 (See page 70 for answers.)

1. In each part, find the limit by inspection.

- (a) $\lim_{x \rightarrow 8} 7 = \underline{\hspace{2cm}}$ (b) $\lim_{y \rightarrow 3^+} 12y = \underline{\hspace{2cm}}$
 (c) $\lim_{x \rightarrow 0^-} \frac{x}{|x|} = \underline{\hspace{2cm}}$ (d) $\lim_{w \rightarrow 5} \frac{w}{|w|} = \underline{\hspace{2cm}}$
 (e) $\lim_{z \rightarrow 1^-} \frac{1}{1-z} = \underline{\hspace{2cm}}$

2. Given that $\lim_{x \rightarrow a} f(x) = 1$ and $\lim_{x \rightarrow a} g(x) = 2$, find the limits.

- (a) $\lim_{x \rightarrow a} [3f(x) + 2g(x)] = \underline{\hspace{2cm}}$
 (b) $\lim_{x \rightarrow a} \frac{2f(x) + 1}{1 - f(x)g(x)} = \underline{\hspace{2cm}}$
 (c) $\lim_{x \rightarrow a} \frac{\sqrt{f(x) + 3}}{g(x)} = \underline{\hspace{2cm}}$

3. Find the limits.

- (a) $\lim_{x \rightarrow -1} (x^3 + x^2 + x)^{101} = \underline{\hspace{2cm}}$
 (b) $\lim_{x \rightarrow 2^-} \frac{(x-1)(x-2)}{x+1} = \underline{\hspace{2cm}}$
 (c) $\lim_{x \rightarrow -1^+} \frac{(x-1)(x-2)}{x+1} = \underline{\hspace{2cm}}$
 (d) $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4} = \underline{\hspace{2cm}}$

4. Let

$$f(x) = \begin{cases} x+1, & x \leq 1 \\ x-1, & x > 1 \end{cases}$$

Find the limits that exist.

- (a) $\lim_{x \rightarrow 1^-} f(x) = \underline{\hspace{2cm}}$
 (b) $\lim_{x \rightarrow 1^+} f(x) = \underline{\hspace{2cm}}$
 (c) $\lim_{x \rightarrow 1} f(x) = \underline{\hspace{2cm}}$

EXERCISE SET 1.2

1. Given that

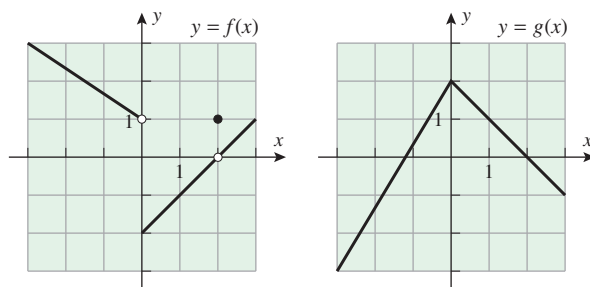
$$\lim_{x \rightarrow a} f(x) = 2, \quad \lim_{x \rightarrow a} g(x) = -4, \quad \lim_{x \rightarrow a} h(x) = 0$$

find the limits.

- (a) $\lim_{x \rightarrow a} [f(x) + 2g(x)]$ (b) $\lim_{x \rightarrow a} [h(x) - 3g(x) + 1]$
 (c) $\lim_{x \rightarrow a} [f(x)g(x)]$ (d) $\lim_{x \rightarrow a} [g(x)]^2$
 (e) $\lim_{x \rightarrow a} \sqrt[3]{6 + f(x)}$ (f) $\lim_{x \rightarrow a} \frac{2}{g(x)}$

2. Use the graphs of f and g in the accompanying figure to find the limits that exist. If the limit does not exist, explain why.

- (a) $\lim_{x \rightarrow 2} [f(x) + g(x)]$ (b) $\lim_{x \rightarrow 0} [f(x) + g(x)]$
 (c) $\lim_{x \rightarrow 0^+} [f(x) + g(x)]$ (d) $\lim_{x \rightarrow 0^-} [f(x) + g(x)]$
 (e) $\lim_{x \rightarrow 2} \frac{f(x)}{1 + g(x)}$ (f) $\lim_{x \rightarrow 2} \frac{1 + g(x)}{f(x)}$
 (g) $\lim_{x \rightarrow 0^+} \sqrt{f(x)}$ (h) $\lim_{x \rightarrow 0^-} \sqrt{f(x)}$



▲ Figure Ex-2

3. $\lim_{x \rightarrow 2} x(x-1)(x+1)$

5. $\lim_{x \rightarrow 3} \frac{x^2 - 2x}{x + 1}$

7. $\lim_{x \rightarrow 1^+} \frac{x^4 - 1}{x - 1}$

9. $\lim_{x \rightarrow -1} \frac{x^2 + 6x + 5}{x^2 - 3x - 4}$

11. $\lim_{x \rightarrow -1} \frac{2x^2 + x - 1}{x + 1}$

13. $\lim_{t \rightarrow 2} \frac{t^3 + 3t^2 - 12t + 4}{t^3 - 4t}$

15. $\lim_{x \rightarrow 3^+} \frac{x}{x - 3}$

17. $\lim_{x \rightarrow 3} \frac{x}{x - 3}$

19. $\lim_{x \rightarrow 2^-} \frac{x}{x^2 - 4}$

21. $\lim_{y \rightarrow 6^+} \frac{y + 6}{y^2 - 36}$

23. $\lim_{y \rightarrow 6} \frac{y + 6}{y^2 - 36}$

25. $\lim_{x \rightarrow 4^-} \frac{3 - x}{x^2 - 2x - 8}$

27. $\lim_{x \rightarrow 2^+} \frac{1}{|2 - x|}$

29. $\lim_{x \rightarrow 9} \frac{x - 9}{\sqrt{x} - 3}$

31. Let

$$f(x) = \begin{cases} x - 1, & x \leq 3 \\ 3x - 7, & x > 3 \end{cases}$$

4. $\lim_{x \rightarrow 3} x^3 - 3x^2 + 9x$

6. $\lim_{x \rightarrow 0} \frac{6x - 9}{x^3 - 12x + 3}$

8. $\lim_{t \rightarrow -2} \frac{t^3 + 8}{t + 2}$

10. $\lim_{x \rightarrow 2} \frac{x^2 - 4x + 4}{x^2 + x - 6}$

12. $\lim_{x \rightarrow 1} \frac{3x^2 - x - 2}{2x^2 + x - 3}$

14. $\lim_{t \rightarrow 1} \frac{t^3 + t^2 - 5t + 3}{t^3 - 3t + 2}$

16. $\lim_{x \rightarrow 3^-} \frac{x}{x - 3}$

18. $\lim_{x \rightarrow 2^+} \frac{x}{x^2 - 4}$

20. $\lim_{x \rightarrow 2} \frac{x}{x^2 - 4}$

22. $\lim_{y \rightarrow 6^-} \frac{y + 6}{y^2 - 36}$

24. $\lim_{x \rightarrow 4^+} \frac{3 - x}{x^2 - 2x - 8}$

26. $\lim_{x \rightarrow 4} \frac{3 - x}{x^2 - 2x - 8}$

28. $\lim_{x \rightarrow 3^-} \frac{1}{|x - 3|}$

30. $\lim_{y \rightarrow 4} \frac{4 - y}{2 - \sqrt{y}}$

Find

(a) $\lim_{x \rightarrow 3^-} f(x)$ (b) $\lim_{x \rightarrow 3^+} f(x)$ (c) $\lim_{x \rightarrow 3} f(x)$.

32. Let

$$g(t) = \begin{cases} t - 2, & t < 0 \\ t^2, & 0 \leq t \leq 2 \\ 2t, & t > 2 \end{cases}$$

Find

(a) $\lim_{t \rightarrow 0} g(t)$ (b) $\lim_{t \rightarrow 1} g(t)$ (c) $\lim_{t \rightarrow 2} g(t)$.

33–36 True–False Determine whether the statement is true or false. Explain your answer. ■

33. If $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist, then so does $\lim_{x \rightarrow a} [f(x) + g(x)]$.

34. If $\lim_{x \rightarrow a} g(x) = 0$ and $\lim_{x \rightarrow a} f(x)$ exists, then $\lim_{x \rightarrow a} [f(x)/g(x)]$ does not exist.

35. If $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ both exist and are equal, then $\lim_{x \rightarrow a} [f(x)/g(x)] = 1$.

36. If $f(x)$ is a rational function and $x = a$ is in the domain of f , then $\lim_{x \rightarrow a} f(x) = f(a)$.

37–38 First rationalize the numerator and then find the limit. ■

37. $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$

38. $\lim_{x \rightarrow 0} \frac{\sqrt{x^2+4} - 2}{x}$

39. Let

$$f(x) = \frac{x^3 - 1}{x - 1}$$

(a) Find $\lim_{x \rightarrow 1} f(x)$.

(b) Sketch the graph of $y = f(x)$.

40. Let

$$f(x) = \begin{cases} \frac{x^2 - 9}{x + 3}, & x \neq -3 \\ k, & x = -3 \end{cases}$$

(a) Find k so that $f(-3) = \lim_{x \rightarrow -3} f(x)$.

(b) With k assigned the value $\lim_{x \rightarrow -3} f(x)$, show that $f(x)$ can be expressed as a polynomial.

FOCUS ON CONCEPTS

41. (a) Explain why the following calculation is incorrect.

$$\begin{aligned} \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{x^2} \right) &= \lim_{x \rightarrow 0^+} \frac{1}{x} - \lim_{x \rightarrow 0^+} \frac{1}{x^2} \\ &= +\infty - (+\infty) = 0 \end{aligned}$$

(b) Show that $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{x^2} \right) = -\infty$.

42. (a) Explain why the following argument is incorrect.

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{2}{x^2 + 2x} \right) &= \lim_{x \rightarrow 0} \frac{1}{x} \left(1 - \frac{2}{x + 2} \right) \\ &= \infty \cdot 0 = 0 \end{aligned}$$

(b) Show that $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{2}{x^2 + 2x} \right) = \frac{1}{2}$.

43. Find all values of a such that

$$\lim_{x \rightarrow 1} \left(\frac{1}{x-1} - \frac{a}{x^2-1} \right)$$

exists and is finite.

44. (a) Explain informally why

$$\lim_{x \rightarrow 0^-} \left(\frac{1}{x} + \frac{1}{x^2} \right) = +\infty$$

(b) Verify the limit in part (a) algebraically.

45. Let $p(x)$ and $q(x)$ be polynomials, with $q(x_0) = 0$. Discuss the behavior of the graph of $y = p(x)/q(x)$ in the vicinity of $x = x_0$. Give examples to support your conclusions.

46. Suppose that f and g are two functions such that $\lim_{x \rightarrow a} f(x)$ exists but $\lim_{x \rightarrow a} [f(x) + g(x)]$ does not exist. Use Theorem 1.2.2. to prove that $\lim_{x \rightarrow a} g(x)$ does not exist.

47. Suppose that f and g are two functions such that both $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} [f(x) + g(x)]$ exist. Use Theorem 1.2.2 to prove that $\lim_{x \rightarrow a} g(x)$ exists.

48. Suppose that f and g are two functions such that

$$\lim_{x \rightarrow a} g(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

exists. Use Theorem 1.2.2 to prove that $\lim_{x \rightarrow a} f(x) = 0$.

49. **Writing** According to Newton's Law of Universal Gravitation, the gravitational force of attraction between two masses is inversely proportional to the square of the distance between them. What results of this section are useful in describing the gravitational force of attraction between the masses as they get closer and closer together?

50. **Writing** Suppose that f and g are two functions that are equal except at a finite number of points and that a denotes a real number. Explain informally why both

$$\lim_{x \rightarrow a} f(x) \quad \text{and} \quad \lim_{x \rightarrow a} g(x)$$

exist and are equal, or why both limits fail to exist. Write a short paragraph that explains the relationship of this result to the use of "algebraic simplification" in the evaluation of a limit.

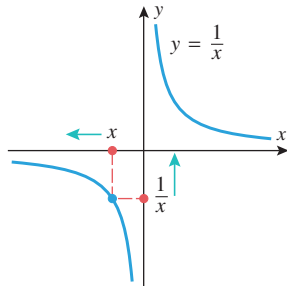
✓ QUICK CHECK ANSWERS 1.2

1. (a) 7 (b) 36 (c) -1 (d) 1 (e) $+\infty$ 2. (a) 7 (b) -3 (c) 1 3. (a) -1 (b) 0 (c) $+\infty$ (d) 8

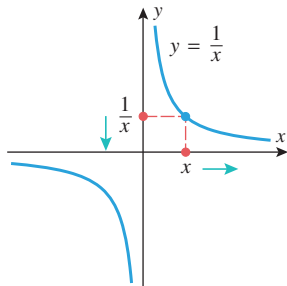
4. (a) 2 (b) 0 (c) does not exist

1.3 LIMITS AT INFINITY; END BEHAVIOR OF A FUNCTION

Up to now we have been concerned with limits that describe the behavior of a function $f(x)$ as x approaches some real number a . In this section we will be concerned with the behavior of $f(x)$ as x increases or decreases without bound.

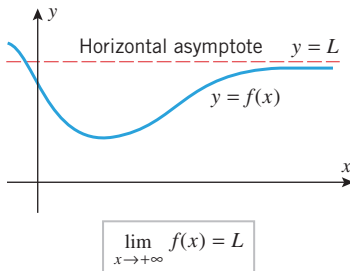


$$\lim_{x \rightarrow -\infty} \frac{1}{x} = 0$$

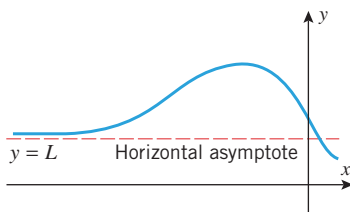


$$\lim_{x \rightarrow +\infty} \frac{1}{x} = 0$$

▲ Figure 1.3.1



$$\lim_{x \rightarrow +\infty} f(x) = L$$



$$\lim_{x \rightarrow -\infty} f(x) = L$$

▲ Figure 1.3.2

LIMITS AT INFINITY AND HORIZONTAL ASYMPTOTES

If the values of a variable x increase without bound, then we write $x \rightarrow +\infty$, and if the values of x decrease without bound, then we write $x \rightarrow -\infty$. The behavior of a function $f(x)$ as x increases without bound or decreases without bound is sometimes called the **end behavior** of the function. For example,

$$\lim_{x \rightarrow -\infty} \frac{1}{x} = 0 \quad \text{and} \quad \lim_{x \rightarrow +\infty} \frac{1}{x} = 0 \quad (1-2)$$

are illustrated numerically in Table 1.3.1 and geometrically in Figure 1.3.1.

Table 1.3.1

	VALUES						CONCLUSION
x	-1	-10	-100	-1000	-10,000	$\cdots$	As $x \rightarrow -\infty$ the value of $1/x$ increases toward zero.
$1/x$	-1	-0.1	-0.01	-0.001	-0.0001	$\cdots$	
x	1	10	100	1000	10,000	$\cdots$	As $x \rightarrow +\infty$ the value of $1/x$ decreases toward zero.
$1/x$	1	0.1	0.01	0.001	0.0001	$\cdots$	

In general, we will use the following notation.

1.3.1 LIMITS AT INFINITY (AN INFORMAL VIEW) If the values of $f(x)$ eventually get as close as we like to a number L as x increases without bound, then we write

$$\lim_{x \rightarrow +\infty} f(x) = L \quad \text{or} \quad f(x) \rightarrow L \text{ as } x \rightarrow +\infty \quad (3)$$

Similarly, if the values of $f(x)$ eventually get as close as we like to a number L as x decreases without bound, then we write

$$\lim_{x \rightarrow -\infty} f(x) = L \quad \text{or} \quad f(x) \rightarrow L \text{ as } x \rightarrow -\infty \quad (4)$$

Figure 1.3.2 illustrates the end behavior of a function f when

$$\lim_{x \rightarrow +\infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

In the first case the graph of f eventually comes as close as we like to the line $y = L$ as x increases without bound, and in the second case it eventually comes as close as we like to the line $y = L$ as x decreases without bound. If either limit holds, we call the line $y = L$ a **horizontal asymptote** for the graph of f .

► **Example 1** It follows from (1) and (2) that $y = 0$ is a horizontal asymptote for the graph of $f(x) = 1/x$ in both the positive and negative directions. This is consistent with the graph of $y = 1/x$ shown in Figure 1.3.1. ◀

■ LIMIT LAWS FOR LIMITS AT INFINITY

It can be shown that the limit laws in Theorem 1.2.2 carry over without change to limits at $+\infty$ and $-\infty$. Moreover, it follows by the same argument used in Section 1.2 that if n is a positive integer, then

$$\lim_{x \rightarrow +\infty} (f(x))^n = \left(\lim_{x \rightarrow +\infty} f(x) \right)^n \quad \lim_{x \rightarrow -\infty} (f(x))^n = \left(\lim_{x \rightarrow -\infty} f(x) \right)^n \quad (5-6)$$

provided the indicated limit of $f(x)$ exists. It also follows that constants can be moved through the limit symbols for limits at infinity:

$$\lim_{x \rightarrow +\infty} kf(x) = k \lim_{x \rightarrow +\infty} f(x) \quad \lim_{x \rightarrow -\infty} kf(x) = k \lim_{x \rightarrow -\infty} f(x) \quad (7-8)$$

provided the indicated limit of $f(x)$ exists.

Finally, if $f(x) = k$ is a constant function, then the values of f do not change as $x \rightarrow +\infty$ or as $x \rightarrow -\infty$, so

$$\lim_{x \rightarrow +\infty} k = k \quad \lim_{x \rightarrow -\infty} k = k \quad (9-10)$$

► **Example 2** It follows from (1), (2), (5), and (6) that if n is a positive integer, then

$$\lim_{x \rightarrow +\infty} \frac{1}{x^n} = \left(\lim_{x \rightarrow +\infty} \frac{1}{x} \right)^n = 0 \quad \text{and} \quad \lim_{x \rightarrow -\infty} \frac{1}{x^n} = \left(\lim_{x \rightarrow -\infty} \frac{1}{x} \right)^n = 0$$

■ INFINITE LIMITS AT INFINITY

Limits at infinity, like limits at a real number a , can fail to exist for various reasons. One such possibility is that the values of $f(x)$ increase or decrease without bound as $x \rightarrow +\infty$ or as $x \rightarrow -\infty$. We will use the following notation to describe this situation.

1.3.2 INFINITE LIMITS AT INFINITY (AN INFORMAL VIEW) If the values of $f(x)$ increase without bound as $x \rightarrow +\infty$ or as $x \rightarrow -\infty$, then we write

$$\lim_{x \rightarrow +\infty} f(x) = +\infty \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = +\infty$$

as appropriate; and if the values of $f(x)$ decrease without bound as $x \rightarrow +\infty$ or as $x \rightarrow -\infty$, then we write

$$\lim_{x \rightarrow +\infty} f(x) = -\infty \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = -\infty$$

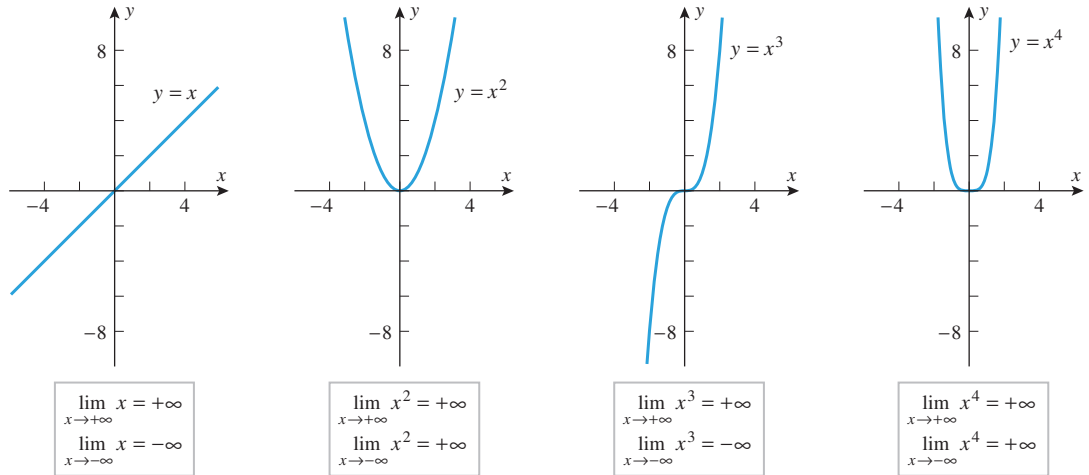
as appropriate.

LIMITS OF x^n AS $x \rightarrow \pm\infty$

Figure 1.3.3 illustrates the end behavior of the polynomials x^n for $n = 1, 2, 3$, and 4. These are special cases of the following general results:

$$\lim_{x \rightarrow +\infty} x^n = +\infty, \quad n = 1, 2, 3, \dots$$

$$\lim_{x \rightarrow -\infty} x^n = \begin{cases} -\infty, & n = 1, 3, 5, \dots \\ +\infty, & n = 2, 4, 6, \dots \end{cases} \quad (11-12)$$



▲ Figure 1.3.3

Multiplying x^n by a positive real number does not affect limits (11) and (12), but multiplying by a negative real number reverses the sign.

► Example 3

$$\begin{aligned} \lim_{x \rightarrow +\infty} 2x^5 &= +\infty, & \lim_{x \rightarrow -\infty} 2x^5 &= -\infty \\ \lim_{x \rightarrow +\infty} -7x^6 &= -\infty, & \lim_{x \rightarrow -\infty} -7x^6 &= -\infty \quad \blacktriangleleft \end{aligned}$$

LIMITS OF POLYNOMIALS AS $x \rightarrow \pm\infty$

There is a useful principle about polynomials which, expressed informally, states:

The end behavior of a polynomial matches the end behavior of its highest degree term.

More precisely, if $c_n \neq 0$, then

$$\lim_{x \rightarrow -\infty} (c_0 + c_1x + \cdots + c_nx^n) = \lim_{x \rightarrow -\infty} c_nx^n \quad (13)$$

$$\lim_{x \rightarrow +\infty} (c_0 + c_1x + \cdots + c_nx^n) = \lim_{x \rightarrow +\infty} c_nx^n \quad (14)$$

We can motivate these results by factoring out the highest power of x from the polynomial and examining the limit of the factored expression. Thus,

$$c_0 + c_1x + \cdots + c_nx^n = x^n \left(\frac{c_0}{x^n} + \frac{c_1}{x^{n-1}} + \cdots + c_n \right)$$

As $x \rightarrow -\infty$ or $x \rightarrow +\infty$, it follows from Example 2 that all of the terms with positive powers of x in the denominator approach 0, so (13) and (14) are certainly plausible.

► Example 4

$$\lim_{x \rightarrow -\infty} (7x^5 - 4x^3 + 2x - 9) = \lim_{x \rightarrow -\infty} 7x^5 = -\infty$$

$$\lim_{x \rightarrow -\infty} (-4x^8 + 17x^3 - 5x + 1) = \lim_{x \rightarrow -\infty} -4x^8 = -\infty \quad \blacktriangleleft$$

■ LIMITS OF RATIONAL FUNCTIONS AS $x \rightarrow \pm\infty$

One technique for determining the end behavior of a rational function is to divide each term in the numerator and denominator by the highest power of x that occurs in the denominator, after which the limiting behavior can be determined using results we have already established. Here are some examples.

► Example 5 Find $\lim_{x \rightarrow +\infty} \frac{3x + 5}{6x - 8}$.

Solution. Divide each term in the numerator and denominator by the highest power of x that occurs in the denominator, namely, $x^1 = x$. We obtain

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{3x + 5}{6x - 8} &= \lim_{x \rightarrow +\infty} \frac{3 + \frac{5}{x}}{6 - \frac{8}{x}} && \text{Divide each term by } x. \\ &= \frac{\lim_{x \rightarrow +\infty} \left(3 + \frac{5}{x} \right)}{\lim_{x \rightarrow +\infty} \left(6 - \frac{8}{x} \right)} && \text{Limit of a quotient is the quotient of the limits.} \\ &= \frac{\lim_{x \rightarrow +\infty} 3 + \lim_{x \rightarrow +\infty} \frac{5}{x}}{\lim_{x \rightarrow +\infty} 6 - \lim_{x \rightarrow +\infty} \frac{8}{x}} && \text{Limit of a sum is the sum of the limits.} \\ &= \frac{3 + 5 \lim_{x \rightarrow +\infty} \frac{1}{x}}{6 - 8 \lim_{x \rightarrow +\infty} \frac{1}{x}} = \frac{3 + 0}{6 - 0} = \frac{1}{2} && \text{A constant can be moved through a limit symbol; Formulas (2) and (9).} \quad \blacktriangleleft \end{aligned}$$

► Example 6 Find

$$(a) \lim_{x \rightarrow -\infty} \frac{4x^2 - x}{2x^3 - 5}$$

$$(b) \lim_{x \rightarrow +\infty} \frac{5x^3 - 2x^2 + 1}{1 - 3x}$$

Solution (a). Divide each term in the numerator and denominator by the highest power of x that occurs in the denominator, namely, x^3 . We obtain

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} \frac{4x^2 - x}{2x^3 - 5} &= \lim_{x \rightarrow -\infty} \frac{\frac{4}{x} - \frac{1}{x^2}}{2 - \frac{5}{x^3}} && \text{Divide each term by } x^3. \\
 &= \frac{\lim_{x \rightarrow -\infty} \left(\frac{4}{x} - \frac{1}{x^2} \right)}{\lim_{x \rightarrow -\infty} \left(2 - \frac{5}{x^3} \right)} && \text{Limit of a quotient is the quotient of the limits.} \\
 &= \frac{\lim_{x \rightarrow -\infty} \frac{4}{x} - \lim_{x \rightarrow -\infty} \frac{1}{x^2}}{\lim_{x \rightarrow -\infty} 2 - \lim_{x \rightarrow -\infty} \frac{5}{x^3}} && \text{Limit of a difference is the difference of the limits.} \\
 &= \frac{4 \lim_{x \rightarrow -\infty} \frac{1}{x} - \lim_{x \rightarrow -\infty} \frac{1}{x^2}}{2 - 5 \lim_{x \rightarrow -\infty} \frac{1}{x^3}} = \frac{0 - 0}{2 - 0} = 0 && \text{A constant can be moved through a limit symbol; Formula (10) and Example 2.}
 \end{aligned}$$

Solution (b). Divide each term in the numerator and denominator by the highest power of x that occurs in the denominator, namely, $x^1 = x$. We obtain

$$\lim_{x \rightarrow +\infty} \frac{5x^3 - 2x^2 + 1}{1 - 3x} = \lim_{x \rightarrow +\infty} \frac{5x^2 - 2x + \frac{1}{x}}{\frac{1}{x} - 3} \quad (15)$$

In this case we cannot argue that the limit of the quotient is the quotient of the limits because the limit of the numerator does not exist. However, we have

$$\lim_{x \rightarrow +\infty} 5x^2 - 2x = +\infty, \quad \lim_{x \rightarrow +\infty} \frac{1}{x} = 0, \quad \lim_{x \rightarrow +\infty} \left(\frac{1}{x} - 3 \right) = -3$$

Thus, the numerator on the right side of (15) approaches $+\infty$ and the denominator has a finite *negative* limit. We conclude from this that the quotient approaches $-\infty$; that is,

$$\lim_{x \rightarrow +\infty} \frac{5x^3 - 2x^2 + 1}{1 - 3x} = \lim_{x \rightarrow +\infty} \frac{5x^2 - 2x + \frac{1}{x}}{\frac{1}{x} - 3} = -\infty \quad \blacktriangleleft$$

■ A QUICK METHOD FOR FINDING LIMITS OF RATIONAL FUNCTIONS AS $x \rightarrow +\infty$ OR $x \rightarrow -\infty$

Since the end behavior of a polynomial matches the end behavior of its highest degree term, one can reasonably conclude:

The end behavior of a rational function matches the end behavior of the quotient of the highest degree term in the numerator divided by the highest degree term in the denominator.

► **Example 7** Use the preceding observation to compute the limits in Examples 5 and 6.

Solution.

$$\begin{aligned}\lim_{x \rightarrow +\infty} \frac{3x+5}{6x-8} &= \lim_{x \rightarrow +\infty} \frac{3x}{6x} = \lim_{x \rightarrow +\infty} \frac{1}{2} = \frac{1}{2} \\ \lim_{x \rightarrow -\infty} \frac{4x^2-x}{2x^3-5} &= \lim_{x \rightarrow -\infty} \frac{4x^2}{2x^3} = \lim_{x \rightarrow -\infty} \frac{2}{x} = 0 \\ \lim_{x \rightarrow +\infty} \frac{5x^3-2x^2+1}{1-3x} &= \lim_{x \rightarrow +\infty} \frac{5x^3}{(-3x)} = \lim_{x \rightarrow +\infty} \left(-\frac{5}{3}x^2\right) = -\infty \quad \blacktriangleleft\end{aligned}$$

■ LIMITS INVOLVING RADICALS

► **Example 8** Find

$$(a) \lim_{x \rightarrow +\infty} \frac{\sqrt{x^2+2}}{3x-6} \quad (b) \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+2}}{3x-6}$$

In both parts it would be helpful to manipulate the function so that the powers of x are transformed to powers of $1/x$. This can be achieved in both cases by dividing the numerator and denominator by $|x|$ and using the fact that $\sqrt{x^2} = |x|$.

Solution (a). As $x \rightarrow +\infty$, the values of x under consideration are positive, so we can replace $|x|$ by x where helpful. We obtain

$$\begin{aligned}\lim_{x \rightarrow +\infty} \frac{\sqrt{x^2+2}}{3x-6} &= \lim_{x \rightarrow +\infty} \frac{\frac{\sqrt{x^2+2}}{|x|}}{\frac{3x-6}{|x|}} = \lim_{x \rightarrow +\infty} \frac{\frac{\sqrt{x^2+2}}{\sqrt{x^2}}}{\frac{3x-6}{x}} \\ &= \lim_{x \rightarrow +\infty} \frac{\sqrt{1+\frac{2}{x^2}}}{3-\frac{6}{x}} = \frac{\lim_{x \rightarrow +\infty} \sqrt{1+\frac{2}{x^2}}}{\lim_{x \rightarrow +\infty} \left(3-\frac{6}{x}\right)} \\ &= \frac{\sqrt{\lim_{x \rightarrow +\infty} \left(1+\frac{2}{x^2}\right)}}{\lim_{x \rightarrow +\infty} \left(3-\frac{6}{x}\right)} = \frac{\sqrt{\left(\lim_{x \rightarrow +\infty} 1\right) + \left(2 \lim_{x \rightarrow +\infty} \frac{1}{x^2}\right)}}{\left(\lim_{x \rightarrow +\infty} 3\right) - \left(6 \lim_{x \rightarrow +\infty} \frac{1}{x}\right)} \\ &= \frac{\sqrt{1+(2 \cdot 0)}}{3-(6 \cdot 0)} = \frac{1}{3}\end{aligned}$$

TECHNOLOGY MASTERY

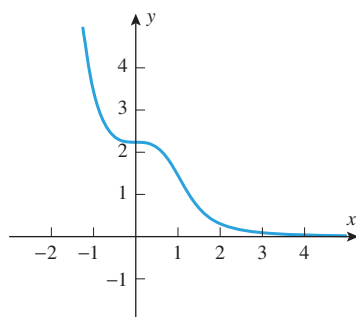
It follows from Example 8 that the function

$$f(x) = \frac{\sqrt{x^2+2}}{3x-6}$$

has an asymptote of $y = \frac{1}{3}$ in the positive direction and an asymptote of $y = -\frac{1}{3}$ in the negative direction. Confirm this using a graphing utility.

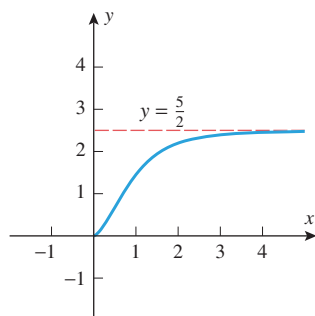
Solution (b). As $x \rightarrow -\infty$, the values of x under consideration are negative, so we can replace $|x|$ by $-x$ where helpful. We obtain

$$\begin{aligned}\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+2}}{3x-6} &= \lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{x^2+2}}{|x|}}{\frac{3x-6}{|x|}} = \lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{x^2+2}}{\sqrt{x^2}}}{\frac{3x-6}{(-x)}} \\ &= \lim_{x \rightarrow -\infty} \frac{\sqrt{1+\frac{2}{x^2}}}{-3+\frac{6}{x}} = -\frac{1}{3} \quad \blacktriangleleft\end{aligned}$$



$$y = \sqrt{x^6 + 5} - x^3$$

(a)

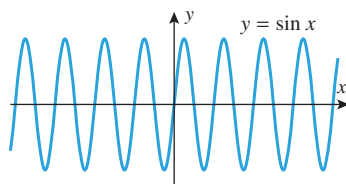


$$y = \sqrt{x^6 + 5x^3} - x^3, x \geq 0$$

(b)

▲ Figure 1.3.4

We noted in Section 1.1 that the standard rules of algebra do not apply to the symbols $+\infty$ and $-\infty$. Part (b) of Example 9 illustrates this. The terms $\sqrt{x^6 + 5x^3}$ and x^3 both approach $+\infty$ as $x \rightarrow +\infty$, but their difference does not approach 0.



There is no limit as $x \rightarrow +\infty$ or $x \rightarrow -\infty$.

▲ Figure 1.3.5

► **Example 9** Find

$$(a) \lim_{x \rightarrow +\infty} (\sqrt{x^6 + 5} - x^3) \quad (b) \lim_{x \rightarrow +\infty} (\sqrt{x^6 + 5x^3} - x^3)$$

Solution. Graphs of the functions $f(x) = \sqrt{x^6 + 5} - x^3$, and $g(x) = \sqrt{x^6 + 5x^3} - x^3$ for $x \geq 0$, are shown in Figure 1.3.4. From the graphs we might conjecture that the requested limits are 0 and $\frac{5}{2}$, respectively. To confirm this, we treat each function as a fraction with a denominator of 1 and rationalize the numerator.

$$\begin{aligned} \lim_{x \rightarrow +\infty} (\sqrt{x^6 + 5} - x^3) &= \lim_{x \rightarrow +\infty} (\sqrt{x^6 + 5} - x^3) \left(\frac{\sqrt{x^6 + 5} + x^3}{\sqrt{x^6 + 5} + x^3} \right) \\ &= \lim_{x \rightarrow +\infty} \frac{(x^6 + 5) - x^6}{\sqrt{x^6 + 5} + x^3} = \lim_{x \rightarrow +\infty} \frac{5}{\sqrt{x^6 + 5} + x^3} \\ &= \lim_{x \rightarrow +\infty} \frac{5}{\sqrt{1 + \frac{5}{x^6}} + 1} \quad \boxed{\sqrt{x^6} = x^3 \text{ for } x > 0} \\ &= \frac{0}{\sqrt{1 + 0} + 1} = 0 \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow +\infty} (\sqrt{x^6 + 5x^3} - x^3) &= \lim_{x \rightarrow +\infty} (\sqrt{x^6 + 5x^3} - x^3) \left(\frac{\sqrt{x^6 + 5x^3} + x^3}{\sqrt{x^6 + 5x^3} + x^3} \right) \\ &= \lim_{x \rightarrow +\infty} \frac{(x^6 + 5x^3) - x^6}{\sqrt{x^6 + 5x^3} + x^3} = \lim_{x \rightarrow +\infty} \frac{5x^3}{\sqrt{x^6 + 5x^3} + x^3} \\ &= \lim_{x \rightarrow +\infty} \frac{5}{\sqrt{1 + \frac{5}{x^3}} + 1} \quad \boxed{\sqrt{x^6} = x^3 \text{ for } x > 0} \\ &= \frac{5}{\sqrt{1 + 0} + 1} = \frac{5}{2} \quad \blacktriangleleft \end{aligned}$$

■ **END BEHAVIOR OF TRIGONOMETRIC FUNCTIONS**

Consider the function $f(x) = \sin x$ that is graphed in Figure 1.3.5. For this function the limits as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$ fail to exist not because $f(x)$ increases or decreases without bound, but rather because the values vary between -1 and 1 without approaching some specific real number. In general, the trigonometric functions fail to have limits as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$ because of periodicity. There is no specific notation to denote this kind of behavior.

✓ **QUICK CHECK EXERCISES 1.3** (See page 80 for answers.)

1. Find the limits.

$$(a) \lim_{x \rightarrow +\infty} (-2x) = \underline{\hspace{2cm}}$$

$$(b) \lim_{x \rightarrow -\infty} \frac{x}{|x|} = \underline{\hspace{2cm}}$$

$$(c) \lim_{x \rightarrow -\infty} (3 - x) = \underline{\hspace{2cm}}$$

$$(d) \lim_{x \rightarrow +\infty} \left(5 - \frac{1}{x} \right) = \underline{\hspace{2cm}}$$

2. Find the limits that exist.

(a) $\lim_{x \rightarrow -\infty} \frac{2x^2 + x}{4x^2 - 3} = \underline{\hspace{2cm}}$

(b) $\lim_{x \rightarrow +\infty} \frac{1}{2 + \sin x} = \underline{\hspace{2cm}}$

3. Given that

$$\lim_{x \rightarrow +\infty} f(x) = 2 \quad \text{and} \quad \lim_{x \rightarrow +\infty} g(x) = -3$$

find the limits that exist.

(a) $\lim_{x \rightarrow +\infty} [3f(x) - g(x)] = \underline{\hspace{2cm}}$

(b) $\lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = \underline{\hspace{2cm}}$

(c) $\lim_{x \rightarrow +\infty} \frac{2f(x) + 3g(x)}{3f(x) + 2g(x)} = \underline{\hspace{2cm}}$

(d) $\lim_{x \rightarrow +\infty} \sqrt{10 - f(x)g(x)} = \underline{\hspace{2cm}}$

4. Consider the graphs of
- $y = 1/(x + 1)$
- ,
- $y = x/(x + 1)$
- , and
- $y = x^2/(x + 1)$
- . Which of these graphs has a horizontal asymptote?

EXERCISE SET 1.3



1–4 In these exercises, make reasonable assumptions about the end behavior of the indicated function. ■

1. For the function
- g
- graphed in the accompanying figure, find

(a) $\lim_{x \rightarrow -\infty} g(x)$ (b) $\lim_{x \rightarrow +\infty} g(x)$.

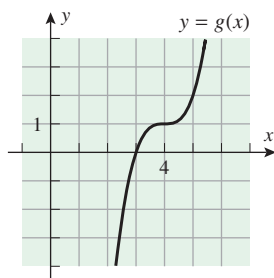


Figure Ex-1

2. For the function
- ϕ
- graphed in the accompanying figure, find

(a) $\lim_{x \rightarrow -\infty} \phi(x)$
(b) $\lim_{x \rightarrow +\infty} \phi(x)$.

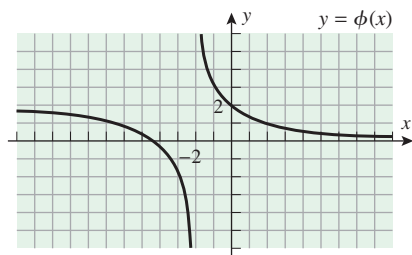


Figure Ex-2

3. For the function
- ϕ
- graphed in the accompanying figure, find

(a) $\lim_{x \rightarrow -\infty} \phi(x)$ (b) $\lim_{x \rightarrow +\infty} \phi(x)$.

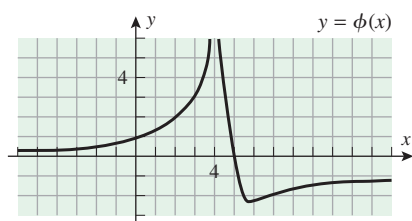


Figure Ex-3

4. For the function
- G
- graphed in the accompanying figure, find

(a) $\lim_{x \rightarrow -\infty} G(x)$ (b) $\lim_{x \rightarrow +\infty} G(x)$.

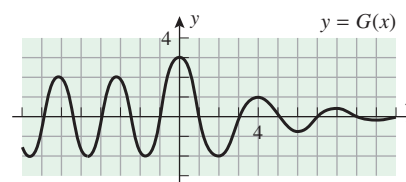


Figure Ex-4

5. Given that

$$\lim_{x \rightarrow +\infty} f(x) = 3, \quad \lim_{x \rightarrow +\infty} g(x) = -5, \quad \lim_{x \rightarrow +\infty} h(x) = 0$$

find the limits that exist. If the limit does not exist, explain why.

(a) $\lim_{x \rightarrow +\infty} [f(x) + 3g(x)]$

(b) $\lim_{x \rightarrow +\infty} [h(x) - 4g(x) + 1]$

(c) $\lim_{x \rightarrow +\infty} [f(x)g(x)]$

(d) $\lim_{x \rightarrow +\infty} [g(x)]^2$

(e) $\lim_{x \rightarrow +\infty} \sqrt[3]{5 + f(x)}$

(f) $\lim_{x \rightarrow +\infty} \frac{3}{g(x)}$

(g) $\lim_{x \rightarrow +\infty} \frac{3h(x) + 4}{x^2}$

(h) $\lim_{x \rightarrow +\infty} \frac{6f(x)}{5f(x) + 3g(x)}$

6. Given that

$$\lim_{x \rightarrow -\infty} f(x) = 7 \quad \text{and} \quad \lim_{x \rightarrow -\infty} g(x) = -6$$

find the limits that exist. If the limit does not exist, explain why.

(a) $\lim_{x \rightarrow -\infty} [2f(x) - g(x)]$

(b) $\lim_{x \rightarrow -\infty} [6f(x) + 7g(x)]$

(c) $\lim_{x \rightarrow -\infty} [x^2 + g(x)]$

(d) $\lim_{x \rightarrow -\infty} [x^2 g(x)]$

(e) $\lim_{x \rightarrow -\infty} \sqrt[3]{f(x)g(x)}$

(f) $\lim_{x \rightarrow -\infty} \frac{g(x)}{f(x)}$

(g) $\lim_{x \rightarrow -\infty} \left[f(x) + \frac{g(x)}{x} \right]$

(h) $\lim_{x \rightarrow -\infty} \frac{xf(x)}{(2x + 3)g(x)}$

7. Complete the table and make a guess about the limit indicated.

$$f(x) = \frac{\sqrt{x^2 + x}}{x + 1} \quad \lim_{x \rightarrow +\infty} f(x)$$

x	10	100	1000	10,000	100,000	1,000,000
$f(x)$						

8. Complete the table and make a guess about the limit indicated.

$$f(x) = \frac{\sqrt{x^2 + x}}{x + 1} \quad \lim_{x \rightarrow -\infty} f(x)$$

x	-10	-100	-1000	-10,000	-100,000	-1,000,000
$f(x)$						

9–32 Find the limits. ■

9. $\lim_{x \rightarrow +\infty} (1 + 2x - 3x^5)$
10. $\lim_{x \rightarrow +\infty} (2x^3 - 100x + 5)$
11. $\lim_{x \rightarrow +\infty} \sqrt{x}$
12. $\lim_{x \rightarrow -\infty} \sqrt{5 - x}$
13. $\lim_{x \rightarrow +\infty} \frac{3x + 1}{2x - 5}$
14. $\lim_{x \rightarrow +\infty} \frac{5x^2 - 4x}{2x^2 + 3}$
15. $\lim_{y \rightarrow -\infty} \frac{3}{y + 4}$
16. $\lim_{x \rightarrow +\infty} \frac{1}{x - 12}$
17. $\lim_{x \rightarrow -\infty} \frac{x - 2}{x^2 + 2x + 1}$
18. $\lim_{x \rightarrow +\infty} \frac{5x^2 + 7}{3x^2 - x}$
19. $\lim_{x \rightarrow +\infty} \frac{7 - 6x^5}{x + 3}$
20. $\lim_{t \rightarrow -\infty} \frac{5 - 2t^3}{t^2 + 1}$
21. $\lim_{t \rightarrow +\infty} \frac{6 - t^3}{7t^3 + 3}$
22. $\lim_{x \rightarrow -\infty} \frac{x + 4x^3}{1 - x^2 + 7x^3}$
23. $\lim_{x \rightarrow +\infty} \sqrt[3]{\frac{2 + 3x - 5x^2}{1 + 8x^2}}$
24. $\lim_{s \rightarrow +\infty} \sqrt[3]{\frac{3s^7 - 4s^5}{2s^7 + 1}}$
25. $\lim_{x \rightarrow -\infty} \frac{\sqrt{5x^2 - 2}}{x + 3}$
26. $\lim_{x \rightarrow +\infty} \frac{\sqrt{5x^2 - 2}}{x + 3}$
27. $\lim_{y \rightarrow -\infty} \frac{2 - y}{\sqrt{7 + 6y^2}}$
28. $\lim_{y \rightarrow +\infty} \frac{2 - y}{\sqrt{7 + 6y^2}}$
29. $\lim_{x \rightarrow -\infty} \frac{\sqrt{3x^4 + x}}{x^2 - 8}$
30. $\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^4 + x}}{x^2 - 8}$
31. $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 3} - x)$
32. $\lim_{x \rightarrow +\infty} (\sqrt{x^2 - 3x} - x)$

33–36 True–False Determine whether the statement is true or false. Explain your answer. ■

33. By subtraction, $\lim_{x \rightarrow +\infty} (x^2 - 1000x) = \infty - \infty = 0$.
34. If $y = L$ is a horizontal asymptote for the curve $y = f(x)$, then

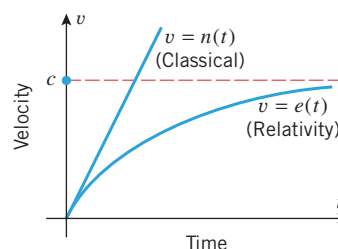
$$\lim_{x \rightarrow -\infty} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow +\infty} f(x) = L$$

35. If $y = L$ is a horizontal asymptote for the curve $y = f(x)$, then it is possible for the graph of f to intersect the line $y = L$ infinitely many times.

36. If a rational function $p(x)/q(x)$ has a horizontal asymptote, then the degree of $p(x)$ must equal the degree of $q(x)$.

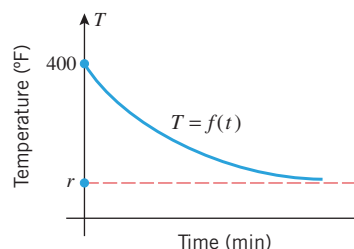
FOCUS ON CONCEPTS

37. Assume that a particle is accelerated by a constant force. The two curves $v = n(t)$ and $v = e(t)$ in the accompanying figure provide velocity versus time curves for the particle as predicted by classical physics and by the special theory of relativity, respectively. The parameter c represents the speed of light. Using the language of limits, describe the differences in the long-term predictions of the two theories.



◀ Figure Ex-37

38. Let $T = f(t)$ denote the temperature of a baked potato t minutes after it has been removed from a hot oven. The accompanying figure shows the temperature versus time curve for the potato, where r is the temperature of the room.
- (a) What is the physical significance of $\lim_{t \rightarrow 0^+} f(t)$?
- (b) What is the physical significance of $\lim_{t \rightarrow +\infty} f(t)$?



◀ Figure Ex-38

39. Let

$$f(x) = \begin{cases} 2x^2 + 5, & x < 0 \\ \frac{3 - 5x^3}{1 + 4x + x^3}, & x \geq 0 \end{cases}$$

Find

- (a) $\lim_{x \rightarrow -\infty} f(x)$ (b) $\lim_{x \rightarrow +\infty} f(x)$.

40. Let

$$g(t) = \begin{cases} \frac{2 + 3t}{5t^2 + 6}, & t < 1,000,000 \\ \frac{\sqrt{36t^2 - 100}}{5 - t}, & t > 1,000,000 \end{cases}$$

Find

- (a) $\lim_{t \rightarrow -\infty} g(t)$ (b) $\lim_{t \rightarrow +\infty} g(t)$.

41. Discuss the limits of $p(x) = (1 - x)^n$ as $x \rightarrow +\infty$ and $x \rightarrow -\infty$ for positive integer values of n .
42. In each part, find examples of polynomials $p(x)$ and $q(x)$ that satisfy the stated condition and such that $p(x) \rightarrow +\infty$ and $q(x) \rightarrow +\infty$ as $x \rightarrow +\infty$.
- (a) $\lim_{x \rightarrow +\infty} \frac{p(x)}{q(x)} = 1$ (b) $\lim_{x \rightarrow +\infty} \frac{p(x)}{q(x)} = 0$
- (c) $\lim_{x \rightarrow +\infty} \frac{p(x)}{q(x)} = +\infty$ (d) $\lim_{x \rightarrow +\infty} [p(x) - q(x)] = 3$
43. (a) Do any of the trigonometric functions $\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, and $\csc x$ have horizontal asymptotes?
 (b) Do any of the trigonometric functions have vertical asymptotes? Where?
44. Find $\lim_{x \rightarrow +\infty} \frac{c_0 + c_1x + \cdots + c_nx^n}{d_0 + d_1x + \cdots + d_mx^m}$
 where $c_n \neq 0$ and $d_m \neq 0$. [Hint: Your answer will depend on whether $m < n$, $m = n$, or $m > n$.]

FOCUS ON CONCEPTS

45–46 These exercises develop some versions of the *substitution principle*, a useful tool for the evaluation of limits.

45. (a) Suppose $g(x) \rightarrow +\infty$ as $x \rightarrow +\infty$. Given any function $f(x)$, explain why we can evaluate $\lim_{x \rightarrow +\infty} f[g(x)]$ by substituting $t = g(x)$ and writing

$$\lim_{x \rightarrow +\infty} f[g(x)] = \lim_{t \rightarrow +\infty} f(t)$$

(Here, “equality” is interpreted to mean that either both limits exist and are equal or that both limits fail to exist.)

- (b) Why does the result in part (a) remain valid if $\lim_{x \rightarrow +\infty}$ is replaced everywhere by one of $\lim_{x \rightarrow -\infty}$, $\lim_{x \rightarrow c}$, $\lim_{x \rightarrow c^-}$, or $\lim_{x \rightarrow c^+}$?

46. (a) Suppose $g(x) \rightarrow -\infty$ as $x \rightarrow +\infty$. Given any function $f(x)$, explain why we can evaluate $\lim_{x \rightarrow +\infty} f[g(x)]$ by substituting $t = g(x)$ and writing

$$\lim_{x \rightarrow +\infty} f[g(x)] = \lim_{t \rightarrow -\infty} f(t)$$

(Here, “equality” is interpreted to mean that either both limits exist and are equal or that both limits fail to exist.)

- (b) Why does the result in part (a) remain valid if $\lim_{x \rightarrow +\infty}$ is replaced everywhere by one of $\lim_{x \rightarrow -\infty}$, $\lim_{x \rightarrow c}$, $\lim_{x \rightarrow c^-}$, or $\lim_{x \rightarrow c^+}$?

47–50 Given that

$$\lim_{x \rightarrow -\infty} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow +\infty} f(x) = +\infty$$

evaluate the limit using an appropriate substitution. ■

47. $\lim_{x \rightarrow 0^+} f(1/x)$ 48. $\lim_{x \rightarrow 0^-} f(1/x)$
 49. $\lim_{x \rightarrow 0^+} f(\csc x)$ 50. $\lim_{x \rightarrow 0^-} f(\csc x)$

 **51–55** The notion of an asymptote can be extended to include curves as well as lines. Specifically, we say that curves $y = f(x)$ and $y = g(x)$ are *asymptotic as $x \rightarrow +\infty$* provided

$$\lim_{x \rightarrow +\infty} [f(x) - g(x)] = 0$$

and are *asymptotic as $x \rightarrow -\infty$* provided

$$\lim_{x \rightarrow -\infty} [f(x) - g(x)] = 0$$

In these exercises, determine a simpler function $g(x)$ such that $y = f(x)$ is asymptotic to $y = g(x)$ as $x \rightarrow +\infty$ or $x \rightarrow -\infty$. Use a graphing utility to generate the graphs of $y = f(x)$ and $y = g(x)$ and identify all vertical asymptotes. ■

51. $f(x) = \frac{x^2 - 2}{x - 2}$ [Hint: Divide $x - 2$ into $x^2 - 2$.]
 52. $f(x) = \frac{x^3 - x + 3}{x}$ 53. $f(x) = \frac{-x^3 + 3x^2 + x - 1}{x - 3}$
 54. $f(x) = \frac{x^5 - x^3 + 3}{x^2 - 1}$ 55. $f(x) = \sin x + \frac{1}{x - 1}$
56. **Writing** In some models for learning a skill (e.g., juggling), it is assumed that the skill level for an individual increases with practice but cannot become arbitrarily high. How do concepts of this section apply to such a model?
57. **Writing** In some population models it is assumed that a given ecological system possesses a *carrying capacity* L . Populations greater than the carrying capacity tend to decline toward L , while populations less than the carrying capacity tend to increase toward L . Explain why these assumptions are reasonable, and discuss how the concepts of this section apply to such a model.

 QUICK CHECK ANSWERS 1.3

1. (a) $-\infty$ (b) -1 (c) $+\infty$ (d) 5 2. (a) $\frac{1}{2}$ (b) does not exist 3. (a) 9 (b) $-\frac{2}{3}$ (c) does not exist (d) 4
 4. The graphs of $y = 1/(x + 1)$ and $y = x/(x + 1)$ have horizontal asymptotes.

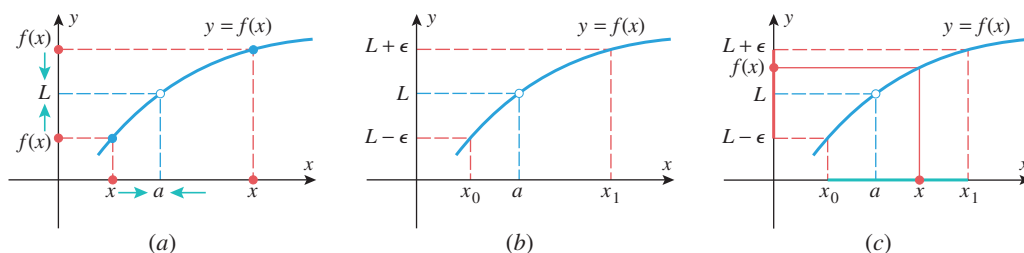
1.4 LIMITS (DISCUSSED MORE RIGOROUSLY)

In the previous sections of this chapter we focused on the discovery of values of limits, either by sampling selected x -values or by applying limit theorems that were stated without proof. Our main goal in this section is to define the notion of a limit precisely, thereby making it possible to establish limits with certainty and to prove theorems about them. This will also provide us with a deeper understanding of some of the more subtle properties of functions.

MOTIVATION FOR THE DEFINITION OF A TWO-SIDED LIMIT

The statement $\lim_{x \rightarrow a} f(x) = L$ can be interpreted informally to mean that we can make the value of $f(x)$ as close as we like to the real number L by making the value of x sufficiently close to a . It is our goal to make the informal phrases “as close as we like to L ” and “sufficiently close to a ” mathematically precise.

To do this, consider the function f graphed in Figure 1.4.1a for which $f(x) \rightarrow L$ as $x \rightarrow a$. For visual simplicity we have drawn the graph of f to be increasing on an open interval containing a , and we have intentionally placed a hole in the graph at $x = a$ to emphasize that f need not be defined at $x = a$ to have a limit there.



▲ Figure 1.4.1

Next, let us choose any positive number ϵ and ask how close x must be to a in order for the values of $f(x)$ to be within ϵ units of L . We can answer this geometrically by drawing horizontal lines from the points $L + \epsilon$ and $L - \epsilon$ on the y -axis until they meet the curve $y = f(x)$, and then drawing vertical lines from those points on the curve to the x -axis (Figure 1.4.1b). As indicated in the figure, let x_0 and x_1 be the points where those vertical lines intersect the x -axis.

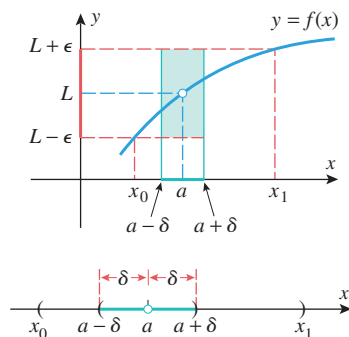
Now imagine that x gets closer and closer to a (from either side). Eventually, x will lie inside the interval (x_0, x_1) , which is marked in green in Figure 1.4.1c; and when this happens, the value of $f(x)$ will fall between $L - \epsilon$ and $L + \epsilon$, marked in red in the figure. Thus, we conclude:

If $f(x) \rightarrow L$ as $x \rightarrow a$, then for any positive number ϵ , we can find an open interval (x_0, x_1) on the x -axis that contains a and has the property that for each x in that interval (except possibly for $x = a$), the value of $f(x)$ is between $L - \epsilon$ and $L + \epsilon$.

What is important about this result is that it holds no matter how small we make ϵ . However, making ϵ smaller and smaller forces $f(x)$ closer and closer to L —which is precisely the concept we were trying to capture mathematically.

Observe that in Figure 1.4.1 the interval (x_0, x_1) extends farther on the right side of a than on the left side. However, for many purposes it is preferable to have an interval that extends the same distance on both sides of a . For this purpose, let us choose any positive number δ that is smaller than both $x_1 - a$ and $a - x_0$, and consider the interval

$$(a - \delta, a + \delta)$$



▲ Figure 1.4.2

The definitions of one-sided limits require minor adjustments to Definition 1.4.1. For example, for a limit from the right we need only assume that $f(x)$ is defined on an interval (a, b) extending to the right of a and that the ϵ condition is met for x in an interval $a < x < a + \delta$ extending to the right of a . A similar adjustment must be made for a limit from the left. (See Exercise 27.)

This interval extends the same distance δ on both sides of a and lies inside of the interval (x_0, x_1) (Figure 1.4.2). Moreover, the condition

$$L - \epsilon < f(x) < L + \epsilon \quad (1)$$

holds for every x in this interval (except possibly $x = a$), since this condition holds on the larger interval (x_0, x_1) .

Since (1) can be expressed as

$$|f(x) - L| < \epsilon$$

and the condition that x lies in the interval $(a - \delta, a + \delta)$, but $x \neq a$, can be expressed as

$$0 < |x - a| < \delta$$

we are led to the following precise definition of a two-sided limit.

1.4.1 LIMIT DEFINITION Let $f(x)$ be defined for all x in some open interval containing the number a , with the possible exception that $f(x)$ need not be defined at a . We will write

$$\lim_{x \rightarrow a} f(x) = L$$

if given any number $\epsilon > 0$ we can find a number $\delta > 0$ such that

$$|f(x) - L| < \epsilon \quad \text{if} \quad 0 < |x - a| < \delta$$

This definition, which is attributed to the German mathematician Karl Weierstrass and is commonly called the “epsilon-delta” definition of a two-sided limit, makes the transition from an informal concept of a limit to a precise definition. Specifically, the informal phrase “as close as we like to L ” is given quantitative meaning by our ability to choose the positive number ϵ arbitrarily, and the phrase “sufficiently close to a ” is quantified by the positive number δ .

In the preceding sections we illustrated various numerical and graphical methods for *guessing* at limits. Now that we have a precise definition to work with, we can actually confirm the validity of those guesses with mathematical proof. Here is a typical example of such a proof.



Karl Weierstrass (1815–1897) Weierstrass, the son of a customs officer, was born in Ostenfelde, Germany. As a youth Weierstrass showed outstanding skills in languages and mathematics. However, at the urging of his dominant father, Weierstrass entered the law and commerce program at the University of Bonn. To the chagrin of his family, the rugged and congenial young man concentrated instead on fencing and beer drinking. Four years later he returned home without a degree. In 1839 Weierstrass entered the Academy of Münster to study for a career in secondary education, and he met and studied under an excellent mathematician named Christof Gudermann. Gudermann’s ideas greatly influenced the work of Weierstrass. After receiving his teaching certificate, Weierstrass spent the next 15 years in secondary education teaching German, geography, and mathematics. In addition, he taught handwriting to small children. During this period much of Weierstrass’s mathematical work

was ignored because he was a secondary schoolteacher and not a college professor. Then, in 1854, he published a paper of major importance that created a sensation in the mathematics world and catapulted him to international fame overnight. He was immediately given an honorary Doctorate at the University of Königsberg and began a new career in college teaching at the University of Berlin in 1856. In 1859 the strain of his mathematical research caused a temporary nervous breakdown and led to spells of dizziness that plagued him for the rest of his life. Weierstrass was a brilliant teacher and his classes overflowed with multitudes of auditors. In spite of his fame, he never lost his early beer-drinking congeniality and was always in the company of students, both ordinary and brilliant. Weierstrass was acknowledged as the leading mathematical analyst in the world. He and his students opened the door to the modern school of mathematical analysis.

[Image: http://commons.wikimedia.org/wiki/File:Karl_Weierstrass.jpg]

► **Example 1** Use Definition 1.4.1 to prove that $\lim_{x \rightarrow 2} (3x - 5) = 1$.

Solution. We must show that given any positive number ϵ , we can find a positive number δ such that

$$\left| \underbrace{(3x - 5)}_{f(x)} - \underbrace{1}_L \right| < \epsilon \quad \text{if} \quad 0 < \underbrace{|x - 2|}_a < \delta \quad (2)$$

There are two things to do. First, we must *discover* a value of δ for which this statement holds, and then we must *prove* that the statement holds for that δ . For the discovery part we begin by simplifying (2) and writing it as

$$|3x - 6| < \epsilon \quad \text{if} \quad 0 < |x - 2| < \delta$$

Next we will rewrite this statement in a form that will facilitate the discovery of an appropriate δ :

$$\begin{aligned} 3|x - 2| &< \epsilon & \text{if} & \quad 0 < |x - 2| < \delta \\ |x - 2| &< \epsilon/3 & \text{if} & \quad 0 < |x - 2| < \delta \end{aligned} \quad (3)$$

It should be self-evident that this last statement holds if $\delta = \epsilon/3$, which completes the discovery portion of our work. Now we need to prove that (2) holds for this choice of δ . However, statement (2) is equivalent to (3), and (3) holds with $\delta = \epsilon/3$, so (2) also holds with $\delta = \epsilon/3$. This proves that $\lim_{x \rightarrow 2} (3x - 5) = 1$. ◀

This example illustrates the general form of a limit proof: We *assume* that we are given a positive number ϵ , and we try to *prove* that we can find a positive number δ such that

$$|f(x) - L| < \epsilon \quad \text{if} \quad 0 < |x - a| < \delta \quad (4)$$

This is done by first discovering δ , and then proving that the discovered δ works. Since the argument has to be general enough to work for all positive values of ϵ , the quantity δ has to be expressed as a function of ϵ . In Example 1 we found the function $\delta = \epsilon/3$ by some simple algebra; however, most limit proofs require a little more algebraic and logical ingenuity. Thus, if you find our ensuing discussion of “ ϵ - δ ” proofs challenging, do not become discouraged; the concepts and techniques are intrinsically difficult. In fact, a precise understanding of limits evaded the finest mathematical minds for more than 150 years after the basic concepts of calculus were discovered.

► **Example 2** Prove that $\lim_{x \rightarrow 0^+} \sqrt{x} = 0$.

Solution. Note that the domain of $\sqrt{x}$ is $0 \leq x$, so it is valid to discuss the limit as $x \rightarrow 0^+$. We must show that given $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|\sqrt{x} - 0| < \epsilon \quad \text{if} \quad 0 < x - 0 < \delta$$

or more simply,

$$\sqrt{x} < \epsilon \quad \text{if} \quad 0 < x < \delta \quad (5)$$

But, by squaring both sides of the inequality $\sqrt{x} < \epsilon$, we can rewrite (5) as

$$x < \epsilon^2 \quad \text{if} \quad 0 < x < \delta \quad (6)$$

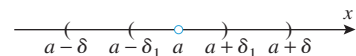
It should be self-evident that (6) is true if $\delta = \epsilon^2$; and since (6) is a reformulation of (5), we have shown that (5) holds with $\delta = \epsilon^2$. This proves that $\lim_{x \rightarrow 0^+} \sqrt{x} = 0$. ◀

In Example 2 the limit from the left and the two-sided limit do not exist at $x = 0$ because $\sqrt{x}$ is defined only for nonnegative values of x .

THE VALUE OF δ IS NOT UNIQUE

In preparation for our next example, we note that the value of δ in Definition 1.4.1 is not unique; once we have found a value of δ that fulfills the requirements of the definition, then any *smaller* positive number δ_1 will also fulfill those requirements. That is, if it is true that

$$|f(x) - L| < \epsilon \quad \text{if} \quad 0 < |x - a| < \delta$$



▲ Figure 1.4.3

then it will also be true that

$$|f(x) - L| < \epsilon \quad \text{if} \quad 0 < |x - a| < \delta_1$$

This is because $\{x : 0 < |x - a| < \delta_1\}$ is a subset of $\{x : 0 < |x - a| < \delta\}$ (Figure 1.4.3), and hence if $|f(x) - L| < \epsilon$ is satisfied for all x in the larger set, then it will automatically be satisfied for all x in the subset. Thus, in Example 1, where we used $\delta = \epsilon/3$, we could have used any smaller value of δ such as $\delta = \epsilon/4$, $\delta = \epsilon/5$, or $\delta = \epsilon/6$.

► **Example 3** Prove that $\lim_{x \rightarrow 3} x^2 = 9$.

Solution. We must show that given any positive number ϵ , we can find a positive number δ such that

$$|x^2 - 9| < \epsilon \quad \text{if} \quad 0 < |x - 3| < \delta \quad (7)$$

Because $|x - 3|$ occurs on the right side of this “if statement,” it will be helpful to factor the left side to introduce a factor of $|x - 3|$. This yields the following alternative form of (7):

$$|x + 3||x - 3| < \epsilon \quad \text{if} \quad 0 < |x - 3| < \delta \quad (8)$$

We wish to bound the factor $|x + 3|$. If we knew, for example, that $\delta \leq 1$, then we would have $-1 < x - 3 < 1$, so $5 < x + 3 < 7$, and consequently $|x + 3| < 7$. Thus, if $\delta \leq 1$ and $0 < |x - 3| < \delta$, then

$$|x + 3||x - 3| < 7\delta$$

It follows that (8) will be satisfied for any positive δ such that $\delta \leq 1$ and $7\delta < \epsilon$. We can achieve this by taking δ to be the minimum of the numbers 1 and $\epsilon/7$, which is sometimes written as $\delta = \min(1, \epsilon/7)$. This proves that $\lim_{x \rightarrow 3} x^2 = 9$. ◀

If you are wondering how we knew to make the restriction $\delta \leq 1$, as opposed to $\delta \leq 5$ or $\delta \leq \frac{1}{2}$, for example, the answer is that 1 is merely a convenient choice—any restriction of the form $\delta \leq c$ would work equally well.

■ LIMITS AS $x \rightarrow \pm\infty$

In Section 1.3 we discussed the limits

$$\lim_{x \rightarrow +\infty} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

from an intuitive point of view. The first limit can be interpreted to mean that we can make the value of $f(x)$ as close as we like to L by taking x sufficiently large, and the second can be interpreted to mean that we can make the value of $f(x)$ as close as we like to L by taking x sufficiently far to the left of 0. These ideas are captured in the following definitions and are illustrated in Figure 1.4.4.

1.4.2 DEFINITION Let $f(x)$ be defined for all x in some infinite open interval extending in the positive x -direction. We will write

$$\lim_{x \rightarrow +\infty} f(x) = L$$

if given any number $\epsilon > 0$, there corresponds a positive number N such that

$$|f(x) - L| < \epsilon \quad \text{if} \quad x > N$$

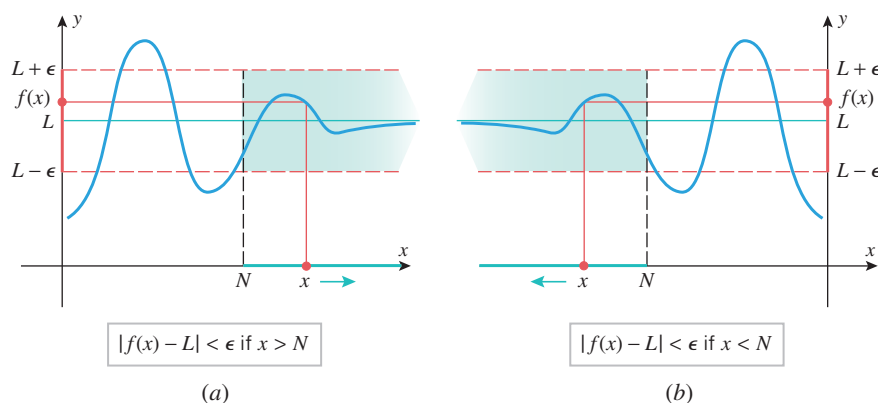
1.4.3 DEFINITION Let $f(x)$ be defined for all x in some infinite open interval extending in the negative x -direction. We will write

$$\lim_{x \rightarrow -\infty} f(x) = L$$

if given any number $\epsilon > 0$, there corresponds a negative number N such that

$$|f(x) - L| < \epsilon \quad \text{if} \quad x < N$$

To see how these definitions relate to our informal concepts of these limits, suppose that $f(x) \rightarrow L$ as $x \rightarrow +\infty$, and for a given ϵ let N be the positive number described in Definition 1.4.2. If x is allowed to increase indefinitely, then eventually x will lie in the interval $(N, +\infty)$, which is marked in green in Figure 1.4.4a; when this happens, the value of $f(x)$ will fall between $L - \epsilon$ and $L + \epsilon$, marked in red in the figure. Since this is true for all positive values of ϵ (no matter how small), we can force the values of $f(x)$ as close as we like to L by making N sufficiently large. This agrees with our informal concept of this limit. Similarly, Figure 1.4.4b illustrates Definition 1.4.3.



▲ Figure 1.4.4

► **Example 4** Prove that $\lim_{x \rightarrow +\infty} \frac{1}{x} = 0$.

Solution. Applying Definition 1.4.2 with $f(x) = 1/x$ and $L = 0$, we must show that given $\epsilon > 0$, we can find a number $N > 0$ such that

$$\left| \frac{1}{x} - 0 \right| < \epsilon \quad \text{if } x > N \quad (9)$$

Because $x \rightarrow +\infty$ we can assume that $x > 0$. Thus, we can eliminate the absolute values in this statement and rewrite it as

$$\frac{1}{x} < \epsilon \quad \text{if } x > N$$

or, on taking reciprocals,

$$x > \frac{1}{\epsilon} \quad \text{if } x > N \quad (10)$$

It is self-evident that $N = 1/\epsilon$ satisfies this requirement, and since (10) and (9) are equivalent for $x > 0$, the proof is complete. ◀

■ INFINITE LIMITS

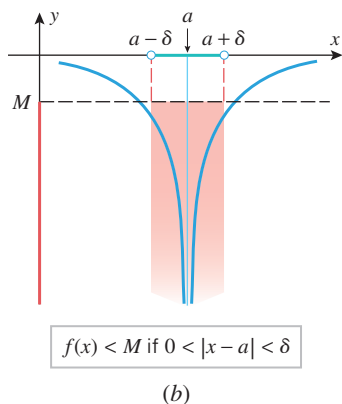
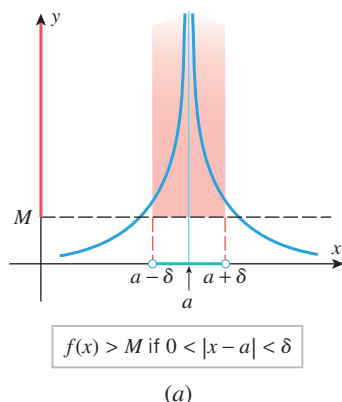
In Section 1.1 we discussed limits of the following type from an intuitive viewpoint:

$$\lim_{x \rightarrow a} f(x) = +\infty, \quad \lim_{x \rightarrow a} f(x) = -\infty \quad (11)$$

$$\lim_{x \rightarrow a^+} f(x) = +\infty, \quad \lim_{x \rightarrow a^+} f(x) = -\infty \quad (12)$$

$$\lim_{x \rightarrow a^-} f(x) = +\infty, \quad \lim_{x \rightarrow a^-} f(x) = -\infty \quad (13)$$

Recall that each of these expressions describes a particular way in which the limit fails to exist. The $+\infty$ indicates that the limit fails to exist because $f(x)$ increases without bound,



▲ Figure 1.4.5

How would you define these limits?

$\lim_{x \rightarrow a^+} f(x) = +\infty$	$\lim_{x \rightarrow a^+} f(x) = -\infty$
$\lim_{x \rightarrow a^-} f(x) = +\infty$	$\lim_{x \rightarrow a^-} f(x) = -\infty$
$\lim_{x \rightarrow +\infty} f(x) = +\infty$	$\lim_{x \rightarrow +\infty} f(x) = -\infty$
$\lim_{x \rightarrow -\infty} f(x) = +\infty$	$\lim_{x \rightarrow -\infty} f(x) = -\infty$

and the $-\infty$ indicates that the limit fails to exist because $f(x)$ decreases without bound. These ideas are captured more precisely in the following definitions and are illustrated in Figure 1.4.5.

1.4.4 DEFINITION Let $f(x)$ be defined for all x in some open interval containing a , except that $f(x)$ need not be defined at a . We will write

$$\lim_{x \rightarrow a} f(x) = +\infty$$

if given any positive number M , we can find a number $\delta > 0$ such that $f(x)$ satisfies

$$f(x) > M \quad \text{if} \quad 0 < |x - a| < \delta$$

1.4.5 DEFINITION Let $f(x)$ be defined for all x in some open interval containing a , except that $f(x)$ need not be defined at a . We will write

$$\lim_{x \rightarrow a} f(x) = -\infty$$

if given any negative number M , we can find a number $\delta > 0$ such that $f(x)$ satisfies

$$f(x) < M \quad \text{if} \quad 0 < |x - a| < \delta$$

To see how these definitions relate to our informal concepts of these limits, suppose that $f(x) \rightarrow +\infty$ as $x \rightarrow a$, and for a given M let δ be the corresponding positive number described in Definition 1.4.4. Next, imagine that x gets closer and closer to a (from either side). Eventually, x will lie in the interval $(a - \delta, a + \delta)$, which is marked in green in Figure 1.4.5a; when this happens the value of $f(x)$ will be greater than M , marked in red in the figure. Since this is true for any positive value of M (no matter how large), we can force the values of $f(x)$ to be as large as we like by making x sufficiently close to a . This agrees with our informal concept of this limit. Similarly, Figure 1.4.5b illustrates Definition 1.4.5.

► **Example 5** Prove that $\lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty$.

Solution. Applying Definition 1.4.4 with $f(x) = 1/x^2$ and $a = 0$, we must show that given a number $M > 0$, we can find a number $\delta > 0$ such that

$$\frac{1}{x^2} > M \quad \text{if} \quad 0 < |x - 0| < \delta \quad (14)$$

or, on taking reciprocals and simplifying,

$$x^2 < \frac{1}{M} \quad \text{if} \quad 0 < |x| < \delta \quad (15)$$

But $x^2 < 1/M$ if $|x| < 1/\sqrt{M}$, so that $\delta = 1/\sqrt{M}$ satisfies (15). Since (14) is equivalent to (15), the proof is complete. ◀

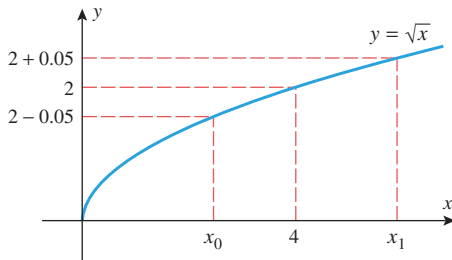
✓ QUICK CHECK EXERCISES 1.4 (See page 90 for answers.)

- The definition of a two-sided limit states: $\lim_{x \rightarrow a} f(x) = L$ if given any number _____ there is a number _____ such that $|f(x) - L| < \epsilon$ if _____.
- Suppose that $f(x)$ is a function such that for any given $\epsilon > 0$, the condition $0 < |x - 1| < \epsilon/2$ guarantees that $|f(x) - 5| < \epsilon$. What limit results from this property?

3. Suppose that ϵ is any positive number. Find the largest value of δ such that $|5x - 10| < \epsilon$ if $0 < |x - 2| < \delta$.
4. The definition of limit at $+\infty$ states: $\lim_{x \rightarrow +\infty} f(x) = L$ if given any number _____ there is a positive number _____ such that $|f(x) - L| < \epsilon$ if _____.
5. Find the smallest positive number N such that for each $x > N$, the value of $f(x) = 1/\sqrt{x}$ is within 0.01 of 0.

EXERCISE SET 1.4 Graphing Utility

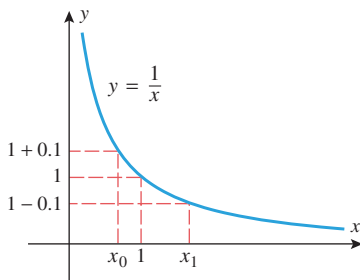
1. (a) Find the largest open interval, centered at the origin on the x -axis, such that for each x in the interval the value of the function $f(x) = x + 2$ is within 0.1 unit of the number $f(0) = 2$.
- (b) Find the largest open interval, centered at $x = 3$, such that for each x in the interval the value of the function $f(x) = 4x - 5$ is within 0.01 unit of the number $f(3) = 7$.
- (c) Find the largest open interval, centered at $x = 4$, such that for each x in the interval the value of the function $f(x) = x^2$ is within 0.001 unit of the number $f(4) = 16$.
2. In each part, find the largest open interval, centered at $x = 0$, such that for each x in the interval the value of $f(x) = 2x + 3$ is within ϵ units of the number $f(0) = 3$.
- (a) $\epsilon = 0.1$ (b) $\epsilon = 0.01$
(c) $\epsilon = 0.0012$
3. (a) Find the values of x_0 and x_1 in the accompanying figure.
- (b) Find a positive number δ such that $|\sqrt{x} - 2| < 0.05$ if $0 < |x - 4| < \delta$.



Not drawn to scale

▲ Figure Ex-3

4. (a) Find the values of x_0 and x_1 in the accompanying figure.
- (b) Find a positive number δ such that $|(1/x) - 1| < 0.1$ if $0 < |x - 1| < \delta$.



Not drawn to scale

◀ Figure Ex-4

-  5. Generate the graph of $f(x) = x^3 - 4x + 5$ with a graphing utility, and use the graph to find a number δ such that $|f(x) - 2| < 0.05$ if $0 < |x - 1| < \delta$. [Hint: Show that the inequality $|f(x) - 2| < 0.05$ can be rewritten as $1.95 < x^3 - 4x + 5 < 2.05$, and estimate the values of x for which $x^3 - 4x + 5 = 1.95$ and $x^3 - 4x + 5 = 2.05$.]
-  6. Use the method of Exercise 5 to find a number δ such that $|\sqrt{5x + 1} - 4| < 0.5$ if $0 < |x - 3| < \delta$.
-  7. Let $f(x) = x + \sqrt{x}$ with $L = \lim_{x \rightarrow 1} f(x)$ and let $\epsilon = 0.2$. Use a graphing utility and its trace feature to find a positive number δ such that $|f(x) - L| < \epsilon$ if $0 < |x - 1| < \delta$.
-  8. Let $f(x) = (\sin 2x)/x$ and use a graphing utility to conjecture the value of $L = \lim_{x \rightarrow 0} f(x)$. Then let $\epsilon = 0.1$ and use the graphing utility and its trace feature to find a positive number δ such that $|f(x) - L| < \epsilon$ if $0 < |x| < \delta$.

9–16 A positive number ϵ and the limit L of a function f at a are given. Find a number δ such that $|f(x) - L| < \epsilon$ if $0 < |x - a| < \delta$. ■

9. $\lim_{x \rightarrow 4} 2x = 8$; $\epsilon = 0.1$ 10. $\lim_{x \rightarrow 3} (5x - 2) = 13$; $\epsilon = 0.01$

11. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6$; $\epsilon = 0.05$

12. $\lim_{x \rightarrow -1/2} \frac{4x^2 - 1}{2x + 1} = -2$; $\epsilon = 0.05$

13. $\lim_{x \rightarrow 2} x^3 = 8$; $\epsilon = 0.001$

14. $\lim_{x \rightarrow 4} \sqrt{x} = 2$; $\epsilon = 0.001$

15. $\lim_{x \rightarrow 5} \frac{1}{x} = \frac{1}{5}$; $\epsilon = 0.05$ 16. $\lim_{x \rightarrow 0} |x| = 0$; $\epsilon = 0.05$

17–26 Use Definition 1.4.1 to prove that the limit is correct. ■

17. $\lim_{x \rightarrow 2} 3 = 3$ 18. $\lim_{x \rightarrow 4} (x + 2) = 6$

19. $\lim_{x \rightarrow 5} 3x = 15$ 20. $\lim_{x \rightarrow -1} (7x + 5) = -2$

21. $\lim_{x \rightarrow 0} \frac{2x^2 + x}{x} = 1$ 22. $\lim_{x \rightarrow -3} \frac{x^2 - 9}{x + 3} = -6$

23. $\lim_{x \rightarrow 1} f(x) = 3$, where $f(x) = \begin{cases} x + 2, & x \neq 1 \\ 10, & x = 1 \end{cases}$

24. $\lim_{x \rightarrow 2} f(x) = 5$, where $f(x) = \begin{cases} 9 - 2x, & x \neq 2 \\ 49, & x = 2 \end{cases}$

25. $\lim_{x \rightarrow 0} |x| = 0$

26. $\lim_{x \rightarrow 2} f(x) = 5$, where $f(x) = \begin{cases} 9 - 2x, & x < 2 \\ 3x - 1, & x > 2 \end{cases}$

FOCUS ON CONCEPTS

27. Give rigorous definitions of $\lim_{x \rightarrow a^+} f(x) = L$ and $\lim_{x \rightarrow a^-} f(x) = L$.

28. Consider the statement that $\lim_{x \rightarrow a} |f(x) - L| = 0$.
 (a) Using Definition 1.4.1, write down precisely what this limit statement means.
 (b) Explain why your answer to part (a) shows that

$$\lim_{x \rightarrow a} |f(x) - L| = 0 \quad \text{if and only if} \quad \lim_{x \rightarrow a} f(x) = L$$

29. (a) Show that

$$|(3x^2 + 2x - 20) - 300| = |3x + 32| \cdot |x - 10|$$

(b) Find an upper bound for $|3x + 32|$ if x satisfies $|x - 10| < 1$.

(c) Fill in the blanks to complete a proof that

$$\lim_{x \rightarrow 10} [3x^2 + 2x - 20] = 300$$

Suppose that $\epsilon > 0$. Set $\delta = \min(1, \text{_____})$ and assume that $0 < |x - 10| < \delta$. Then

$$\begin{aligned} |(3x^2 + 2x - 20) - 300| &= |3x + 32| \cdot |x - 10| \\ &< \text{_____} \cdot |x - 10| \\ &< \text{_____} \cdot \text{_____} \\ &= \epsilon \end{aligned}$$

30. (a) Show that

$$\left| \frac{28}{3x+1} - 4 \right| = \left| \frac{12}{3x+1} \right| \cdot |x-2|$$

(b) Is $|12/(3x+1)|$ bounded if $|x-2| < 4$? If not, explain; if so, give a bound.

(c) Is $|12/(3x+1)|$ bounded if $|x-2| < 1$? If not, explain; if so, give a bound.

(d) Fill in the blanks to complete a proof that

$$\lim_{x \rightarrow 2} \left[\frac{28}{3x+1} \right] = 4$$

Suppose that $\epsilon > 0$. Set $\delta = \min(1, \text{_____})$ and assume that $0 < |x - 2| < \delta$. Then

$$\begin{aligned} \left| \frac{28}{3x+1} - 4 \right| &= \left| \frac{12}{3x+1} \right| \cdot |x-2| \\ &< \text{_____} \cdot |x-2| \\ &< \text{_____} \cdot \text{_____} \\ &= \epsilon \end{aligned}$$

31–36 Use Definition 1.4.1 to prove that the stated limit is correct. In each case, to show that $\lim_{x \rightarrow a} f(x) = L$, factor $|f(x) - L|$ in the form

$$|f(x) - L| = |\text{“something”}| \cdot |x - a|$$

and then bound the size of |“something”| by putting restrictions on the size of δ . ■

31. $\lim_{x \rightarrow 1} 2x^2 = 2$ [Hint: Assume $\delta \leq 1$.]

32. $\lim_{x \rightarrow 3} (x^2 + x) = 12$ [Hint: Assume $\delta \leq 1$.]

33. $\lim_{x \rightarrow -2} \frac{1}{x+1} = -1$

34. $\lim_{x \rightarrow 1/2} \frac{2x+3}{x} = 8$

35. $\lim_{x \rightarrow 4} \sqrt{x} = 2$

36. $\lim_{x \rightarrow 2} x^3 = 8$

37. Let

$$f(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ x, & \text{if } x \text{ is irrational} \end{cases}$$

Use Definition 1.4.1 to prove that $\lim_{x \rightarrow 0} f(x) = 0$.

38. Let

$$f(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ 1, & \text{if } x \text{ is irrational} \end{cases}$$

Use Definition 1.4.1 to prove that $\lim_{x \rightarrow 0} f(x)$ does not exist. [Hint: Assume $\lim_{x \rightarrow 0} f(x) = L$ and apply Definition 1.4.1 with $\epsilon = \frac{1}{2}$ to conclude that $|1 - L| < \frac{1}{2}$ and $|L| = |0 - L| < \frac{1}{2}$. Then show $1 \leq |1 - L| + |L|$ and derive a contradiction.]

39. (a) Find the smallest positive number N such that for each x in the interval $(N, +\infty)$, the value of the function $f(x) = 1/x^2$ is within 0.1 unit of $L = 0$.

(b) Find the smallest positive number N such that for each x in the interval $(N, +\infty)$, the value of $f(x) = x/(x+1)$ is within 0.01 unit of $L = 1$.

(c) Find the largest negative number N such that for each x in the interval $(-\infty, N)$, the value of the function $f(x) = 1/x^3$ is within 0.001 unit of $L = 0$.

(d) Find the largest negative number N such that for each x in the interval $(-\infty, N)$, the value of the function $f(x) = x/(x+1)$ is within 0.01 unit of $L = 1$.

40. In each part, find the smallest positive value of N such that for each x in the interval $(N, +\infty)$, the function $f(x) = 1/x^3$ is within ϵ units of the number $L = 0$.

(a) $\epsilon = 0.1$ (b) $\epsilon = 0.01$ (c) $\epsilon = 0.001$

41. (a) Find the values of x_1 and x_2 in the accompanying figure.

(b) Find a positive number N such that

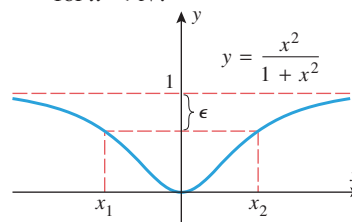
$$\left| \frac{x^2}{1+x^2} - 1 \right| < \epsilon$$

for $x > N$.

(c) Find a negative number N such that

$$\left| \frac{x^2}{1+x^2} - 1 \right| < \epsilon$$

for $x < N$.



Not drawn to scale

◀ Figure Ex-41

42. (a) Find the values of x_1 and x_2 in the accompanying figure.
 (b) Find a positive number N such that

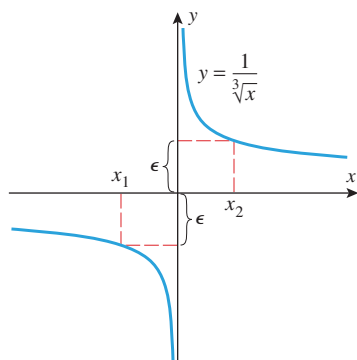
$$\left| \frac{1}{\sqrt[3]{x}} - 0 \right| = \left| \frac{1}{\sqrt[3]{x}} \right| < \epsilon$$

for $x > N$.

- (c) Find a negative number N such that

$$\left| \frac{1}{\sqrt[3]{x}} - 0 \right| = \left| \frac{1}{\sqrt[3]{x}} \right| < \epsilon$$

for $x < N$.



◀ Figure Ex-42

43–46 A positive number ϵ and the limit L of a function f at $+\infty$ are given. Find a positive number N such that $|f(x) - L| < \epsilon$ if $x > N$. ■

43. $\lim_{x \rightarrow +\infty} \frac{1}{x^2} = 0$; $\epsilon = 0.01$

44. $\lim_{x \rightarrow +\infty} \frac{1}{x+2} = 0$; $\epsilon = 0.005$

45. $\lim_{x \rightarrow +\infty} \frac{x}{x+1} = 1$; $\epsilon = 0.001$

46. $\lim_{x \rightarrow +\infty} \frac{4x-1}{2x+5} = 2$; $\epsilon = 0.1$

47–50 A positive number ϵ and the limit L of a function f at $-\infty$ are given. Find a negative number N such that $|f(x) - L| < \epsilon$ if $x < N$. ■

47. $\lim_{x \rightarrow -\infty} \frac{1}{x+2} = 0$; $\epsilon = 0.005$

48. $\lim_{x \rightarrow -\infty} \frac{1}{x^2} = 0$; $\epsilon = 0.01$

49. $\lim_{x \rightarrow -\infty} \frac{4x-1}{2x+5} = 2$; $\epsilon = 0.1$

50. $\lim_{x \rightarrow -\infty} \frac{x}{x+1} = 1$; $\epsilon = 0.001$

51–56 Use Definition 1.4.2 or 1.4.3 to prove that the stated limit is correct. ■

51. $\lim_{x \rightarrow +\infty} \frac{1}{x^2} = 0$

52. $\lim_{x \rightarrow +\infty} \frac{1}{x+2} = 0$

53. $\lim_{x \rightarrow -\infty} \frac{4x-1}{2x+5} = 2$

54. $\lim_{x \rightarrow -\infty} \frac{x}{x+1} = 1$

55. $\lim_{x \rightarrow +\infty} \frac{2\sqrt{x}}{\sqrt{x}-1} = 2$

56. $\lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x}}{\sqrt[3]{x}+2} = 1$

57. (a) Find the largest open interval, centered at the origin on the x -axis, such that for each x in the interval, other than the center, the values of $f(x) = 1/x^2$ are greater than 100.
 (b) Find the largest open interval, centered at $x = 1$, such that for each x in the interval, other than the center, the values of the function $f(x) = 1/|x-1|$ are greater than 1000.
 (c) Find the largest open interval, centered at $x = 3$, such that for each x in the interval, other than the center, the values of the function $f(x) = -1/(x-3)^2$ are less than -1000 .
 (d) Find the largest open interval, centered at the origin on the x -axis, such that for each x in the interval, other than the center, the values of $f(x) = -1/x^4$ are less than $-10,000$.

58. In each part, find the largest open interval centered at $x = 1$, such that for each x in the interval, other than the center, the value of $f(x) = 1/(x-1)^2$ is greater than M .

(a) $M = 10$ (b) $M = 1000$ (c) $M = 100,000$

59–64 Use Definition 1.4.4 or 1.4.5 to prove that the stated limit is correct. ■

59. $\lim_{x \rightarrow 3} \frac{1}{(x-3)^2} = +\infty$

60. $\lim_{x \rightarrow 3} \frac{-1}{(x-3)^2} = -\infty$

61. $\lim_{x \rightarrow 0} \frac{1}{|x|} = +\infty$

62. $\lim_{x \rightarrow 1} \frac{1}{|x-1|} = +\infty$

63. $\lim_{x \rightarrow 0} \left(-\frac{1}{x^4}\right) = -\infty$

64. $\lim_{x \rightarrow 0} \frac{1}{x^4} = +\infty$

65–70 Use the definitions in Exercise 27 to prove that the stated one-sided limit is correct. ■

65. $\lim_{x \rightarrow 2^+} (x+1) = 3$

66. $\lim_{x \rightarrow 1^-} (3x+2) = 5$

67. $\lim_{x \rightarrow 4^+} \sqrt{x-4} = 0$

68. $\lim_{x \rightarrow 0^-} \sqrt{-x} = 0$

69. $\lim_{x \rightarrow 2^+} f(x) = 2$, where $f(x) = \begin{cases} x, & x > 2 \\ 3x, & x \leq 2 \end{cases}$

70. $\lim_{x \rightarrow 2^-} f(x) = 6$, where $f(x) = \begin{cases} x, & x > 2 \\ 3x, & x \leq 2 \end{cases}$

71–74 Write out the definition for the corresponding limit in the marginal note on page 86, and use your definition to prove that the stated limit is correct. ■

71. (a) $\lim_{x \rightarrow 1^+} \frac{1}{1-x} = -\infty$ (b) $\lim_{x \rightarrow 1^-} \frac{1}{1-x} = +\infty$

72. (a) $\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$

(b) $\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$

73. (a) $\lim_{x \rightarrow +\infty} (x+1) = +\infty$

(b) $\lim_{x \rightarrow -\infty} (x+1) = -\infty$

74. (a) $\lim_{x \rightarrow +\infty} (x^2-3) = +\infty$

(b) $\lim_{x \rightarrow -\infty} (x^3+5) = -\infty$

75. According to Ohm's law, when a voltage of V volts is applied across a resistor with a resistance of R ohms, a current of $I = V/R$ amperes flows through the resistor. (cont.)

- How much current flows if a voltage of 3.0 volts is applied across a resistance of 7.5 ohms?
- If the resistance varies by ± 0.1 ohm, and the voltage remains constant at 3.0 volts, what is the resulting range of values for the current?
- If temperature variations cause the resistance to vary by $\pm \delta$ from its value of 7.5 ohms, and the voltage remains constant at 3.0 volts, what is the resulting range of values for the current?
- If the current is not allowed to vary by more than $\epsilon = \pm 0.001$ ampere at a voltage of 3.0 volts, what variation of $\pm \delta$ from the value of 7.5 ohms is allowable?
- Certain alloys become **superconductors** as their temperature approaches absolute zero (-273°C), meaning that their resistance approaches zero. If the voltage remains constant, what happens to the current in a superconductor as $R \rightarrow 0^+$?

- Writing** Compare informal Definition 1.1.1 with Definition 1.4.1.
 - What portions of Definition 1.4.1 correspond to the expression “values of $f(x)$ can be made as close as we like to L ” in Definition 1.1.1? Explain.
 - What portions of Definition 1.4.1 correspond to the expression “taking values of x sufficiently close to a (but not equal to a)” in Definition 1.1.1? Explain.
- Writing** Compare informal Definition 1.3.1 with Definition 1.4.2.
 - What portions of Definition 1.4.2 correspond to the expression “values of $f(x)$ eventually get as close as we like to a number L ” in Definition 1.3.1? Explain.
 - What portions of Definition 1.4.2 correspond to the expression “as x increases without bound” in Definition 1.3.1? Explain.

✓ QUICK CHECK ANSWERS 1.4

1. $\epsilon > 0$; $\delta > 0$; $0 < |x - a| < \delta$ 2. $\lim_{x \rightarrow 1} f(x) = 5$ 3. $\delta = \epsilon/5$ 4. $\epsilon > 0$; N ; $x > N$ 5. $N = 10,000$

1.5 CONTINUITY

A thrown baseball cannot vanish at some point and reappear someplace else to continue its motion. Thus, we perceive the path of the ball as an unbroken curve. In this section, we translate “unbroken curve” into a precise mathematical formulation called continuity, and develop some fundamental properties of continuous curves.



Joseph Helfenberger/iStockphoto

A baseball moves along a “continuous” trajectory after leaving the pitcher’s hand.

■ DEFINITION OF CONTINUITY

Intuitively, the graph of a function can be described as a “continuous curve” if it has no breaks or holes. To make this idea more precise we need to understand what properties of a function can cause breaks or holes. Referring to Figure 1.5.1, we see that the graph of a function has a break or hole if any of the following conditions occur:

- The function f is undefined at c (Figure 1.5.1a).
- The limit of $f(x)$ does not exist as x approaches c (Figures 1.5.1b, 1.5.1c).
- The value of the function and the value of the limit at c are different (Figure 1.5.1d).

This suggests the following definition.

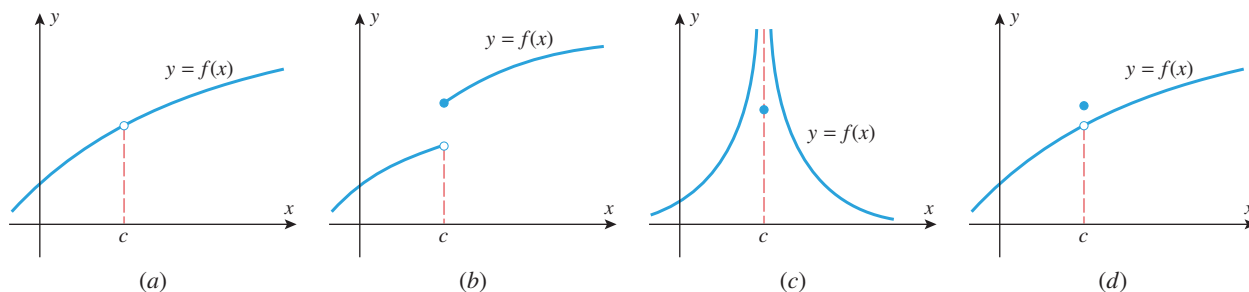
The third condition in Definition 1.5.1 actually implies the first two, since it is tacitly understood in the statement

$$\lim_{x \rightarrow c} f(x) = f(c)$$

that the limit exists and the function is defined at c . Thus, when we want to establish continuity at c our usual procedure will be to verify the third condition only.

1.5.1 DEFINITION A function f is said to be **continuous at $x = c$** provided the following conditions are satisfied:

- $f(c)$ is defined.
- $\lim_{x \rightarrow c} f(x)$ exists.
- $\lim_{x \rightarrow c} f(x) = f(c)$.



▲ Figure 1.5.1

If one or more of the conditions of this definition fails to hold, then we will say that f has a **discontinuity at $x = c$** . Each function drawn in Figure 1.5.1 illustrates a discontinuity at $x = c$. In Figure 1.5.1a, the function is not defined at c , violating the first condition of Definition 1.5.1. In Figure 1.5.1b, the one-sided limits of $f(x)$ as x approaches c both exist but are not equal. Thus, $\lim_{x \rightarrow c} f(x)$ does not exist, and this violates the second condition of Definition 1.5.1. We will say that a function like that in Figure 1.5.1b has a **jump discontinuity** at c . In Figure 1.5.1c, the one-sided limits of $f(x)$ as x approaches c are infinite. Thus, $\lim_{x \rightarrow c} f(x)$ does not exist, and this violates the second condition of Definition 1.5.1. We will say that a function like that in Figure 1.5.1c has an **infinite discontinuity** at c . In Figure 1.5.1d, the function is defined at c and $\lim_{x \rightarrow c} f(x)$ exists, but these two values are not equal, violating the third condition of Definition 1.5.1. We will say that a function like that in Figure 1.5.1d has a **removable discontinuity** at c . Exercises 33 and 34 help to explain why discontinuities of this type are given this name.

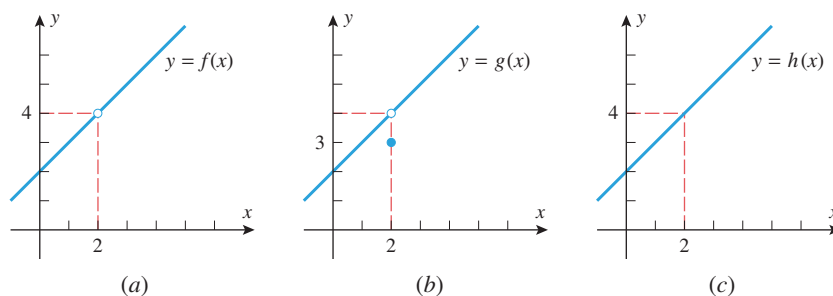
► **Example 1** Determine whether the following functions are continuous at $x = 2$.

$$f(x) = \frac{x^2 - 4}{x - 2}, \quad g(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & x \neq 2 \\ 3, & x = 2, \end{cases} \quad h(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & x \neq 2 \\ 4, & x = 2 \end{cases}$$

Solution. In each case we must determine whether the limit of the function as $x \rightarrow 2$ is the same as the value of the function at $x = 2$. In all three cases the functions are identical, except at $x = 2$, and hence all three have the same limit at $x = 2$, namely,

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} g(x) = \lim_{x \rightarrow 2} h(x) = \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = 4$$

The function f is undefined at $x = 2$, and hence is not continuous at $x = 2$ (Figure 1.5.2a). The function g is defined at $x = 2$, but its value there is $g(2) = 3$, which is not the same as the limit as x approaches 2; hence, g is also not continuous at $x = 2$ (Figure 1.5.2b). The value of the function h at $x = 2$ is $h(2) = 4$, which is the same as the limit as x approaches 2; hence, h is continuous at $x = 2$ (Figure 1.5.2c). (Note that the function h could have been written more simply as $h(x) = x + 2$, but we wrote it in piecewise form to emphasize its relationship to f and g .) ◀

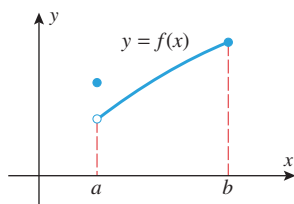


► Figure 1.5.2



Chris Hondros/Getty Images

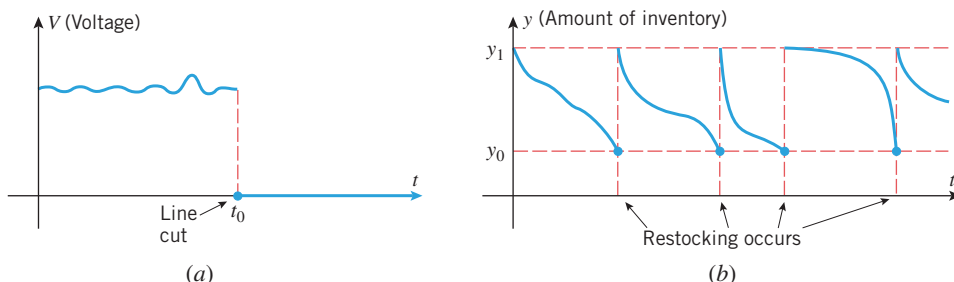
A poor connection in a transmission cable can cause a discontinuity in the electrical signal it carries.



▲ Figure 1.5.4

CONTINUITY IN APPLICATIONS

In applications, discontinuities often signal the occurrence of important physical events. For example, Figure 1.5.3a is a graph of voltage versus time for an underground cable that is accidentally cut by a work crew at time $t = t_0$ (the voltage drops to zero when the line is cut). Figure 1.5.3b shows the graph of inventory versus time for a company that restocks its warehouse to y_1 units when the inventory falls to y_0 units. The discontinuities occur at those times when restocking occurs.



▲ Figure 1.5.3

CONTINUITY ON AN INTERVAL

If a function f is continuous at each number in an open interval (a, b) , then we say that f is **continuous on (a, b)** . This definition applies to infinite open intervals of the form $(a, +\infty)$, $(-\infty, b)$, and $(-\infty, +\infty)$. In the case where f is continuous on $(-\infty, +\infty)$, we will say that f is **continuous everywhere**.

Because Definition 1.5.1 involves a two-sided limit, that definition does not generally apply at the endpoints of a closed interval $[a, b]$ or at the endpoint of an interval of the form $[a, b)$, $(a, b]$, $(-\infty, b]$, or $[a, +\infty)$. To remedy this problem, we will agree that a function is continuous at an endpoint of an interval if its value at the endpoint is equal to the appropriate one-sided limit at that endpoint. For example, the function graphed in Figure 1.5.4 is continuous at the right endpoint of the interval $[a, b]$ because

$$\lim_{x \rightarrow b^-} f(x) = f(b)$$

but it is not continuous at the left endpoint because

$$\lim_{x \rightarrow a^+} f(x) \neq f(a)$$

In general, we will say a function f is **continuous from the left** at c if

$$\lim_{x \rightarrow c^-} f(x) = f(c)$$

and is **continuous from the right** at c if

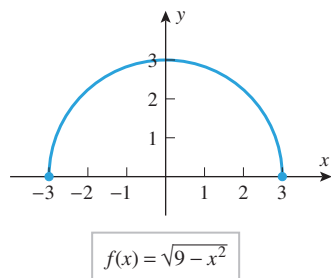
$$\lim_{x \rightarrow c^+} f(x) = f(c)$$

Using this terminology we define continuity on a closed interval as follows.

1.5.2 DEFINITION A function f is said to be **continuous on a closed interval $[a, b]$** if the following conditions are satisfied:

1. f is continuous on (a, b) .
2. f is continuous from the right at a .
3. f is continuous from the left at b .

Modify Definition 1.5.2 appropriately so that it applies to intervals of the form $[a, +\infty)$, $(-\infty, b]$, $(a, b]$, and $[a, b)$.



▲ Figure 1.5.5

► **Example 2** What can you say about the continuity of the function $f(x) = \sqrt{9 - x^2}$?

Solution. Because the natural domain of this function is the closed interval $[-3, 3]$, we will need to investigate the continuity of f on the open interval $(-3, 3)$ and at the two endpoints. If c is any point in the interval $(-3, 3)$, then it follows from Theorem 1.2.2(e) that

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} \sqrt{9 - x^2} = \sqrt{\lim_{x \rightarrow c} (9 - x^2)} = \sqrt{9 - c^2} = f(c)$$

which proves f is continuous at each point in the interval $(-3, 3)$. The function f is also continuous at the endpoints since

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} \sqrt{9 - x^2} = \sqrt{\lim_{x \rightarrow 3^-} (9 - x^2)} = 0 = f(3)$$

$$\lim_{x \rightarrow -3^+} f(x) = \lim_{x \rightarrow -3^+} \sqrt{9 - x^2} = \sqrt{\lim_{x \rightarrow -3^+} (9 - x^2)} = 0 = f(-3)$$

Thus, f is continuous on the closed interval $[-3, 3]$ (Figure 1.5.5). ◀

SOME PROPERTIES OF CONTINUOUS FUNCTIONS

The following theorem, which is a consequence of Theorem 1.2.2, will enable us to reach conclusions about the continuity of functions that are obtained by adding, subtracting, multiplying, and dividing continuous functions.

1.5.3 THEOREM If the functions f and g are continuous at c , then

- (a) $f + g$ is continuous at c .
- (b) $f - g$ is continuous at c .
- (c) fg is continuous at c .
- (d) f/g is continuous at c if $g(c) \neq 0$ and has a discontinuity at c if $g(c) = 0$.

We will prove part (d). The remaining proofs are similar and will be left to the exercises.

PROOF First, consider the case where $g(c) = 0$. In this case $f(c)/g(c)$ is undefined, so the function f/g has a discontinuity at c .

Next, consider the case where $g(c) \neq 0$. To prove that f/g is continuous at c , we must show that

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{f(c)}{g(c)} \quad (1)$$

Since f and g are continuous at c ,

$$\lim_{x \rightarrow c} f(x) = f(c) \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = g(c)$$

Thus, by Theorem 1.2.2(d)

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)} = \frac{f(c)}{g(c)}$$

which proves (1). ■

CONTINUITY OF POLYNOMIALS AND RATIONAL FUNCTIONS

The general procedure for showing that a function is continuous everywhere is to show that it is continuous at an *arbitrary* point. For example, we know from Theorem 1.2.3 that if

$p(x)$ is a polynomial and a is any real number, then

$$\lim_{x \rightarrow a} p(x) = p(a)$$

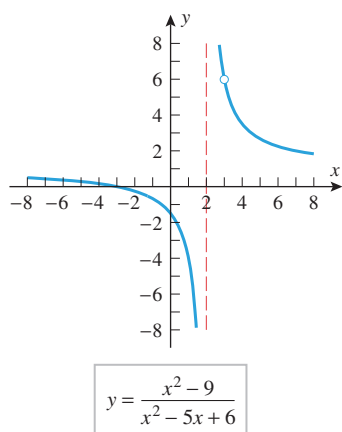
This shows that polynomials are continuous everywhere. Moreover, since rational functions are ratios of polynomials, it follows from part (d) of Theorem 1.5.3 that rational functions are continuous at points other than the zeros of the denominator, and at these zeros they have discontinuities. Thus, we have the following result.

1.5.4 THEOREM

- (a) A polynomial is continuous everywhere.
- (b) A rational function is continuous at every point where the denominator is nonzero, and has discontinuities at the points where the denominator is zero.

TECHNOLOGY MASTERY

If you use a graphing utility to generate the graph of the equation in Example 3, there is a good chance you will see the discontinuity at $x = 2$ but not at $x = 3$. Try it, and explain what you think is happening.



▲ Figure 1.5.6

► **Example 3** For what values of x is there a discontinuity in the graph of

$$y = \frac{x^2 - 9}{x^2 - 5x + 6}?$$

Solution. The function being graphed is a rational function, and hence is continuous at every number where the denominator is nonzero. Solving the equation

$$x^2 - 5x + 6 = 0$$

yields discontinuities at $x = 2$ and at $x = 3$ (Figure 1.5.6). ◀

► **Example 4** Show that $|x|$ is continuous everywhere (Figure 0.1.9).

Solution. We can write $|x|$ as

$$|x| = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -x & \text{if } x < 0 \end{cases}$$

so $|x|$ is the same as the polynomial x on the interval $(0, +\infty)$ and is the same as the polynomial $-x$ on the interval $(-\infty, 0)$. But polynomials are continuous everywhere, so $x = 0$ is the only possible discontinuity for $|x|$. Since $|0| = 0$, to prove the continuity at $x = 0$ we must show that

$$\lim_{x \rightarrow 0} |x| = 0 \quad (2)$$

Because the piecewise formula for $|x|$ changes at 0, it will be helpful to consider the one-sided limits at 0 rather than the two-sided limit. We obtain

$$\lim_{x \rightarrow 0^+} |x| = \lim_{x \rightarrow 0^+} x = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^-} |x| = \lim_{x \rightarrow 0^-} (-x) = 0$$

Thus, (2) holds and $|x|$ is continuous at $x = 0$. ◀

CONTINUITY OF COMPOSITIONS

The following theorem, whose proof is given in Appendix D, will be useful for calculating limits of compositions of functions.

In words, Theorem 1.5.5 states that a limit symbol can be moved through a function sign provided the limit of the expression inside the function sign exists and the function is continuous at this limit.

1.5.5 THEOREM If $\lim_{x \rightarrow c} g(x) = L$ and if the function f is continuous at L , then $\lim_{x \rightarrow c} f(g(x)) = f(L)$. That is,

$$\lim_{x \rightarrow c} f(g(x)) = f\left(\lim_{x \rightarrow c} g(x)\right)$$

This equality remains valid if $\lim_{x \rightarrow c}$ is replaced everywhere by one of $\lim_{x \rightarrow c^+}$, $\lim_{x \rightarrow c^-}$, $\lim_{x \rightarrow +\infty}$, or $\lim_{x \rightarrow -\infty}$.

In the special case of this theorem where $f(x) = |x|$, the fact that $|x|$ is continuous everywhere allows us to write

$$\lim_{x \rightarrow c} |g(x)| = \left| \lim_{x \rightarrow c} g(x) \right| \quad (3)$$

provided $\lim_{x \rightarrow c} g(x)$ exists. Thus, for example,

$$\lim_{x \rightarrow 3} |5 - x^2| = \left| \lim_{x \rightarrow 3} (5 - x^2) \right| = |-4| = 4$$

The following theorem is concerned with the continuity of compositions of functions; the first part deals with continuity at a specific number and the second with continuity everywhere.

1.5.6 THEOREM

- (a) If the function g is continuous at c , and the function f is continuous at $g(c)$, then the composition $f \circ g$ is continuous at c .
- (b) If the function g is continuous everywhere and the function f is continuous everywhere, then the composition $f \circ g$ is continuous everywhere.

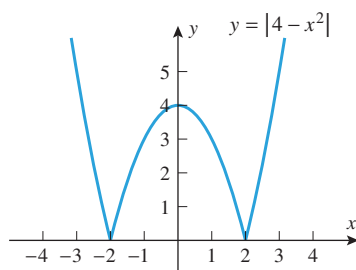
PROOF We will prove part (a) only; the proof of part (b) can be obtained by applying part (a) at an arbitrary number c . To prove that $f \circ g$ is continuous at c , we must show that the value of $f \circ g$ and the value of its limit are the same at $x = c$. But this is so, since we can write

$$\lim_{x \rightarrow c} (f \circ g)(x) = \lim_{x \rightarrow c} f(g(x)) = f\left(\lim_{x \rightarrow c} g(x)\right) = f(g(c)) = (f \circ g)(c) \quad \blacksquare$$

Theorem 1.5.5

g is continuous at c .

Can the absolute value of a function that is not continuous everywhere be continuous everywhere? Justify your answer.



▲ Figure 1.5.7

We know from Example 4 that the function $|x|$ is continuous everywhere. Thus, if $g(x)$ is continuous at c , then by part (a) of Theorem 1.5.6, the function $|g(x)|$ must also be continuous at c ; and, more generally, if $g(x)$ is continuous everywhere, then so is $|g(x)|$. Stated informally:

The absolute value of a continuous function is continuous.

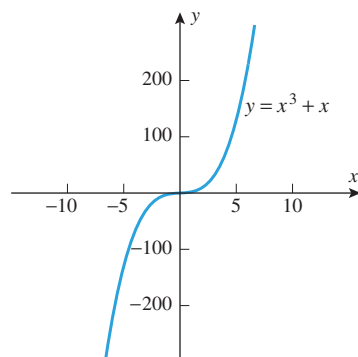
For example, the polynomial $g(x) = 4 - x^2$ is continuous everywhere, so we can conclude that the function $|4 - x^2|$ is also continuous everywhere (Figure 1.5.7).

CONTINUITY OF INVERSE FUNCTIONS

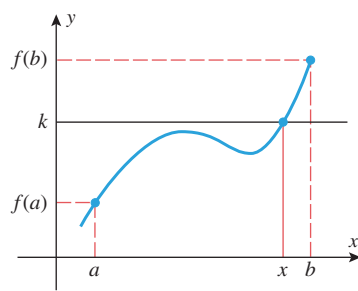
Since the graphs of a one-to-one function f and its inverse f^{-1} are reflections of one another about the line $y = x$, it is clear geometrically that if the graph of f has no breaks or holes in it, then neither does the graph of f^{-1} . This, and the fact that the range of f is the domain of f^{-1} , suggests the following result, which we state without formal proof.

To paraphrase Theorem 1.5.7, the inverse of a continuous function is continuous.

1.5.7 THEOREM If f is a one-to-one function that is continuous at each point of its domain, then f^{-1} is continuous at each point of its domain; that is, f^{-1} is continuous at each point in the range of f .



▲ Figure 1.5.8



▲ Figure 1.5.9

► **Example 5** Show that the function $f(x) = x^3 + x$ has an inverse. Is f^{-1} continuous at each point in $(-\infty, +\infty)$?

Solution. Note that f is increasing for all x (why?), so f is one-to-one and has an inverse. Also notice that f is continuous everywhere since it is a polynomial. From Figure 1.5.8 we infer that the range of f is $(-\infty, +\infty)$, so the domain of f^{-1} is also $(-\infty, +\infty)$. Although a formula for $f^{-1}(x)$ cannot be found easily, we can use Theorem 1.5.7 to conclude that f^{-1} is continuous on $(-\infty, +\infty)$. ◀

THE INTERMEDIATE-VALUE THEOREM

Figure 1.5.9 shows the graph of a function that is continuous on the closed interval $[a, b]$. The figure suggests that if we draw any horizontal line $y = k$, where k is between $f(a)$ and $f(b)$, then that line will cross the curve $y = f(x)$ at least once over the interval $[a, b]$. Stated in numerical terms, if f is continuous on $[a, b]$, then the function f must take on every value k between $f(a)$ and $f(b)$ at least once as x varies from a to b . For example, the polynomial $p(x) = x^5 - x + 3$ has a value of 3 at $x = 1$ and a value of 33 at $x = 2$. Thus, it follows from the continuity of p that the equation $x^5 - x + 3 = k$ has at least one solution in the interval $[1, 2]$ for every value of k between 3 and 33. This idea is stated more precisely in the following theorem.

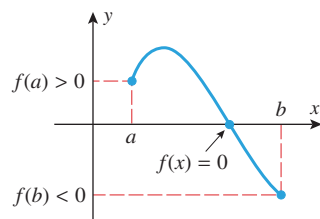
1.5.8 THEOREM (Intermediate-Value Theorem) If f is continuous on a closed interval $[a, b]$ and k is any number between $f(a)$ and $f(b)$, inclusive, then there is at least one number x in the interval $[a, b]$ such that $f(x) = k$.

Although this theorem is intuitively obvious, its proof depends on a mathematically precise development of the real number system, which is beyond the scope of this text.

APPROXIMATING ROOTS USING THE INTERMEDIATE-VALUE THEOREM

A variety of problems can be reduced to solving an equation $f(x) = 0$ for its roots. Sometimes it is possible to solve for the roots exactly using algebra, but often this is not possible and one must settle for decimal approximations of the roots. One procedure for approximating roots is based on the following consequence of the Intermediate-Value Theorem.

1.5.9 THEOREM If f is continuous on $[a, b]$, and if $f(a)$ and $f(b)$ are nonzero and have opposite signs, then there is at least one solution of the equation $f(x) = 0$ in the interval (a, b) .



▲ Figure 1.5.10

This result, which is illustrated in Figure 1.5.10, can be proved as follows.

PROOF Since $f(a)$ and $f(b)$ have opposite signs, 0 is between $f(a)$ and $f(b)$. Thus, by the Intermediate-Value Theorem there is at least one number x in the interval $[a, b]$ such that $f(x) = 0$. However, $f(a)$ and $f(b)$ are nonzero, so x must lie in the interval (a, b) , which completes the proof. ■

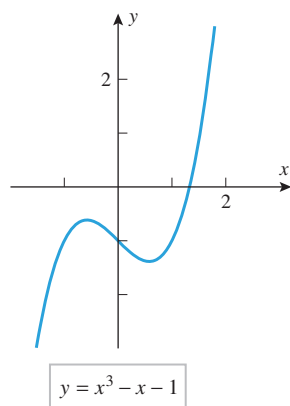
Before we illustrate how this theorem can be used to approximate roots, it will be helpful to discuss some standard terminology for describing errors in approximations. If x is an approximation to a quantity x_0 , then we call

$$\epsilon = |x - x_0|$$

the **absolute error** or (less precisely) the **error** in the approximation. The terminology in Table 1.5.1 is used to describe the size of such errors.

Table 1.5.1

ERROR	DESCRIPTION
$ x - x_0 \leq 0.1$	x approximates x_0 with an error of at most 0.1.
$ x - x_0 \leq 0.01$	x approximates x_0 with an error of at most 0.01.
$ x - x_0 \leq 0.001$	x approximates x_0 with an error of at most 0.001.
$ x - x_0 \leq 0.0001$	x approximates x_0 with an error of at most 0.0001.
$ x - x_0 \leq 0.5$	x approximates x_0 to the nearest integer.
$ x - x_0 \leq 0.05$	x approximates x_0 to 1 decimal place (i.e., to the nearest tenth).
$ x - x_0 \leq 0.005$	x approximates x_0 to 2 decimal places (i.e., to the nearest hundredth).
$ x - x_0 \leq 0.0005$	x approximates x_0 to 3 decimal places (i.e., to the nearest thousandth).



▲ Figure 1.5.11

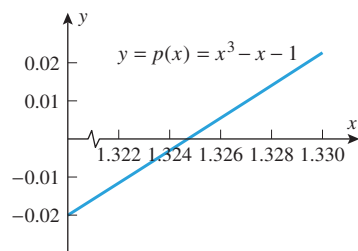
► **Example 6** The equation

$$x^3 - x - 1 = 0$$

cannot be solved algebraically very easily because the left side has no simple factors. However, if we graph $p(x) = x^3 - x - 1$ with a graphing utility (Figure 1.5.11), then we are led to conjecture that there is one real root and that this root lies inside the interval $[1, 2]$. The existence of a root in this interval is also confirmed by Theorem 1.5.9, since $p(1) = -1$ and $p(2) = 5$ have opposite signs. Approximate this root to two decimal-place accuracy.

Solution. Our objective is to approximate the unknown root x_0 with an error of at most 0.005. It follows that if we can find an interval of length 0.01 that contains the root, then the midpoint of that interval will approximate the root with an error of at most $\frac{1}{2}(0.01) = 0.005$, which will achieve the desired accuracy.

We know that the root x_0 lies in the interval $[1, 2]$. However, this interval has length 1, which is too large. We can pinpoint the location of the root more precisely by dividing the interval $[1, 2]$ into 10 equal parts and evaluating p at the points of subdivision using a calculating utility (Table 1.5.2). In this table $p(1.3)$ and $p(1.4)$ have opposite signs, so we know that the root lies in the interval $[1.3, 1.4]$. This interval has length 0.1, which is still too large, so we repeat the process by dividing the interval $[1.3, 1.4]$ into 10 parts and evaluating p at the points of subdivision; this yields Table 1.5.3, which tells us that the root is inside the interval $[1.32, 1.33]$ (Figure 1.5.12). Since this interval has length 0.01, its midpoint 1.325 will approximate the root with an error of at most 0.005. Thus, $x_0 \approx 1.325$ to two decimal-place accuracy. ◀



▲ Figure 1.5.12

Table 1.5.2

x	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
$p(x)$	-1	-0.77	-0.47	-0.10	0.34	0.88	1.50	2.21	3.03	3.96	5

Table 1.5.3

x	1.3	1.31	1.32	1.33	1.34	1.35	1.36	1.37	1.38	1.39	1.4
$p(x)$	-0.103	-0.062	-0.020	0.023	0.066	0.110	0.155	0.201	0.248	0.296	0.344

REMARK

To say that x approximates x_0 to n decimal places does *not* mean that the first n decimal places of x and x_0 will be the same when the numbers are rounded to n decimal places. For example, $x = 1.084$ approximates $x_0 = 1.087$ to two decimal places because $|x - x_0| = 0.003 (< 0.005)$. However, if we round these values to two decimal places, then we obtain $x \approx 1.08$ and $x_0 \approx 1.09$. Thus, if you approximate a number to n decimal places, then you should display that approximation to at least $n + 1$ decimal places to preserve the accuracy.

TECHNOLOGY MASTERY

Use a graphing or calculating utility to show that the root x_0 in Example 6 can be approximated as $x_0 \approx 1.3245$ to three decimal-place accuracy.

✓ QUICK CHECK EXERCISES 1.5 (See page 101 for answers.)

- What three conditions are satisfied if f is continuous at $x = c$?
- Suppose that f and g are continuous functions such that $f(2) = 1$ and $\lim_{x \rightarrow 2} [f(x) + 4g(x)] = 13$. Find
 - $g(2)$
 - $\lim_{x \rightarrow 2} g(x)$.
- Suppose that f and g are continuous functions such that $\lim_{x \rightarrow 3} g(x) = 5$ and $f(3) = -2$. Find $\lim_{x \rightarrow 3} [f(x)/g(x)]$.

- For what values of x , if any, is the function

$$f(x) = \frac{x^2 - 16}{x^2 - 5x + 4}$$

discontinuous?

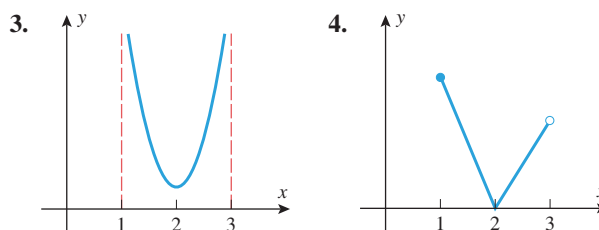
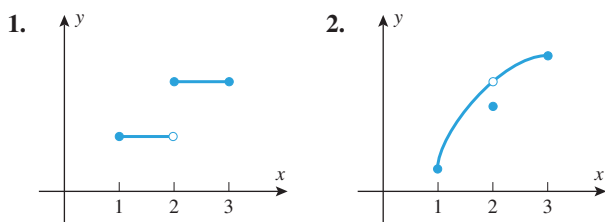
- Suppose that a function f is continuous everywhere and that $f(-2) = 3$, $f(-1) = -1$, $f(0) = -4$, $f(1) = 1$, and $f(2) = 5$. Does the Intermediate-Value Theorem guarantee that f has a root on the following intervals?
 - $[-2, -1]$
 - $[-1, 0]$
 - $[-1, 1]$
 - $[0, 2]$

EXERCISE SET 1.5 Graphing Utility

1–4 Let f be the function whose graph is shown. On which of the following intervals, if any, is f continuous?

- $[1, 3]$
- $(1, 3)$
- $[1, 2]$
- $(1, 2)$
- $[2, 3]$
- $(2, 3)$

For each interval on which f is not continuous, indicate which conditions for the continuity of f do not hold. ■



- Consider the functions

$$f(x) = \begin{cases} 1, & x \neq 4 \\ -1, & x = 4 \end{cases} \quad \text{and} \quad g(x) = \begin{cases} 4x - 10, & x \neq 4 \\ -6, & x = 4 \end{cases}$$

In each part, is the given function continuous at $x = 4$?

- $f(x)$
- $g(x)$
- $-g(x)$
- $|f(x)|$
- $f(x)g(x)$
- $g(f(x))$
- $g(x) - 6f(x)$

6. Consider the functions

$$f(x) = \begin{cases} 1, & 0 \leq x \\ 0, & x < 0 \end{cases} \quad \text{and} \quad g(x) = \begin{cases} 0, & 0 \leq x \\ 1, & x < 0 \end{cases}$$

In each part, is the given function continuous at $x = 0$?

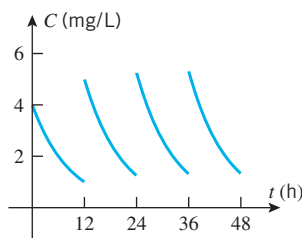
- (a) $f(x)$ (b) $g(x)$ (c) $f(-x)$ (d) $|g(x)|$
 (e) $f(x)g(x)$ (f) $g(f(x))$ (g) $f(x) + g(x)$

FOCUS ON CONCEPTS

7. In each part sketch the graph of a function
- f
- that satisfies the stated conditions.

- (a) f is continuous everywhere except at $x = 3$, at which point it is continuous from the right.
 (b) f has a two-sided limit at $x = 3$, but it is not continuous at $x = 3$.
 (c) f is not continuous at $x = 3$, but if its value at $x = 3$ is changed from $f(3) = 1$ to $f(3) = 0$, it becomes continuous at $x = 3$.
 (d) f is continuous on the interval $[0, 3)$ and is defined on the closed interval $[0, 3]$; but f is not continuous on the interval $[0, 3]$.

8. The accompanying figure models the concentration
- C
- of medication in the bloodstream of a patient over a 48-hour period of time. Discuss the significance of the discontinuities in the graph.



◀ Figure Ex-8

9. A student parking lot at a university charges \$2.00 for the first half hour (or any part) and \$1.00 for each subsequent half hour (or any part) up to a daily maximum of \$10.00.

- (a) Sketch a graph of cost as a function of the time parked.
 (b) Discuss the significance of the discontinuities in the graph to a student who parks there.

10. In each part determine whether the function is continuous or not, and explain your reasoning.

- (a) The Earth's population as a function of time.
 (b) Your exact height as a function of time.
 (c) The cost of a taxi ride in your city as a function of the distance traveled.
 (d) The volume of a melting ice cube as a function of time.

- 11–22 Find values of
- x
- , if any, at which
- f
- is not continuous.

11. $f(x) = 5x^4 - 3x + 7$ 12. $f(x) = \sqrt[3]{x-8}$

13. $f(x) = \frac{x+2}{x^2+4}$

14. $f(x) = \frac{x+2}{x^2-4}$

15. $f(x) = \frac{x}{2x^2+x}$

16. $f(x) = \frac{2x+1}{4x^2+4x+5}$

17. $f(x) = \frac{3}{x} + \frac{x-1}{x^2-1}$

18. $f(x) = \frac{5}{x} + \frac{2x}{x+4}$

19. $f(x) = \frac{x^2+6x+9}{|x|+3}$

20. $f(x) = \left| 4 - \frac{8}{x^4+x} \right|$

21. $f(x) = \begin{cases} 2x+3, & x \leq 4 \\ 7 + \frac{16}{x}, & x > 4 \end{cases}$

22. $f(x) = \begin{cases} \frac{3}{x-1}, & x \neq 1 \\ 3, & x = 1 \end{cases}$

23–28 True–False Determine whether the statement is true or false. Explain your answer. ■23. If $f(x)$ is continuous at $x = c$, then so is $|f(x)|$.24. If $|f(x)|$ is continuous at $x = c$, then so is $f(x)$.25. If f and g are discontinuous at $x = c$, then so is $f + g$.26. If f and g are discontinuous at $x = c$, then so is fg .27. If $\sqrt{f(x)}$ is continuous at $x = c$, then so is $f(x)$.28. If $f(x)$ is continuous at $x = c$, then so is $\sqrt{f(x)}$.**29–30** Find a value of the constant k , if possible, that will make the function continuous everywhere. ■

29. (a) $f(x) = \begin{cases} 7x-2, & x \leq 1 \\ kx^2, & x > 1 \end{cases}$

(b) $f(x) = \begin{cases} kx^2, & x \leq 2 \\ 2x+k, & x > 2 \end{cases}$

30. (a) $f(x) = \begin{cases} 9-x^2, & x \geq -3 \\ k/x^2, & x < -3 \end{cases}$

(b) $f(x) = \begin{cases} 9-x^2, & x \geq 0 \\ k/x^2, & x < 0 \end{cases}$

31. Find values of the constants
- k
- and
- m
- , if possible, that will make the function
- f
- continuous everywhere.

$$f(x) = \begin{cases} x^2+5, & x > 2 \\ m(x+1)+k, & -1 < x \leq 2 \\ 2x^3+x+7, & x \leq -1 \end{cases}$$

32. On which of the following intervals is

$$f(x) = \frac{1}{\sqrt{x-2}}$$

continuous?

- (a)
- $[2, +\infty)$
- (b)
- $(-\infty, +\infty)$
- (c)
- $(2, +\infty)$
- (d)
- $[1, 2)$

33–36 A function f is said to have a **removable discontinuity** at $x = c$ if $\lim_{x \rightarrow c} f(x)$ exists but f is not continuous at $x = c$, either because f is not defined at c or because the definition for $f(c)$ differs from the value of the limit. This terminology will be needed in these exercises. ■

33. (a) Sketch the graph of a function with a removable discontinuity at $x = c$ for which $f(c)$ is undefined.
 (b) Sketch the graph of a function with a removable discontinuity at $x = c$ for which $f(c)$ is defined.
34. (a) The terminology *removable discontinuity* is appropriate because a removable discontinuity of a function f at $x = c$ can be “removed” by redefining the value of f appropriately at $x = c$. What value for $f(c)$ removes the discontinuity?
 (b) Show that the following functions have removable discontinuities at $x = 1$, and sketch their graphs.

$$f(x) = \frac{x^2 - 1}{x - 1} \quad \text{and} \quad g(x) = \begin{cases} 1, & x > 1 \\ 0, & x = 1 \\ 1, & x < 1 \end{cases}$$

- (c) What values should be assigned to $f(1)$ and $g(1)$ to remove the discontinuities?

35–36 Find the values of x (if any) at which f is not continuous, and determine whether each such value is a removable discontinuity. ■

35. (a) $f(x) = \frac{|x|}{x}$ (b) $f(x) = \frac{x^2 + 3x}{x + 3}$

(c) $f(x) = \frac{x - 2}{|x| - 2}$

36. (a) $f(x) = \frac{x^2 - 4}{x^3 - 8}$ (b) $f(x) = \begin{cases} 2x - 3, & x \leq 2 \\ x^2, & x > 2 \end{cases}$

(c) $f(x) = \begin{cases} 3x^2 + 5, & x \neq 1 \\ 6, & x = 1 \end{cases}$

37. (a) Use a graphing utility to generate the graph of the function $f(x) = (x + 3)/(2x^2 + 5x - 3)$, and then use the graph to make a conjecture about the number and locations of all discontinuities.
 (b) Check your conjecture by factoring the denominator.

38. (a) Use a graphing utility to generate the graph of the function $f(x) = x/(x^3 - x + 2)$, and then use the graph to make a conjecture about the number and locations of all discontinuities.
 (b) Use the Intermediate-Value Theorem to approximate the locations of all discontinuities to two decimal places.

39. Prove that $f(x) = x^{3/5}$ is continuous everywhere, carefully justifying each step.

40. Prove that $f(x) = 1/\sqrt{x^4 + 7x^2 + 1}$ is continuous everywhere, carefully justifying each step.

41. Prove:

- (a) part (a) of Theorem 1.5.3
 (b) part (b) of Theorem 1.5.3
 (c) part (c) of Theorem 1.5.3.

42. Prove part (b) of Theorem 1.5.4.

43. (a) Use Theorem 1.5.5 to prove that if f is continuous at $x = c$, then $\lim_{h \rightarrow 0} f(c + h) = f(c)$.

- (b) Prove that if $\lim_{h \rightarrow 0} f(c + h) = f(c)$, then f is continuous at $x = c$. [Hint: What does this limit tell you about the continuity of $g(h) = f(c + h)$?]
 (c) Conclude from parts (a) and (b) that f is continuous at $x = c$ if and only if $\lim_{h \rightarrow 0} f(c + h) = f(c)$.

44. Prove: If f and g are continuous on $[a, b]$, and $f(a) > g(a)$, $f(b) < g(b)$, then there is at least one solution of the equation $f(x) = g(x)$ in (a, b) . [Hint: Consider $f(x) - g(x)$.]

FOCUS ON CONCEPTS

45. Give an example of a function f that is defined on a closed interval, and whose values at the endpoints have opposite signs, but for which the equation $f(x) = 0$ has no solution in the interval.

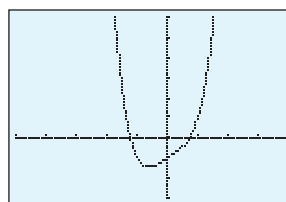
46. Let f be the function whose graph is shown in Exercise 2. For each interval, determine (i) whether the hypothesis of the Intermediate-Value Theorem is satisfied, and (ii) whether the conclusion of the Intermediate-Value Theorem is satisfied.

- (a) $[1, 2]$ (b) $[2, 3]$ (c) $[1, 3]$

47. Show that the equation $x^3 + x^2 - 2x = 1$ has at least one solution in the interval $[-1, 1]$.

48. Prove: If $p(x)$ is a polynomial of odd degree, then the equation $p(x) = 0$ has at least one real solution.

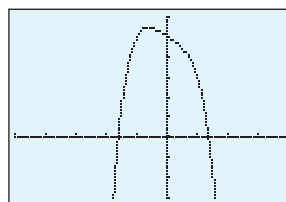
49. The accompanying figure shows the graph of the equation $y = x^4 + x - 1$. Use the method of Example 6 to approximate the x -intercepts with an error of at most 0.05.



$[-5, 4] \times [-3, 6]$
 $x\text{Scl} = 1, y\text{Scl} = 1$

◀ Figure Ex-49

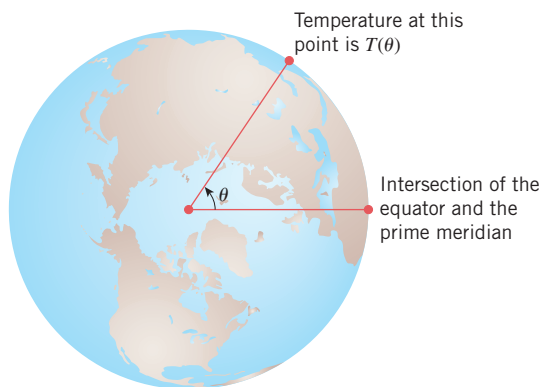
50. The accompanying figure shows the graph of the equation $y = 5 - x - x^4$. Use the method of Example 6 to approximate the roots of the equation $5 - x - x^4 = 0$ to two decimal-place accuracy.



$[-5, 4] \times [-3, 6]$
 $x\text{Scl} = 1, y\text{Scl} = 1$

◀ Figure Ex-50

51. Use the fact that $\sqrt{5}$ is a solution of $x^2 - 5 = 0$ to approximate $\sqrt{5}$ with an error of at most 0.005.
52. A sprinter, who is timed with a stopwatch, runs a hundred yard dash in 10 s. The stopwatch is reset to 0, and the sprinter is timed jogging back to the starting block. Show that there is at least one point on the track at which the reading on the stopwatch during the sprint is the same as the reading during the return jog. [Hint: Use the result in Exercise 44.]
53. Prove that there exist points on opposite sides of the equator that are at the same temperature. [Hint: Consider the accompanying figure, which shows a view of the equator from a point above the North Pole. Assume that the temperature $T(\theta)$ is a continuous function of the angle θ , and consider the function $f(\theta) = T(\theta + \pi) - T(\theta)$.]



▲ Figure Ex-53

54. Let R denote an elliptical region in the xy -plane, and define $f(z)$ to be the area within R that is on, or to the left of, the vertical line $x = z$. Prove that f is a continuous function of z . [Hint: Assume the ellipse is between the horizontal lines $y = a$ and $y = b$, $a < b$. Argue that $|f(z_1) - f(z_2)| \leq (b - a) \cdot |z_1 - z_2|$.]

55. Let R denote an elliptical region in the plane. For any line L , prove there is a line perpendicular to L that divides R in half by area. [Hint: Introduce coordinates so that L is the x -axis. Use the result in Exercise 54 and the Intermediate-Value Theorem.]
56. Suppose that f is continuous on the interval $[0, 1]$ and that $0 \leq f(x) \leq 1$ for all x in this interval.
- Sketch the graph of $y = x$ together with a possible graph for f over the interval $[0, 1]$.
 - Use the Intermediate-Value Theorem to help prove that there is at least one number c in the interval $[0, 1]$ such that $f(c) = c$.
57. Let $f(x) = x^6 + 3x + 5$, $x \geq 0$. Show that f is an invertible function and that f^{-1} is continuous on $[5, +\infty)$.
58. Suppose that f is an invertible function, $f(0) = 0$, f is continuous at 0, and $\lim_{x \rightarrow 0} f(x)/x$ exists. Given that $L = \lim_{x \rightarrow 0} f(x)/x$, show

$$\lim_{x \rightarrow 0} \frac{x}{f^{-1}(x)} = L$$

[Hint: Apply Theorem 1.5.5 to the composition $h \circ g$, where

$$h(x) = \begin{cases} f(x)/x, & x \neq 0 \\ L, & x = 0 \end{cases}$$

and $g(x) = f^{-1}(x)$.]

59. **Writing** It is often assumed that changing physical quantities such as the height of a falling object or the weight of a melting snowball, are continuous functions of time. Use specific examples to discuss the merits of this assumption.
60. **Writing** The Intermediate-Value Theorem is an example of what is known as an “existence theorem.” In your own words, describe how to recognize an existence theorem, and discuss some of the ways in which an existence theorem can be useful.

✓ QUICK CHECK ANSWERS 1.5

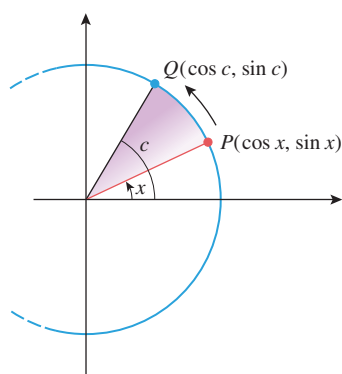
1. $f(c)$ is defined; $\lim_{x \rightarrow c} f(x)$ exists; $\lim_{x \rightarrow c} f(x) = f(c)$ 2. (a) 3 (b) 3 3. $-2/5$ 4. $x = 1, 4$
5. (a) yes (b) no (c) yes (d) yes

1.6 CONTINUITY OF TRIGONOMETRIC FUNCTIONS

In this section we will discuss the continuity properties of trigonometric functions. We will also discuss some important limits involving such functions.

■ CONTINUITY OF TRIGONOMETRIC FUNCTIONS

Recall from trigonometry that the graphs of $\sin x$ and $\cos x$ are drawn as continuous curves. We will not formally prove that these functions are continuous, but we can motivate this fact by letting c be a fixed angle in radian measure and x a variable angle in radian measure. If, as illustrated in Figure 1.6.1, the angle x approaches the angle c , then the point $P(\cos x, \sin x)$



▲ Figure 1.6.1

Theorem 1.6.1 implies that the six basic trigonometric functions are continuous on their domains. In particular, $\sin x$ and $\cos x$ are continuous everywhere.

moves along the unit circle toward $Q(\cos c, \sin c)$, and the coordinates of P approach the corresponding coordinates of Q . This implies that

$$\lim_{x \rightarrow c} \sin x = \sin c \quad \text{and} \quad \lim_{x \rightarrow c} \cos x = \cos c \quad (1)$$

Thus, $\sin x$ and $\cos x$ are continuous at the arbitrary point c ; that is, these functions are continuous everywhere.

The formulas in (1) can be used to find limits of the remaining trigonometric functions by expressing them in terms of $\sin x$ and $\cos x$; for example, if $\cos c \neq 0$, then

$$\lim_{x \rightarrow c} \tan x = \lim_{x \rightarrow c} \frac{\sin x}{\cos x} = \frac{\sin c}{\cos c} = \tan c$$

Thus, we are led to the following theorem.

1.6.1 THEOREM If c is any number in the natural domain of the stated trigonometric function, then

$$\begin{array}{lll} \lim_{x \rightarrow c} \sin x = \sin c & \lim_{x \rightarrow c} \cos x = \cos c & \lim_{x \rightarrow c} \tan x = \tan c \\ \lim_{x \rightarrow c} \csc x = \csc c & \lim_{x \rightarrow c} \sec x = \sec c & \lim_{x \rightarrow c} \cot x = \cot c \end{array}$$

► **Example 1** Find the limit

$$\lim_{x \rightarrow 1} \cos \left(\frac{x^2 - 1}{x - 1} \right)$$

Solution. Since the cosine function is continuous everywhere, it follows from Theorem 1.5.5 that

$$\lim_{x \rightarrow 1} \cos(g(x)) = \cos \left(\lim_{x \rightarrow 1} g(x) \right)$$

provided $\lim_{x \rightarrow 1} g(x)$ exists. Thus,

$$\lim_{x \rightarrow 1} \cos \left(\frac{x^2 - 1}{x - 1} \right) = \lim_{x \rightarrow 1} \cos(x + 1) = \cos \left(\lim_{x \rightarrow 1} (x + 1) \right) = \cos 2 \quad \blacktriangleleft$$

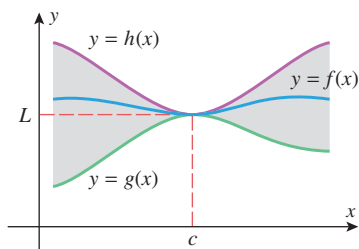
■ OBTAINING LIMITS BY SQUEEZING

In Section 1.1 we used numerical evidence to conjecture that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad (2)$$

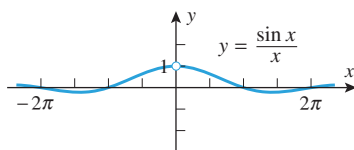
However, this limit is not easy to establish with certainty. The limit is an indeterminate form of type $0/0$, and there is no simple algebraic manipulation that one can perform to obtain the limit. Later in the text we will develop general methods for finding limits of indeterminate forms, but in this particular case we can use a technique called *squeezing*.

The method of squeezing is used to prove that $f(x) \rightarrow L$ as $x \rightarrow c$ by “trapping” or “squeezing” f between two functions, g and h , whose limits as $x \rightarrow c$ are known with certainty to be L . As illustrated in Figure 1.6.2, this forces f to have a limit of L as well. This is the idea behind the following theorem, which we state without proof.

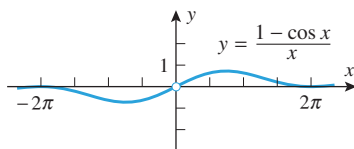


▲ Figure 1.6.2

The Squeezing Theorem also holds for one-sided limits and limits at $+\infty$ and $-\infty$. How do you think the hypotheses would change in those cases?



$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$



$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

▲ Figure 1.6.3

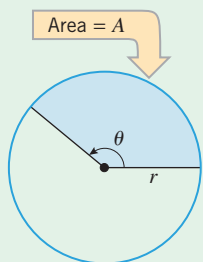
Recall that the area A of a sector of radius r and central angle θ is

$$A = \frac{1}{2} r^2 \theta$$

This can be derived from the relationship

$$\frac{A}{\pi r^2} = \frac{\theta}{2\pi}$$

which states that the area of the sector is to the area of the circle as the central angle of the sector is to the central angle of the circle.



1.6.2 THEOREM (The Squeezing Theorem) Let f , g , and h be functions satisfying

$$g(x) \leq f(x) \leq h(x)$$

for all x in some open interval containing the number c , with the possible exception that the inequalities need not hold at c . If g and h have the same limit as x approaches c , say

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L$$

then f also has this limit as x approaches c , that is,

$$\lim_{x \rightarrow c} f(x) = L$$

To illustrate how the Squeezing Theorem works, we will prove the following results, which are illustrated in Figure 1.6.3.

1.6.3 THEOREM

$$(a) \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad (b) \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

PROOF (a) In this proof we will interpret x as an angle in radian measure, and we will assume to start that $0 < x < \pi/2$. As illustrated in Figure 1.6.4, the area of a sector with central angle x and radius 1 lies between the areas of two triangles, one with area $\frac{1}{2} \tan x$ and the other with area $\frac{1}{2} \sin x$. Since the sector has area $\frac{1}{2}x$ (see marginal note), it follows that

$$\frac{1}{2} \tan x \geq \frac{1}{2}x \geq \frac{1}{2} \sin x$$

Multiplying through by $2/(\sin x)$ and using the fact that $\sin x > 0$ for $0 < x < \pi/2$, we obtain

$$\frac{1}{\cos x} \geq \frac{x}{\sin x} \geq 1$$

Next, taking reciprocals reverses the inequalities, so we obtain

$$\cos x \leq \frac{\sin x}{x} \leq 1 \quad (3)$$

which squeezes the function $(\sin x)/x$ between the functions $\cos x$ and 1. Although we derived these inequalities by assuming that $0 < x < \pi/2$, they also hold for $-\pi/2 < x < 0$ [since replacing x by $-x$ and using the identities $\sin(-x) = -\sin x$, and $\cos(-x) = \cos x$ leaves (3) unchanged]. Finally, since

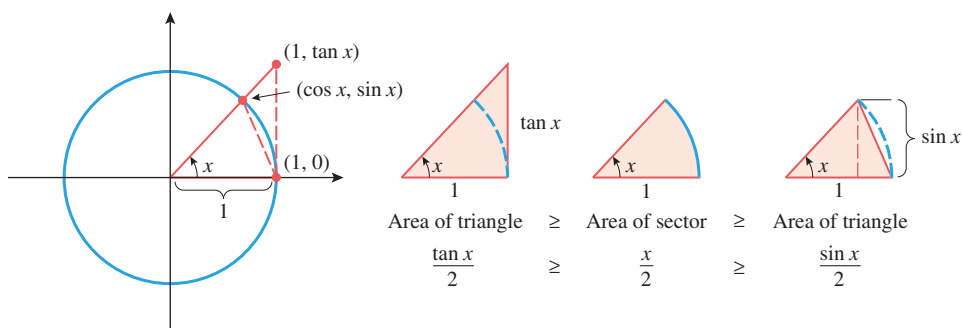
$$\lim_{x \rightarrow 0} \cos x = 1 \quad \text{and} \quad \lim_{x \rightarrow 0} 1 = 1$$

the Squeezing Theorem implies that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

PROOF (b) For this proof we will use the limit in part (a), the continuity of the sine function, and the trigonometric identity $\sin^2 x = 1 - \cos^2 x$. We obtain

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} &= \lim_{x \rightarrow 0} \left[\frac{1 - \cos x}{x} \cdot \frac{1 + \cos x}{1 + \cos x} \right] = \lim_{x \rightarrow 0} \frac{\sin^2 x}{(1 + \cos x)x} \\ &= \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right) \left(\lim_{x \rightarrow 0} \frac{\sin x}{1 + \cos x} \right) = (1) \left(\frac{0}{1 + 1} \right) = 0 \quad \blacksquare \end{aligned}$$



► Figure 1.6.4

► **Example 2** Find

$$(a) \lim_{x \rightarrow 0} \frac{\tan x}{x} \quad (b) \lim_{\theta \rightarrow 0} \frac{\sin 2\theta}{\theta} \quad (c) \lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x}$$

Solution (a).

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \cdot \frac{1}{\cos x} \right) = \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right) \left(\lim_{x \rightarrow 0} \frac{1}{\cos x} \right) = (1)(1) = 1$$

Solution (b). The trick is to multiply and divide by 2, which will make the denominator the same as the argument of the sine function [just as in Theorem 1.6.3(a)]:

$$\lim_{\theta \rightarrow 0} \frac{\sin 2\theta}{\theta} = \lim_{\theta \rightarrow 0} 2 \cdot \frac{\sin 2\theta}{2\theta} = 2 \lim_{\theta \rightarrow 0} \frac{\sin 2\theta}{2\theta}$$

Now make the substitution $x = 2\theta$, and use the fact that $x \rightarrow 0$ as $\theta \rightarrow 0$. This yields

$$\lim_{\theta \rightarrow 0} \frac{\sin 2\theta}{\theta} = 2 \lim_{\theta \rightarrow 0} \frac{\sin 2\theta}{2\theta} = 2 \lim_{x \rightarrow 0} \frac{\sin x}{x} = 2(1) = 2$$

TECHNOLOGY MASTERY

Use a graphing utility to confirm the limits in Example 2, and if you have a CAS, use it to obtain the limits.

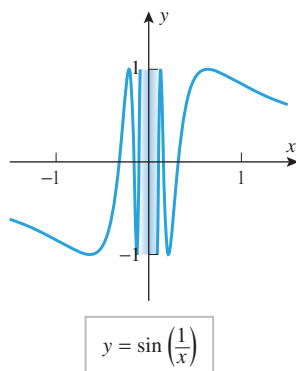
Solution (c).

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x} = \lim_{x \rightarrow 0} \frac{\frac{\sin 3x}{x}}{\frac{\sin 5x}{x}} = \lim_{x \rightarrow 0} \frac{3 \cdot \frac{\sin 3x}{3x}}{5 \cdot \frac{\sin 5x}{5x}} = \frac{3 \cdot 1}{5 \cdot 1} = \frac{3}{5} \quad \blacktriangleleft$$

► **Example 3** Discuss the limits

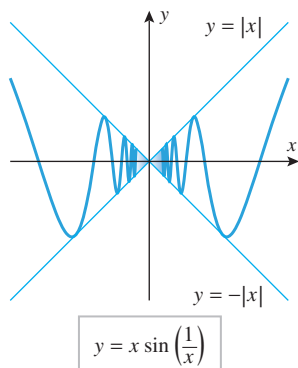
$$(a) \lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right) \quad (b) \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right)$$

Solution (a). Let us view $1/x$ as an angle in radian measure. As $x \rightarrow 0^+$, the angle $1/x$ approaches $+\infty$, so the values of $\sin(1/x)$ keep oscillating between -1 and 1 without approaching a limit. Similarly, as $x \rightarrow 0^-$, the angle $1/x$ approaches $-\infty$, so again the values of $\sin(1/x)$ keep oscillating between -1 and 1 without approaching a limit. These conclusions are consistent with the graph shown in Figure 1.6.5. Note that the oscillations become more and more rapid as $x \rightarrow 0$ because $1/x$ increases (or decreases) more and more rapidly as x approaches 0.



▲ Figure 1.6.5

Confirm (4) by considering the cases $x > 0$ and $x < 0$ separately.



▲ Figure 1.6.6

Solution (b). Since

$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$$

it follows that if $x \neq 0$, then

$$-|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x| \quad (4)$$

Since $|x| \rightarrow 0$ as $x \rightarrow 0$, the inequalities in (4) and the Squeezing Theorem imply that

$$\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$$

This is consistent with the graph shown in Figure 1.6.6. ◀

REMARK

It follows from part (b) of this example that the function

$$f(x) = \begin{cases} x \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is continuous at $x = 0$, since the value of the function and the value of the limit are the same at 0. This shows that the behavior of a function can be very complex in the vicinity of $x = c$, even though the function is continuous at c .

QUICK CHECK EXERCISES 1.6 (See page 107 for answers.)

1. In each part, is the given function continuous on the interval $[0, \pi/2)$?

(a) $\sin x$ (b) $\cos x$ (c) $\tan x$ (d) $\csc x$

2. Evaluate

(a) $\lim_{x \rightarrow 0} \frac{\sin x}{x}$
 (b) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$.

3. Suppose a function f has the property that for all real numbers x

$$3 - |x| \leq f(x) \leq 3 + |x|$$

From this we can conclude that $f(x) \rightarrow$ _____ as $x \rightarrow$ _____.

EXERCISE SET 1.6



Graphing Utility

- 1–8** Find the discontinuities, if any. ■

1. $f(x) = \sin(x^2 - 2)$ 2. $f(x) = \cos\left(\frac{x}{x - \pi}\right)$
 3. $f(x) = |\cot x|$ 4. $f(x) = \sec x$
 5. $f(x) = \csc x$ 6. $f(x) = \frac{1}{1 + \sin^2 x}$
 7. $f(x) = \frac{1}{1 - 2 \sin x}$ 8. $f(x) = \sqrt{2 + \tan^2 x}$

- 9–10** In each part, use Theorem 1.5.6(b) to show that the function is continuous everywhere. ■

9. (a) $\sin(x^3 + 7x + 1)$ (b) $|\sin x|$
 (c) $\cos^3(x + 1)$

10. (a) $|3 + \sin 2x|$ (b) $\sin(\sin x)$
 (c) $\cos^5 x - 2 \cos^3 x + 1$

- 11–32** Find the limits. ■

11. $\lim_{x \rightarrow +\infty} \cos\left(\frac{1}{x}\right)$ 12. $\lim_{x \rightarrow +\infty} \sin\left(\frac{\pi x}{2 - 3x}\right)$
 13. $\lim_{\theta \rightarrow 0} \frac{\sin 3\theta}{\theta}$ 14. $\lim_{h \rightarrow 0} \frac{\sin h}{2h}$
 15. $\lim_{x \rightarrow 0} \frac{x^2 - 3 \sin x}{x}$ 16. $\lim_{x \rightarrow 0} \frac{2 - \cos 3x - \cos 4x}{x}$
 17. $\lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta^2}$ 18. $\lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\theta}$

19. $\lim_{x \rightarrow 0} \frac{\tan 7x}{\sin 3x}$

21. $\lim_{x \rightarrow 0^+} \frac{\sin x}{5\sqrt{x}}$

23. $\lim_{x \rightarrow 0} \frac{\sin x^2}{x}$

25. $\lim_{t \rightarrow 0} \frac{t^2}{1 - \cos^2 t}$

27. $\lim_{\theta \rightarrow 0} \frac{\theta^2}{1 - \cos \theta}$

29. $\lim_{x \rightarrow 0^+} \sin\left(\frac{1}{x}\right)$

31. $\lim_{x \rightarrow 0} \frac{\tan ax}{\sin bx}, \quad (a \neq 0, b \neq 0)$

32. $\lim_{x \rightarrow 0} \frac{\sin^2(kx)}{x}, \quad k \neq 0$

20. $\lim_{x \rightarrow 0} \frac{\sin 6x}{\sin 8x}$

22. $\lim_{x \rightarrow 0} \frac{\sin^2 x}{3x^2}$

24. $\lim_{h \rightarrow 0} \frac{\sin h}{1 - \cos h}$

26. $\lim_{x \rightarrow 0} \frac{x}{\cos\left(\frac{1}{2}\pi - x\right)}$

28. $\lim_{h \rightarrow 0} \frac{1 - \cos 3h}{\cos^2 5h - 1}$

30. $\lim_{x \rightarrow 0} \frac{\tan 3x^2 + \sin^2 5x}{x^2}$

33–34 (a) Complete the table and make a guess about the limit indicated. (b) Find the exact value of the limit. ■

33. $f(x) = \frac{\sin(x-5)}{x^2-25}; \lim_{x \rightarrow 5} f(x)$

x	4	4.5	4.9	5.1	5.5	6
$f(x)$						

◀ Table Ex-33

34. $f(x) = \frac{\sin(x^2 + 3x + 2)}{x + 2}; \lim_{x \rightarrow -2} f(x)$

x	-2.1	-2.01	-2.001	-1.999	-1.99	-1.9
$f(x)$						

▲ Table Ex-34

35–38 True–False Determine whether the statement is true or false. Explain your answer. ■

35. Suppose that for all real numbers x , a function f satisfies

$$|f(x) + 5| \leq |x + 1|$$

Then $\lim_{x \rightarrow -1} f(x) = -5$.

36. For $0 < x < \pi/2$, the graph of $y = \sin x$ lies below the graph of $y = x$ and above the graph of $y = x \cos x$.

37. The function

$$f(x) = \begin{cases} x \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is continuous everywhere.

38. Suppose that M is a positive number and that for all real numbers x , a function f satisfies

$$-M \leq f(x) \leq M$$

Then

$$\lim_{x \rightarrow 0} xf(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow +\infty} \frac{f(x)}{x} = 0$$

FOCUS ON CONCEPTS

39. In an attempt to verify that $\lim_{x \rightarrow 0} (\sin x)/x = 1$, a student constructs the accompanying table.

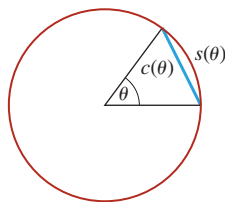
(a) What mistake did the student make?

(b) What is the exact value of the limit illustrated by this table?

x	-0.01	-0.001	0.001	0.01
$\sin x/x$	0.017453	0.017453	0.017453	0.017453

▲ Table Ex-39

40. In the circle in the accompanying figure, a central angle of measure θ radians subtends a chord of length $c(\theta)$ and a circular arc of length $s(\theta)$. Based on your intuition, what would you conjecture is the value of $\lim_{\theta \rightarrow 0^+} c(\theta)/s(\theta)$? Verify your conjecture by computing the limit.



◀ Figure Ex-40

41. Find a nonzero value for the constant k that makes

$$f(x) = \begin{cases} \frac{\tan kx}{x}, & x < 0 \\ 3x + 2k^2, & x \geq 0 \end{cases}$$

continuous at $x = 0$.

42. Is

$$f(x) = \begin{cases} \frac{\sin x}{|x|}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

continuous at $x = 0$? Explain.

43. In parts (a)–(c), find the limit by making the indicated substitution.

(a) $\lim_{x \rightarrow +\infty} x \sin \frac{1}{x}; \quad t = \frac{1}{x}$

(b) $\lim_{x \rightarrow -\infty} x \left(1 - \cos \frac{1}{x}\right); \quad t = \frac{1}{x}$

(c) $\lim_{x \rightarrow \pi} \frac{\pi - x}{\sin x}; \quad t = \pi - x$

44. Find $\lim_{x \rightarrow 2} \frac{\cos(\pi/x)}{x-2}$. [Hint: Let $t = \frac{\pi}{2} - \frac{\pi}{x}$.]

45. Find $\lim_{x \rightarrow 1} \frac{\sin(\pi x)}{x-1}$. 46. Find $\lim_{x \rightarrow \pi/4} \frac{\tan x - 1}{x - \pi/4}$.

FOCUS ON CONCEPTS

47. Use the Squeezing Theorem to show that

$$\lim_{x \rightarrow 0} x \cos \frac{50\pi}{x} = 0$$

and illustrate the principle involved by using a graphing utility to graph the equations $y = |x|$, $y = -|x|$, and $y = x \cos(50\pi/x)$ on the same screen in the window $[-1, 1] \times [-1, 1]$.

48. Use the Squeezing Theorem to show that

$$\lim_{x \rightarrow 0} x^2 \sin \left(\frac{50\pi}{\sqrt[3]{x}} \right) = 0$$

and illustrate the principle involved by using a graphing utility to graph the equations $y = x^2$, $y = -x^2$, and $y = x^2 \sin(50\pi/\sqrt[3]{x})$ on the same screen in the window $[-0.5, 0.5] \times [-0.25, 0.25]$.

49. In Example 3 we used the Squeezing Theorem to prove that

$$\lim_{x \rightarrow 0} x \sin \left(\frac{1}{x} \right) = 0$$

Why couldn't we have obtained the same result by writing

$$\begin{aligned} \lim_{x \rightarrow 0} x \sin \left(\frac{1}{x} \right) &= \lim_{x \rightarrow 0} x \cdot \lim_{x \rightarrow 0} \sin \left(\frac{1}{x} \right) \\ &= 0 \cdot \lim_{x \rightarrow 0} \sin \left(\frac{1}{x} \right) = 0? \end{aligned}$$

50. Sketch the graphs of the curves $y = 1 - x^2$, $y = \cos x$, and $y = f(x)$, where f is a function that satisfies the inequalities

$$1 - x^2 \leq f(x) \leq \cos x$$

for all x in the interval $(-\pi/2, \pi/2)$. What can you say about the limit of $f(x)$ as $x \rightarrow 0$? Explain.

51. Sketch the graphs of the curves $y = 1/x$, $y = -1/x$, and $y = f(x)$, where f is a function that satisfies the inequalities

$$-\frac{1}{x} \leq f(x) \leq \frac{1}{x}$$

for all x in the interval $[1, +\infty)$. What can you say about the limit of $f(x)$ as $x \rightarrow +\infty$? Explain your reasoning.

52. Draw pictures analogous to Figure 1.6.2 that illustrate the Squeezing Theorem for limits of the forms $\lim_{x \rightarrow +\infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$.

53. (a) Use the Intermediate-Value Theorem to show that the equation $x = \cos x$ has at least one solution in the interval $[0, \pi/2]$.

(b) Show graphically that there is exactly one solution in the interval.

(c) Approximate the solution to three decimal places.

54. (a) Use the Intermediate-Value Theorem to show that the equation $x + \sin x = 1$ has at least one solution in the interval $[0, \pi/6]$.

(b) Show graphically that there is exactly one solution in the interval.

(c) Approximate the solution to three decimal places.

55. In the study of falling objects near the surface of the Earth, the **acceleration g due to gravity** is commonly taken to be a constant 9.8 m/s^2 . However, the elliptical shape of the Earth and other factors cause variations in this value that depend on latitude. The following formula, known as the World Geodetic System 1984 (WGS 84) Ellipsoidal Gravity Formula, is used to predict the value of g at a latitude of ϕ degrees (either north or south of the equator):

$$g = 9.7803253359 \frac{1 + 0.0019318526461 \sin^2 \phi}{\sqrt{1 - 0.0066943799901 \sin^2 \phi}} \text{ m/s}^2$$

- (a) Use a graphing utility to graph the curve $y = g(\phi)$ for $0^\circ \leq \phi \leq 90^\circ$. What do the values of g at $\phi = 0^\circ$ and at $\phi = 90^\circ$ tell you about the WGS 84 ellipsoid model for the Earth?

(b) Show that $g = 9.8 \text{ m/s}^2$ somewhere between latitudes of 38° and 39° .

56. **Writing** In your own words, explain the *practical value* of the Squeezing Theorem.

57. **Writing** A careful examination of the proof of Theorem 1.6.3 raises the issue of whether the proof might actually be a circular argument! Read the article "A Circular Argument" by Fred Richman in the March 1993 issue of *The College Mathematics Journal*, and write a short report on the author's principal points.

QUICK CHECK ANSWERS 1.6

1. (a) yes (b) yes (c) yes (d) no 2. (a) 1 (b) 0 3. 3; 0

CHAPTER 1 REVIEW EXERCISES



1. For the function f graphed in the accompanying figure, find the limit if it exists.

- | | | |
|-------------------------------------|---|---|
| (a) $\lim_{x \rightarrow 1} f(x)$ | (b) $\lim_{x \rightarrow 2} f(x)$ | (c) $\lim_{x \rightarrow 3} f(x)$ |
| (d) $\lim_{x \rightarrow 4} f(x)$ | (e) $\lim_{x \rightarrow +\infty} f(x)$ | (f) $\lim_{x \rightarrow -\infty} f(x)$ |
| (g) $\lim_{x \rightarrow 3^+} f(x)$ | (h) $\lim_{x \rightarrow 3^-} f(x)$ | (i) $\lim_{x \rightarrow 0} f(x)$ |

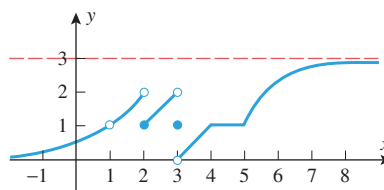


Figure Ex-1

2. In each part, complete the table and make a conjecture about the value of the limit indicated. Confirm your conjecture by finding the limit analytically.

(a) $f(x) = \frac{x-2}{x^2-4}$; $\lim_{x \rightarrow 2^+} f(x)$

x	2.00001	2.0001	2.001	2.01	2.1	2.5
$f(x)$						

(b) $f(x) = \frac{\tan 4x}{x}$; $\lim_{x \rightarrow 0} f(x)$

x	-0.01	-0.001	-0.0001	0.0001	0.001	0.01
$f(x)$						

3–8 Find the limits. ■

3. $\lim_{x \rightarrow -1} \frac{x^3 - x^2}{x - 1}$

4. $\lim_{x \rightarrow 1} \frac{x^3 - x^2}{x - 1}$

5. $\lim_{x \rightarrow -3} \frac{3x + 9}{x^2 + 4x + 3}$

6. $\lim_{x \rightarrow 2^-} \frac{x + 2}{x - 2}$

7. $\lim_{x \rightarrow +\infty} \frac{(2x - 1)^5}{(3x^2 + 2x - 7)(x^3 - 9x)}$

8. $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 4} - 2}{x^2}$

9. In each part, find the horizontal asymptotes, if any.

(a) $y = \frac{2x - 7}{x^2 - 4x}$

(b) $y = \frac{x^3 - x^2 + 10}{3x^2 - 4x}$

(c) $y = \frac{2x^2 - 6}{x^2 + 5x}$

10. In each part, find $\lim_{x \rightarrow a} f(x)$, if it exists, where a is replaced by $0, 5^+, -5^-, -5, 5, -\infty$, and $+\infty$.

(a) $f(x) = \sqrt{5 - x}$

(b) $f(x) = \begin{cases} (x - 5)/|x - 5|, & x \neq 5 \\ 0, & x = 5 \end{cases}$

11–15 Find the limits. ■

11. $\lim_{x \rightarrow 0} \frac{\sin 3x}{\tan 3x}$

12. $\lim_{x \rightarrow 0} \frac{x \sin x}{1 - \cos x}$

13. $\lim_{x \rightarrow 0} \frac{3x - \sin(kx)}{x}, \quad k \neq 0$

14. $\lim_{\theta \rightarrow 0} \tan\left(\frac{1 - \cos \theta}{\theta}\right)$

15. $\lim_{x \rightarrow -1} \frac{\sin(x + 1)}{x^2 - 1}$

16. (a) Write a paragraph or two that describes how the limit of a function can fail to exist at $x = a$, and accompany your description with some specific examples.
 (b) Write a paragraph or two that describes how the limit of a function can fail to exist as $x \rightarrow +\infty$ or $x \rightarrow -\infty$, and accompany your description with some specific examples.
 (c) Write a paragraph or two that describes how a function can fail to be continuous at $x = a$, and accompany your description with some specific examples.

17. (a) Find a formula for a rational function that has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$.
 (b) Check your work by using a graphing utility to graph the function.

18. Paraphrase the ϵ - δ definition for $\lim_{x \rightarrow a} f(x) = L$ in terms of a graphing utility viewing window centered at the point (a, L) .

19. Suppose that $f(x)$ is a function and that for any given $\epsilon > 0$, the condition $0 < |x - 2| < \frac{3}{4}\epsilon$ guarantees that $|f(x) - 5| < \epsilon$.

(a) What limit is described by this statement?

(b) Find a value of δ such that $0 < |x - 2| < \delta$ guarantees that $|8f(x) - 40| < 0.048$.

20. The limit

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

ensures that there is a number δ such that

$$\left| \frac{\sin x}{x} - 1 \right| < 0.001$$

if $0 < |x| < \delta$. Estimate the largest such δ .

21. In each part, a positive number ϵ and the limit L of a function f at a are given. Find a number δ such that $|f(x) - L| < \epsilon$ if $0 < |x - a| < \delta$.

(a) $\lim_{x \rightarrow 2} (4x - 7) = 1$; $\epsilon = 0.01$

(b) $\lim_{x \rightarrow 3/2} \frac{4x^2 - 9}{2x - 3} = 6$; $\epsilon = 0.05$

(c) $\lim_{x \rightarrow 4} x^2 = 16$; $\epsilon = 0.001$

22. Use Definition 1.4.1 to prove the stated limits are correct.

(a) $\lim_{x \rightarrow 2} (4x - 7) = 1$

(b) $\lim_{x \rightarrow 3/2} \frac{4x^2 - 9}{2x - 3} = 6$

23. Suppose that f is continuous at x_0 and that $f(x_0) > 0$. Give either an ϵ - δ proof or a convincing verbal argument to show that there must be an open interval containing x_0 on which $f(x) > 0$.

24. (a) Let

$$f(x) = \frac{\sin x - \sin 1}{x - 1}$$

Approximate $\lim_{x \rightarrow 1} f(x)$ by graphing f and calculating values for some appropriate choices of x .

- (b) Use the identity

$$\sin \alpha - \sin \beta = 2 \sin \frac{\alpha - \beta}{2} \cos \frac{\alpha + \beta}{2}$$

to find the exact value of $\lim_{x \rightarrow 1} f(x)$.

25. Find values of x , if any, at which the given function is not continuous.

(a) $f(x) = \frac{x}{x^2 - 1}$

(b) $f(x) = |x^3 - 2x^2|$

(c) $f(x) = \frac{x + 3}{|x^2 + 3x|}$

26. Determine where f is continuous.

(a) $f(x) = \frac{x}{|x| - 3}$ (b) $f(x) = \cos^{-1}\left(\frac{1}{x}\right)$

(c) $f(x) = \frac{2x - 1}{2x^2 + 3x - 2}$

27. Suppose that

$$f(x) = \begin{cases} -x^4 + 3, & x \leq 2 \\ x^2 + 9, & x > 2 \end{cases}$$

Is f continuous everywhere? Justify your conclusion.

28. One dictionary describes a continuous function as “one whose value at each point is closely approached by its values at neighboring points.”

- (a) How would you explain the meaning of the terms “neighboring points” and “closely approached” to a nonmathematician?
- (b) Write a paragraph that explains why the dictionary definition is consistent with Definition 1.5.1.

29. Show that the conclusion of the Intermediate-Value Theorem may be false if f is not continuous on the interval $[a, b]$.

30. Suppose that f is continuous on the interval $[0, 1]$, that $f(0) = 2$, and that f has no zeros in the interval. Prove that $f(x) > 0$ for all x in $[0, 1]$.

31. Show that the equation $x^4 + 5x^3 + 5x - 1 = 0$ has at least two real solutions in the interval $[-6, 2]$.

CHAPTER 1 MAKING CONNECTIONS

In Section 1.1 we developed the notion of a tangent line to a graph at a given point by considering it as a limiting position of secant lines through that point (Figure 1.1.4a). In these exercises we will develop an analogous idea in which secant lines are replaced by “secant circles” and the tangent line is replaced by a “tangent circle” (called the *osculating circle*). We begin with the graph of $y = x^2$.

1. Recall that there is a unique circle through any three non-collinear points in the plane. For any positive real number x , consider the unique “secant circle” that passes through the fixed point $O(0, 0)$ and the variable points $Q(-x, x^2)$ and $P(x, x^2)$ (see the accompanying figure). Use plane geometry to explain why the center of this circle is the intersection of the y -axis and the perpendicular bisector of segment OP .

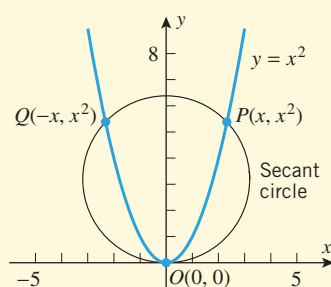


Figure Ex-1

2. (a) Let $(0, C(x))$ denote the center of the circle in Exercise 1 and show that

$$C(x) = \frac{1}{2}x^2 + \frac{1}{2}$$

(b) Show that as $x \rightarrow 0^+$, the secant circles approach a limiting position given by the circle that passes through the origin and is centered at $(0, \frac{1}{2})$. This circle is the osculating circle to the graph of $y = x^2$ at the origin. (Figure Ex-2).

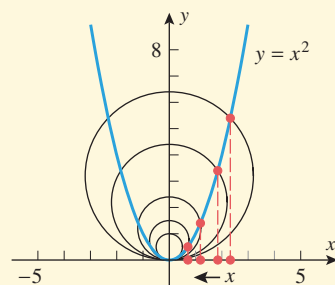


Figure Ex-2

3. Show that if we replace the curve $y = x^2$ by the curve $y = f(x)$, where f is an even function, then the formula for $C(x)$ becomes

$$C(x) = \frac{1}{2} \left[f(0) + f(x) + \frac{x^2}{f(x) - f(0)} \right]$$

[Here we assume that $f(x) \neq f(0)$ for positive values of x close to 0.] If $\lim_{x \rightarrow 0^+} C(x) = L \neq f(0)$, then we define the osculating circle to the curve $y = f(x)$ at $(0, f(0))$ to be the unique circle through $(0, f(0))$ with center $(0, L)$. If $C(x)$ does not have a finite limit different from $f(0)$ as $x \rightarrow 0^+$, then we say that the curve has no osculating circle at $(0, f(0))$.

4. In each part, determine the osculating circle to the curve $y = f(x)$ at $(0, f(0))$, if it exists.

(a) $f(x) = 4x^2$ (b) $f(x) = x^2 \cos x$

(c) $f(x) = |x|$ (d) $f(x) = x \sin x$

(e) $f(x) = \cos x$

(f) $f(x) = x^2 g(x)$, where $g(x)$ is an even continuous function with $g(0) \neq 0$

(g) $f(x) = x^4$

2

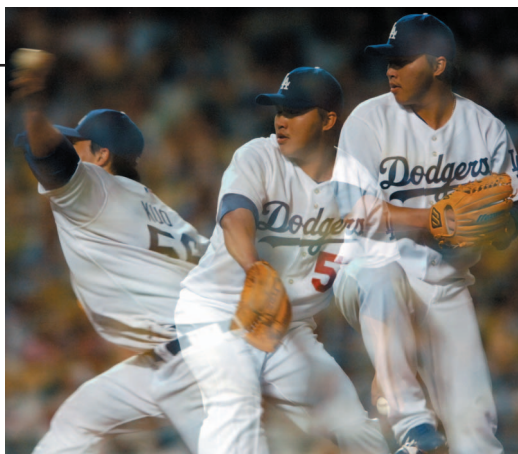


Photo by Kirby Lee/WireImage/Getty Images

THE DERIVATIVE

One of the crowning achievements of calculus is its ability to capture continuous motion mathematically, allowing that motion to be analyzed instant by instant.

Many real-world phenomena involve changing quantities—the speed of a rocket, the inflation of currency, the number of bacteria in a culture, the shock intensity of an earthquake, the voltage of an electrical signal, and so forth. In this chapter we will develop the concept of a “derivative,” which is the mathematical tool for studying the rate at which one quantity changes relative to another. The study of rates of change is closely related to the geometric concept of a tangent line to a curve, so we will also be discussing the general definition of a tangent line and methods for finding its slope and equation. Later in the chapter, we will consider some applications of the derivative. These will include ways in which different rates of change can be related as well as the use of linear functions to approximate nonlinear functions.

2.1 TANGENT LINES AND RATES OF CHANGE

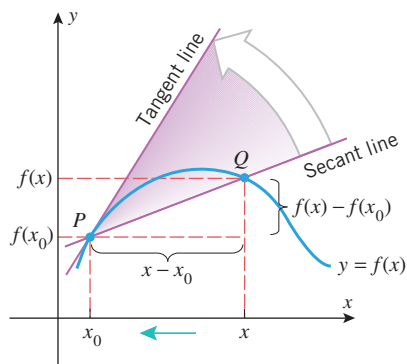
In this section we will discuss three ideas: tangent lines to curves, the velocity of an object moving along a line, and the rate at which one variable changes relative to another. Our goal is to show how these seemingly unrelated ideas are, in actuality, closely linked.

TANGENT LINES

In Example 1 of Section 1.1, we showed how the notion of a limit could be used to find an equation of a tangent line to a curve. At that stage in the text we did not have precise definitions of tangent lines and limits to work with, so the argument was intuitive and informal. However, now that limits have been defined precisely, we are in a position to give a mathematical definition of the tangent line to a curve $y = f(x)$ at a point $P(x_0, f(x_0))$ on the curve. As illustrated in Figure 2.1.1, consider a point $Q(x, f(x))$ on the curve that is distinct from P , and compute the slope m_{PQ} of the secant line through P and Q :

$$m_{PQ} = \frac{f(x) - f(x_0)}{x - x_0}$$

If we let x approach x_0 , then the point Q will move along the curve and approach the point P . If the secant line through P and Q approaches a limiting position as $x \rightarrow x_0$, then we will regard that position to be the position of the tangent line at P . Stated another way, if the slope m_{PQ} of the secant line through P and Q approaches a limit as $x \rightarrow x_0$, then we regard that limit to be the slope m_{tan} of the tangent line at P . Thus, we make the following definition.



► Figure 2.1.1

2.1.1 DEFINITION Suppose that x_0 is in the domain of the function f . The **tangent line** to the curve $y = f(x)$ at the point $P(x_0, f(x_0))$ is the line with equation

$$y - f(x_0) = m_{\tan}(x - x_0)$$

where

$$m_{\tan} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \quad (1)$$

provided the limit exists. For simplicity, we will also call this the tangent line to $y = f(x)$ at x_0 .

► **Example 1** Use Definition 2.1.1 to find an equation for the tangent line to the parabola $y = x^2$ at the point $P(1, 1)$, and confirm the result agrees with that obtained in Example 1 of Section 1.1.

Solution. Applying Formula (1) with $f(x) = x^2$ and $x_0 = 1$, we have

$$\begin{aligned} m_{\tan} &= \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} \\ &= \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} \\ &= \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 2 \end{aligned}$$

Thus, the tangent line to $y = x^2$ at $(1, 1)$ has equation

$$y - 1 = 2(x - 1) \quad \text{or equivalently} \quad y = 2x - 1$$

which agrees with Example 1 of Section 1.1. ◀

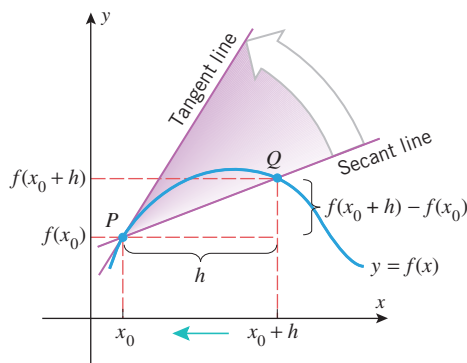
There is an alternative way of expressing Formula (1) that is commonly used. If we let h denote the difference

$$h = x - x_0$$

then the statement that $x \rightarrow x_0$ is equivalent to the statement $h \rightarrow 0$, so we can rewrite (1) in terms of x_0 and h as

$$m_{\tan} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \quad (2)$$

Figure 2.1.2 shows how Formula (2) expresses the slope of the tangent line as a limit of slopes of secant lines.



► Figure 2.1.2

► **Example 2** Compute the slope in Example 1 using Formula (2).

Formulas (1) and (2) for $m_{\tan}$ usually lead to indeterminate forms of type $0/0$, so you will generally need to perform algebraic simplifications or use other methods to determine limits of such indeterminate forms.

Solution. Applying Formula (2) with $f(x) = x^2$ and $x_0 = 1$, we obtain

$$\begin{aligned} m_{\tan} &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(1+h)^2 - 1^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{1 + 2h + h^2 - 1}{h} = \lim_{h \rightarrow 0} (2 + h) = 2 \end{aligned}$$

which agrees with the slope found in Example 1. ◀

► **Example 3** Find an equation for the tangent line to the curve $y = 2/x$ at the point $(2, 1)$ on this curve.

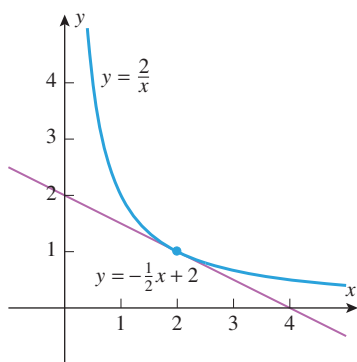
Solution. First, we will find the slope of the tangent line by applying Formula (2) with $f(x) = 2/x$ and $x_0 = 2$. This yields

$$\begin{aligned} m_{\tan} &= \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{2}{2+h} - 1}{h} = \lim_{h \rightarrow 0} \frac{\left(\frac{2 - (2+h)}{2+h}\right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h}{h(2+h)} = -\left(\lim_{h \rightarrow 0} \frac{1}{2+h}\right) = -\frac{1}{2} \end{aligned}$$

Thus, an equation of the tangent line at $(2, 1)$ is

$$y - 1 = -\frac{1}{2}(x - 2) \quad \text{or equivalently} \quad y = -\frac{1}{2}x + 2$$

(see Figure 2.1.3). ◀



▲ Figure 2.1.3

► **Example 4** Find the slopes of the tangent lines to the curve $y = \sqrt{x}$ at $x_0 = 1$, $x_0 = 4$, and $x_0 = 9$.

Solution. We could compute each of these slopes separately, but it will be more efficient to find the slope for a general value of x_0 and then substitute the specific numerical values. Proceeding in this way we obtain

$$\begin{aligned} m_{\tan} &= \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{x_0 + h} - \sqrt{x_0}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{x_0 + h} - \sqrt{x_0}}{h} \cdot \frac{\sqrt{x_0 + h} + \sqrt{x_0}}{\sqrt{x_0 + h} + \sqrt{x_0}} \\ &= \lim_{h \rightarrow 0} \frac{x_0 + h - x_0}{h(\sqrt{x_0 + h} + \sqrt{x_0})} \\ &= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x_0 + h} + \sqrt{x_0})} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x_0 + h} + \sqrt{x_0}} = \frac{1}{2\sqrt{x_0}} \end{aligned}$$

Rationalize the numerator to help eliminate the indeterminate form of the limit.

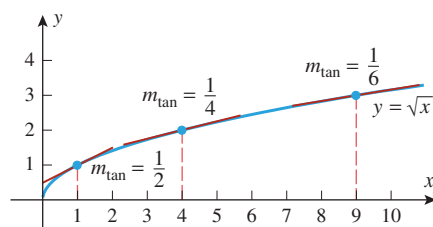
The slopes at $x_0 = 1, 4$, and 9 can now be obtained by substituting these values into our general formula for $m_{\tan}$. Thus,

$$\text{slope at } x_0 = 1: \frac{1}{2\sqrt{1}} = \frac{1}{2}$$

$$\text{slope at } x_0 = 4: \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

$$\text{slope at } x_0 = 9: \frac{1}{2\sqrt{9}} = \frac{1}{6}$$

(see Figure 2.1.4). ◀



► Figure 2.1.4



Carlos Santa Maria/iStockphoto

The velocity of an airplane describes its speed and direction.

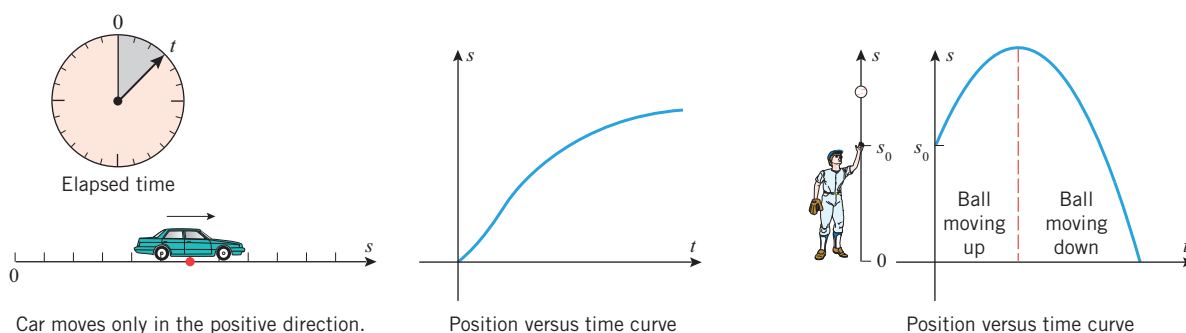
VELOCITY

One of the important themes in calculus is the study of motion. To describe the motion of an object completely, one must specify its **speed** (how fast it is going) and the direction in which it is moving. The speed and the direction of motion together comprise what is called the **velocity** of the object. For example, knowing that the speed of an aircraft is 500 mi/h tells us how fast it is going, but not which way it is moving. In contrast, knowing that the velocity of the aircraft is 500 mi/h *due south* pins down the speed and the direction of motion.

Later, we will study the motion of objects that move along curves in two- or three-dimensional space, but for now we will only consider motion along a line; this is called **rectilinear motion**. Some examples are a piston moving up and down in a cylinder, a race

car moving along a straight track, an object dropped from the top of a building and falling straight down, a ball thrown straight up and then falling down along the same line, and so forth.

For computational purposes, we will assume that a particle in rectilinear motion moves along a coordinate line, which we will call the s -axis. A graphical description of rectilinear motion along an s -axis can be obtained by making a plot of the s -coordinate of the particle versus the elapsed time t from starting time $t = 0$. This is called the **position versus time curve** for the particle. Figure 2.1.5 shows two typical position versus time curves. The first is for a car that starts at the origin and moves only in the positive direction of the s -axis. In this case s increases as t increases. The second is for a ball that is thrown straight up in the positive direction of an s -axis from some initial height s_0 and then falls straight down in the negative direction. In this case s increases as the ball moves up and decreases as it moves down.



▲ Figure 2.1.5

If a particle in rectilinear motion moves along an s -axis so that its position coordinate function of the elapsed time t is

$$s = f(t) \quad (3)$$

then f is called the **position function of the particle**; the graph of (3) is the position versus time curve. The **average velocity** of the particle over a time interval $[t_0, t_0 + h]$, $h > 0$, is defined to be

$$v_{\text{ave}} = \frac{\text{change in position}}{\text{time elapsed}} = \frac{f(t_0 + h) - f(t_0)}{h} \quad (4)$$

Show that (4) is also correct for a time interval $[t_0 + h, t_0]$, $h < 0$.

The change in position

$$f(t_0 + h) - f(t_0)$$

is also called the **displacement** of the particle over the time interval between t_0 and $t_0 + h$.

► **Example 5** Suppose that $s = f(t) = 1 + 5t - 2t^2$ is the position function of a particle, where s is in meters and t is in seconds. Find the average velocities of the particle over the time intervals (a) $[0, 2]$ and (b) $[2, 3]$.

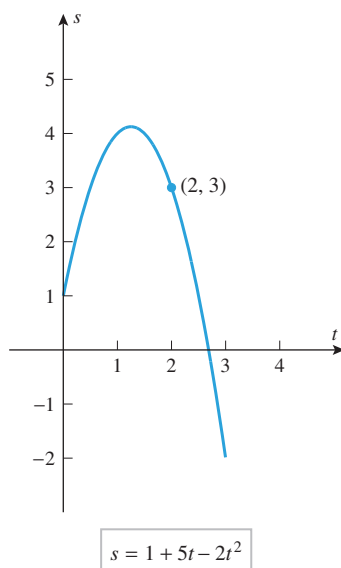
Solution (a). Applying (4) with $t_0 = 0$ and $h = 2$, we see that the average velocity is

$$v_{\text{ave}} = \frac{f(t_0 + h) - f(t_0)}{h} = \frac{f(2) - f(0)}{2} = \frac{3 - 1}{2} = \frac{2}{2} = 1 \text{ m/s}$$

Solution (b). Applying (4) with $t_0 = 2$ and $h = 1$, we see that the average velocity is

$$v_{\text{ave}} = \frac{f(t_0 + h) - f(t_0)}{h} = \frac{f(3) - f(2)}{1} = \frac{-2 - 3}{1} = \frac{-5}{1} = -5 \text{ m/s} \quad \blacktriangleleft$$

For a particle in rectilinear motion, average velocity describes its behavior over an *interval* of time. We are interested in the particle's “instantaneous velocity,” which describes



▲ Figure 2.1.6

Table 2.1.1

TIME INTERVAL	AVERAGE VELOCITY (m/s)
$2.0 \leq t \leq 3.0$	-5
$2.0 \leq t \leq 2.1$	-3.2
$2.0 \leq t \leq 2.01$	-3.02
$2.0 \leq t \leq 2.001$	-3.002
$2.0 \leq t \leq 2.0001$	-3.0002

Note the negative values for the velocities in Example 6. This is consistent with the fact that the object is moving in the negative direction along the s -axis.

its behavior at a specific *instant* in time. Formula (4) is not directly applicable for computing instantaneous velocity because the “time elapsed” at a specific instant is zero, so (4) is undefined. One way to circumvent this problem is to compute average velocities for small time intervals between $t = t_0$ and $t = t_0 + h$. These average velocities may be viewed as approximations to the “instantaneous velocity” of the particle at time t_0 . If these average velocities have a limit as h approaches zero, then we can take that limit to be the *instantaneous velocity* of the particle at time t_0 . Here is an example.

► **Example 6** Consider the particle in Example 5, whose position function is

$$s = f(t) = 1 + 5t - 2t^2$$

The position of the particle at time $t = 2$ s is $s = 3$ m (Figure 2.1.6). Find the particle’s instantaneous velocity at time $t = 2$ s.

Solution. As a first approximation to the particle’s instantaneous velocity at time $t = 2$ s, let us recall from Example 5(b) that the average velocity over the time interval from $t = 2$ to $t = 3$ is $v_{\text{ave}} = -5$ m/s. To improve on this initial approximation we will compute the average velocity over a succession of smaller and smaller time intervals. We leave it to you to verify the results in Table 2.1.1. The average velocities in this table appear to be approaching a limit of -3 m/s, providing strong evidence that the instantaneous velocity at time $t = 2$ s is -3 m/s. To confirm this analytically, we start by computing the object’s average velocity over a general time interval between $t = 2$ and $t = 2 + h$ using Formula (4):

$$v_{\text{ave}} = \frac{f(2+h) - f(2)}{h} = \frac{[1 + 5(2+h) - 2(2+h)^2] - 3}{h}$$

The object’s instantaneous velocity at time $t = 2$ is calculated as a limit as $h \rightarrow 0$:

$$\begin{aligned} \text{instantaneous velocity} &= \lim_{h \rightarrow 0} \frac{[1 + 5(2+h) - 2(2+h)^2] - 3}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2 + (10 + 5h) - (8 + 8h + 2h^2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-3h - 2h^2}{h} = \lim_{h \rightarrow 0} (-3 - 2h) = -3 \end{aligned}$$

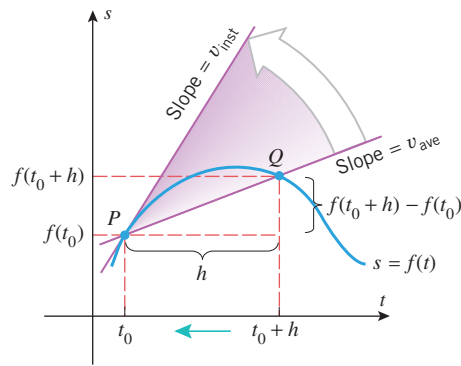
This confirms our numerical conjecture that the instantaneous velocity after 2 s is -3 m/s. ◀

Consider a particle in rectilinear motion with position function $s = f(t)$. Motivated by Example 6, we define the instantaneous velocity v_{inst} of the particle at time t_0 to be the limit as $h \rightarrow 0$ of its average velocities v_{ave} over time intervals between $t = t_0$ and $t = t_0 + h$. Thus, from (4) we obtain

$$v_{\text{inst}} = \lim_{h \rightarrow 0} \frac{f(t_0 + h) - f(t_0)}{h} \quad (5)$$

Geometrically, the average velocity v_{ave} between $t = t_0$ and $t = t_0 + h$ is the slope of the secant line through points $P(t_0, f(t_0))$ and $Q(t_0 + h, f(t_0 + h))$ on the position versus time curve, and the instantaneous velocity v_{inst} at time t_0 is the slope of the tangent line to the position versus time curve at the point $P(t_0, f(t_0))$ (Figure 2.1.7).

Confirm the solution to Example 5(b) by computing the slope of an appropriate secant line.

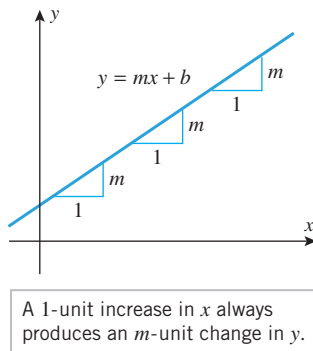


► Figure 2.1.7

■ SLOPES AND RATES OF CHANGE

Velocity can be viewed as *rate of change*—the rate of change of position with respect to time. Rates of change occur in other applications as well. For example:

- A microbiologist might be interested in the rate at which the number of bacteria in a colony changes with time.
- An engineer might be interested in the rate at which the length of a metal rod changes with temperature.
- An economist might be interested in the rate at which production cost changes with the quantity of a product that is manufactured.
- A medical researcher might be interested in the rate at which the radius of an artery changes with the concentration of alcohol in the bloodstream.



▲ Figure 2.1.8

Our next objective is to define precisely what is meant by the “rate of change of y with respect to x ” when y is a function of x . In the case where y is a linear function of x , say $y = mx + b$, the slope m is the natural measure of the rate of change of y with respect to x . As illustrated in Figure 2.1.8, each 1-unit increase in x anywhere along the line produces an m -unit change in y , so we see that y changes at a constant rate with respect to x along the line and that m measures this rate of change.

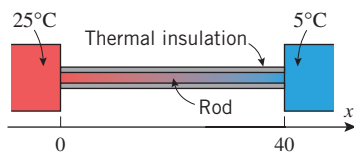
► **Example 7** Find the rate of change of y with respect to x if

$$(a) \ y = 2x - 1 \quad (b) \ y = -5x + 1$$

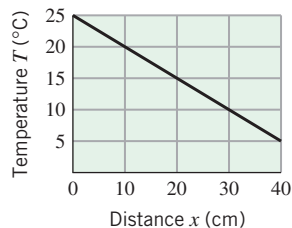
Solution. In part (a) the rate of change of y with respect to x is $m = 2$, so each 1-unit increase in x produces a 2-unit increase in y . In part (b) the rate of change of y with respect to x is $m = -5$, so each 1-unit increase in x produces a 5-unit decrease in y . ◀

In applied problems, changing the units of measurement can change the slope of a line, so it is essential to include the units when calculating the slope and describing rates of change. The following example illustrates this.

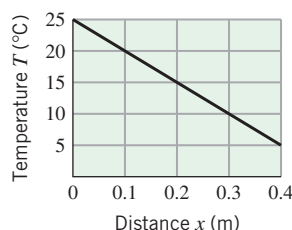
► **Example 8** Suppose that a uniform rod of length 40 cm ($= 0.4$ m) is thermally insulated around the lateral surface and that the exposed ends of the rod are held at constant temperatures of 25°C and 5°C , respectively (Figure 2.1.9a). It is shown in physics that under appropriate conditions the graph of the temperature T versus the distance x from the left-hand end of the rod will be a straight line. Parts (b) and (c) of Figure 2.1.9 show two



(a)

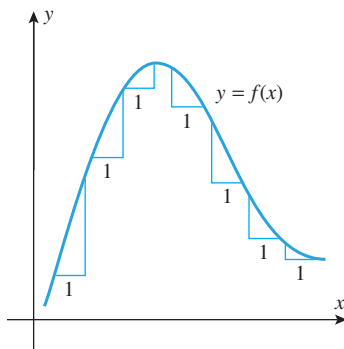


(b)



(c)

▲ Figure 2.1.9



▲ Figure 2.1.10

such graphs: one in which x is measured in centimeters and one in which it is measured in meters. The slopes in the two cases are

$$m = \frac{5 - 25}{40 - 0} = \frac{-20}{40} = -0.5 \quad (6)$$

$$m = \frac{5 - 25}{0.4 - 0} = \frac{-20}{0.4} = -50 \quad (7)$$

The slope in (6) implies that the temperature *decreases* at a rate of 0.5°C per centimeter of distance from the left end of the rod, and the slope in (7) implies that the temperature decreases at a rate of 50°C per meter of distance from the left end of the rod. The two statements are equivalent physically, even though the slopes differ. ◀

Although the rate of change of y with respect to x is constant along a nonvertical line $y = mx + b$, this is not true for a general curve $y = f(x)$. For example, in Figure 2.1.10 the change in y that results from a 1-unit increase in x tends to have greater magnitude in regions where the curve rises or falls rapidly than in regions where it rises or falls slowly. As with velocity, we will distinguish between the average rate of change over an interval and the instantaneous rate of change at a specific point.

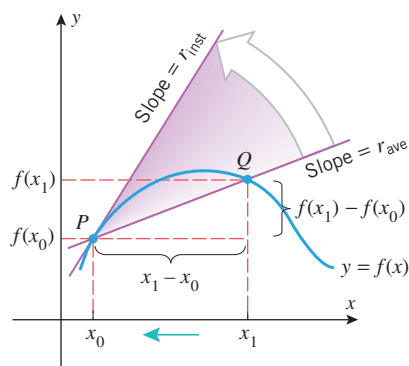
If $y = f(x)$, then we define the **average rate of change of y with respect to x over the interval $[x_0, x_1]$** to be

$$r_{\text{ave}} = \frac{f(x_1) - f(x_0)}{x_1 - x_0} \quad (8)$$

and we define the **instantaneous rate of change of y with respect to x at x_0** to be

$$r_{\text{inst}} = \lim_{x_1 \rightarrow x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0} \quad (9)$$

Geometrically, the average rate of change of y with respect to x over the interval $[x_0, x_1]$ is the slope of the secant line through the points $P(x_0, f(x_0))$ and $Q(x_1, f(x_1))$ (Figure 2.1.11), and the instantaneous rate of change of y with respect to x at x_0 is the slope of the tangent line at the point $P(x_0, f(x_0))$ (since it is the limit of the slopes of the secant lines through P).



► Figure 2.1.11

If desired, we can let $h = x_1 - x_0$, and rewrite (8) and (9) as

$$r_{\text{ave}} = \frac{f(x_0 + h) - f(x_0)}{h} \quad (10)$$

$$r_{\text{inst}} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \quad (11)$$

► **Example 9** Let $y = x^2 + 1$.

- (a) Find the average rate of change of y with respect to x over the interval $[3, 5]$.
 (b) Find the instantaneous rate of change of y with respect to x when $x = -4$.

Solution (a). We will apply Formula (8) with $f(x) = x^2 + 1$, $x_0 = 3$, and $x_1 = 5$. This yields

$$r_{\text{ave}} = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{f(5) - f(3)}{5 - 3} = \frac{26 - 10}{2} = 8$$

Thus, y increases an average of 8 units per unit increase in x over the interval $[3, 5]$.

Solution (b). We will apply Formula (9) with $f(x) = x^2 + 1$ and $x_0 = -4$. This yields

$$\begin{aligned} r_{\text{inst}} &= \lim_{x_1 \rightarrow x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \lim_{x_1 \rightarrow -4} \frac{f(x_1) - f(-4)}{x_1 - (-4)} = \lim_{x_1 \rightarrow -4} \frac{(x_1^2 + 1) - 17}{x_1 + 4} \\ &= \lim_{x_1 \rightarrow -4} \frac{x_1^2 - 16}{x_1 + 4} = \lim_{x_1 \rightarrow -4} \frac{(x_1 + 4)(x_1 - 4)}{x_1 + 4} = \lim_{x_1 \rightarrow -4} (x_1 - 4) = -8 \end{aligned}$$

Thus, a small increase in x from $x = -4$ will produce approximately an 8-fold decrease in y . ◀

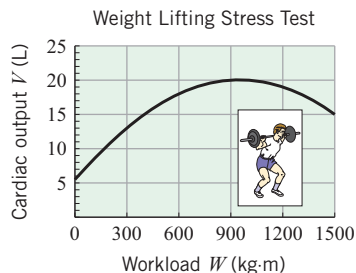
Perform the calculations in Example 9 using Formulas (10) and (11).

RATES OF CHANGE IN APPLICATIONS

In applied problems, average and instantaneous rates of change must be accompanied by appropriate units. In general, the units for a rate of change of y with respect to x are obtained by “dividing” the units of y by the units of x and then simplifying according to the standard rules of algebra. Here are some examples:

- If y is in degrees Fahrenheit ($^{\circ}\text{F}$) and x is in inches (in), then a rate of change of y with respect to x has units of degrees Fahrenheit per inch ($^{\circ}\text{F/in}$).
- If y is in feet per second (ft/s) and x is in seconds (s), then a rate of change of y with respect to x has units of feet per second per second (ft/s/s), which would usually be written as ft/s^2 .
- If y is in newton-meters (N·m) and x is in meters (m), then a rate of change of y with respect to x has units of newtons (N), since $\text{N}\cdot\text{m}/\text{m} = \text{N}$.
- If y is in foot-pounds (ft·lb) and x is in hours (h), then a rate of change of y with respect to x has units of foot-pounds per hour (ft·lb/h).

► **Example 10** The limiting factor in athletic endurance is cardiac output, that is, the volume of blood that the heart can pump per unit of time during an athletic competition. Figure 2.1.12 shows a stress-test graph of cardiac output V in liters (L) of blood versus workload W in kilogram-meters (kg·m) for 1 minute of weight lifting. This graph illustrates the known medical fact that cardiac output increases with the workload, but after reaching a peak value begins to decrease.



▲ Figure 2.1.12

- (a) Use the secant line shown in Figure 2.1.13a to estimate the average rate of change of cardiac output with respect to workload as the workload increases from 300 to 1200 kg·m.
 (b) Use the line segment shown in Figure 2.1.13b to estimate the instantaneous rate of change of cardiac output with respect to workload at the point where the workload is 300 kg·m.

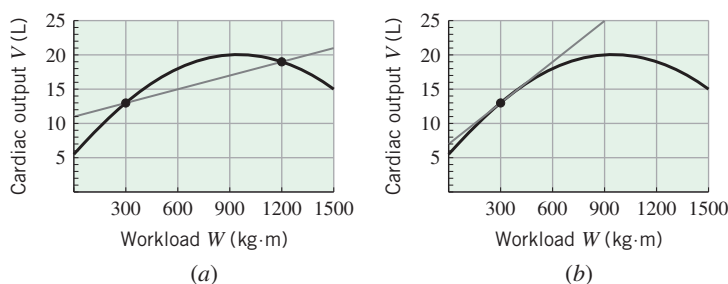
Solution (a). Using the estimated points (300, 13) and (1200, 19) to find the slope of the secant line, we obtain

$$r_{\text{ave}} \approx \frac{19 - 13}{1200 - 300} \approx 0.0067 \frac{\text{L}}{\text{kg} \cdot \text{m}}$$

This means that on average a 1-unit increase in workload produced a 0.0067 L increase in cardiac output over the interval.

Solution (b). We estimate the slope of the cardiac output curve at $W = 300$ by sketching a line that appears to meet the curve at $W = 300$ with slope equal to that of the curve (Figure 2.1.13b). Estimating points (0, 7) and (900, 25) on this line, we obtain

$$r_{\text{inst}} \approx \frac{25 - 7}{900 - 0} = 0.02 \frac{\text{L}}{\text{kg} \cdot \text{m}} \quad \blacktriangleleft$$



► Figure 2.1.13

QUICK CHECK EXERCISES 2.1 (See page 122 for answers.)

1. The slope m_{tan} of the tangent line to the curve $y = f(x)$ at the point $P(x_0, f(x_0))$ is given by

$$m_{\text{tan}} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

2. The tangent line to the curve $y = (x - 1)^2$ at the point $(-1, 4)$ has equation $4x + y = 0$. Thus, the value of the limit

$$\lim_{x \rightarrow -1} \frac{x^2 - 2x - 3}{x + 1}$$

is _____.

3. A particle is moving along an s -axis, where s is in feet. During the first 5 seconds of motion, the position of the particle is given by

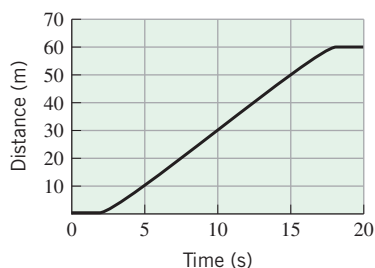
$$s = 10 - (3 - t)^2, \quad 0 \leq t \leq 5$$

Use this position function to complete each part.

- (a) Initially, the particle moves a distance of _____ ft in the (positive/negative) _____ direction; then it reverses direction, traveling a distance of _____ ft during the remainder of the 5-second period.
- (b) The average velocity of the particle over the 5-second period is _____.
4. Let $s = f(t)$ be the equation of a position versus time curve for a particle in rectilinear motion, where s is in meters and t is in seconds. Assume that $s = -1$ when $t = 2$ and that the instantaneous velocity of the particle at this instant is 3 m/s. The equation of the tangent line to the position versus time curve at time $t = 2$ is _____.
5. Suppose that $y = x^2 + x$.
- (a) The average rate of change of y with respect to x over the interval $2 \leq x \leq 5$ is _____.
- (b) The instantaneous rate of change of y with respect to x at $x = 2$, r_{inst} , is given by the limit _____.

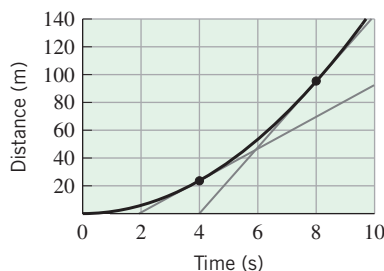
EXERCISE SET 2.1

1. The accompanying figure on the next page shows the position versus time curve for an elevator that moves upward a distance of 60 m and then discharges its passengers.
- (a) Estimate the instantaneous velocity of the elevator at $t = 10$ s.
- (b) Sketch a velocity versus time curve for the motion of the elevator for $0 \leq t \leq 20$.



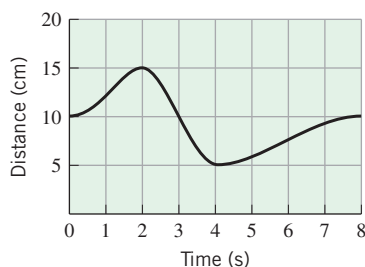
◀ Figure Ex-1

2. The accompanying figure shows the position versus time curve for an automobile over a period of time of 10 s. Use the line segments shown in the figure to estimate the instantaneous velocity of the automobile at time $t = 4$ s and again at time $t = 8$ s.



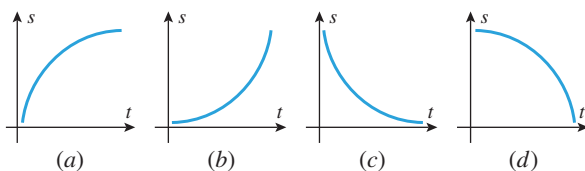
◀ Figure Ex-2

3. The accompanying figure shows the position versus time curve for a certain particle moving along a straight line. Estimate each of the following from the graph:
- the average velocity over the interval $0 \leq t \leq 3$
 - the values of t at which the instantaneous velocity is zero
 - the values of t at which the instantaneous velocity is either a maximum or a minimum
 - the instantaneous velocity when $t = 3$ s.



◀ Figure Ex-3

4. The accompanying figure shows the position versus time curves of four different particles moving on a straight line. For each particle, determine whether its instantaneous velocity is increasing or decreasing with time.



▲ Figure Ex-4

FOCUS ON CONCEPTS

- If a particle moves at constant velocity, what can you say about its position versus time curve?
- An automobile, initially at rest, begins to move along a straight track. The velocity increases steadily until suddenly the driver sees a concrete barrier in the road and applies the brakes sharply at time t_0 . The car decelerates rapidly, but it is too late—the car crashes into the barrier at time t_1 and instantaneously comes to rest. Sketch a position versus time curve that might represent the motion of the car. Indicate how characteristics of your curve correspond to the events of this scenario.

7–10 For each exercise, sketch a curve and a line L satisfying the stated conditions. ■

- L is tangent to the curve and intersects the curve in at least two points.
- L intersects the curve in exactly one point, but L is not tangent to the curve.
- L is tangent to the curve at two different points.
- L is tangent to the curve at two different points and intersects the curve at a third point.

11–14 A function $y = f(x)$ and values of x_0 and x_1 are given.

- Find the average rate of change of y with respect to x over the interval $[x_0, x_1]$.
- Find the instantaneous rate of change of y with respect to x at the specified value of x_0 .
- Find the instantaneous rate of change of y with respect to x at an arbitrary value of x_0 .
- The average rate of change in part (a) is the slope of a certain secant line, and the instantaneous rate of change in part (b) is the slope of a certain tangent line. Sketch the graph of $y = f(x)$ together with those two lines. ■

11. $y = 2x^2$; $x_0 = 0$, $x_1 = 1$ **12.** $y = x^3$; $x_0 = 1$, $x_1 = 2$

13. $y = 1/x$; $x_0 = 2$, $x_1 = 3$ **14.** $y = 1/x^2$; $x_0 = 1$, $x_1 = 2$

15–18 A function $y = f(x)$ and an x -value x_0 are given.

- Find a formula for the slope of the tangent line to the graph of f at a general point $x = x_0$.
- Use the formula obtained in part (a) to find the slope of the tangent line for the given value of x_0 . ■

15. $f(x) = x^2 - 1$; $x_0 = -1$

16. $f(x) = x^2 + 3x + 2$; $x_0 = 2$

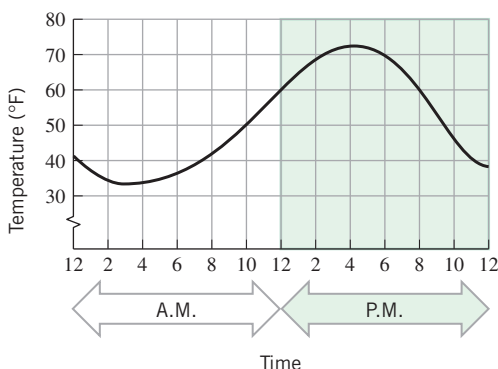
17. $f(x) = x + \sqrt{x}$; $x_0 = 1$

18. $f(x) = 1/\sqrt{x}$; $x_0 = 4$

19–22 True–False Determine whether the statement is true or false. Explain your answer. ■

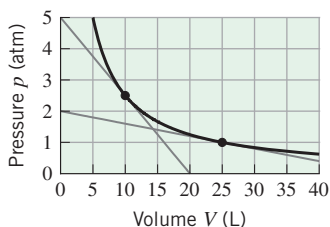
19. If $\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = 3$, then $\lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = 3$.

20. A tangent line to a curve $y = f(x)$ is a particular kind of secant line to the curve.
21. The velocity of an object represents a change in the object's position.
22. A 50-foot horizontal metal beam is supported on either end by concrete pillars and a weight is placed on the middle of the beam. If $f(x)$ models how many inches the center of the beam sags when the weight measures x tons, then the units of the rate of change of $y = f(x)$ with respect to x are inches/ton.
23. Suppose that the outside temperature versus time curve over a 24-hour period is as shown in the accompanying figure.
- Estimate the maximum temperature and the time at which it occurs.
 - The temperature rise is fairly linear from 8 A.M. to 2 P.M. Estimate the rate at which the temperature is increasing during this time period.
 - Estimate the time at which the temperature is decreasing most rapidly. Estimate the instantaneous rate of change of temperature with respect to time at this instant.



▲ Figure Ex-23

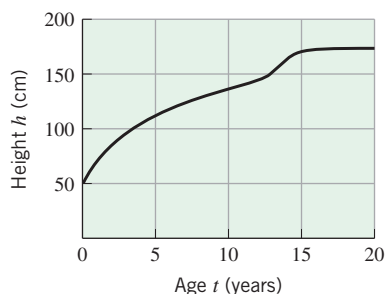
24. The accompanying figure shows the graph of the pressure p in atmospheres (atm) versus the volume V in liters (L) of 1 mole of an ideal gas at a constant temperature of 300 K (kelvins). Use the line segments shown in the figure to estimate the rate of change of pressure with respect to volume at the points where $V = 10$ L and $V = 25$ L.



◀ Figure Ex-24

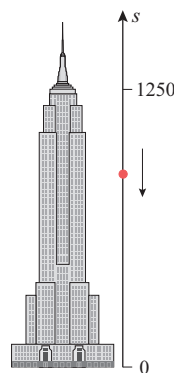
25. The accompanying figure shows the graph of the height h in centimeters versus the age t in years of an individual from birth to age 20.

- When is the growth rate greatest?
- Estimate the growth rate at age 5.
- At approximately what age between 10 and 20 is the growth rate greatest? Estimate the growth rate at this age.
- Draw a rough graph of the growth rate versus age.



◀ Figure Ex-25

26. An object is released from rest (its initial velocity is zero) from the Empire State Building at a height of 1250 ft above street level (Figure Ex-26). The height of the object can be modeled by the position function $s = f(t) = 1250 - 16t^2$.
- Verify that the object is still falling at $t = 5$ s.
 - Find the average velocity of the object over the time interval from $t = 5$ to $t = 6$ s.
 - Find the object's instantaneous velocity at time $t = 5$ s.



◀ Figure Ex-26

27. During the first 40 s of a rocket flight, the rocket is propelled straight up so that in t seconds it reaches a height of $s = 0.3t^3$ ft.
- How high does the rocket travel in 40 s?
 - What is the average velocity of the rocket during the first 40 s?
 - What is the average velocity of the rocket during the first 1000 ft of its flight?
 - What is the instantaneous velocity of the rocket at the end of 40 s?

28. An automobile is driven down a straight highway such that after $0 \leq t \leq 12$ seconds it is $s = 4.5t^2$ feet from its initial position.

(cont.)

- (a) Find the average velocity of the car over the interval $[0, 12]$.
 (b) Find the instantaneous velocity of the car at $t = 6$.
29. A robot moves in the positive direction along a straight line so that after t minutes its distance is $s = 6t^4$ feet from the origin.
 (a) Find the average velocity of the robot over the interval $[2, 4]$.
 (b) Find the instantaneous velocity at $t = 2$.

30. **Writing** Discuss how the tangent line to the graph of a function $y = f(x)$ at a point $P(x_0, f(x_0))$ is defined in terms of secant lines to the graph through point P .
31. **Writing** A particle is in rectilinear motion during the time interval $0 \leq t \leq 2$. Explain the connection between the instantaneous velocity of the particle at time $t = 1$ and the average velocities of the particle during portions of the interval $0 \leq t \leq 2$.

✓ QUICK CHECK ANSWERS 2.1

1. $\frac{f(x) - f(x_0)}{x - x_0}$; $\frac{f(x_0 + h) - f(x_0)}{h}$ 2. -4 3. (a) 9; positive; 4 (b) 1 ft/s 4. $s = 3t - 7$
5. (a) 8 (b) $\lim_{x \rightarrow 2} \frac{(x^2 + x) - 6}{x - 2}$ or $\lim_{h \rightarrow 0} \frac{[(2 + h)^2 + (2 + h)] - 6}{h}$

2.2 THE DERIVATIVE FUNCTION

In this section we will discuss the concept of a “derivative,” which is the primary mathematical tool that is used to calculate and study rates of change.

■ DEFINITION OF THE DERIVATIVE FUNCTION

In the last section we showed that if the limit

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

exists, then it can be interpreted either as the slope of the tangent line to the curve $y = f(x)$ at $x = x_0$ or as the instantaneous rate of change of y with respect to x at $x = x_0$ [see Formulas (2) and (11) of that section]. This limit is so important that it has a special notation:

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \quad (1)$$

You can think of f' (read “ f prime”) as a function whose input is x_0 and whose output is the number $f'(x_0)$ that represents either the slope of the tangent line to $y = f(x)$ at $x = x_0$ or the instantaneous rate of change of y with respect to x at $x = x_0$. To emphasize this function point of view, we will replace x_0 by x in (1) and make the following definition.

The expression

$$\frac{f(x + h) - f(x)}{h}$$

that appears in (2) is commonly called the **difference quotient**.

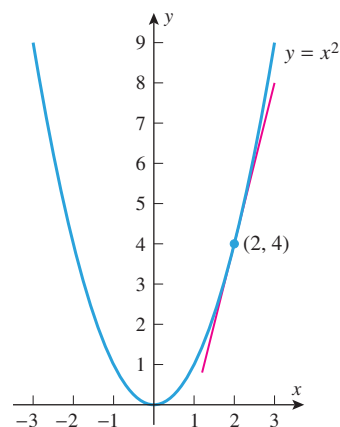
2.2.1 DEFINITION The function f' defined by the formula

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} \quad (2)$$

is called the **derivative of f with respect to x** . The domain of f' consists of all x in the domain of f for which the limit exists.

The term “derivative” is used because the function f' is *derived* from the function f by a limiting process.

► **Example 1** Find the derivative with respect to x of $f(x) = x^2$, and use it to find the equation of the tangent line to $y = x^2$ at $x = 2$.



▲ Figure 2.2.1

Solution. It follows from (2) that

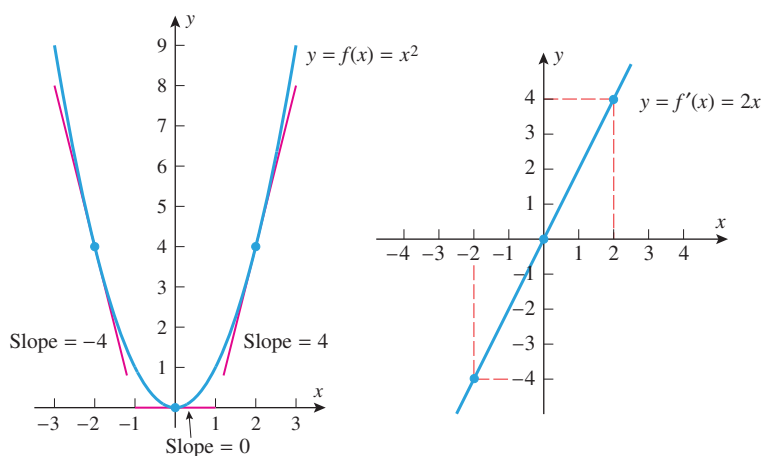
$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\ &= \lim_{h \rightarrow 0} (2x + h) = 2x \end{aligned}$$

Thus, the slope of the tangent line to $y = x^2$ at $x = 2$ is $f'(2) = 4$. Since $y = 4$ if $x = 2$, the point-slope form of the tangent line is

$$y - 4 = 4(x - 2)$$

which we can rewrite in slope-intercept form as $y = 4x - 4$ (Figure 2.2.1). ◀

You can think of f' as a “slope-producing function” in the sense that the value of $f'(x)$ at $x = x_0$ is the slope of the tangent line to the graph of f at $x = x_0$. This aspect of the derivative is illustrated in Figure 2.2.2, which shows the graphs of $f(x) = x^2$ and its derivative $f'(x) = 2x$ (obtained in Example 1). The figure illustrates that the values of $f'(x) = 2x$ at $x = -2, 0$, and 2 correspond to the slopes of the tangent lines to the graph of $f(x) = x^2$ at those values of x .



► Figure 2.2.2

In general, if $f'(x)$ is defined at $x = x_0$, then the point-slope form of the equation of the tangent line to the graph of $y = f(x)$ at $x = x_0$ may be found using the following steps.

Finding an Equation for the Tangent Line to $y = f(x)$ at $x = x_0$.

Step 1. Evaluate $f(x_0)$; the point of tangency is $(x_0, f(x_0))$.

Step 2. Find $f'(x)$ and evaluate $f'(x_0)$, which is the slope m of the line.

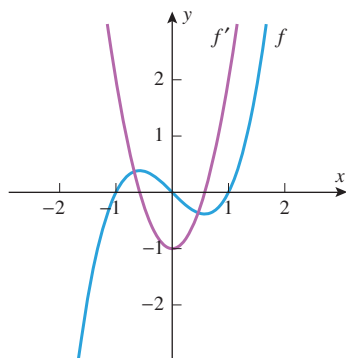
Step 3. Substitute the value of the slope m and the point $(x_0, f(x_0))$ into the point-slope form of the line

$$y - f(x_0) = f'(x_0)(x - x_0)$$

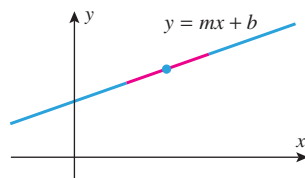
or, equivalently,

$$y = f(x_0) + f'(x_0)(x - x_0) \quad (3)$$

In Solution (a), the binomial formula is used to expand $(x + h)^3$. This formula may be found on the front endpaper.



▲ Figure 2.2.3



At each value of x the tangent line has slope m .

▲ Figure 2.2.4

The result in Example 3 is consistent with our earlier observation that the rate of change of y with respect to x along a line $y = mx + b$ is constant and that constant is m .

► Example 2

- (a) Find the derivative with respect to x of $f(x) = x^3 - x$.
 (b) Graph f and f' together, and discuss the relationship between the two graphs.

Solution (a).

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[(x+h)^3 - (x+h)] - [x^3 - x]}{h} \\ &= \lim_{h \rightarrow 0} \frac{[x^3 + 3x^2h + 3xh^2 + h^3 - x - h] - [x^3 - x]}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3 - h}{h} \\ &= \lim_{h \rightarrow 0} [3x^2 + 3xh + h^2 - 1] = 3x^2 - 1 \end{aligned}$$

Solution (b). Since $f'(x)$ can be interpreted as the slope of the tangent line to the graph of $y = f(x)$ at x , it follows that $f'(x)$ is positive where the tangent line has positive slope, is negative where the tangent line has negative slope, and is zero where the tangent line is horizontal. We leave it for you to verify that this is consistent with the graphs of $f(x) = x^3 - x$ and $f'(x) = 3x^2 - 1$ shown in Figure 2.2.3. ◀

► Example 3 At each value of x , the tangent line to a line $y = mx + b$ coincides with the line itself (Figure 2.2.4), and hence all tangent lines have slope m . This suggests geometrically that if $f(x) = mx + b$, then $f'(x) = m$ for all x . This is confirmed by the following computations:

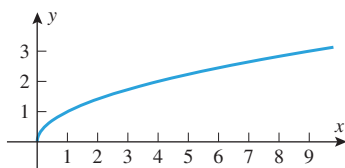
$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[m(x+h) + b] - [mx + b]}{h} \\ &= \lim_{h \rightarrow 0} \frac{mh}{h} = \lim_{h \rightarrow 0} m = m \quad \blacktriangleleft \end{aligned}$$

► Example 4

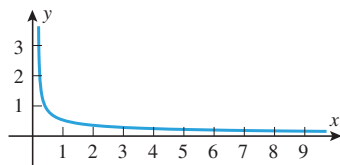
- (a) Find the derivative with respect to x of $f(x) = \sqrt{x}$.
 (b) Find the slope of the tangent line to $y = \sqrt{x}$ at $x = 9$.
 (c) Find the limits of $f'(x)$ as $x \rightarrow 0^+$ and as $x \rightarrow +\infty$, and explain what those limits say about the graph of f .

Solution (a). Recall from Example 4 of Section 2.1 that the slope of the tangent line to $y = \sqrt{x}$ at $x = x_0$ is given by $m_{\text{tan}} = 1/(2\sqrt{x_0})$. Thus, $f'(x) = 1/(2\sqrt{x})$.

Solution (b). The slope of the tangent line at $x = 9$ is $f'(9)$. From part (a), this slope is $f'(9) = 1/(2\sqrt{9}) = \frac{1}{6}$.



$$y = f(x) = \sqrt{x}$$



$$y = f'(x) = \frac{1}{2\sqrt{x}}$$

▲ Figure 2.2.5

Solution (c). The graphs of $f(x) = \sqrt{x}$ and $f'(x) = 1/(2\sqrt{x})$ are shown in Figure 2.2.5. Observe that $f'(x) > 0$ if $x > 0$, which means that all tangent lines to the graph of $y = \sqrt{x}$ have positive slope at all points in this interval. Since

$$\lim_{x \rightarrow 0^+} \frac{1}{2\sqrt{x}} = +\infty \quad \text{and} \quad \lim_{x \rightarrow +\infty} \frac{1}{2\sqrt{x}} = 0$$

the graph of f becomes more and more vertical as $x \rightarrow 0^+$ and more and more horizontal as $x \rightarrow +\infty$. ◀

■ COMPUTING INSTANTANEOUS VELOCITY

It follows from Formula (5) of Section 2.1 (with t replacing t_0) that if $s = f(t)$ is the position function of a particle in rectilinear motion, then the instantaneous velocity at an arbitrary time t is given by

$$v_{\text{inst}} = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}$$

Since the right side of this equation is the derivative of the function f (with t rather than x as the independent variable), it follows that if $f(t)$ is the position function of a particle in rectilinear motion, then the function

$$v(t) = f'(t) = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h} \quad (4)$$

represents the instantaneous velocity of the particle at time t . Accordingly, we call (4) the *instantaneous velocity function* or, more simply, the *velocity function* of the particle.

► **Example 5** Recall the particle from Example 5 of Section 2.1 with position function $s = f(t) = 1 + 5t - 2t^2$. Here $f(t)$ is measured in meters and t is measured in seconds. Find the velocity function of the particle.

Solution. It follows from (4) that the velocity function is

$$\begin{aligned} v(t) &= \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h} = \lim_{h \rightarrow 0} \frac{[1 + 5(t+h) - 2(t+h)^2] - [1 + 5t - 2t^2]}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2[t^2 + 2th + h^2 - t^2] + 5h}{h} = \lim_{h \rightarrow 0} \frac{-4th - 2h^2 + 5h}{h} \\ &= \lim_{h \rightarrow 0} (-4t - 2h + 5) = 5 - 4t \end{aligned}$$

where the units of velocity are meters per second. ◀

■ DIFFERENTIABILITY

It is possible that the limit that defines the derivative of a function f may not exist at certain points in the domain of f . At such points the derivative is undefined. To account for this possibility we make the following definition.

2.2.2 DEFINITION A function f is said to be *differentiable at x_0* if the limit

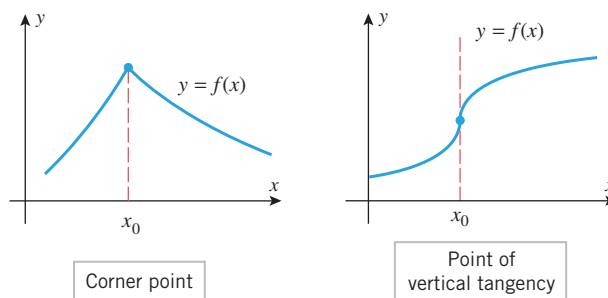
$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \quad (5)$$

exists. If f is differentiable at each point of the open interval (a, b) , then we say that it is *differentiable on (a, b)* , and similarly for open intervals of the form $(a, +\infty)$, $(-\infty, b)$, and $(-\infty, +\infty)$. In the last case we say that f is *differentiable everywhere*.

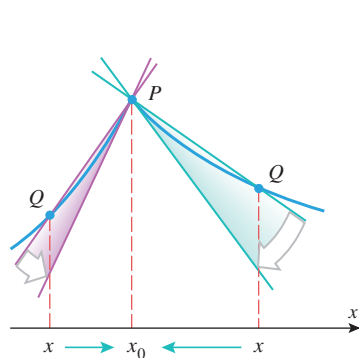
Geometrically, a function f is differentiable at x_0 if the graph of f has a tangent line at x_0 . Thus, f is not differentiable at any point x_0 where the secant lines from $P(x_0, f(x_0))$ to points $Q(x, f(x))$ distinct from P do not approach a unique *nonvertical* limiting position as $x \rightarrow x_0$. Figure 2.2.6 illustrates two common ways in which a function that is continuous at x_0 can fail to be differentiable at x_0 . These can be described informally as

- corner points
- points of vertical tangency

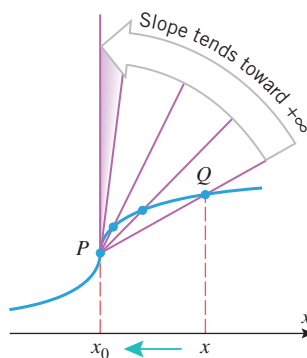
At a corner point, the slopes of the secant lines have different limits from the left and from the right, and hence the *two-sided* limit that defines the derivative does not exist (Figure 2.2.7). At a point of vertical tangency the slopes of the secant lines approach $+\infty$ or $-\infty$ from the left and from the right (Figure 2.2.8), so again the limit that defines the derivative does not exist.



► Figure 2.2.6



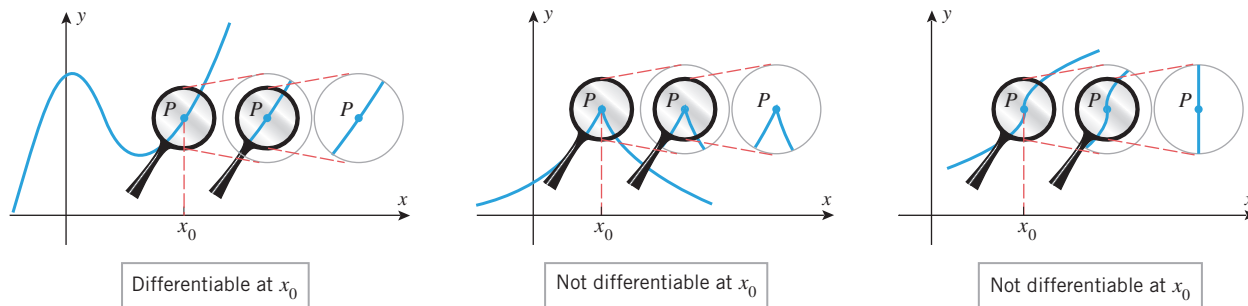
▲ Figure 2.2.7



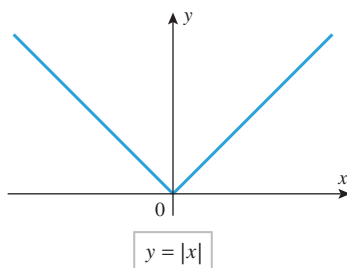
▲ Figure 2.2.8

There are other less obvious circumstances under which a function may fail to be differentiable. (See Exercise 49, for example.)

Differentiability at x_0 can also be described informally in terms of the behavior of the graph of f under increasingly stronger magnification at the point $P(x_0, f(x_0))$ (Figure 2.2.9). If f is differentiable at x_0 , then under sufficiently strong magnification at P the



▲ Figure 2.2.9



▲ Figure 2.2.10

► **Example 6** The graph of $y = |x|$ in Figure 2.2.10 has a corner at $x = 0$, which implies that $f(x) = |x|$ is not differentiable at $x = 0$.

- (a) Prove that $f(x) = |x|$ is not differentiable at $x = 0$ by showing that the limit in Definition 2.2.2 does not exist at $x = 0$.
 (b) Find a formula for $f'(x)$.

Solution (a). From Formula (5) with $x_0 = 0$, the value of $f'(0)$, if it were to exist, would be given by

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{|h| - |0|}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h} \quad (6)$$

But

$$\frac{|h|}{h} = \begin{cases} 1, & h > 0 \\ -1, & h < 0 \end{cases}$$

so that

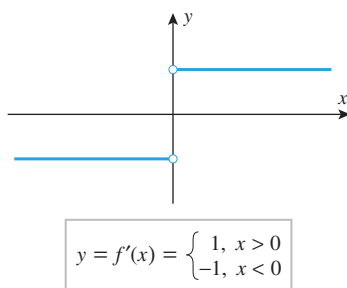
$$\lim_{h \rightarrow 0^-} \frac{|h|}{h} = -1 \quad \text{and} \quad \lim_{h \rightarrow 0^+} \frac{|h|}{h} = 1$$

Since these one-sided limits are not equal, the two-sided limit in (5) does not exist, and hence f is not differentiable at $x = 0$.

Solution (b). A formula for the derivative of $f(x) = |x|$ can be obtained by writing $|x|$ in piecewise form and treating the cases $x > 0$ and $x < 0$ separately. If $x > 0$, then $f(x) = x$ and $f'(x) = 1$; if $x < 0$, then $f(x) = -x$ and $f'(x) = -1$. Thus,

$$f'(x) = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

The graph of f' is shown in Figure 2.2.11. Observe that f' is not continuous at $x = 0$, so this example shows that a function that is continuous everywhere may have a derivative that fails to be continuous everywhere. ◀



▲ Figure 2.2.11

A theorem that says “If statement A is true, then statement B is true” is equivalent to the theorem that says “If statement B is not true, then statement A is not true.” The two theorems are called **contrapositive forms** of one another. Thus, Theorem 2.2.3 can be rewritten in contrapositive form as “If a function f is not continuous at x_0 , then f is not differentiable at x_0 .”

THE RELATIONSHIP BETWEEN DIFFERENTIABILITY AND CONTINUITY

We already know that functions are not differentiable at corner points and points of vertical tangency. The next theorem shows that functions are not differentiable at points of discontinuity. We will do this by proving that if f is differentiable at a point, then it must be continuous at that point.

2.2.3 THEOREM If a function f is differentiable at x_0 , then f is continuous at x_0 .

PROOF We are given that f is differentiable at x_0 , so it follows from (5) that $f'(x_0)$ exists and is given by

$$f'(x_0) = \lim_{h \rightarrow 0} \left[\frac{f(x_0 + h) - f(x_0)}{h} \right] \quad (7)$$

To show that f is continuous at x_0 , we must show that $\lim_{x \rightarrow x_0} f(x) = f(x_0)$ or, equivalently,

$$\lim_{x \rightarrow x_0} [f(x) - f(x_0)] = 0$$

Expressing this in terms of the variable $h = x - x_0$, we must prove that

$$\lim_{h \rightarrow 0} [f(x_0 + h) - f(x_0)] = 0$$

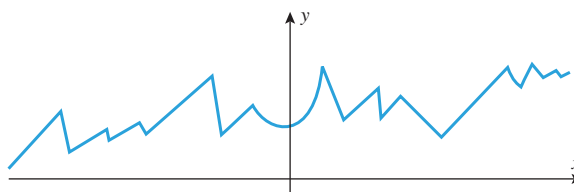
However, this can be proved using (7) as follows:

$$\begin{aligned} \lim_{h \rightarrow 0} [f(x_0 + h) - f(x_0)] &= \lim_{h \rightarrow 0} \left[\frac{f(x_0 + h) - f(x_0)}{h} \cdot h \right] \\ &= \lim_{h \rightarrow 0} \left[\frac{f(x_0 + h) - f(x_0)}{h} \right] \cdot \lim_{h \rightarrow 0} h \\ &= f'(x_0) \cdot 0 = 0 \quad \blacksquare \end{aligned}$$

WARNING

The converse of Theorem 2.2.3 is false; that is, *a function may be continuous at a point but not differentiable at that point*. This occurs, for example, at corner points of continuous functions. For instance, $f(x) = |x|$ is continuous at $x = 0$ but not differentiable there (Example 6).

The relationship between continuity and differentiability was of great historical significance in the development of calculus. In the early nineteenth century mathematicians believed that if a continuous function had many points of nondifferentiability, these points, like the tips of a sawblade, would have to be separated from one another and joined by smooth curve segments (Figure 2.2.12). This misconception was corrected by a series of discoveries beginning in 1834. In that year a Bohemian priest, philosopher, and mathematician named Bernhard Bolzano discovered a procedure for constructing a continuous function that is not differentiable at any point. Later, in 1860, the great German mathematician Karl Weierstrass (biography on p. 82) produced the first formula for such a function. The graphs of such functions are impossible to draw; it is as if the corners are so numerous that any segment of the curve, when suitably enlarged, reveals more corners. The discovery of these functions was important in that it made mathematicians distrustful of their geometric intuition and more reliant on precise mathematical proof. Recently, such functions have started to play a fundamental role in the study of geometric objects called *fractals*. Fractals have revealed an order to natural phenomena that were previously dismissed as random and chaotic.



► Figure 2.2.12



Bernhard Bolzano (1781–1848) Bolzano, the son of an art dealer, was born in Prague, Bohemia (Czech Republic). He was educated at the University of Prague, and eventually won enough mathematical fame to be recommended for a mathematics chair there. However, Bolzano became an ordained Roman Catholic priest, and in 1805

he was appointed to a chair of Philosophy at the University of Prague. Bolzano was a man of great human compassion; he spoke out for educational reform, he voiced the right of individual conscience over government demands, and he lectured on the absurdity

of war and militarism. His views so disenchanted Emperor Franz I of Austria that the emperor pressed the Archbishop of Prague to have Bolzano recant his statements. Bolzano refused and was then forced to retire in 1824 on a small pension. Bolzano's main contribution to mathematics was philosophical. His work helped convince mathematicians that sound mathematics must ultimately rest on rigorous proof rather than intuition. In addition to his work in mathematics, Bolzano investigated problems concerning space, force, and wave propagation.

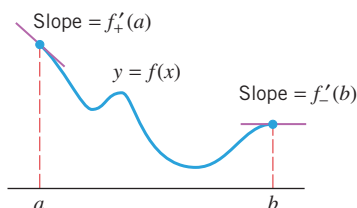
[Image: http://en.wikipedia.org/wiki/File:Bernard_Bolzano.jpg]

DERIVATIVES AT THE ENDPOINTS OF AN INTERVAL

If a function f is defined on a closed interval $[a, b]$ but not outside that interval, then f' is not defined at the endpoints of the interval because derivatives are two-sided limits. To deal with this we define **left-hand derivatives** and **right-hand derivatives** by

$$f'_-(x) = \lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h} \quad \text{and} \quad f'_+(x) = \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h}$$

respectively. These are called **one-sided derivatives**. Geometrically, $f'_-(x)$ is the limit of the slopes of the secant lines as x is approached from the left and $f'_+(x)$ is the limit of the slopes of the secant lines as x is approached from the right. For a closed interval $[a, b]$, we will understand the derivative at the left endpoint to be $f'_+(a)$ and at the right endpoint to be $f'_-(b)$ (Figure 2.2.13).



▲ Figure 2.2.13

In general, we will say that f is **differentiable** on an interval of the form $[a, b]$, $[a, +\infty)$, $(-\infty, b]$, $[a, b)$, or $(a, b]$ if it is differentiable at all points inside the interval and the appropriate one-sided derivative exists at each included endpoint.

It can be proved that a function f is continuous from the left at those points where the left-hand derivative exists and is continuous from the right at those points where the right-hand derivative exists.

OTHER DERIVATIVE NOTATIONS

The process of finding a derivative is called **differentiation**. You can think of differentiation as an *operation* on functions that associates a function f' with a function f . When the independent variable is x , the differentiation operation is also commonly denoted by

$$f'(x) = \frac{d}{dx}[f(x)] \quad \text{or} \quad f'(x) = D_x[f(x)]$$

In the case where there is a dependent variable $y = f(x)$, the derivative is also commonly denoted by

$$f'(x) = y'(x) \quad \text{or} \quad f'(x) = \frac{dy}{dx}$$

With the above notations, the value of the derivative at a point x_0 can be expressed as

$$f'(x_0) = \frac{d}{dx}[f(x)] \Big|_{x=x_0}, \quad f'(x_0) = D_x[f(x)] \Big|_{x=x_0}, \quad f'(x_0) = y'(x_0), \quad f'(x_0) = \frac{dy}{dx} \Big|_{x=x_0}$$

If a variable w changes from some initial value w_0 to some final value w_1 , then the final value minus the initial value is called an **increment** in w and is denoted by

$$\Delta w = w_1 - w_0 \quad (8)$$

Increments can be positive or negative, depending on whether the final value is larger or smaller than the initial value. The increment symbol in (8) should not be interpreted as a product; rather, Δw should be regarded as a single symbol representing the change in the value of w .

It is common to regard the variable h in the derivative formula

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (9)$$

as an increment Δx in x and write (9) as

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \quad (10)$$

Later, the symbols dy and dx will be given specific meanings. However, for the time being do not regard dy/dx as a ratio, but rather as a single symbol denoting the derivative.

Moreover, if $y = f(x)$, then the numerator in (10) can be regarded as the increment

$$\Delta y = f(x + \Delta x) - f(x) \quad (11)$$

in which case

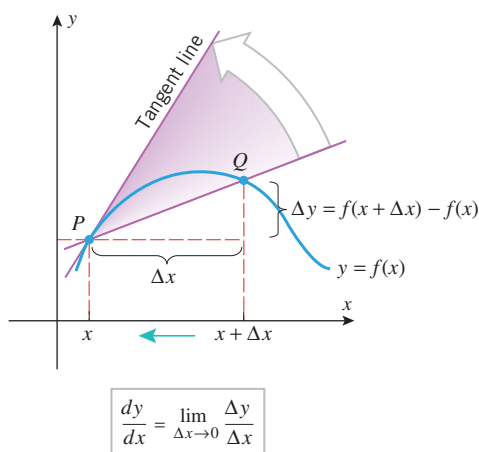
$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \quad (12)$$

The geometric interpretations of Δx and Δy are shown in Figure 2.2.14.

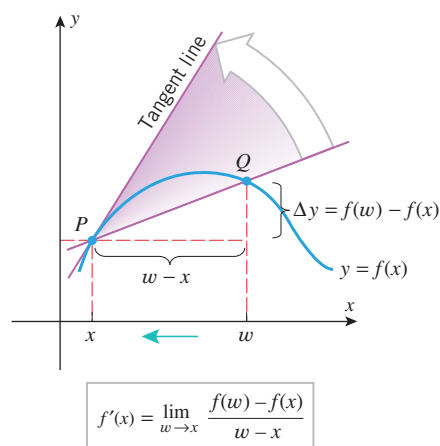
Sometimes it is desirable to express derivatives in a form that does not use increments at all. For example, if we let $w = x + h$ in Formula (9), then $w \rightarrow x$ as $h \rightarrow 0$, so we can rewrite that formula as

$$f'(x) = \lim_{w \rightarrow x} \frac{f(w) - f(x)}{w - x} \quad (13)$$

(Compare Figures 2.2.14 and 2.2.15.)



▲ Figure 2.2.14



▲ Figure 2.2.15

When letters other than x and y are used for the independent and dependent variables, the derivative notations must be adjusted accordingly. Thus, for example, if $s = f(t)$ is the position function for a particle in rectilinear motion, then the velocity function $v(t)$ in (4) can be expressed as

$$v(t) = \frac{ds}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t} \quad (14)$$

✓ QUICK CHECK EXERCISES 2.2 (See page 134 for answers.)

1. The function $f'(x)$ is defined by the formula

$$f'(x) = \lim_{h \rightarrow 0} \underline{\hspace{2cm}}$$

2. (a) The derivative of $f(x) = x^2$ is $f'(x) = \underline{\hspace{2cm}}$.
 (b) The derivative of $f(x) = \sqrt{x}$ is $f'(x) = \underline{\hspace{2cm}}$.

3. Suppose that the line $2x + 3y = 5$ is tangent to the graph of $y = f(x)$ at $x = 1$. The value of $f(1)$ is $\underline{\hspace{2cm}}$ and the value of $f'(1)$ is $\underline{\hspace{2cm}}$.

4. Which theorem guarantees us that if

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

exists, then $\lim_{x \rightarrow x_0} f(x) = f(x_0)$?

EXERCISE SET 2.2



Graphing Utility

1. Use the graph of $y = f(x)$ in the accompanying figure to estimate the value of $f'(1)$, $f'(3)$, $f'(5)$, and $f'(6)$.

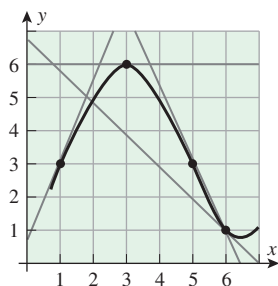


Figure Ex-1

2. For the function graphed in the accompanying figure, arrange the numbers 0, $f'(-3)$, $f'(0)$, $f'(2)$, and $f'(4)$ in increasing order.

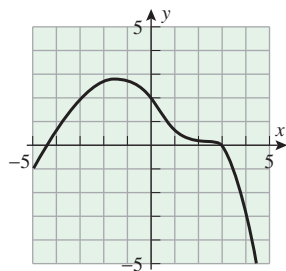


Figure Ex-2

FOCUS ON CONCEPTS

3. (a) If you are given an equation for the tangent line at the point $(a, f(a))$ on a curve $y = f(x)$, how would you go about finding $f'(a)$?
- (b) Given that the tangent line to the graph of $y = f(x)$ at the point $(2, 5)$ has the equation $y = 3x - 1$, find $f'(2)$.
- (c) For the function $y = f(x)$ in part (b), what is the instantaneous rate of change of y with respect to x at $x = 2$?
4. Given that the tangent line to $y = f(x)$ at the point $(1, 2)$ passes through the point $(-1, -1)$, find $f'(1)$.
5. Sketch the graph of a function f for which $f(0) = -1$, $f'(0) = 0$, $f'(x) < 0$ if $x < 0$, and $f'(x) > 0$ if $x > 0$.
6. Sketch the graph of a function f for which $f(0) = 0$, $f'(0) = 0$, and $f'(x) > 0$ if $x < 0$ or $x > 0$.
7. Given that $f(3) = -1$ and $f'(3) = 5$, find an equation for the tangent line to the graph of $y = f(x)$ at $x = 3$.
8. Given that $f(-2) = 3$ and $f'(-2) = -4$, find an equation for the tangent line to the graph of $y = f(x)$ at $x = -2$.

- 9–14 Use Definition 2.2.1 to find $f'(x)$, and then find the tangent line to the graph of $y = f(x)$ at $x = a$. ■

9. $f(x) = 2x^2$; $a = 1$
10. $f(x) = 1/x^2$; $a = -1$
11. $f(x) = x^3$; $a = 0$
12. $f(x) = 2x^3 + 1$; $a = -1$
13. $f(x) = \sqrt{x+1}$; $a = 8$
14. $f(x) = \sqrt{2x+1}$; $a = 4$

- 15–20 Use Formula (12) to find dy/dx . ■

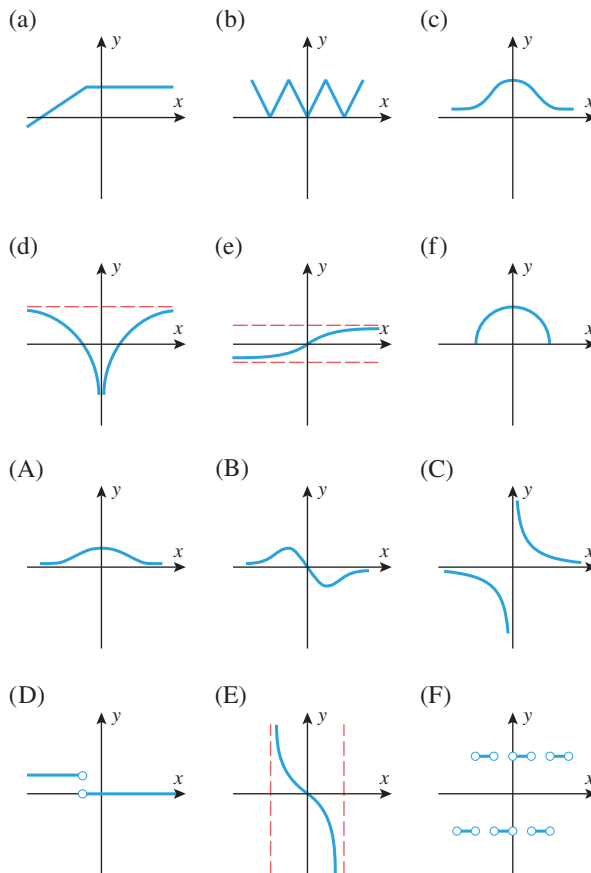
15. $y = \frac{1}{x}$
16. $y = \frac{1}{x+1}$
17. $y = x^2 - x$
18. $y = x^4$
19. $y = \frac{1}{\sqrt{x}}$
20. $y = \frac{1}{\sqrt{x-1}}$

- 21–22 Use Definition 2.2.1 (with appropriate change in notation) to obtain the derivative requested. ■

21. Find $f'(t)$ if $f(t) = 4t^2 + t$.
22. Find dV/dr if $V = \frac{4}{3}\pi r^3$.

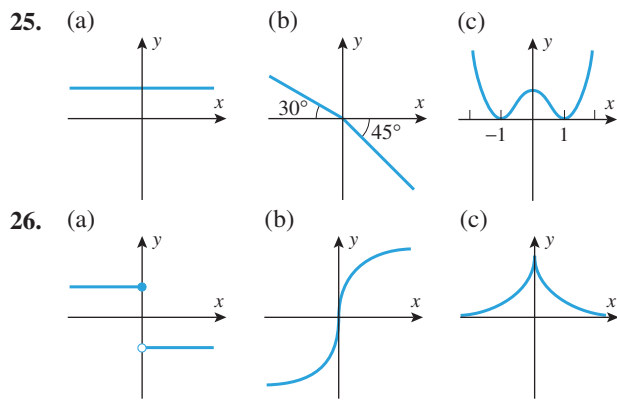
FOCUS ON CONCEPTS

23. Match the graphs of the functions shown in (a)–(f) with the graphs of their derivatives in (A)–(F).



24. Let $f(x) = \sqrt{1-x^2}$. Use a geometric argument to find $f'(\sqrt{2}/2)$.

25–26 Sketch the graph of the derivative of the function whose graph is shown. ■



27–30 True–False Determine whether the statement is true or false. Explain your answer. ■

27. If a curve $y = f(x)$ has a horizontal tangent line at $x = a$, then $f'(a)$ is not defined.
28. If the tangent line to the graph of $y = f(x)$ at $x = -2$ has negative slope, then $f'(-2) < 0$.
29. If a function f is continuous at $x = 0$, then f is differentiable at $x = 0$.
30. If a function f is differentiable at $x = 0$, then f is continuous at $x = 0$.

31–32 The given limit represents $f'(a)$ for some function f and some number a . Find $f(x)$ and a in each case. ■

31. (a) $\lim_{\Delta x \rightarrow 0} \frac{\sqrt{1+\Delta x} - 1}{\Delta x}$ (b) $\lim_{x_1 \rightarrow 3} \frac{x_1^2 - 9}{x_1 - 3}$

32. (a) $\lim_{h \rightarrow 0} \frac{\cos(\pi + h) + 1}{h}$ (b) $\lim_{x \rightarrow 1} \frac{x^7 - 1}{x - 1}$

33. Find $dy/dx|_{x=1}$, given that $y = 1 - x^2$.

34. Find $dy/dx|_{x=-2}$, given that $y = (x + 2)/x$.

 35. Find an equation for the line that is tangent to the curve $y = x^3 - 2x + 1$ at the point $(0, 1)$, and use a graphing utility to graph the curve and its tangent line on the same screen.

 36. Use a graphing utility to graph the following on the same screen: the curve $y = x^2/4$, the tangent line to this curve at $x = 1$, and the secant line joining the points $(0, 0)$ and $(2, 1)$ on this curve.

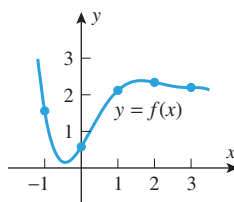
 37. Let $f(x) = 2^x$. Estimate $f'(1)$ by

(a) using a graphing utility to zoom in at an appropriate point until the graph looks like a straight line, and then estimating the slope

(b) using a calculating utility to estimate the limit in Formula (13) by making a table of values for a succession of values of w approaching 1.

-  38. Let $f(x) = \sin x$. Estimate $f'(\pi/4)$ by
- (a) using a graphing utility to zoom in at an appropriate point until the graph looks like a straight line, and then estimating the slope
- (b) using a calculating utility to estimate the limit in Formula (13) by making a table of values for a succession of values of w approaching $\pi/4$.

39–40 The function f whose graph is shown below has values as given in the accompanying table.



x	-1	0	1	2	3
$f(x)$	1.56	0.58	2.12	2.34	2.2

39. (a) Use data from the table to calculate the difference quotients

$$\frac{f(3) - f(1)}{3 - 1}, \quad \frac{f(2) - f(1)}{2 - 1}, \quad \frac{f(2) - f(0)}{2 - 0}$$

(b) Using the graph of $y = f(x)$, indicate which difference quotient in part (a) best approximates $f'(1)$ and which difference quotient gives the worst approximation to $f'(1)$.

40. Use data from the table to approximate the derivative values.

(a) $f'(0.5)$ (b) $f'(2.5)$

FOCUS ON CONCEPTS

41. Suppose that the cost of drilling x feet for an oil well is $C = f(x)$ dollars.

- (a) What are the units of $f'(x)$?
- (b) In practical terms, what does $f'(x)$ mean in this case?
- (c) What can you say about the sign of $f'(x)$?
- (d) Estimate the cost of drilling an additional foot, starting at a depth of 300 ft, given that $f'(300) = 1000$.

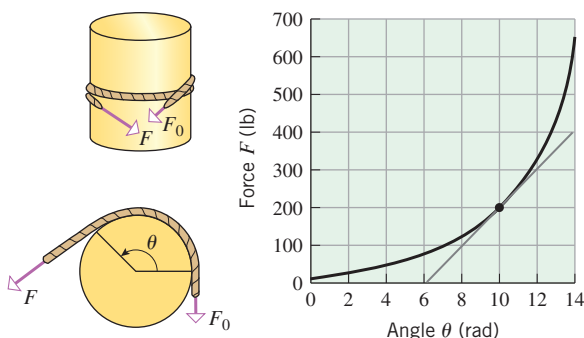
42. A paint manufacturing company estimates that it can sell $g = f(p)$ gallons of paint at a price of p dollars per gallon.

- (a) What are the units of dg/dp ?
- (b) In practical terms, what does dg/dp mean in this case?
- (c) What can you say about the sign of dg/dp ?
- (d) Given that $dg/dp|_{p=10} = -100$, what can you say about the effect of increasing the price from \$10 per gallon to \$11 per gallon?

43. It is a fact that when a flexible rope is wrapped around a rough cylinder, a small force of magnitude F_0 at one end can resist a large force of magnitude F at the other end. The size of F depends on the angle θ through which the rope is wrapped around the cylinder (see the

accompanying figure). The figure shows the graph of F (in pounds) versus θ (in radians), where F is the magnitude of the force that can be resisted by a force with magnitude $F_0 = 10$ lb for a certain rope and cylinder.

- (a) Estimate the values of F and $dF/d\theta$ when the angle $\theta = 10$ radians.
- (b) It can be shown that the force F satisfies the equation $dF/d\theta = \mu F$, where the constant μ is called the **coefficient of friction**. Use the results in part (a) to estimate the value of μ .

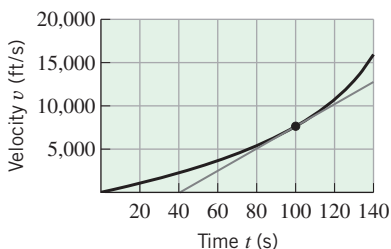


▲ Figure Ex-43

44. The accompanying figure shows the velocity versus time curve for a rocket in outer space where the only significant force on the rocket is from its engines. It can be shown that the mass $M(t)$ (in slugs) of the rocket at time t seconds satisfies the equation

$$M(t) = \frac{T}{dv/dt}$$

where T is the thrust (in lb) of the rocket's engines and v is the velocity (in ft/s) of the rocket. The thrust of the first stage of a *Saturn V* rocket is $T = 7,680,982$ lb. Use this value of T and the line segment in the figure to estimate the mass of the rocket at time $t = 100$.



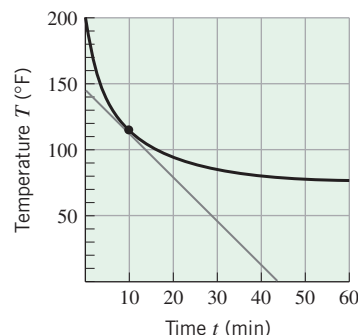
▲ Figure Ex-44

45. According to **Newton's Law of Cooling**, the rate of change of an object's temperature is proportional to the difference between the temperature of the object and that of the surrounding medium. The accompanying figure shows the graph of the temperature T (in degrees Fahrenheit) versus time t (in minutes) for a cup of coffee, initially with a temperature of 200°F , that is allowed to cool in a room with a constant temperature of 75°F .
- (a) Estimate T and dT/dt when $t = 10$ min.

- (b) Newton's Law of Cooling can be expressed as

$$\frac{dT}{dt} = k(T - T_0)$$

where k is the constant of proportionality and T_0 is the temperature (assumed constant) of the surrounding medium. Use the results in part (a) to estimate the value of k .



▲ Figure Ex-45

46. Show that $f(x)$ is continuous but not differentiable at the indicated point. Sketch the graph of f .

(a) $f(x) = \sqrt[3]{x}$, $x = 0$

(b) $f(x) = \sqrt[3]{(x-2)^2}$, $x = 2$

47. Show that

$$f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ 2x, & x > 1 \end{cases}$$

is continuous and differentiable at $x = 1$. Sketch the graph of f .

48. Show that

$$f(x) = \begin{cases} x^2 + 2, & x \leq 1 \\ x + 2, & x > 1 \end{cases}$$

is continuous but not differentiable at $x = 1$. Sketch the graph of f .

49. Show that

$$f(x) = \begin{cases} x \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is continuous but not differentiable at $x = 0$. Sketch the graph of f near $x = 0$. (See Figure 1.6.6 and the remark following Example 3 in Section 1.6.)

50. Show that

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is continuous and differentiable at $x = 0$. Sketch the graph of f near $x = 0$.

FOCUS ON CONCEPTS

51. Suppose that a function f is differentiable at x_0 and that $f'(x_0) > 0$. Prove that there exists an open interval containing x_0 such that if x_1 and x_2 are any two points in this interval with $x_1 < x_0 < x_2$, then $f(x_1) < f(x_0) < f(x_2)$.

52. Suppose that a function f is differentiable at x_0 and define $g(x) = f(mx + b)$, where m and b are constants. Prove that if x_1 is a point at which $mx_1 + b = x_0$, then $g(x)$ is differentiable at x_1 and $g'(x_1) = mf'(x_0)$.
53. Suppose that a function f is differentiable at $x = 0$ with $f(0) = f'(0) = 0$, and let $y = mx$, $m \neq 0$, denote any line of nonzero slope through the origin.
- Prove that there exists an open interval containing 0 such that for all nonzero x in this interval $|f(x)| < |\frac{1}{2}mx|$. [Hint: Let $\epsilon = \frac{1}{2}|m|$ and apply Definition 1.4.1 to (5) with $x_0 = 0$.]
 - Conclude from part (a) and the triangle inequality that there exists an open interval containing 0 such that $|f(x)| < |f(x) - mx|$ for all x in this interval.
 - Explain why the result obtained in part (b) may be interpreted to mean that the tangent line to the graph

of f at the origin is the best linear approximation to f at that point.

54. Suppose that f is differentiable at x_0 . Modify the argument of Exercise 53 to prove that the tangent line to the graph of f at the point $P(x_0, f(x_0))$ provides the best linear approximation to f at P . [Hint: Suppose that $y = f(x_0) + m(x - x_0)$ is any line through $P(x_0, f(x_0))$ with slope $m \neq f'(x_0)$. Apply Definition 1.4.1 to (5) with $x = x_0 + h$ and $\epsilon = \frac{1}{2}|f'(x_0) - m|$.]

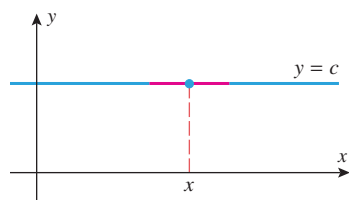
55. **Writing** Write a paragraph that explains what it means for a function to be differentiable. Include examples of functions that are not differentiable as well as examples of functions that are differentiable.
56. **Writing** Explain the relationship between continuity and differentiability.

✓ QUICK CHECK ANSWERS 2.2

- $\frac{f(x+h) - f(x)}{h}$
- (a) $2x$ (b) $\frac{1}{2\sqrt{x}}$
- 1; $-\frac{2}{3}$
- Theorem 2.2.3: If f is differentiable at x_0 , then f is continuous at x_0 .

2.3 INTRODUCTION TO TECHNIQUES OF DIFFERENTIATION

In the last section we defined the derivative of a function f as a limit, and we used that limit to calculate a few simple derivatives. In this section we will develop some important theorems that will enable us to calculate derivatives more efficiently.



The tangent line to the graph of $f(x) = c$ has slope 0 for all x .

▲ Figure 2.3.1

■ DERIVATIVE OF A CONSTANT

The simplest kind of function is a constant function $f(x) = c$. Since the graph of f is a horizontal line of slope 0, the tangent line to the graph of f has slope 0 for every x ; and hence we can see geometrically that $f'(x) = 0$ (Figure 2.3.1). We can also see this algebraically since

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} 0 = 0$$

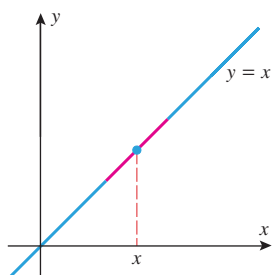
Thus, we have established the following result.

2.3.1 THEOREM The derivative of a constant function is 0; that is, if c is any real number, then

$$\frac{d}{dx}[c] = 0 \quad (1)$$

► Example 1

$$\frac{d}{dx}[1] = 0, \quad \frac{d}{dx}[-3] = 0, \quad \frac{d}{dx}[\pi] = 0, \quad \frac{d}{dx}[-\sqrt{2}] = 0 \quad \blacktriangleleft$$



The tangent line to the graph of $f(x) = x$ has slope 1 for all x .

▲ Figure 2.3.2

DERIVATIVES OF POWER FUNCTIONS

The simplest power function is $f(x) = x$. Since the graph of f is a line of slope 1, it follows from Example 3 of Section 2.2 that $f'(x) = 1$ for all x (Figure 2.3.2). In other words,

$$\frac{d}{dx}[x] = 1 \quad (2)$$

Example 1 of Section 2.2 shows that the power function $f(x) = x^2$ has derivative $f'(x) = 2x$. From Example 2 in that section one can infer that the power function $f(x) = x^3$ has derivative $f'(x) = 3x^2$. That is,

$$\frac{d}{dx}[x^2] = 2x \quad \text{and} \quad \frac{d}{dx}[x^3] = 3x^2 \quad (3-4)$$

These results are special cases of the following more general result.

2.3.2 THEOREM (The Power Rule) If n is a positive integer, then

$$\frac{d}{dx}[x^n] = nx^{n-1} \quad (5)$$

Verify that Formulas (2), (3), and (4) are the special cases of (5) in which $n = 1, 2$, and 3 .

The binomial formula can be found on the front endpaper of the text. Replacing y by h in this formula yields the identity used in the proof of Theorem 2.3.2.

PROOF Let $f(x) = x^n$. Thus, from the definition of a derivative and the binomial formula for expanding the expression $(x + h)^n$, we obtain

$$\begin{aligned} \frac{d}{dx}[x^n] &= f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{\left[x^n + nx^{n-1}h + \frac{n(n-1)}{2!}x^{n-2}h^2 + \cdots + nxh^{n-1} + h^n \right] - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{nx^{n-1}h + \frac{n(n-1)}{2!}x^{n-2}h^2 + \cdots + nxh^{n-1} + h^n}{h} \\ &= \lim_{h \rightarrow 0} \left[nx^{n-1} + \frac{n(n-1)}{2!}x^{n-2}h + \cdots + nxh^{n-2} + h^{n-1} \right] \\ &= nx^{n-1} + 0 + \cdots + 0 + 0 \\ &= nx^{n-1} \quad \blacksquare \end{aligned}$$

► Example 2

$$\frac{d}{dx}[x^4] = 4x^3, \quad \frac{d}{dx}[x^5] = 5x^4, \quad \frac{d}{dt}[t^{12}] = 12t^{11} \quad \blacktriangleleft$$

Although our proof of the power rule in Formula (5) applies only to *positive* integer powers of x , it is not difficult to show that the same formula holds for all integer powers of x (Exercise 82). Also, we saw in Example 4 of Section 2.2 that

$$\frac{d}{dx}[\sqrt{x}] = \frac{1}{2\sqrt{x}} \quad (6)$$

which can be expressed as

$$\frac{d}{dx}[x^{1/2}] = \frac{1}{2}x^{-1/2} = \frac{1}{2}x^{(1/2)-1}$$

Thus, Formula (5) is valid for $n = \frac{1}{2}$, as well. In fact, it can be shown that this formula holds for any real exponent. We state this more general result for our use now; the proof will follow in stages: Exercises 25–26 in Section 2.7 will show that the formula holds for rational exponents, and we will justify it for real exponents in Chapter 6.

2.3.3 THEOREM (Extended Power Rule) If r is any real number, then

$$\frac{d}{dx}[x^r] = rx^{r-1} \quad (7)$$

In words, to differentiate a power function, decrease the constant exponent by one and multiply the resulting power function by the original exponent.

► **Example 3**

$$\frac{d}{dx}[x^\pi] = \pi x^{\pi-1}$$

$$\frac{d}{dx}\left[\frac{1}{x}\right] = \frac{d}{dx}[x^{-1}] = (-1)x^{-1-1} = -x^{-2} = -\frac{1}{x^2}$$

$$\frac{d}{dw}\left[\frac{1}{w^{100}}\right] = \frac{d}{dw}[w^{-100}] = -100w^{-101} = -\frac{100}{w^{101}}$$

$$\frac{d}{dx}[x^{4/5}] = \frac{4}{5}x^{(4/5)-1} = \frac{4}{5}x^{-1/5}$$

$$\frac{d}{dx}[\sqrt[3]{x}] = \frac{d}{dx}[x^{1/3}] = \frac{1}{3}x^{-2/3} = \frac{1}{3\sqrt[3]{x^2}} \quad \blacktriangleleft$$

■ **DERIVATIVE OF A CONSTANT TIMES A FUNCTION**

Formula (8) can also be expressed in function notation as

$$(cf)' = cf'$$

2.3.4 THEOREM (Constant Multiple Rule) If f is differentiable at x and c is any real number, then cf is also differentiable at x and

$$\frac{d}{dx}[cf(x)] = c \frac{d}{dx}[f(x)] \quad (8)$$

PROOF

$$\begin{aligned} \frac{d}{dx}[cf(x)] &= \lim_{h \rightarrow 0} \frac{cf(x+h) - cf(x)}{h} \\ &= \lim_{h \rightarrow 0} c \left[\frac{f(x+h) - f(x)}{h} \right] \\ &= c \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= c \frac{d}{dx}[f(x)] \quad \blacksquare \end{aligned}$$

A constant factor can be moved through a limit sign.

In words, a constant factor can be moved through a derivative sign.

► **Example 4**

$$\frac{d}{dx}[4x^8] = 4 \frac{d}{dx}[x^8] = 4[8x^7] = 32x^7$$

$$\frac{d}{dx}[-x^{12}] = (-1) \frac{d}{dx}[x^{12}] = -12x^{11}$$

$$\frac{d}{dx}\left[\frac{\pi}{x}\right] = \pi \frac{d}{dx}[x^{-1}] = \pi(-x^{-2}) = -\frac{\pi}{x^2} \quad \blacktriangleleft$$

DERIVATIVES OF SUMS AND DIFFERENCES

Formulas (9) and (10) can also be expressed as

$$(f + g)' = f' + g'$$

$$(f - g)' = f' - g'$$

2.3.5 THEOREM (Sum and Difference Rules) If f and g are differentiable at x , then so are $f + g$ and $f - g$ and

$$\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}[f(x)] + \frac{d}{dx}[g(x)] \quad (9)$$

$$\frac{d}{dx}[f(x) - g(x)] = \frac{d}{dx}[f(x)] - \frac{d}{dx}[g(x)] \quad (10)$$

PROOF Formula (9) can be proved as follows:

$$\begin{aligned} \frac{d}{dx}[f(x) + g(x)] &= \lim_{h \rightarrow 0} \frac{[f(x+h) + g(x+h)] - [f(x) + g(x)]}{h} \\ &= \lim_{h \rightarrow 0} \frac{[f(x+h) - f(x)] + [g(x+h) - g(x)]}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= \frac{d}{dx}[f(x)] + \frac{d}{dx}[g(x)] \end{aligned}$$

The limit of a sum is the sum of the limits.

Formula (10) can be proved in a similar manner or, alternatively, by writing $f(x) - g(x)$ as $f(x) + (-1)g(x)$ and then applying Formulas (8) and (9). ■

In words, the derivative of a sum equals the sum of the derivatives, and the derivative of a difference equals the difference of the derivatives.

► **Example 5**

$$\frac{d}{dx}[2x^6 + x^{-9}] = \frac{d}{dx}[2x^6] + \frac{d}{dx}[x^{-9}] = 12x^5 + (-9)x^{-10} = 12x^5 - 9x^{-10}$$

$$\begin{aligned} \frac{d}{dx}\left[\frac{\sqrt{x} - 2x}{\sqrt{x}}\right] &= \frac{d}{dx}[1 - 2\sqrt{x}] \\ &= \frac{d}{dx}[1] - \frac{d}{dx}[2\sqrt{x}] = 0 - 2\left(\frac{1}{2\sqrt{x}}\right) = -\frac{1}{\sqrt{x}} \end{aligned}$$

See Formula (6). ◀

Although Formulas (9) and (10) are stated for sums and differences of two functions, they can be extended to any finite number of functions. For example, by grouping and applying Formula (9) twice we obtain

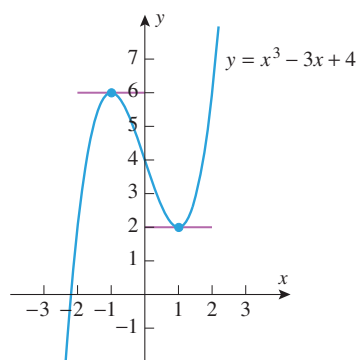
$$(f + g + h)' = [(f + g) + h]' = (f + g)' + h' = f' + g' + h'$$

As illustrated in the following example, the constant multiple rule together with the extended versions of the sum and difference rules can be used to differentiate any polynomial.

► **Example 6** Find dy/dx if $y = 3x^8 - 2x^5 + 6x + 1$.

Solution.

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}[3x^8 - 2x^5 + 6x + 1] \\ &= \frac{d}{dx}[3x^8] - \frac{d}{dx}[2x^5] + \frac{d}{dx}[6x] + \frac{d}{dx}[1] \\ &= 24x^7 - 10x^4 + 6 \quad \blacktriangleleft\end{aligned}$$



▲ Figure 2.3.3

► **Example 7** At what points, if any, does the graph of $y = x^3 - 3x + 4$ have a horizontal tangent line?

Solution. Horizontal tangent lines have slope zero, so we must find those values of x for which $y'(x) = 0$. Differentiating yields

$$y'(x) = \frac{d}{dx}[x^3 - 3x + 4] = 3x^2 - 3$$

Thus, horizontal tangent lines occur at those values of x for which $3x^2 - 3 = 0$, that is, if $x = -1$ or $x = 1$. The corresponding points on the curve $y = x^3 - 3x + 4$ are $(-1, 6)$ and $(1, 2)$ (see Figure 2.3.3). ◀

► **Example 8** Find the area of the triangle formed from the coordinate axes and the tangent line to the curve $y = 5x^{-1} - \frac{1}{5}x$ at the point $(5, 0)$.

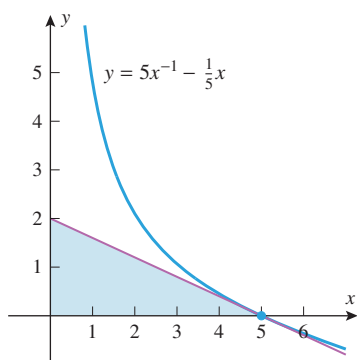
Solution. Since the derivative of y with respect to x is

$$y'(x) = \frac{d}{dx}\left[5x^{-1} - \frac{1}{5}x\right] = \frac{d}{dx}[5x^{-1}] - \frac{d}{dx}\left[\frac{1}{5}x\right] = -5x^{-2} - \frac{1}{5}$$

the slope of the tangent line at the point $(5, 0)$ is $y'(5) = -\frac{2}{5}$. Thus, the equation of the tangent line at this point is

$$y - 0 = -\frac{2}{5}(x - 5) \quad \text{or equivalently} \quad y = -\frac{2}{5}x + 2$$

Since the y -intercept of this line is 2, the right triangle formed from the coordinate axes and the tangent line has legs of length 5 and 2, so its area is $\frac{1}{2}(5)(2) = 5$ (Figure 2.3.4). ◀



▲ Figure 2.3.4

HIGHER DERIVATIVES

The derivative f' of a function f is itself a function and hence may have a derivative of its own. If f' is differentiable, then its derivative is denoted by f'' and is called the **second derivative** of f . As long as we have differentiability, we can continue the process

of differentiating to obtain third, fourth, fifth, and even higher derivatives of f . These successive derivatives are denoted by

$$f', \quad f'' = (f')', \quad f''' = (f'')', \quad f^{(4)} = (f''')', \quad f^{(5)} = (f^{(4)})', \dots$$

If $y = f(x)$, then successive derivatives can also be denoted by

$$y', \quad y'', \quad y''', \quad y^{(4)}, \quad y^{(5)}, \dots$$

Other common notations are

$$\begin{aligned} y' &= \frac{dy}{dx} = \frac{d}{dx} [f(x)] \\ y'' &= \frac{d^2 y}{dx^2} = \frac{d}{dx} \left[\frac{d}{dx} [f(x)] \right] = \frac{d^2}{dx^2} [f(x)] \\ y''' &= \frac{d^3 y}{dx^3} = \frac{d}{dx} \left[\frac{d^2}{dx^2} [f(x)] \right] = \frac{d^3}{dx^3} [f(x)] \\ \vdots & \qquad \qquad \qquad \vdots \end{aligned}$$

These are called, in succession, the *first derivative*, the *second derivative*, the *third derivative*, and so forth. The number of times that f is differentiated is called the **order** of the derivative. A general n th order derivative can be denoted by

$$\frac{d^n y}{dx^n} = f^{(n)}(x) = \frac{d^n}{dx^n} [f(x)] \quad (11)$$

and the value of a general n th order derivative at a specific point $x = x_0$ can be denoted by

$$\left. \frac{d^n y}{dx^n} \right|_{x=x_0} = f^{(n)}(x_0) = \left. \frac{d^n}{dx^n} [f(x)] \right|_{x=x_0} \quad (12)$$

► **Example 9** If $f(x) = 3x^4 - 2x^3 + x^2 - 4x + 2$, then

$$f'(x) = 12x^3 - 6x^2 + 2x - 4$$

$$f''(x) = 36x^2 - 12x + 2$$

$$f'''(x) = 72x - 12$$

$$f^{(4)}(x) = 72$$

$$f^{(5)}(x) = 0$$

$$\vdots$$

$$f^{(n)}(x) = 0 \quad (n \geq 5) \quad \blacktriangleleft$$

We will discuss the significance of second derivatives and those of higher order in later sections.

✓ QUICK CHECK EXERCISES 2.3 (See page 142 for answers.)

1. In each part, determine $f'(x)$.

(a) $f(x) = \sqrt{6}$

(c) $f(x) = 6\sqrt{x}$

(b) $f(x) = \sqrt{6}x$

(d) $f(x) = \sqrt{6}x$

2. In parts (a)–(d), determine $f'(x)$.

(a) $f(x) = x^3 + 5$

(b) $f(x) = x^2(x^3 + 5)$

(c) $f(x) = \frac{x^3 + 5}{2}$

(d) $f(x) = \frac{x^3 + 5}{x^2}$

3. The slope of the tangent line to the curve $y = x^2 + 4x + 7$ at $x = 1$ is _____.

4. If $f(x) = 3x^3 - 3x^2 + x + 1$, then $f''(x) = \underline{\hspace{2cm}}$.

EXERCISE SET 2.3



Graphing Utility

1–8 Find dy/dx . ■

1. $y = 4x^7$
2. $y = -3x^{12}$
3. $y = 3x^8 + 2x + 1$
4. $y = \frac{1}{2}(x^4 + 7)$
5. $y = \pi^3$
6. $y = \sqrt{2}x + (1/\sqrt{2})$
7. $y = -\frac{1}{3}(x^7 + 2x - 9)$
8. $y = \frac{x^2 + 1}{5}$

9–16 Find $f'(x)$. ■

9. $f(x) = x^{-3} + \frac{1}{x^7}$
10. $f(x) = \sqrt{x} + \frac{1}{x}$
11. $f(x) = -3x^{-8} + 2\sqrt{x}$
12. $f(x) = 7x^{-6} - 5\sqrt{x}$
13. $f(x) = x^\pi + \frac{1}{x\sqrt{10}}$
14. $f(x) = \sqrt[3]{\frac{8}{x}}$
15. $f(x) = (3x^2 + 1)^2$
16. $f(x) = ax^3 + bx^2 + cx + d$ (a, b, c, d constant)

17–18 Find $y'(1)$. ■

17. $y = 5x^2 - 3x + 1$
18. $y = \frac{x^{3/2} + 2}{x}$

19–20 Find dx/dt . ■

19. $x = t^2 - t$
20. $x = \frac{t^2 + 1}{3t}$

21–24 Find $dy/dx|_{x=1}$. ■

21. $y = 1 + x + x^2 + x^3 + x^4 + x^5$
22. $y = \frac{1 + x + x^2 + x^3 + x^4 + x^5 + x^6}{x^3}$
23. $y = (1 - x)(1 + x)(1 + x^2)(1 + x^4)$
24. $y = x^{24} + 2x^{12} + 3x^8 + 4x^6$

25–26 Approximate $f'(1)$ by considering the difference quotient

$$\frac{f(1+h) - f(1)}{h}$$

for values of h near 0, and then find the exact value of $f'(1)$ by differentiating. ■

25. $f(x) = x^3 - 3x + 1$
26. $f(x) = \frac{1}{x^2}$

27–28 Use a graphing utility to estimate the value of $f'(1)$ by zooming in on the graph of f , and then compare your estimate to the exact value obtained by differentiating. ■

27. $f(x) = \frac{x^2 + 1}{x}$
28. $f(x) = \frac{x + 2x^{3/2}}{\sqrt{x}}$

29–32 Find the indicated derivative. ■

29. $\frac{d}{dt}[16t^2]$
30. $\frac{dC}{dr}$, where $C = 2\pi r$

31. $V'(r)$, where $V = \pi r^3$
32. $\frac{d}{d\alpha}[2\alpha^{-1} + \alpha]$

33–36 True–False Determine whether the statement is true or false. Explain your answer. ■

33. If f and g are differentiable at $x = 2$, then

$$\left. \frac{d}{dx}[f(x) - 8g(x)] \right|_{x=2} = f'(2) - 8g'(2)$$

34. If $f(x)$ is a cubic polynomial, then $f'(x)$ is a quadratic polynomial.35. If $f'(2) = 5$, then

$$\left. \frac{d}{dx}[4f(x) + x^3] \right|_{x=2} = \left. \frac{d}{dx}[4f(x) + 8] \right|_{x=2} = 4f'(2) = 20$$

36. If $f(x) = x^2(x^4 - x)$, then

$$f''(x) = \frac{d}{dx}[x^2] \cdot \frac{d}{dx}[x^4 - x] = 2x(4x^3 - 1)$$

37. A spherical balloon is being inflated.

- (a) Find a general formula for the instantaneous rate of change of the volume V with respect to the radius r , given that $V = \frac{4}{3}\pi r^3$.
- (b) Find the rate of change of V with respect to r at the instant when the radius is $r = 5$.

38. Find $\frac{d}{d\lambda} \left[\frac{\lambda\lambda_0 + \lambda^6}{2 - \lambda_0} \right]$ (λ_0 is constant).39. Find an equation of the tangent line to the graph of $y = f(x)$ at $x = -3$ if $f(-3) = 2$ and $f'(-3) = 5$.40. Find an equation of the tangent line to the graph of $y = f(x)$ at $x = 2$ if $f(2) = -2$ and $f'(2) = -1$.41–42 Find d^2y/dx^2 . ■

41. (a) $y = 7x^3 - 5x^2 + x$
- (b) $y = 12x^2 - 2x + 3$
- (c) $y = \frac{x+1}{x}$
- (d) $y = (5x^2 - 3)(7x^3 + x)$
42. (a) $y = 4x^7 - 5x^3 + 2x$
- (b) $y = 3x + 2$
- (c) $y = \frac{3x-2}{5x}$
- (d) $y = (x^3 - 5)(2x + 3)$

43–44 Find y''' . ■

43. (a) $y = x^{-5} + x^5$
- (b) $y = 1/x$
- (c) $y = ax^3 + bx + c$ (a, b, c constant)
44. (a) $y = 5x^2 - 4x + 7$
- (b) $y = 3x^{-2} + 4x^{-1} + x$
- (c) $y = ax^4 + bx^2 + c$ (a, b, c constant)

45. Find

- (a) $f'''(2)$, where $f(x) = 3x^2 - 2$
- (b) $\left. \frac{d^2y}{dx^2} \right|_{x=1}$, where $y = 6x^5 - 4x^2$
- (c) $\left. \frac{d^4}{dx^4}[x^{-3}] \right|_{x=1}$.

46. Find

(a) $y'''(0)$, where $y = 4x^4 + 2x^3 + 3$

(b) $\left. \frac{d^4 y}{dx^4} \right|_{x=1}$, where $y = \frac{6}{x^4}$.

47. Show that $y = x^3 + 3x + 1$ satisfies $y''' + xy'' - 2y' = 0$.48. Show that if $x \neq 0$, then $y = 1/x$ satisfies the equation $x^3 y'' + x^2 y' - xy = 0$.

 **49–50** Use a graphing utility to make rough estimates of the locations of all horizontal tangent lines, and then find their exact locations by differentiating. ■

49. $y = \frac{1}{3}x^3 - \frac{3}{2}x^2 + 2x$ 50. $y = \frac{x^2 + 9}{x}$

FOCUS ON CONCEPTS51. Find a function $y = ax^2 + bx + c$ whose graph has an x -intercept of 1, a y -intercept of -2 , and a tangent line with a slope of -1 at the y -intercept.52. Find k if the curve $y = x^2 + k$ is tangent to the line $y = 2x$.53. Find the x -coordinate of the point on the graph of $y = x^2$ where the tangent line is parallel to the secant line that cuts the curve at $x = -1$ and $x = 2$.54. Find the x -coordinate of the point on the graph of $y = \sqrt{x}$ where the tangent line is parallel to the secant line that cuts the curve at $x = 1$ and $x = 4$.55. Find the coordinates of all points on the graph of $y = 1 - x^2$ at which the tangent line passes through the point $(2, 0)$.56. Show that any two tangent lines to the parabola $y = ax^2$, $a \neq 0$, intersect at a point that is on the vertical line halfway between the points of tangency.57. Suppose that L is the tangent line at $x = x_0$ to the graph of the cubic equation $y = ax^3 + bx$. Find the x -coordinate of the point where L intersects the graph a second time.58. Show that the segment of the tangent line to the graph of $y = 1/x$ that is cut off by the coordinate axes is bisected by the point of tangency.59. Show that the triangle that is formed by any tangent line to the graph of $y = 1/x$, $x > 0$, and the coordinate axes has an area of 2 square units.60. Find conditions on a , b , c , and d so that the graph of the polynomial $f(x) = ax^3 + bx^2 + cx + d$ has

- (a) exactly two horizontal tangents
 (b) exactly one horizontal tangent
 (c) no horizontal tangents.

61. Newton's Law of Universal Gravitation states that the magnitude F of the force exerted by a point with mass M on apoint with mass m is

$$F = \frac{GmM}{r^2}$$

where G is a constant and r is the distance between the bodies. Assuming that the points are moving, find a formula for the instantaneous rate of change of F with respect to r .

62. In the temperature range between 0°C and 700°C the resistance R [in ohms (Ω)] of a certain platinum resistance thermometer is given by

$$R = 10 + 0.04124T - 1.779 \times 10^{-5}T^2$$

where T is the temperature in degrees Celsius. Where in the interval from 0°C to 700°C is the resistance of the thermometer most sensitive and least sensitive to temperature changes? [Hint: Consider the size of dR/dT in the interval $0 \leq T \leq 700$.]

 **63–64** Use a graphing utility to make rough estimates of the intervals on which $f'(x) > 0$, and then find those intervals exactly by differentiating. ■

63. $f(x) = x - \frac{1}{x}$

64. $f(x) = x^3 - 3x$

65–68 You are asked in these exercises to determine whether a piecewise-defined function f is differentiable at a value $x = x_0$, where f is defined by different formulas on different sides of x_0 . You may use without proof the following result, which is a consequence of the Mean-Value Theorem (discussed in Section 3.8). **Theorem.** Let f be continuous at x_0 and suppose that $\lim_{x \rightarrow x_0} f'(x)$ exists. Then f is differentiable at x_0 , and $f'(x_0) = \lim_{x \rightarrow x_0} f'(x)$. ■

65. Show that

$$f(x) = \begin{cases} x^2 + x + 1, & x \leq 1 \\ 3x, & x > 1 \end{cases}$$

is continuous at $x = 1$. Determine whether f is differentiable at $x = 1$. If so, find the value of the derivative there. Sketch the graph of f .

66. Let

$$f(x) = \begin{cases} x^2 - 16x, & x < 9 \\ \sqrt{x}, & x \geq 9 \end{cases}$$

Is f continuous at $x = 9$? Determine whether f is differentiable at $x = 9$. If so, find the value of the derivative there.

67. Let

$$f(x) = \begin{cases} x^2, & x \leq 1 \\ \sqrt{x}, & x > 1 \end{cases}$$

Determine whether f is differentiable at $x = 1$. If so, find the value of the derivative there.

68. Let

$$f(x) = \begin{cases} x^3 + \frac{1}{16}, & x < \frac{1}{2} \\ \frac{3}{4}x^2, & x \geq \frac{1}{2} \end{cases}$$

Determine whether f is differentiable at $x = \frac{1}{2}$. If so, find the value of the derivative there.

69. Find all points where f fails to be differentiable. Justify your answer.

(a) $f(x) = |3x - 2|$

(b) $f(x) = |x^2 - 4|$

70. In each part, compute f' , f'' , f''' , and then state the formula for $f^{(n)}$.

(a) $f(x) = 1/x$ (b) $f(x) = 1/x^2$

[Hint: The expression $(-1)^n$ has a value of 1 if n is even and -1 if n is odd. Use this expression in your answer.]

71. (a) Prove:

$$\frac{d^2}{dx^2}[cf(x)] = c \frac{d^2}{dx^2}[f(x)]$$

$$\frac{d^2}{dx^2}[f(x) + g(x)] = \frac{d^2}{dx^2}[f(x)] + \frac{d^2}{dx^2}[g(x)]$$

(b) Do the results in part (a) generalize to n th derivatives? Justify your answer.

72. Let $f(x) = x^8 - 2x + 3$; find

$$\lim_{w \rightarrow 2} \frac{f'(w) - f'(2)}{w - 2}$$

73. (a) Find $f^{(n)}(x)$ if $f(x) = x^n$, $n = 1, 2, 3, \dots$

(b) Find $f^{(n)}(x)$ if $f(x) = x^k$ and $n > k$, where k is a positive integer.

(c) Find $f^{(n)}(x)$ if

$$f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

74. (a) Prove: If $f''(x)$ exists for each x in (a, b) , then both f and f' are continuous on (a, b) .

(b) What can be said about the continuity of f and its derivatives if $f^{(n)}(x)$ exists for each x in (a, b) ?

75. Let $f(x) = (mx + b)^n$, where m and b are constants and n is an integer. Use the result of Exercise 52 in Section 2.2 to prove that $f'(x) = nm(mx + b)^{n-1}$.

76–77 Verify the result of Exercise 75 for $f(x)$. ■

76. $f(x) = (2x + 3)^2$ 77. $f(x) = (3x - 1)^3$

78–81 Use the result of Exercise 75 to compute the derivative of the given function $f(x)$. ■

78. $f(x) = \frac{1}{x-1}$

79. $f(x) = \frac{3}{(2x+1)^2}$

80. $f(x) = \frac{x}{x+1}$

81. $f(x) = \frac{2x^2 + 4x + 3}{x^2 + 2x + 1}$

82. The purpose of this exercise is to extend the power rule (Theorem 2.3.2) to any integer exponent. Let $f(x) = x^n$, where n is any integer. If $n > 0$, then $f'(x) = nx^{n-1}$ by Theorem 2.3.2.

(a) Show that the conclusion of Theorem 2.3.2 holds in the case $n = 0$.

(b) Suppose that $n < 0$ and set $m = -n$ so that

$$f(x) = x^n = x^{-m} = \frac{1}{x^m}$$

Use Definition 2.2.1 and Theorem 2.3.2 to show that

$$\frac{d}{dx} \left[\frac{1}{x^m} \right] = -mx^{m-1} \cdot \frac{1}{x^{2m}}$$

and conclude that $f'(x) = nx^{n-1}$.

✓ QUICK CHECK ANSWERS 2.3

1. (a) 0 (b) $\sqrt{6}$ (c) $3/\sqrt{x}$ (d) $\sqrt{6}/(2\sqrt{x})$ 2. (a) $3x^2$ (b) $5x^4 + 10x$ (c) $\frac{3}{2}x^2$ (d) $1 - 10x^{-3}$ 3. 6 4. $18x - 6$

2.4 THE PRODUCT AND QUOTIENT RULES

In this section we will develop techniques for differentiating products and quotients of functions whose derivatives are known.

■ DERIVATIVE OF A PRODUCT

You might be tempted to conjecture that the derivative of a product of two functions is the product of their derivatives. However, a simple example will show this to be false. Consider the functions

$$f(x) = x \quad \text{and} \quad g(x) = x^2$$

The product of their derivatives is

$$f'(x)g'(x) = (1)(2x) = 2x$$

but their product is $h(x) = f(x)g(x) = x^3$, so the derivative of the product is

$$h'(x) = 3x^2$$

Thus, the derivative of the product is not equal to the product of the derivatives. The correct relationship, which is credited to Leibniz, is given by the following theorem.

2.4.1 THEOREM (The Product Rule) *If f and g are differentiable at x , then so is the product $f \cdot g$, and*

$$\frac{d}{dx}[f(x)g(x)] = f(x)\frac{d}{dx}[g(x)] + g(x)\frac{d}{dx}[f(x)] \quad (1)$$

Formula (1) can also be expressed as

$$(f \cdot g)' = f \cdot g' + g \cdot f'$$

PROOF Whereas the proofs of the derivative rules in the last section were straightforward applications of the derivative definition, a key step in this proof involves adding and subtracting the quantity $f(x+h)g(x)$ to the numerator in the derivative definition. This yields

$$\begin{aligned} \frac{d}{dx}[f(x)g(x)] &= \lim_{h \rightarrow 0} \frac{f(x+h) \cdot g(x+h) - f(x) \cdot g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[f(x+h) \cdot \frac{g(x+h) - g(x)}{h} + g(x) \cdot \frac{f(x+h) - f(x)}{h} \right] \\ &= \lim_{h \rightarrow 0} f(x+h) \cdot \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} + \lim_{h \rightarrow 0} g(x) \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \left[\lim_{h \rightarrow 0} f(x+h) \right] \frac{d}{dx}[g(x)] + \left[\lim_{h \rightarrow 0} g(x) \right] \frac{d}{dx}[f(x)] \\ &= f(x) \frac{d}{dx}[g(x)] + g(x) \frac{d}{dx}[f(x)] \end{aligned}$$

[Note: In the last step $f(x+h) \rightarrow f(x)$ as $h \rightarrow 0$ because f is continuous at x by Theorem 2.2.3. Also, $g(x) \rightarrow g(x)$ as $h \rightarrow 0$ because $g(x)$ does not involve h and hence is treated as constant for the limit.] ■

In words, *the derivative of a product of two functions is the first function times the derivative of the second plus the second function times the derivative of the first.*

► **Example 1** Find dy/dx if $y = (4x^2 - 1)(7x^3 + x)$.

Solution. There are two methods that can be used to find dy/dx . We can either use the product rule or we can multiply out the factors in y and then differentiate. We will give both methods.

Method 1. (Using the Product Rule)

$$\begin{aligned}
\frac{dy}{dx} &= \frac{d}{dx}[(4x^2 - 1)(7x^3 + x)] \\
&= (4x^2 - 1) \frac{d}{dx}[7x^3 + x] + (7x^3 + x) \frac{d}{dx}[4x^2 - 1] \\
&= (4x^2 - 1)(21x^2 + 1) + (7x^3 + x)(8x) = 140x^4 - 9x^2 - 1
\end{aligned}$$

Method 2. (Multiplying First)

$$y = (4x^2 - 1)(7x^3 + x) = 28x^5 - 3x^3 - x$$

Thus,

$$\frac{dy}{dx} = \frac{d}{dx}[28x^5 - 3x^3 - x] = 140x^4 - 9x^2 - 1$$

which agrees with the result obtained using the product rule. ◀

► Example 2 Find ds/dt if $s = (1 + t)\sqrt{t}$.**Solution.** Applying the product rule yields

$$\begin{aligned}
\frac{ds}{dt} &= \frac{d}{dt}[(1 + t)\sqrt{t}] \\
&= (1 + t) \frac{d}{dt}[\sqrt{t}] + \sqrt{t} \frac{d}{dt}[1 + t] \\
&= \frac{1 + t}{2\sqrt{t}} + \sqrt{t} = \frac{1 + 3t}{2\sqrt{t}} \quad \blacktriangleleft
\end{aligned}$$

DERIVATIVE OF A QUOTIENT

Just as the derivative of a product is not generally the product of the derivatives, so the derivative of a quotient is not generally the quotient of the derivatives. The correct relationship is given by the following theorem.

2.4.2 THEOREM (The Quotient Rule) If f and g are both differentiable at x and if $g(x) \neq 0$, then f/g is differentiable at x and

Formula (2) can also be expressed as

$$\left(\frac{f}{g}\right)' = \frac{g \cdot f' - f \cdot g'}{g^2}$$

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \frac{d}{dx}[f(x)] - f(x) \frac{d}{dx}[g(x)]}{[g(x)]^2} \quad (2)$$

PROOF

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \lim_{h \rightarrow 0} \frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} = \lim_{h \rightarrow 0} \frac{f(x+h) \cdot g(x) - f(x) \cdot g(x+h)}{h \cdot g(x) \cdot g(x+h)}$$

Adding and subtracting $f(x) \cdot g(x)$ in the numerator yields

$$\begin{aligned}
 \frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] &= \lim_{h \rightarrow 0} \frac{f(x+h) \cdot g(x) - f(x) \cdot g(x) - f(x) \cdot g(x+h) + f(x) \cdot g(x)}{h \cdot g(x) \cdot g(x+h)} \\
 &= \lim_{h \rightarrow 0} \frac{\left[g(x) \cdot \frac{f(x+h) - f(x)}{h} \right] - \left[f(x) \cdot \frac{g(x+h) - g(x)}{h} \right]}{g(x) \cdot g(x+h)} \\
 &= \frac{\lim_{h \rightarrow 0} g(x) \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} - \lim_{h \rightarrow 0} f(x) \cdot \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}}{\lim_{h \rightarrow 0} g(x) \cdot \lim_{h \rightarrow 0} g(x+h)} \\
 &= \frac{\left[\lim_{h \rightarrow 0} g(x) \right] \cdot \frac{d}{dx} [f(x)] - \left[\lim_{h \rightarrow 0} f(x) \right] \cdot \frac{d}{dx} [g(x)]}{\lim_{h \rightarrow 0} g(x) \cdot \lim_{h \rightarrow 0} g(x+h)} \\
 &= \frac{g(x) \frac{d}{dx} [f(x)] - f(x) \frac{d}{dx} [g(x)]}{[g(x)]^2}
 \end{aligned}$$

[See the note at the end of the proof of Theorem 2.4.1 for an explanation of the last step.]

In words, *the derivative of a quotient of two functions is the denominator times the derivative of the numerator minus the numerator times the derivative of the denominator, all divided by the denominator squared.*

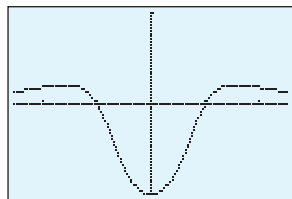
Sometimes it is better to simplify a function first than to apply the quotient rule immediately. For example, it is easier to differentiate

$$f(x) = \frac{x^{3/2} + x}{\sqrt{x}}$$

by rewriting it as

$$f(x) = x + \sqrt{x}$$

as opposed to using the quotient rule.



$[-2.5, 2.5] \times [-1, 1]$
xScl = 1, yScl = 1

$$y = \frac{x^2 - 1}{x^4 + 1}$$

▲ Figure 2.4.1

► **Example 3** Find $y'(x)$ for $y = \frac{x^3 + 2x^2 - 1}{x + 5}$.

Solution. Applying the quotient rule yields

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} \left[\frac{x^3 + 2x^2 - 1}{x + 5} \right] = \frac{(x + 5) \frac{d}{dx} [x^3 + 2x^2 - 1] - (x^3 + 2x^2 - 1) \frac{d}{dx} [x + 5]}{(x + 5)^2} \\
 &= \frac{(x + 5)(3x^2 + 4x) - (x^3 + 2x^2 - 1)(1)}{(x + 5)^2} \\
 &= \frac{(3x^3 + 19x^2 + 20x) - (x^3 + 2x^2 - 1)}{(x + 5)^2} \\
 &= \frac{2x^3 + 17x^2 + 20x + 1}{(x + 5)^2} \quad \blacktriangleleft
 \end{aligned}$$

► **Example 4** Let $f(x) = \frac{x^2 - 1}{x^4 + 1}$.

- Graph $y = f(x)$, and use your graph to make rough estimates of the locations of all horizontal tangent lines.
- By differentiating, find the exact locations of the horizontal tangent lines.

Solution (a). In Figure 2.4.1 we have shown the graph of the equation $y = f(x)$ in the window $[-2.5, 2.5] \times [-1, 1]$. This graph suggests that horizontal tangent lines occur at $x = 0$, $x \approx 1.5$, and $x \approx -1.5$.

Solution (b). To find the exact locations of the horizontal tangent lines, we must find the points where $dy/dx = 0$. We start by finding dy/dx :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left[\frac{x^2 - 1}{x^4 + 1} \right] = \frac{(x^4 + 1) \frac{d}{dx} [x^2 - 1] - (x^2 - 1) \frac{d}{dx} [x^4 + 1]}{(x^4 + 1)^2} \\ &= \frac{(x^4 + 1)(2x) - (x^2 - 1)(4x^3)}{(x^4 + 1)^2} \\ &= \frac{-2x^5 + 4x^3 + 2x}{(x^4 + 1)^2} = -\frac{2x(x^4 - 2x^2 - 1)}{(x^4 + 1)^2}\end{aligned}$$

The differentiation is complete.
The rest is simplification.

Now we will set $dy/dx = 0$ and solve for x . We obtain

$$-\frac{2x(x^4 - 2x^2 - 1)}{(x^4 + 1)^2} = 0$$

The solutions of this equation are the values of x for which the numerator is 0, that is,

$$2x(x^4 - 2x^2 - 1) = 0$$

The first factor yields the solution $x = 0$. Other solutions can be found by solving the equation

$$x^4 - 2x^2 - 1 = 0$$

This can be treated as a quadratic equation in x^2 and solved by the quadratic formula. This yields

$$x^2 = \frac{2 \pm \sqrt{8}}{2} = 1 \pm \sqrt{2}$$

The minus sign yields imaginary values of x , which we ignore since they are not relevant to the problem. The plus sign yields the solutions

$$x = \pm \sqrt{1 + \sqrt{2}}$$

In summary, horizontal tangent lines occur at

$$x = 0, \quad x = \sqrt{1 + \sqrt{2}} \approx 1.55, \quad \text{and} \quad x = -\sqrt{1 + \sqrt{2}} \approx -1.55$$

which is consistent with the rough estimates that we obtained graphically in part (a). ◀

Derive the following rule for differentiating a reciprocal:

$$\left(\frac{1}{g} \right)' = -\frac{g'}{g^2}$$

Use it to find the derivative of

$$f(x) = \frac{1}{x^2 + 1}$$

SUMMARY OF DIFFERENTIATION RULES

The following table summarizes the differentiation rules that we have encountered thus far.

Table 2.4.1

RULES FOR DIFFERENTIATION

$\frac{d}{dx}[c] = 0$	$(f + g)' = f' + g'$	$(f \cdot g)' = f \cdot g' + g \cdot f'$	$\left(\frac{1}{g} \right)' = -\frac{g'}{g^2}$
$(cf)' = cf'$	$(f - g)' = f' - g'$	$\left(\frac{f}{g} \right)' = \frac{g \cdot f' - f \cdot g'}{g^2}$	$\frac{d}{dx}[x^r] = rx^{r-1}$



QUICK CHECK EXERCISES 2.4

(See page 148 for answers.)

1. (a) $\frac{d}{dx}[x^2 f(x)] = \underline{\hspace{2cm}}$ (b) $\frac{d}{dx} \left[\frac{f(x)}{x^2 + 1} \right] = \underline{\hspace{2cm}}$

(c) $\frac{d}{dx} \left[\frac{x^2 + 1}{f(x)} \right] = \underline{\hspace{2cm}}$

2. Find $F'(1)$ given that $f(1) = -1$, $f'(1) = 2$, $g(1) = 3$, and $g'(1) = -1$.

- (a) $F(x) = 2f(x) - 3g(x)$ (b) $F(x) = [f(x)]^2$
(c) $F(x) = f(x)g(x)$ (d) $F(x) = f(x)/g(x)$

EXERCISE SET 2.4



Graphing Utility

1–4 Compute the derivative of the given function $f(x)$ by (a) multiplying and then differentiating and (b) using the product rule. Verify that (a) and (b) yield the same result. ■

1. $f(x) = (x + 1)(2x - 1)$ 2. $f(x) = (3x^2 - 1)(x^2 + 2)$
3. $f(x) = (x^2 + 1)(x^2 - 1)$
4. $f(x) = (x + 1)(x^2 - x + 1)$

5–20 Find $f'(x)$. ■

5. $f(x) = (3x^2 + 6)(2x - \frac{1}{4})$
6. $f(x) = (2 - x - 3x^3)(7 + x^5)$
7. $f(x) = (x^3 + 7x^2 - 8)(2x^{-3} + x^{-4})$
8. $f(x) = \left(\frac{1}{x} + \frac{1}{x^2}\right)(3x^3 + 27)$
9. $f(x) = (x - 2)(x^2 + 2x + 4)$
10. $f(x) = (x^2 + x)(x^2 - x)$
11. $f(x) = \frac{3x + 4}{x^2 + 1}$ 12. $f(x) = \frac{x - 2}{x^4 + x + 1}$
13. $f(x) = \frac{x^2}{3x - 4}$ 14. $f(x) = \frac{2x^2 + 5}{3x - 4}$
15. $f(x) = \frac{(2\sqrt{x} + 1)(x - 1)}{x + 3}$
16. $f(x) = (2\sqrt{x} + 1)\left(\frac{2 - x}{x^2 + 3x}\right)$
17. $f(x) = (2x + 1)\left(1 + \frac{1}{x}\right)(x^{-3} + 7)$
18. $f(x) = x^{-5}(x^2 + 2x)(4 - 3x)(2x^9 + 1)$
19. $f(x) = (x^7 + 2x - 3)^3$ 20. $f(x) = (x^2 + 1)^4$

21–24 Find $dy/dx|_{x=1}$. ■

21. $y = \frac{2x - 1}{x + 3}$ 22. $y = \frac{4x + 1}{x^2 - 5}$
23. $y = \left(\frac{3x + 2}{x}\right)(x^{-5} + 1)$ 24. $y = (2x^7 - x^2)\left(\frac{x - 1}{x + 1}\right)$

25–26 Use a graphing utility to estimate the value of $f'(1)$ by zooming in on the graph of f , and then compare your estimate to the exact value obtained by differentiating. ■

25. $f(x) = \frac{x}{x^2 + 1}$ 26. $f(x) = \frac{x^2 - 1}{x^2 + 1}$
27. Find $g'(4)$ given that $f(4) = 3$ and $f'(4) = -5$.
(a) $g(x) = \sqrt{x}f(x)$ (b) $g(x) = \frac{f(x)}{x}$
28. Find $g'(3)$ given that $f(3) = -2$ and $f'(3) = 4$.
(a) $g(x) = 3x^2 - 5f(x)$ (b) $g(x) = \frac{2x + 1}{f(x)}$
29. In parts (a)–(d), $F(x)$ is expressed in terms of $f(x)$ and $g(x)$. Find $F'(2)$ given that $f(2) = -1$, $f'(2) = 4$, $g(2) = 1$, and $g'(2) = -5$.

- (a) $F(x) = 5f(x) + 2g(x)$ (b) $F(x) = f(x) - 3g(x)$
- (c) $F(x) = f(x)g(x)$ (d) $F(x) = f(x)/g(x)$

30. Find $F'(\pi)$ given that $f(\pi) = 10$, $f'(\pi) = -1$, $g(\pi) = -3$, and $g'(\pi) = 2$.

- (a) $F(x) = 6f(x) - 5g(x)$ (b) $F(x) = x(f(x) + g(x))$
- (c) $F(x) = 2f(x)g(x)$ (d) $F(x) = \frac{f(x)}{4 + g(x)}$

31–36 Find all values of x at which the tangent line to the given curve satisfies the stated property. ■

31. $y = \frac{x^2 - 1}{x + 2}$; horizontal 32. $y = \frac{x^2 + 1}{x - 1}$; horizontal
33. $y = \frac{x^2 + 1}{x + 1}$; parallel to the line $y = x$
34. $y = \frac{x + 3}{x + 2}$; perpendicular to the line $y = x$
35. $y = \frac{1}{x + 4}$; passes through the origin
36. $y = \frac{2x + 5}{x + 2}$; y-intercept 2

FOCUS ON CONCEPTS

37. (a) What should it mean to say that two curves intersect at right angles?
(b) Show that the curves $y = 1/x$ and $y = 1/(2 - x)$ intersect at right angles.
38. Find all values of a such that the curves $y = a/(x - 1)$ and $y = x^2 - 2x + 1$ intersect at right angles.
39. Find a general formula for $F''(x)$ if $F(x) = xf(x)$ and f and f' are differentiable at x .
40. Suppose that the function f is differentiable everywhere and $F(x) = xf(x)$.
(a) Express $F'''(x)$ in terms of x and derivatives of f .
(b) For $n \geq 2$, conjecture a formula for $F^{(n)}(x)$.

41. A manufacturer of athletic footwear finds that the sales of their ZipStride brand running shoes is a function $f(p)$ of the selling price p (in dollars) for a pair of shoes. Suppose that $f(120) = 9000$ pairs of shoes and $f'(120) = -60$ pairs of shoes per dollar. The revenue that the manufacturer will receive for selling $f(p)$ pairs of shoes at p dollars per pair is $R(p) = p \cdot f(p)$. Find $R'(120)$. What impact would a small increase in price have on the manufacturer's revenue?
42. Solve the problem in Exercise 41 under the assumption that $f(120) = 9000$ and $f'(120) = -80$.
43. Use the quotient rule (Theorem 2.4.2) to derive the formula for the derivative of $f(x) = x^{-n}$, where n is a positive integer.

✓ QUICK CHECK ANSWERS 2.4

1. (a) $x^2 f'(x) + 2x f(x)$ (b) $\frac{(x^2 + 1)f'(x) - 2x f(x)}{(x^2 + 1)^2}$ (c) $\frac{2x f(x) - (x^2 + 1)f'(x)}{[f(x)^2]}$ 2. (a) 7 (b) -4 (c) 7 (d) $\frac{5}{9}$

2.5 DERIVATIVES OF TRIGONOMETRIC FUNCTIONS

The main objective of this section is to obtain formulas for the derivatives of the six basic trigonometric functions. If needed, you will find a review of trigonometric functions in Appendix B.

We will assume in this section that the variable x in the trigonometric functions $\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, and $\csc x$ is measured in radians. Also, we will need the limits in Theorem 1.6.3, but restated as follows using h rather than x as the variable:

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1 \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{1 - \cos h}{h} = 0 \quad (1-2)$$

Let us start with the problem of differentiating $f(x) = \sin x$. Using the definition of the derivative we obtain

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} && \text{By the addition formula for sine} \\ &= \lim_{h \rightarrow 0} \left[\sin x \left(\frac{\cos h - 1}{h} \right) + \cos x \left(\frac{\sin h}{h} \right) \right] \\ &= \lim_{h \rightarrow 0} \left[\cos x \left(\frac{\sin h}{h} \right) - \sin x \left(\frac{1 - \cos h}{h} \right) \right] && \text{Algebraic reorganization} \\ &= \lim_{h \rightarrow 0} \cos x \cdot \lim_{h \rightarrow 0} \frac{\sin h}{h} - \lim_{h \rightarrow 0} \sin x \cdot \lim_{h \rightarrow 0} \frac{1 - \cos h}{h} \\ &= \left(\lim_{h \rightarrow 0} \cos x \right) (1) - \left(\lim_{h \rightarrow 0} \sin x \right) (0) && \text{Formulas (1) and (2)} \\ &= \lim_{h \rightarrow 0} \cos x = \cos x && \begin{array}{l} \cos x \text{ does not involve the variable } h \text{ and hence} \\ \text{is treated as a constant in the limit computation.} \end{array} \end{aligned}$$

Formulas (1) and (2) and the derivation of Formulas (3) and (4) are only valid if h and x are in radians. See Exercise 49 for how Formulas (3) and (4) change when x is measured in degrees.

Thus, we have shown that

$$\frac{d}{dx} [\sin x] = \cos x \quad (3)$$

In the exercises we will ask you to use the same method to derive the following formula for the derivative of $\cos x$:

$$\frac{d}{dx} [\cos x] = -\sin x \quad (4)$$

► **Example 1** Find dy/dx if $y = x \sin x$.

Solution. Using Formula (3) and the product rule we obtain

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}[x \sin x] \\ &= x \frac{d}{dx}[\sin x] + \sin x \frac{d}{dx}[x] \\ &= x \cos x + \sin x \quad \blacktriangleleft\end{aligned}$$

► **Example 2** Find dy/dx if $y = \frac{\sin x}{1 + \cos x}$.

Solution. Using the quotient rule together with Formulas (3) and (4) we obtain

$$\begin{aligned}\frac{dy}{dx} &= \frac{(1 + \cos x) \cdot \frac{d}{dx}[\sin x] - \sin x \cdot \frac{d}{dx}[1 + \cos x]}{(1 + \cos x)^2} \\ &= \frac{(1 + \cos x)(\cos x) - (\sin x)(-\sin x)}{(1 + \cos x)^2} \\ &= \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2} = \frac{\cos x + 1}{(1 + \cos x)^2} = \frac{1}{1 + \cos x} \quad \blacktriangleleft\end{aligned}$$

Since Formulas (3) and (4) are only valid if x is in radians, the same is true for Formulas (5)–(8).

The derivatives of the remaining trigonometric functions are

$$\frac{d}{dx}[\tan x] = \sec^2 x \qquad \frac{d}{dx}[\sec x] = \sec x \tan x \qquad (5-6)$$

$$\frac{d}{dx}[\cot x] = -\csc^2 x \qquad \frac{d}{dx}[\csc x] = -\csc x \cot x \qquad (7-8)$$

These can all be obtained using the definition of the derivative, but it is easier to use Formulas (3) and (4) and apply the quotient rule to the relationships

$$\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}, \quad \sec x = \frac{1}{\cos x}, \quad \csc x = \frac{1}{\sin x}$$

For example,

$$\begin{aligned}\frac{d}{dx}[\tan x] &= \frac{d}{dx}\left[\frac{\sin x}{\cos x}\right] = \frac{\cos x \cdot \frac{d}{dx}[\sin x] - \sin x \cdot \frac{d}{dx}[\cos x]}{\cos^2 x} \\ &= \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x\end{aligned}$$

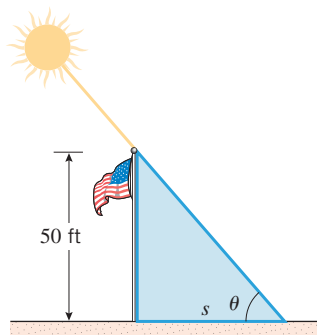
When finding the value of a derivative at a specific point $x = x_0$, it is important to substitute x_0 *after* the derivative is obtained. Thus, in Example 3 we made the substitution $x = \pi/4$ after f'' was calculated. What would have happened had we *incorrectly* substituted $x = \pi/4$ into $f'(x)$ before calculating f'' ?

► **Example 3** Find $f''(\pi/4)$ if $f(x) = \sec x$.

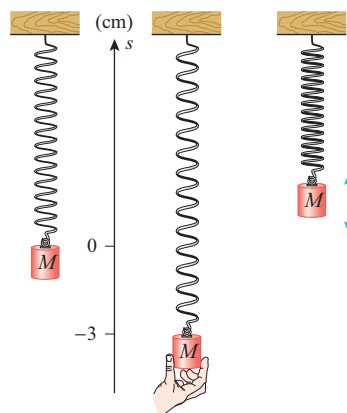
$$\begin{aligned}f'(x) &= \sec x \tan x \\ f''(x) &= \sec x \cdot \frac{d}{dx}[\tan x] + \tan x \cdot \frac{d}{dx}[\sec x] \\ &= \sec x \cdot \sec^2 x + \tan x \cdot \sec x \tan x \\ &= \sec^3 x + \sec x \tan^2 x\end{aligned}$$

Thus,

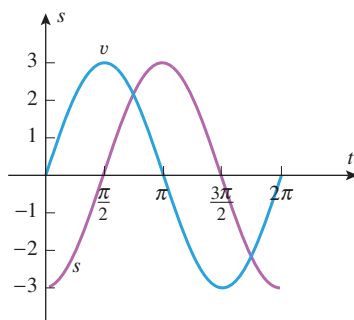
$$\begin{aligned} f''(\pi/4) &= \sec^3(\pi/4) + \sec(\pi/4) \tan^2(\pi/4) \\ &= (\sqrt{2})^3 + (\sqrt{2})(1)^2 = 3\sqrt{2} \quad \blacktriangleleft \end{aligned}$$



▲ Figure 2.5.1



▲ Figure 2.5.2



▲ Figure 2.5.3

In Example 5, the top of the mass has its maximum speed when it passes through its rest position. Why? What is that maximum speed?

► **Example 4** On a sunny day, a 50 ft flagpole casts a shadow that changes with the angle of elevation of the Sun. Let s be the length of the shadow and θ the angle of elevation of the Sun (Figure 2.5.1). Find the rate at which the length of the shadow is changing with respect to θ when $\theta = 45^\circ$. Express your answer in units of feet/degree.

Solution. The variables s and θ are related by $\tan \theta = 50/s$ or, equivalently,

$$s = 50 \cot \theta \quad (9)$$

If θ is measured in radians, then Formula (7) is applicable, which yields

$$\frac{ds}{d\theta} = -50 \csc^2 \theta$$

which is the rate of change of shadow length with respect to the elevation angle θ in units of feet/radian. When $\theta = 45^\circ$ (or equivalently $\theta = \pi/4$ radians), we obtain

$$\left. \frac{ds}{d\theta} \right|_{\theta=\pi/4} = -50 \csc^2(\pi/4) = -100 \text{ feet/radian}$$

Converting radians (rad) to degrees (deg) yields

$$-100 \frac{\text{ft}}{\text{rad}} \cdot \frac{\pi}{180} \frac{\text{rad}}{\text{deg}} = -\frac{5}{9} \pi \frac{\text{ft}}{\text{deg}} \approx -1.75 \text{ ft/deg}$$

Thus, when $\theta = 45^\circ$, the shadow length is decreasing (because of the minus sign) at an approximate rate of 1.75 ft/deg increase in the angle of elevation. ◀

► **Example 5** As illustrated in Figure 2.5.2, suppose that a spring with an attached mass is stretched 3 cm beyond its rest position and released at time $t = 0$. Assuming that the position function of the top of the attached mass is

$$s = -3 \cos t \quad (10)$$

where s is in centimeters and t is in seconds, find the velocity function and discuss the motion of the attached mass.

Solution. The velocity function is

$$v = \frac{ds}{dt} = \frac{d}{dt}[-3 \cos t] = 3 \sin t$$

Figure 2.5.3 shows the graphs of the position and velocity functions. The position function tells us that the top of the mass oscillates between a low point of $s = -3$ and a high point of $s = 3$ with one complete oscillation occurring every 2π seconds [the period of (10)]. The top of the mass is moving up (the positive s -direction) when v is positive, is moving down when v is negative, and is at a high or low point when $v = 0$. Thus, for example, the top of the mass moves up from time $t = 0$ to time $t = \pi$, at which time it reaches the high point $s = 3$ and then moves down until time $t = 2\pi$, at which time it reaches the low point of $s = -3$. The motion then repeats periodically. ◀

 QUICK CHECK EXERCISES 2.5 (See page 153 for answers.)

- Find dy/dx .
 - $y = \sin x$
 - $y = \cos x$
 - $y = \tan x$
 - $y = \sec x$
- Find $f'(x)$ and $f'(\pi/3)$ if $f(x) = \sin x \cos x$.
- Use a derivative to evaluate each limit.
 - $\lim_{h \rightarrow 0} \frac{\sin(\frac{\pi}{2} + h) - 1}{h}$
 - $\lim_{h \rightarrow 0} \frac{\csc(x + h) - \csc x}{h}$

EXERCISE SET 2.5 Graphing Utility

1–18 Find $f'(x)$. ■

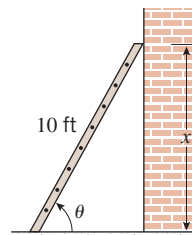
- $f(x) = 4 \cos x + 2 \sin x$
- $f(x) = \frac{5}{x^2} + \sin x$
- $f(x) = -4x^2 \cos x$
- $f(x) = 2 \sin^2 x$
- $f(x) = \frac{5 - \cos x}{5 + \sin x}$
- $f(x) = \frac{\sin x}{x^2 + \sin x}$
- $f(x) = \sec x - \sqrt{2} \tan x$
- $f(x) = (x^2 + 1) \sec x$
- $f(x) = 4 \csc x - \cot x$
- $f(x) = \cos x - x \csc x$
- $f(x) = \sec x \tan x$
- $f(x) = \csc x \cot x$
- $f(x) = \frac{\cot x}{1 + \csc x}$
- $f(x) = \frac{\sec x}{1 + \tan x}$
- $f(x) = \sin^2 x + \cos^2 x$
- $f(x) = \sec^2 x - \tan^2 x$
- $f(x) = \frac{\sin x \sec x}{1 + x \tan x}$
- $f(x) = \frac{(x^2 + 1) \cot x}{3 - \cos x \csc x}$

19–24 Find d^2y/dx^2 . ■

- $y = x \cos x$
- $y = \csc x$
- $y = x \sin x - 3 \cos x$
- $y = x^2 \cos x + 4 \sin x$
- $y = \sin x \cos x$
- $y = \tan x$
- Find the equation of the line tangent to the graph of $\tan x$ at
 - $x = 0$
 - $x = \pi/4$
 - $x = -\pi/4$
- Find the equation of the line tangent to the graph of $\sin x$ at
 - $x = 0$
 - $x = \pi$
 - $x = \pi/4$
- Show that $y = x \sin x$ is a solution to $y'' + y = 2 \cos x$.
 - Show that $y = x \sin x$ is a solution of the equation $y^{(4)} + y'' = -2 \cos x$.
- Show that $y = \cos x$ and $y = \sin x$ are solutions of the equation $y'' + y = 0$.
 - Show that $y = A \sin x + B \cos x$ is a solution of the equation $y'' + y = 0$ for all constants A and B .
- Find all values in the interval $[-2\pi, 2\pi]$ at which the graph of f has a horizontal tangent line.
 - $f(x) = \sin x$
 - $f(x) = x + \cos x$
 - $f(x) = \tan x$
 - $f(x) = \sec x$

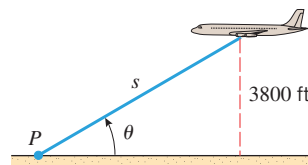
-  **30.** (a) Use a graphing utility to make rough estimates of the values in the interval $[0, 2\pi]$ at which the graph of $y = \sin x \cos x$ has a horizontal tangent line.
 (b) Find the exact locations of the points where the graph has a horizontal tangent line.

- 31.** A 10 ft ladder leans against a wall at an angle θ with the horizontal, as shown in the accompanying figure. The top of the ladder is x feet above the ground. If the bottom of the ladder is pushed toward the wall, find the rate at which x changes with respect to θ when $\theta = 60^\circ$. Express the answer in units of feet/degree.



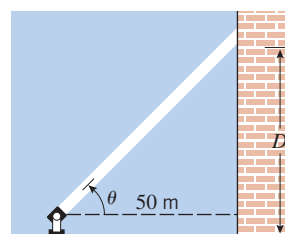
◀ Figure Ex-31

- 32.** An airplane is flying on a horizontal path at a height of 3800 ft, as shown in the accompanying figure. At what rate is the distance s between the airplane and the fixed point P changing with respect to θ when $\theta = 30^\circ$? Express the answer in units of feet/degree.



◀ Figure Ex-32

- 33.** A searchlight is trained on the side of a tall building. As the light rotates, the spot it illuminates moves up and down the side of the building. That is, the distance D between ground level and the illuminated spot on the side of the building is a function of the angle θ formed by the light beam and the horizontal (see the accompanying figure). If the searchlight is located 50 m from the building, find the rate at which D is changing with respect to θ when $\theta = 45^\circ$. Express your answer in units of meters/degree.



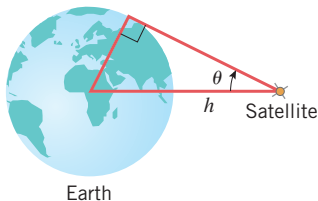
◀ Figure Ex-33

34. An Earth-observing satellite can see only a portion of the Earth's surface. The satellite has horizon sensors that can detect the angle θ shown in the accompanying figure. Let r be the radius of the Earth (assumed spherical) and h the distance of the satellite from the Earth's surface.

(a) Show that $h = r(\csc \theta - 1)$.

(b) Using $r = 6378$ km, find the rate at which h is changing with respect to θ when $\theta = 30^\circ$. Express the answer in units of kilometers/degree.

Source: Adapted from *Space Mathematics*, NASA, 1985.



◀ Figure Ex-34

35–38 True–False Determine whether the statement is true or false. Explain your answer. ■

35. If $g(x) = f(x) \sin x$, then $g'(x) = f'(x) \cos x$.
 36. If $g(x) = f(x) \sin x$, then $g'(0) = f(0)$.
 37. If $f(x) \cos x = \sin x$, then $f'(x) = \sec^2 x$.
 38. Suppose that $g(x) = f(x) \sec x$, where $f(0) = 8$ and $f'(0) = -2$. Then

$$\begin{aligned} g'(0) &= \lim_{h \rightarrow 0} \frac{f(h) \sec h - f(0)}{h} = \lim_{h \rightarrow 0} \frac{8(\sec h - 1)}{h} \\ &= 8 \cdot \frac{d}{dx} [\sec x] \Big|_{x=0} = 8 \sec 0 \tan 0 = 0 \end{aligned}$$

39–40 Make a conjecture about the derivative by calculating the first few derivatives and observing the resulting pattern. ■

39. $\frac{d^{87}}{dx^{87}} [\sin x]$

40. $\frac{d^{100}}{dx^{100}} [\cos x]$

41. Let $f(x) = \cos x$. Find all positive integers n for which $f^{(n)}(x) = \sin x$.
 42. Let $f(x) = \sin x$. Find all positive integers n for which $f^{(n)}(x) = \sin x$.

FOCUS ON CONCEPTS

43. In each part, determine where f is differentiable.

(a) $f(x) = \sin x$

(b) $f(x) = \cos x$

(c) $f(x) = \tan x$

(d) $f(x) = \cot x$

(e) $f(x) = \sec x$

(f) $f(x) = \csc x$

(g) $f(x) = \frac{1}{1 + \cos x}$

(h) $f(x) = \frac{1}{\sin x \cos x}$

(i) $f(x) = \frac{\cos x}{2 - \sin x}$

44. (a) Derive Formula (4) using the definition of a derivative.
 (b) Use Formulas (3) and (4) to obtain (7).
 (c) Use Formula (4) to obtain (6).
 (d) Use Formula (3) to obtain (8).

45. Use Formula (1), the alternative form for the definition of derivative given in Formula (13) of Section 2.2, that is,

$$f'(x) = \lim_{w \rightarrow x} \frac{f(w) - f(x)}{w - x}$$

and the difference identity

$$\sin \alpha - \sin \beta = 2 \sin \left(\frac{\alpha - \beta}{2} \right) \cos \left(\frac{\alpha + \beta}{2} \right)$$

to show that $\frac{d}{dx} [\sin x] = \cos x$.

46. Follow the directions of Exercise 45 using the difference identity

$$\cos \alpha - \cos \beta = -2 \sin \left(\frac{\alpha - \beta}{2} \right) \sin \left(\frac{\alpha + \beta}{2} \right)$$

to show that $\frac{d}{dx} [\cos x] = -\sin x$.

47. (a) Show that $\lim_{h \rightarrow 0} \frac{\tan h}{h} = 1$.
 (b) Use the result in part (a) to help derive the formula for the derivative of $\tan x$ directly from the definition of a derivative.

48. Without using any trigonometric identities, find

$$\lim_{x \rightarrow 0} \frac{\tan(x + y) - \tan y}{x}$$

[Hint: Relate the given limit to the definition of the derivative of an appropriate function of y .]

49. The derivative formulas for $\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, and $\csc x$ were obtained under the assumption that x is measured in radians. If x is measured in degrees, then

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{\pi}{180}$$

(See Exercise 39 of Section 1.6). Use this result to prove that if x is measured in degrees, then

(a) $\frac{d}{dx} [\sin x] = \frac{\pi}{180} \cos x$

(b) $\frac{d}{dx} [\cos x] = -\frac{\pi}{180} \sin x$.

50. **Writing** Suppose that f is a function that is differentiable everywhere. Explain the relationship, if any, between the periodicity of f and that of f' . That is, if f is periodic, must f' also be periodic? If f' is periodic, must f also be periodic?

✓ QUICK CHECK ANSWERS 2.5

1. (a) $\cos x$ (b) $-\sin x$ (c) $\sec^2 x$ (d) $\sec x \tan x$ 2. $f'(x) = \cos^2 x - \sin^2 x$, $f'(\pi/3) = -\frac{1}{2}$
 3. (a) $\left. \frac{d}{dx} [\sin x] \right|_{x=\pi/2} = 0$ (b) $\frac{d}{dx} [\csc x] = -\csc x \cot x$

2.6 THE CHAIN RULE

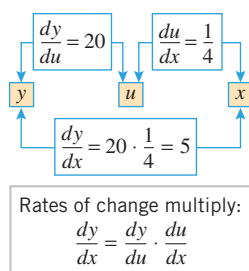
In this section we will derive a formula that expresses the derivative of a composition $f \circ g$ in terms of the derivatives of f and g . This formula will enable us to differentiate complicated functions using known derivatives of simpler functions.

DERIVATIVES OF COMPOSITIONS



Mike Brinson/Getty Images

The cost of a car trip is a combination of fuel efficiency and the cost of gasoline.



▲ Figure 2.6.1

Suppose you are traveling to school in your car, which gets 20 miles per gallon of gasoline. The number of miles you can travel in your car without refueling is a function of the number of gallons of gas you have in the gas tank. In symbols, if y is the number of miles you can travel and u is the number of gallons of gas you have initially, then y is a function of u , or $y = f(u)$. As you continue your travels, you note that your local service station is selling gasoline for \$4 per gallon. The number of gallons of gas you have initially is a function of the amount of money you spend for that gas. If x is the number of dollars you spend on gas, then $u = g(x)$. Now 20 miles per gallon is the rate at which your mileage changes with respect to the amount of gasoline you use, so

$$f'(u) = \frac{dy}{du} = 20 \text{ miles per gallon}$$

Similarly, since gasoline costs \$4 per gallon, each dollar you spend will give you $1/4$ of a gallon of gas, and

$$g'(x) = \frac{du}{dx} = \frac{1}{4} \text{ gallons per dollar}$$

Notice that the number of miles you can travel is also a function of the number of dollars you spend on gasoline. This fact is expressible as the composition of functions

$$y = f(u) = f(g(x))$$

You might be interested in how many miles you can travel per dollar, which is dy/dx . Intuition suggests that rates of change multiply in this case (see Figure 2.6.1), so

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{20 \text{ miles}}{1 \text{ gallon}} \cdot \frac{1 \text{ gallon}}{4 \text{ dollars}} = \frac{20 \text{ miles}}{4 \text{ dollars}} = 5 \text{ miles per dollar}$$

The following theorem, the proof of which is given in Appendix D, formalizes the preceding ideas.

2.6.1 THEOREM (The Chain Rule) If g is differentiable at x and f is differentiable at $g(x)$, then the composition $f \circ g$ is differentiable at x . Moreover, if

$$y = f(g(x)) \quad \text{and} \quad u = g(x)$$

then $y = f(u)$ and

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad (1)$$

The name “chain rule” is appropriate because the desired derivative is obtained by a two-link “chain” of simpler derivatives.

Formula (1) is easy to remember because the left side is exactly what results if we “cancel” the du 's on the right side. This “canceling” device provides a good way of deducing the correct form of the chain rule when different variables are used. For example, if w is a function of x and x is a function of t , then the chain rule takes the form

$$\frac{dw}{dt} = \frac{dw}{dx} \cdot \frac{dx}{dt}$$

► **Example 1** Find dy/dx if $y = \cos(x^3)$.

Solution. Let $u = x^3$ and express y as $y = \cos u$. Applying Formula (1) yields

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= \frac{d}{du}[\cos u] \cdot \frac{d}{dx}[x^3] \\ &= (-\sin u) \cdot (3x^2) \\ &= (-\sin(x^3)) \cdot (3x^2) = -3x^2 \sin(x^3) \quad \blacktriangleleft\end{aligned}$$

► **Example 2** Find dw/dt if $w = \tan x$ and $x = 4t^3 + t$.

Solution. In this case the chain rule computations take the form

$$\begin{aligned}\frac{dw}{dt} &= \frac{dw}{dx} \cdot \frac{dx}{dt} \\ &= \frac{d}{dx}[\tan x] \cdot \frac{d}{dt}[4t^3 + t] \\ &= (\sec^2 x) \cdot (12t^2 + 1) \\ &= [\sec^2(4t^3 + t)] \cdot (12t^2 + 1) = (12t^2 + 1) \sec^2(4t^3 + t) \quad \blacktriangleleft\end{aligned}$$

■ AN ALTERNATIVE VERSION OF THE CHAIN RULE

Formula (1) for the chain rule can be unwieldy in some problems because it involves so many variables. As you become more comfortable with the chain rule, you may want to dispense with writing out the dependent variables by expressing (1) in the form

Confirm that (2) is an alternative version of (1) by letting $y = f(g(x))$ and $u = g(x)$.

$$\frac{d}{dx}[f(g(x))] = (f \circ g)'(x) = f'(g(x))g'(x) \quad (2)$$

A convenient way to remember this formula is to call f the “outside function” and g the “inside function” in the composition $f(g(x))$ and then express (2) in words as:

The derivative of $f(g(x))$ is the derivative of the outside function evaluated at the inside function times the derivative of the inside function.

$$\frac{d}{dx}[f(g(x))] = \underbrace{f'(g(x))}_{\text{Derivative of the outside function evaluated at the inside function}} \cdot \underbrace{g'(x)}_{\text{Derivative of the inside function}}$$

Derivative of the outside function evaluated at the inside function

Derivative of the inside function

► **Example 3** (Example 1 revisited) Find $h'(x)$ if $h(x) = \cos(x^3)$.

Solution. We can think of h as a composition $f(g(x))$ in which $g(x) = x^3$ is the inside function and $f(x) = \cos x$ is the outside function. Thus, Formula (2) yields

$$\begin{aligned}
 h'(x) &= \underbrace{f'(g(x))}_{\substack{\text{Derivative of the outside} \\ \text{function evaluated at the} \\ \text{inside function}}} \cdot \underbrace{g'(x)}_{\substack{\text{Derivative of the} \\ \text{inside function}}} \\
 &= f'(x^3) \cdot 3x^2 \\
 &= -\sin(x^3) \cdot 3x^2 = -3x^2 \sin(x^3)
 \end{aligned}$$

which agrees with the result obtained in Example 1. ◀

► **Example 4**

$$\begin{aligned}
 \frac{d}{dx} [\tan^2 x] &= \frac{d}{dx} [(\tan x)^2] = \underbrace{(2 \tan x)}_{\substack{\text{Derivative of the outside} \\ \text{function evaluated at the} \\ \text{inside function}}} \cdot \underbrace{(\sec^2 x)}_{\substack{\text{Derivative of the} \\ \text{inside function}}} = 2 \tan x \sec^2 x \\
 \\
 \frac{d}{dx} [\sqrt{x^2 + 1}] &= \frac{1}{\underbrace{2\sqrt{x^2 + 1}}_{\substack{\text{Derivative of the outside} \\ \text{function evaluated at the} \\ \text{inside function}}}} \cdot \underbrace{2x}_{\substack{\text{Derivative of the} \\ \text{inside function}}} = \frac{x}{\sqrt{x^2 + 1}} \quad \text{See Formula (6) of Section 2.3.} \quad \blacktriangleleft
 \end{aligned}$$

GENERALIZED DERIVATIVE FORMULAS

There is a useful third variation of the chain rule that strikes a middle ground between Formulas (1) and (2). If we let $u = g(x)$ in (2), then we can rewrite that formula as

$$\frac{d}{dx} [f(u)] = f'(u) \frac{du}{dx} \quad (3)$$

This result, called the **generalized derivative formula** for f , provides a way of using the derivative of $f(x)$ to produce the derivative of $f(u)$, where u is a function of x . Table 2.6.1 gives some examples of this formula.

Table 2.6.1
GENERALIZED DERIVATIVE FORMULAS

$\frac{d}{dx} [u^r] = ru^{r-1} \frac{du}{dx}$	
$\frac{d}{dx} [\sin u] = \cos u \frac{du}{dx}$	$\frac{d}{dx} [\cos u] = -\sin u \frac{du}{dx}$
$\frac{d}{dx} [\tan u] = \sec^2 u \frac{du}{dx}$	$\frac{d}{dx} [\cot u] = -\csc^2 u \frac{du}{dx}$
$\frac{d}{dx} [\sec u] = \sec u \tan u \frac{du}{dx}$	$\frac{d}{dx} [\csc u] = -\csc u \cot u \frac{du}{dx}$

► **Example 5** Find

$$\begin{array}{lll} \text{(a)} \quad \frac{d}{dx}[\sin(2x)] & \text{(b)} \quad \frac{d}{dx}[\tan(x^2 + 1)] & \text{(c)} \quad \frac{d}{dx}[\sqrt{x^3 + \csc x}] \\ \text{(d)} \quad \frac{d}{dx}[x^2 - x + 2]^{3/4} & \text{(e)} \quad \frac{d}{dx}[(1 + x^5 \cot x)^{-8}] \end{array}$$

Solution (a). Taking $u = 2x$ in the generalized derivative formula for $\sin u$ yields

$$\frac{d}{dx}[\sin(2x)] = \frac{d}{dx}[\sin u] = \cos u \frac{du}{dx} = \cos 2x \cdot \frac{d}{dx}[2x] = \cos 2x \cdot 2 = 2 \cos 2x$$

Solution (b). Taking $u = x^2 + 1$ in the generalized derivative formula for $\tan u$ yields

$$\begin{aligned} \frac{d}{dx}[\tan(x^2 + 1)] &= \frac{d}{dx}[\tan u] = \sec^2 u \frac{du}{dx} \\ &= \sec^2(x^2 + 1) \cdot \frac{d}{dx}[x^2 + 1] = \sec^2(x^2 + 1) \cdot 2x \\ &= 2x \sec^2(x^2 + 1) \end{aligned}$$

Solution (c). Taking $u = x^3 + \csc x$ in the generalized derivative formula for $\sqrt{u}$ yields

$$\begin{aligned} \frac{d}{dx}[\sqrt{x^3 + \csc x}] &= \frac{d}{dx}[\sqrt{u}] = \frac{1}{2\sqrt{u}} \frac{du}{dx} = \frac{1}{2\sqrt{x^3 + \csc x}} \cdot \frac{d}{dx}[x^3 + \csc x] \\ &= \frac{1}{2\sqrt{x^3 + \csc x}} \cdot (3x^2 - \csc x \cot x) = \frac{3x^2 - \csc x \cot x}{2\sqrt{x^3 + \csc x}} \end{aligned}$$

Solution (d). Taking $u = x^2 - x + 2$ in the generalized derivative formula for $u^{3/4}$ yields

$$\begin{aligned} \frac{d}{dx}[x^2 - x + 2]^{3/4} &= \frac{d}{dx}[u^{3/4}] = \frac{3}{4}u^{-1/4} \frac{du}{dx} \\ &= \frac{3}{4}(x^2 - x + 2)^{-1/4} \cdot \frac{d}{dx}[x^2 - x + 2] \\ &= \frac{3}{4}(x^2 - x + 2)^{-1/4}(2x - 1) \end{aligned}$$

Solution (e). Taking $u = 1 + x^5 \cot x$ in the generalized derivative formula for u^{-8} yields

$$\begin{aligned} \frac{d}{dx}[(1 + x^5 \cot x)^{-8}] &= \frac{d}{dx}[u^{-8}] = -8u^{-9} \frac{du}{dx} \\ &= -8(1 + x^5 \cot x)^{-9} \cdot \frac{d}{dx}[1 + x^5 \cot x] \\ &= -8(1 + x^5 \cot x)^{-9} \cdot [x^5(-\csc^2 x) + 5x^4 \cot x] \\ &= (8x^5 \csc^2 x - 40x^4 \cot x)(1 + x^5 \cot x)^{-9} \quad \blacktriangleleft \end{aligned}$$

Sometimes you will have to make adjustments in notation or apply the chain rule more than once to calculate a derivative.

► **Example 6** Find

$$\begin{array}{ll} \text{(a)} \quad \frac{d}{dx}[\sin(\sqrt{1 + \cos x})] & \text{(b)} \quad \frac{d\mu}{dt} \text{ if } \mu = \sec \sqrt{\omega t} \quad (\omega \text{ constant}) \end{array}$$

Solution (a). Taking $u = \sqrt{1 + \cos x}$ in the generalized derivative formula for $\sin u$ yields

$$\begin{aligned}\frac{d}{dx} [\sin(\sqrt{1 + \cos x})] &= \frac{d}{dx} [\sin u] = \cos u \frac{du}{dx} \\ &= \cos(\sqrt{1 + \cos x}) \cdot \frac{d}{dx} [\sqrt{1 + \cos x}] \\ &= \cos(\sqrt{1 + \cos x}) \cdot \frac{-\sin x}{2\sqrt{1 + \cos x}} \\ &= -\frac{\sin x \cos(\sqrt{1 + \cos x})}{2\sqrt{1 + \cos x}}\end{aligned}$$

We used the generalized derivative formula for $\sqrt{u}$ with $u = 1 + \cos x$.

Solution (b).

$$\begin{aligned}\frac{d\mu}{dt} &= \frac{d}{dt} [\sec \sqrt{\omega t}] = \sec \sqrt{\omega t} \tan \sqrt{\omega t} \frac{d}{dt} [\sqrt{\omega t}] \\ &= \sec \sqrt{\omega t} \tan \sqrt{\omega t} \frac{\omega}{2\sqrt{\omega t}}\end{aligned}$$

We used the generalized derivative formula for $\sec u$ with $u = \sqrt{\omega t}$.

We used the generalized derivative formula for $\sqrt{u}$ with $u = \omega t$.

DIFFERENTIATING USING COMPUTER ALGEBRA SYSTEMS

Even with the chain rule and other differentiation rules, some derivative computations can be tedious to perform. For complicated derivatives, engineers and scientists often use computer algebra systems such as *Mathematica*, *Maple*, or *Sage*. For example, although we have all the mathematical tools to compute

$$\frac{d}{dx} \left[\frac{(x^2 + 1)^{10} \sin^3(\sqrt{x})}{\sqrt{1 + \csc x}} \right] \quad (4)$$

by hand, the computation is sufficiently involved that it may be more efficient (and less error-prone) to use a computer algebra system.

TECHNOLOGY MASTERY

If you have a CAS, use it to perform the differentiation in (4).

QUICK CHECK EXERCISES 2.6 (See page 160 for answers.)

- The chain rule states that the derivative of the composition of two functions is the derivative of the _____ function evaluated at the _____ function times the derivative of the _____ function.
- If y is a differentiable function of u , and u is a differentiable function of x , then

$$\frac{dy}{dx} = \underline{\hspace{2cm}} \cdot \underline{\hspace{2cm}}$$

- Find dy/dx .
(a) $y = (x^2 + 5)^{10}$ (b) $y = \sqrt{1 + 6x}$
- Find dy/dx .
(a) $y = \sin(3x + 2)$ (b) $y = (x^2 \tan x)^4$
- Suppose that $f(2) = 3$, $f'(2) = 4$, $g(3) = 6$, and $g'(3) = -5$. Evaluate
(a) $h'(2)$, where $h(x) = g(f(x))$
(b) $k'(3)$, where $k(x) = f(\frac{1}{3}g(x))$.

EXERCISE SET 2.6



Graphing Utility



CAS

- Given that
 $f'(0) = 2$, $g(0) = 0$ and $g'(0) = 3$
find $(f \circ g)'(0)$.
- Given that
 $f'(9) = 5$, $g(2) = 9$ and $g'(2) = -3$
find $(f \circ g)'(2)$.

- Let $f(x) = x^5$ and $g(x) = 2x - 3$.
(a) Find $(f \circ g)(x)$ and $(f \circ g)'(x)$.
(b) Find $(g \circ f)(x)$ and $(g \circ f)'(x)$.
- Let $f(x) = 5\sqrt{x}$ and $g(x) = 4 + \cos x$.
(a) Find $(f \circ g)(x)$ and $(f \circ g)'(x)$.
(b) Find $(g \circ f)(x)$ and $(g \circ f)'(x)$.

FOCUS ON CONCEPTS

5. Given the following table of values, find the indicated derivatives in parts (a) and (b).

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
3	5	-2	5	7
5	3	-1	12	4

(a) $F'(3)$, where $F(x) = f(g(x))$

(b) $G'(3)$, where $G(x) = g(f(x))$

6. Given the following table of values, find the indicated derivatives in parts (a) and (b).

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
-1	2	3	2	-3
2	0	4	1	-5

(a) $F'(-1)$, where $F(x) = f(g(x))$

(b) $G'(-1)$, where $G(x) = g(f(x))$

7-26 Find $f'(x)$. ■

7. $f(x) = (x^3 + 2x)^{37}$ 8. $f(x) = (3x^2 + 2x - 1)^6$

9. $f(x) = \left(x^3 - \frac{7}{x}\right)^{-2}$ 10. $f(x) = \frac{1}{(x^5 - x + 1)^9}$

11. $f(x) = \frac{4}{(3x^2 - 2x + 1)^3}$ 12. $f(x) = \sqrt{x^3 - 2x + 5}$

13. $f(x) = \sqrt{4 + \sqrt{3x}}$ 14. $f(x) = \sqrt[3]{12 + \sqrt{x}}$

15. $f(x) = \sin\left(\frac{1}{x^2}\right)$ 16. $f(x) = \tan \sqrt{x}$

17. $f(x) = 4 \cos^5 x$ 18. $f(x) = 4x + 5 \sin^4 x$

19. $f(x) = \cos^2(3\sqrt{x})$ 20. $f(x) = \tan^4(x^3)$

21. $f(x) = 2 \sec^2(x^7)$ 22. $f(x) = \cos^3\left(\frac{x}{x+1}\right)$

23. $f(x) = \sqrt{\cos(5x)}$ 24. $f(x) = \sqrt{3x - \sin^2(4x)}$

25. $f(x) = [x + \csc(x^3 + 3)]^{-3}$

26. $f(x) = [x^4 - \sec(4x^2 - 2)]^{-4}$

27-40 Find dy/dx . ■

27. $y = x^3 \sin^2(5x)$ 28. $y = \sqrt{x} \tan^3(\sqrt{x})$

29. $y = x^5 \sec(1/x)$ 30. $y = \frac{\sin x}{\sec(3x + 1)}$

31. $y = \cos(\cos x)$ 32. $y = \sin(\tan 3x)$

33. $y = \cos^3(\sin 2x)$ 34. $y = \frac{1 + \csc(x^2)}{1 - \cot(x^2)}$

35. $y = (5x + 8)^7 (1 - \sqrt{x})^6$ 36. $y = (x^2 + x)^5 \sin^8 x$

37. $y = \left(\frac{x-5}{2x+1}\right)^3$ 38. $y = \left(\frac{1+x^2}{1-x^2}\right)^{17}$

39. $y = \frac{(2x+3)^3}{(4x^2-1)^8}$ 40. $y = [1 + \sin^3(x^5)]^{12}$

C 41-42 Use a CAS to find dy/dx . ■

41. $y = [x \sin 2x + \tan^4(x^7)]^5$

42. $y = \tan^4\left(2 + \frac{(7-x)\sqrt{3x^2+5}}{x^3 + \sin x}\right)$

43-50 Find an equation for the tangent line to the graph at the specified value of x . ■

43. $y = x \cos 3x$, $x = \pi$

44. $y = \sin(1 + x^3)$, $x = -3$

45. $y = \sec^3\left(\frac{\pi}{2} - x\right)$, $x = -\frac{\pi}{2}$

46. $y = \left(x - \frac{1}{x}\right)^3$, $x = 2$ 47. $y = \tan(4x^2)$, $x = \sqrt{\pi}$

48. $y = 3 \cot^4 x$, $x = \frac{\pi}{4}$ 49. $y = x^2 \sqrt{5 - x^2}$, $x = 1$

50. $y = \frac{x}{\sqrt{1-x^2}}$, $x = 0$

51-54 Find d^2y/dx^2 . ■

51. $y = x \cos(5x) - \sin^2 x$ 52. $y = \sin(3x^2)$

53. $y = \frac{1+x}{1-x}$ 54. $y = x \tan\left(\frac{1}{x}\right)$

55-58 Find the indicated derivative. ■

55. $y = \cot^3(\pi - \theta)$; find $\frac{dy}{d\theta}$.

56. $\lambda = \left(\frac{au+b}{cu+d}\right)^6$; find $\frac{d\lambda}{du}$ (a, b, c, d constants).

57. $\frac{d}{d\omega}[a \cos^2 \pi\omega + b \sin^2 \pi\omega]$ (a, b constants)

58. $x = \csc^2\left(\frac{\pi}{3} - y\right)$; find $\frac{dx}{dy}$.

59. (a) Use a graphing utility to obtain the graph of the function $f(x) = x\sqrt{4-x^2}$.

(b) Use the graph in part (a) to make a rough sketch of the graph of f' .

(c) Find $f'(x)$, and then check your work in part (b) by using the graphing utility to obtain the graph of f' .

(d) Find the equation of the tangent line to the graph of f at $x = 1$, and graph f and the tangent line together.

60. (a) Use a graphing utility to obtain the graph of the function $f(x) = \sin x^2 \cos x$ over the interval $[-\pi/2, \pi/2]$.

(b) Use the graph in part (a) to make a rough sketch of the graph of f' over the interval.

(c) Find $f'(x)$, and then check your work in part (b) by using the graphing utility to obtain the graph of f' over the interval.

(d) Find the equation of the tangent line to the graph of f at $x = 1$, and graph f and the tangent line together over the interval.

61–64 True–False Determine whether the statement is true or false. Explain your answer. ■

61. If $y = f(x)$, then $\frac{d}{dx}[\sqrt{y}] = \sqrt{f'(x)}$.
62. If $y = f(u)$ and $u = g(x)$, then $dy/dx = f'(x) \cdot g'(x)$.
63. If $y = \cos[g(x)]$, then $dy/dx = -\sin[g'(x)]$.
64. If $y = \sin^3(3x^3)$, then $dy/dx = 27x^2 \sin^2(3x^3) \cos(3x^3)$.
65. If an object suspended from a spring is displaced vertically from its equilibrium position by a small amount and released, and if the air resistance and the mass of the spring are ignored, then the resulting oscillation of the object is called **simple harmonic motion**. Under appropriate conditions the displacement y from equilibrium in terms of time t is given by

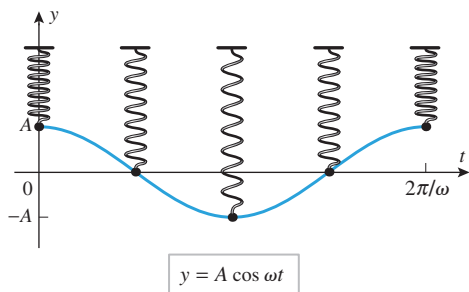
$$y = A \cos \omega t$$

where A is the initial displacement at time $t = 0$, and ω is a constant that depends on the mass of the object and the stiffness of the spring (see the accompanying figure). The constant $|A|$ is called the **amplitude** of the motion and ω the **angular frequency**.

(a) Show that

$$\frac{d^2 y}{dt^2} = -\omega^2 y$$

- (b) The **period** T is the time required to make one complete oscillation. Show that $T = 2\pi/\omega$.
- (c) The **frequency** f of the vibration is the number of oscillations per unit time. Find f in terms of the period T .
- (d) Find the amplitude, period, and frequency of an object that is executing simple harmonic motion given by $y = 0.6 \cos 15t$, where t is in seconds and y is in centimeters.



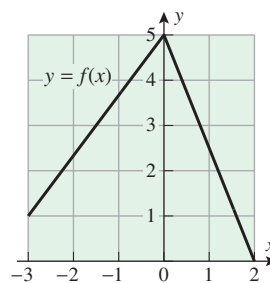
▲ Figure Ex-65

66. Find the value of the constant A so that $y = A \sin 3t$ satisfies the equation
- $$\frac{d^2 y}{dt^2} + 2y = 4 \sin 3t$$

FOCUS ON CONCEPTS

67. Use the graph of the function f in the accompanying figure to evaluate

$$\left. \frac{d}{dx} [\sqrt{x + f(x)}] \right|_{x=-1}$$

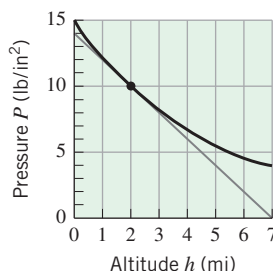


◀ Figure Ex-67

68. Using the function f in Exercise 67, evaluate

$$\left. \frac{d}{dx} [f(2 \sin x)] \right|_{x=\pi/6}$$

69. The accompanying figure shows the graph of atmospheric pressure p (lb/in²) versus the altitude h (mi) above sea level.
- (a) From the graph and the tangent line at $h = 2$ shown on the graph, estimate the values of p and dp/dh at an altitude of 2 mi.
- (b) If the altitude of a space vehicle is increasing at the rate of 0.3 mi/s at the instant when it is 2 mi above sea level, how fast is the pressure changing with time at this instant?



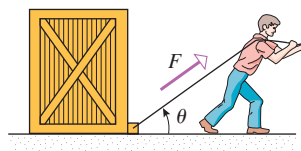
◀ Figure Ex-69

70. The force F (in pounds) acting at an angle θ with the horizontal that is needed to drag a crate weighing W pounds along a horizontal surface at a constant velocity is given by

$$F = \frac{\mu W}{\cos \theta + \mu \sin \theta}$$

where μ is a constant called the **coefficient of sliding friction** between the crate and the surface (see the accompanying figure). Suppose that the crate weighs 150 lb and that $\mu = 0.3$.

- (a) Find $dF/d\theta$ when $\theta = 30^\circ$. Express the answer in units of pounds/degree.
- (b) Find dF/dt when $\theta = 30^\circ$ if θ is decreasing at the rate of $0.5^\circ/\text{s}$ at this instant.



◀ Figure Ex-70

71. Recall that

$$\frac{d}{dx}(|x|) = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

Use this result and the chain rule to find

$$\frac{d}{dx}(|\sin x|)$$

for nonzero x in the interval $(-\pi, \pi)$.

72. Use the derivative formula for $\sin x$ and the identity

$$\cos x = \sin\left(\frac{\pi}{2} - x\right)$$

to obtain the derivative formula for $\cos x$.

73. Let

$$f(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

- Show that f is continuous at $x = 0$.
- Use Definition 2.2.1 to show that $f'(0)$ does not exist.
- Find $f'(x)$ for $x \neq 0$.
- Determine whether $\lim_{x \rightarrow 0} f'(x)$ exists.

74. Let

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

- Show that f is continuous at $x = 0$.
- Use Definition 2.2.1 to find $f'(0)$.
- Find $f'(x)$ for $x \neq 0$.
- Show that f' is not continuous at $x = 0$.

75. Given the following table of values, find the indicated derivatives in parts (a) and (b).

x	$f(x)$	$f'(x)$
2	1	7
8	5	-3

- $g'(2)$, where $g(x) = [f(x)]^3$
- $h'(2)$, where $h(x) = f(x^3)$

76. Given that $f'(x) = \sqrt{3x+4}$ and $g(x) = x^2 - 1$, find $F'(x)$ if $F(x) = f(g(x))$.

77. Given that $f'(x) = \frac{x}{x^2+1}$ and $g(x) = \sqrt{3x-1}$, find $F'(x)$ if $F(x) = f(g(x))$.

78. Find $f'(x^2)$ if $\frac{d}{dx}[f(x^2)] = x^2$.

79. Find $\frac{d}{dx}[f(x)]$ if $\frac{d}{dx}[f(3x)] = 6x$.

80. Recall that a function f is **even** if $f(-x) = f(x)$ and **odd** if $f(-x) = -f(x)$, for all x in the domain of f . Assuming that f is differentiable, prove:

- f' is odd if f is even
- f' is even if f is odd.

81. Draw some pictures to illustrate the results in Exercise 80, and write a paragraph that gives an informal explanation of why the results are true.

82. Let $y = f_1(u)$, $u = f_2(v)$, $v = f_3(w)$, and $w = f_4(x)$. Express dy/dx in terms of dy/du , dw/dx , du/dv , and dv/dw .

83. Find a formula for

$$\frac{d}{dx}[f(g(h(x)))]$$

84. **Writing** The “co” in “cosine” comes from “complementary,” since the cosine of an angle is the sine of the complementary angle, and vice versa:

$$\cos x = \sin\left(\frac{\pi}{2} - x\right) \quad \text{and} \quad \sin x = \cos\left(\frac{\pi}{2} - x\right)$$

Suppose that we define a function g to be a *cofunction* of a function f if

$$g(x) = f\left(\frac{\pi}{2} - x\right) \quad \text{for all } x$$

Thus, cosine and sine are cofunctions of each other, as are cotangent and tangent, and also cosecant and secant. If g is the cofunction of f , state a formula that relates g' and the cofunction of f' . Discuss how this relationship is exhibited by the derivatives of the cosine, cotangent, and cosecant functions.

✓ QUICK CHECK ANSWERS 2.6

- outside; inside; inside
- $\frac{dy}{du} \cdot \frac{du}{dx}$
- (a) $10(x^2 + 5)^9 \cdot 2x = 20x(x^2 + 5)^9$ (b) $\frac{1}{2\sqrt{1+6x}} \cdot 6 = \frac{3}{\sqrt{1+6x}}$
- (a) $3 \cos(3x + 2)$ (b) $4(x^2 \tan x)^3(2x \tan x + x^2 \sec^2 x)$
- (a) $g'(f(2))f'(2) = -20$ (b) $f'\left(\frac{1}{3}g(3)\right) \cdot \frac{1}{3}g'(3) = -\frac{20}{3}$

2.7 IMPLICIT DIFFERENTIATION

Up to now we have been concerned with differentiating functions that are given by equations of the form $y = f(x)$. In this section we will consider methods for differentiating functions for which it is inconvenient or impossible to express them in this form.

FUNCTIONS DEFINED EXPLICITLY AND IMPLICITLY

An equation of the form $y = f(x)$ is said to define y **explicitly** as a function of x because the variable y appears alone on one side of the equation and does not appear at all on the other side. However, sometimes functions are defined by equations in which y is not alone on one side; for example, the equation

$$yx + y + 1 = x \quad (1)$$

is not of the form $y = f(x)$, but it still defines y as a function of x since it can be rewritten as

$$y = \frac{x - 1}{x + 1}$$

Thus, we say that (1) defines y **implicitly** as a function of x , the function being

$$f(x) = \frac{x - 1}{x + 1}$$

An equation in x and y can implicitly define more than one function of x . This can occur when the graph of the equation fails the vertical line test, so it is not the graph of a function of x . For example, if we solve the equation of the circle

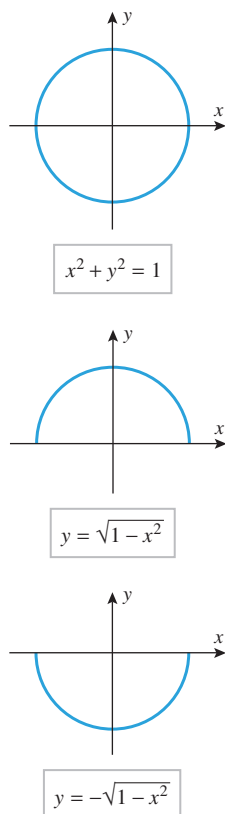
$$x^2 + y^2 = 1 \quad (2)$$

for y in terms of x , we obtain $y = \pm\sqrt{1 - x^2}$, so we have found two functions that are defined implicitly by (2), namely,

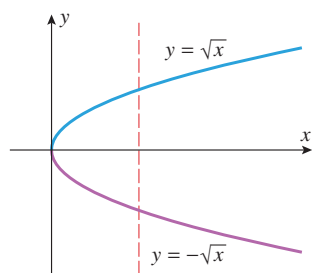
$$f_1(x) = \sqrt{1 - x^2} \quad \text{and} \quad f_2(x) = -\sqrt{1 - x^2} \quad (3)$$

The graphs of these functions are the upper and lower semicircles of the circle $x^2 + y^2 = 1$ (Figure 2.7.1). This leads us to the following definition.

2.7.1 DEFINITION We will say that a given equation in x and y defines the function f **implicitly** if the graph of $y = f(x)$ coincides with a portion of the graph of the equation.



▲ Figure 2.7.1



▲ Figure 2.7.2 The graph of $x = y^2$ does not pass the vertical line test, but the graphs of $y = \sqrt{x}$ and $y = -\sqrt{x}$ do.

► **Example 1** The graph of $x = y^2$ is not the graph of a function of x , since it does not pass the vertical line test (Figure 2.7.2). However, if we solve this equation for y in terms of x , we obtain the equations $y = \sqrt{x}$ and $y = -\sqrt{x}$, whose graphs pass the vertical line test and are portions of the graph of $x = y^2$ (Figure 2.7.2). Thus, the equation $x = y^2$ implicitly defines the functions

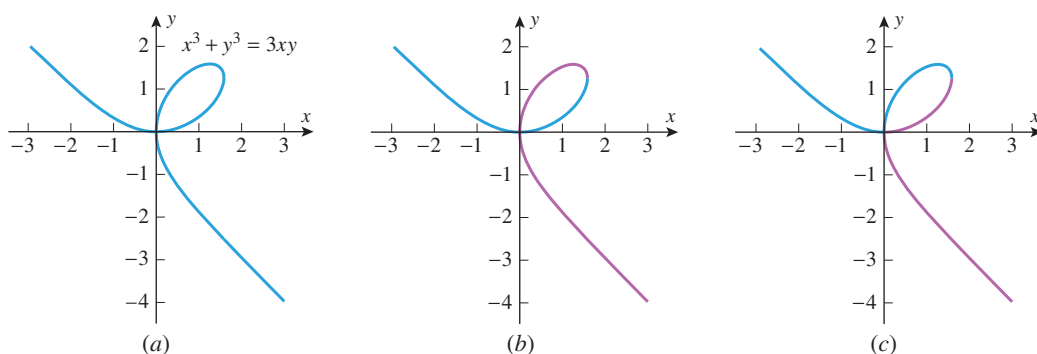
$$f_1(x) = \sqrt{x} \quad \text{and} \quad f_2(x) = -\sqrt{x} \quad \blacktriangleleft$$

Although it was a trivial matter in the last example to solve the equation $x = y^2$ for y in terms of x , it is difficult or impossible to do this for some equations. For example, the equation

$$x^3 + y^3 = 3xy \quad (4)$$

can be solved for y in terms of x , but the resulting formulas are too complicated to be practical. Other equations, such as $\sin(xy) = y$, cannot be solved for y by any elementary method. Thus, even though an equation may define one or more functions of x , it may not be possible or practical to find explicit formulas for those functions.

Fortunately, CAS programs, such as *Mathematica* and *Maple*, have “implicit plotting” capabilities that can graph equations such as (4). The graph of this equation, which is called the **Folium of Descartes**, is shown in Figure 2.7.3a. Parts (b) and (c) of the figure show the graphs (in blue) of two functions that are defined implicitly by (4).



▲ Figure 2.7.3

■ IMPLICIT DIFFERENTIATION

In general, it is not necessary to solve an equation for y in terms of x in order to differentiate the functions defined implicitly by the equation. To illustrate this, let us consider the simple equation

$$xy = 1 \quad (5)$$

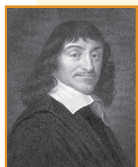
One way to find dy/dx is to rewrite this equation as

$$y = \frac{1}{x} \quad (6)$$

from which it follows that

$$\frac{dy}{dx} = -\frac{1}{x^2} \quad (7)$$

Another way to obtain this derivative is to differentiate both sides of (5) *before* solving for y in terms of x , treating y as a (temporarily unspecified) differentiable function of x . With



René Descartes (1596–1650) Descartes, a French aristocrat, was the son of a government official. He graduated from the University of Poitiers with a law degree at age 20. After a brief probe into the pleasures of Paris he became a military engineer, first for the Dutch Prince of Nassau and then for the German Duke of Bavaria. It was during his service as a soldier that Descartes began to pursue mathematics seriously and develop his analytic geometry. After the wars, he returned to Paris where he stalked the city as an eccentric, wearing a sword in his belt and a plumed hat. He lived in leisure, seldom

arose before 11 A.M., and dabbled in the study of human physiology, philosophy, glaciers, meteors, and rainbows. He eventually moved to Holland, where he published his *Discourse on the Method*, and finally to Sweden where he died while serving as tutor to Queen Christina. Descartes is regarded as a genius of the first magnitude. In addition to major contributions in mathematics and philosophy he is considered, along with William Harvey, to be a founder of modern physiology.

[Image: http://en.wikipedia.org/wiki/File:Frans_Hals_-_Portret_van_Ren%C3%A9_Descartes.jpg]

this approach we obtain

$$\begin{aligned}\frac{d}{dx}[xy] &= \frac{d}{dx}[1] \\ x \frac{d}{dx}[y] + y \frac{d}{dx}[x] &= 0 \\ x \frac{dy}{dx} + y &= 0 \\ \frac{dy}{dx} &= -\frac{y}{x}\end{aligned}$$

If we now substitute (6) into the last expression, we obtain

$$\frac{dy}{dx} = -\frac{1}{x^2}$$

which agrees with Equation (7). This method of obtaining derivatives is called **implicit differentiation**.

► **Example 2** Use implicit differentiation to find dy/dx if $5y^2 + \sin y = x^2$.

$$\begin{aligned}\frac{d}{dx}[5y^2 + \sin y] &= \frac{d}{dx}[x^2] \\ 5 \frac{d}{dx}[y^2] + \frac{d}{dx}[\sin y] &= 2x \\ 5 \left(2y \frac{dy}{dx} \right) + (\cos y) \frac{dy}{dx} &= 2x \\ 10y \frac{dy}{dx} + (\cos y) \frac{dy}{dx} &= 2x\end{aligned}$$

The chain rule was used here because y is a function of x .

Solving for dy/dx we obtain

$$\frac{dy}{dx} = \frac{2x}{10y + \cos y} \quad (8)$$

Note that this formula involves both x and y . In order to obtain a formula for dy/dx that involves x alone, we would have to solve the original equation for y in terms of x and then substitute in (8). However, it is impossible to do this, so we are forced to leave the formula for dy/dx in terms of x and y . ◀

► **Example 3** Use implicit differentiation to find d^2y/dx^2 if $4x^2 - 2y^2 = 9$.

Solution. Differentiating both sides of $4x^2 - 2y^2 = 9$ with respect to x yields

$$8x - 4y \frac{dy}{dx} = 0$$

from which we obtain

$$\frac{dy}{dx} = \frac{2x}{y} \quad (9)$$

Differentiating both sides of (9) yields

$$\frac{d^2y}{dx^2} = \frac{(y)(2) - (2x)(dy/dx)}{y^2} \quad (10)$$

Substituting (9) into (10) and simplifying using the original equation, we obtain

$$\frac{d^2y}{dx^2} = \frac{2y - 2x(2x/y)}{y^2} = \frac{2y^2 - 4x^2}{y^3} = -\frac{9}{y^3} \quad \blacktriangleleft$$

In Examples 2 and 3, the resulting formulas for dy/dx involved both x and y . Although it is usually more desirable to have the formula for dy/dx expressed in terms of x alone, having the formula in terms of x and y is not an impediment to finding slopes and equations of tangent lines provided the x - and y -coordinates of the point of tangency are known. This is illustrated in the following example.

► **Example 4** Find the slopes of the tangent lines to the curve $y^2 - x + 1 = 0$ at the points $(2, -1)$ and $(2, 1)$.

Solution. We could proceed by solving the equation for y in terms of x , and then evaluating the derivative of $y = \sqrt{x-1}$ at $(2, 1)$ and the derivative of $y = -\sqrt{x-1}$ at $(2, -1)$ (Figure 2.7.4). However, implicit differentiation is more efficient since it can be used for the slopes of *both* tangent lines. Differentiating implicitly yields

$$\begin{aligned}\frac{d}{dx}[y^2 - x + 1] &= \frac{d}{dx}[0] \\ \frac{d}{dx}[y^2] - \frac{d}{dx}[x] + \frac{d}{dx}[1] &= \frac{d}{dx}[0] \\ 2y \frac{dy}{dx} - 1 &= 0 \\ \frac{dy}{dx} &= \frac{1}{2y}\end{aligned}$$

At $(2, -1)$ we have $y = -1$, and at $(2, 1)$ we have $y = 1$, so the slopes of the tangent lines to the curve at those points are

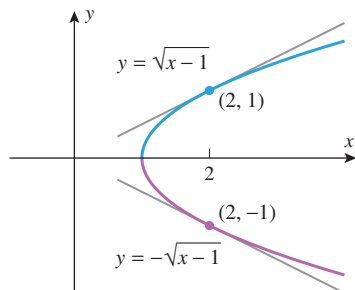
$$\left. \frac{dy}{dx} \right|_{\substack{x=2 \\ y=-1}} = -\frac{1}{2} \quad \text{and} \quad \left. \frac{dy}{dx} \right|_{\substack{x=2 \\ y=1}} = \frac{1}{2} \quad \blacktriangleleft$$

► **Example 5**

- Use implicit differentiation to find dy/dx for the Folium of Descartes $x^3 + y^3 = 3xy$.
- Find an equation for the tangent line to the Folium of Descartes at the point $(\frac{3}{2}, \frac{3}{2})$.
- At what point(s) in the first quadrant is the tangent line to the Folium of Descartes horizontal?

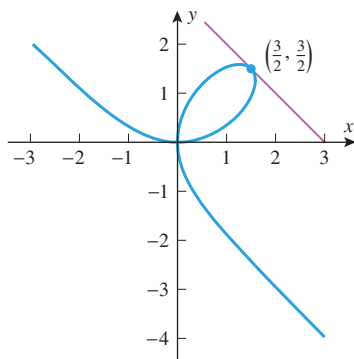
Solution (a). Differentiating implicitly yields

$$\begin{aligned}\frac{d}{dx}[x^3 + y^3] &= \frac{d}{dx}[3xy] \\ 3x^2 + 3y^2 \frac{dy}{dx} &= 3x \frac{dy}{dx} + 3y \\ x^2 + y^2 \frac{dy}{dx} &= x \frac{dy}{dx} + y \\ (y^2 - x) \frac{dy}{dx} &= y - x^2 \\ \frac{dy}{dx} &= \frac{y - x^2}{y^2 - x}\end{aligned} \tag{11}$$



▲ Figure 2.7.4

Formula (11) cannot be evaluated at $(0, 0)$ and hence provides no information about the nature of the Folium of Descartes at the origin. Based on the graphs in Figure 2.7.3, what can you say about the differentiability of the implicitly defined functions graphed in blue in parts (b) and (c) of the figure?



▲ Figure 2.7.5

Solution (b). At the point $(\frac{3}{2}, \frac{3}{2})$, we have $x = \frac{3}{2}$ and $y = \frac{3}{2}$, so from (11) the slope $m_{\tan}$ of the tangent line at this point is

$$m_{\tan} = \left. \frac{dy}{dx} \right|_{x=3/2, y=3/2} = \frac{(3/2) - (3/2)^2}{(3/2)^2 - (3/2)} = -1$$

Thus, the equation of the tangent line at the point $(\frac{3}{2}, \frac{3}{2})$ is

$$y - \frac{3}{2} = -1 \left(x - \frac{3}{2} \right) \quad \text{or} \quad x + y = 3$$

which is consistent with Figure 2.7.5.

Solution (c). The tangent line is horizontal at the points where $dy/dx = 0$, and from (11) this occurs only where $y - x^2 = 0$ or

$$y = x^2 \tag{12}$$

Substituting this expression for y in the equation $x^3 + y^3 = 3xy$ for the curve yields

$$x^3 + (x^2)^3 = 3x^3$$

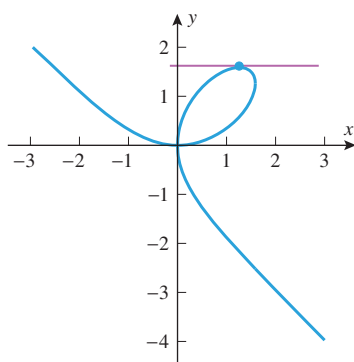
$$x^6 - 2x^3 = 0$$

$$x^3(x^3 - 2) = 0$$

whose solutions are $x = 0$ and $x = 2^{1/3}$. From (12), the solutions $x = 0$ and $x = 2^{1/3}$ yield the points $(0, 0)$ and $(2^{1/3}, 2^{2/3})$, respectively. Of these two, only $(2^{1/3}, 2^{2/3})$ is in the first quadrant. Substituting $x = 2^{1/3}$, $y = 2^{2/3}$ into (11) yields

$$\left. \frac{dy}{dx} \right|_{x=2^{1/3}, y=2^{2/3}} = \frac{0}{2^{4/3} - 2^{2/3}} = 0$$

We conclude that $(2^{1/3}, 2^{2/3}) \approx (1.26, 1.59)$ is the only point on the Folium of Descartes in the first quadrant at which the tangent line is horizontal (Figure 2.7.6). ◀



▲ Figure 2.7.6

■ DIFFERENTIABILITY OF FUNCTIONS DEFINED IMPLICITLY

When differentiating implicitly, it is assumed that y represents a differentiable function of x . If this is not so, then the resulting calculations may be nonsense. For example, if we differentiate the equation

$$x^2 + y^2 + 1 = 0 \tag{13}$$

we obtain

$$2x + 2y \frac{dy}{dx} = 0 \quad \text{or} \quad \frac{dy}{dx} = -\frac{x}{y}$$

However, this derivative is meaningless because there are no real values of x and y that satisfy (13) (why?); and hence (13) does not define any real functions implicitly.

The nonsensical conclusion of these computations conveys the importance of knowing whether an equation in x and y that is to be differentiated implicitly actually defines some differentiable function of x implicitly. Unfortunately, this can be a difficult problem, so we will leave the discussion of such matters for more advanced courses in analysis.

✓ QUICK CHECK EXERCISES 2.7 (See page 167 for answers.)

1. The equation $xy + 2y = 1$ defines implicitly the function $y = \underline{\hspace{2cm}}$.
2. Use implicit differentiation to find dy/dx for $x^2 - y^3 = xy$.
3. The slope of the tangent line to the graph of $x + y + xy = 3$ at $(1, 1)$ is $\underline{\hspace{2cm}}$.
4. Use implicit differentiation to find d^2y/dx^2 for $\sin y = x$.

EXERCISE SET 2.7 C CAS

1–2

- (a) Find dy/dx by differentiating implicitly.
 (b) Solve the equation for y as a function of x , and find dy/dx from that equation.
 (c) Confirm that the two results are consistent by expressing the derivative in part (a) as a function of x alone. ■

1. $x + xy - 2x^3 = 2$ 2. $\sqrt{y} - \sin x = 2$

3–12 Find dy/dx by implicit differentiation. ■

3. $x^2 + y^2 = 100$ 4. $x^3 + y^3 = 3xy^2$
 5. $x^2y + 3xy^3 - x = 3$ 6. $x^3y^2 - 5x^2y + x = 1$
 7. $\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{y}} = 1$ 8. $x^2 = \frac{x+y}{x-y}$
 9. $\sin(x^2y^2) = x$ 10. $\cos(xy^2) = y$
 11. $\tan^3(xy^2 + y) = x$ 12. $\frac{xy^3}{1 + \sec y} = 1 + y^4$

13–18 Find d^2y/dx^2 by implicit differentiation. ■

13. $2x^2 - 3y^2 = 4$ 14. $x^3 + y^3 = 1$
 15. $x^3y^3 - 4 = 0$ 16. $xy + y^2 = 2$
 17. $y + \sin y = x$ 18. $x \cos y = y$

19–20 Find the slope of the tangent line to the curve at the given points in two ways: first by solving for y in terms of x and differentiating and then by implicit differentiation. ■

19. $x^2 + y^2 = 1$; $(1/2, \sqrt{3}/2)$, $(1/2, -\sqrt{3}/2)$
 20. $y^2 - x + 1 = 0$; $(10, 3)$, $(10, -3)$

21–24 True–False Determine whether the statement is true or false. Explain your answer. ■

21. If an equation in x and y defines a function $y = f(x)$ implicitly, then the graph of the equation and the graph of f are identical.
 22. The function

$$f(x) = \begin{cases} \sqrt{1-x^2}, & 0 < x \leq 1 \\ -\sqrt{1-x^2}, & -1 \leq x \leq 0 \end{cases}$$

is defined implicitly by the equation $x^2 + y^2 = 1$.

23. The function $|x|$ is not defined implicitly by the equation $(x+y)(x-y) = 0$.
 24. If y is defined implicitly as a function of x by the equation $x^2 + y^2 = 1$, then $dy/dx = -x/y$.
 25. This exercise extends the Power Rule (Theorem 2.3.2) from positive integer exponents to arbitrary integer exponents. Suppose that $m < 0$ is an integer. Use Theorem 2.3.2 and implicit differentiation to prove that

$$\frac{d}{dx}(x^m) = mx^{m-1}$$

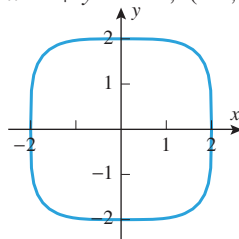
26. This exercise extends the Power Rule for integer exponents (Exercise 25) to rational exponents. Suppose that $r = m/n$

where m and n are integers with $n > 0$. Use implicit differentiation and the result of Exercise 25 to prove that

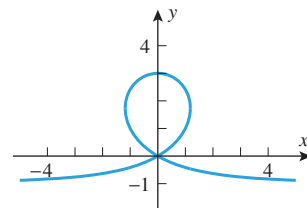
$$\frac{d}{dx}(x^r) = rx^{r-1}$$

27–30 Use implicit differentiation to find the slope of the tangent line to the curve at the specified point, and check that your answer is consistent with the accompanying graph. ■

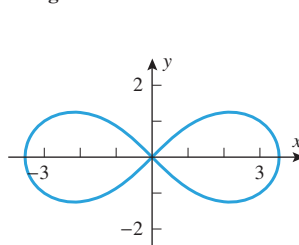
27. $x^4 + y^4 = 16$; $(1, \sqrt[4]{15})$ [Lamé's special quartic]
 28. $y^3 + yx^2 + x^2 - 3y^2 = 0$; $(0, 3)$ [trisectrix]
 29. $2(x^2 + y^2)^2 = 25(x^2 - y^2)$; $(3, 1)$ [lemniscate]
 30. $x^{2/3} + y^{2/3} = 4$; $(-1, 3\sqrt{3})$ [four-cusped hypocycloid]



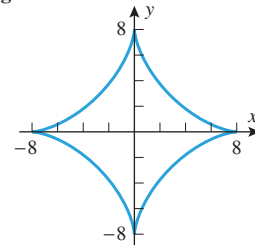
▲ Figure Ex-27



▲ Figure Ex-28



▲ Figure Ex-29



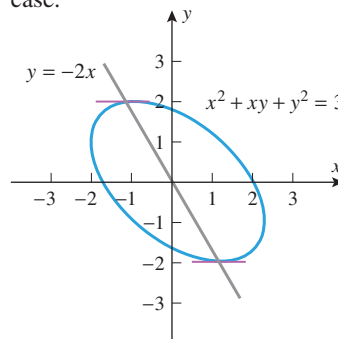
▲ Figure Ex-30

31–34 Use implicit differentiation to find the specified derivative. ■

31. $a^4 - t^4 = 6a^2t$; da/dt 32. $\sqrt{u} + \sqrt{v} = 5$; du/dv
 33. $a^2\omega^2 + b^2\lambda^2 = 1$ (a, b constants); $d\omega/d\lambda$
 34. $y = \sin x$; dx/dy

FOCUS ON CONCEPTS

35. In the accompanying figure, it appears that the ellipse $x^2 + xy + y^2 = 3$ has horizontal tangent lines at the points of intersection of the ellipse and the line $y = -2x$. Use implicit differentiation to explain why this is the case.



▲ Figure Ex-35

36. (a) A student claims that the ellipse $x^2 - xy + y^2 = 1$ has a horizontal tangent line at the point $(1, 1)$. Without doing any computations, explain why the student's claim must be incorrect.
 (b) Find all points on the ellipse $x^2 - xy + y^2 = 1$ at which the tangent line is horizontal.

37. (a) Use the implicit plotting capability of a CAS to graph the equation $y^4 + y^2 = x(x - 1)$.
 (b) Use implicit differentiation to help explain why the graph in part (a) has no horizontal tangent lines.
 (c) Solve the equation $y^4 + y^2 = x(x - 1)$ for x in terms of y and explain why the graph in part (a) consists of two parabolas.
38. Use implicit differentiation to find all points on the graph of $y^4 + y^2 = x(x - 1)$ at which the tangent line is vertical.
39. Find the values of a and b for the curve $x^2y + ay^2 = b$ if the point $(1, 1)$ is on its graph and the tangent line at $(1, 1)$ has the equation $4x + 3y = 7$.
40. At what point(s) is the tangent line to the curve $y^3 = 2x^2$ perpendicular to the line $x + 2y - 2 = 0$?

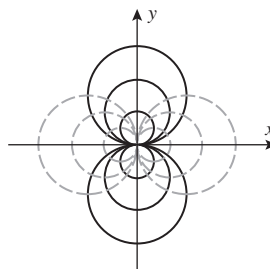
41–42 These exercises deal with the rotated ellipse C whose equation is $x^2 - xy + y^2 = 4$. ■

41. Show that the line $y = x$ intersects C at two points P and Q and that the tangent lines to C at P and Q are parallel.
42. Prove that if $P(a, b)$ is a point on C , then so is $Q(-a, -b)$ and that the tangent lines to C through P and through Q are parallel.

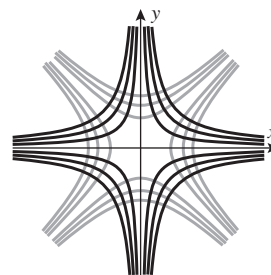
43–44 Two curves are said to be *orthogonal* if their tangent lines are perpendicular at each point of intersection, and two families of curves are said to be *orthogonal trajectories* of one another if each member of one family is orthogonal to each member of the other family. This terminology is used in these exercises. ■

43. The accompanying figure shows some typical members of the families of circles $x^2 + (y - c)^2 = c^2$ (black curves) and $(x - k)^2 + y^2 = k^2$ (gray curves). Show that these families are orthogonal trajectories of one another. [Hint: For the tangent lines to be perpendicular at a point of intersection, the slopes of those tangent lines must be negative reciprocals of one another.]

44. The accompanying figure shows some typical members of the families of hyperbolas $xy = c$ (black curves) and $x^2 - y^2 = k$ (gray curves), where $c \neq 0$ and $k \neq 0$. Use the hint in Exercise 43 to show that these families are orthogonal trajectories of one another.



▲ Figure Ex-43



▲ Figure Ex-44

45. (a) Use the implicit plotting capability of a CAS to graph the curve C whose equation is $x^3 - 2xy + y^3 = 0$.
 (b) Use the graph in part (a) to estimate the x -coordinates of a point in the first quadrant that is on C and at which the tangent line to C is parallel to the x -axis.
 (c) Find the exact value of the x -coordinate in part (b).
46. (a) Use the implicit plotting capability of a CAS to graph the curve C whose equation is $x^3 - 2xy + y^3 = 0$.
 (b) Use the graph to guess the coordinates of a point in the first quadrant that is on C and at which the tangent line to C is parallel to the line $y = -x$.
 (c) Use implicit differentiation to verify your conjecture in part (b).
47. Find dy/dx if

$$2y^3t + t^3y = 1 \quad \text{and} \quad \frac{dt}{dx} = \frac{1}{\cos t}$$

48. Find equations for two lines through the origin that are tangent to the ellipse $2x^2 - 4x + y^2 + 1 = 0$.
49. **Writing** Write a paragraph that compares the concept of an *explicit* definition of a function with that of an *implicit* definition of a function.
50. **Writing** A student asks: "Suppose implicit differentiation yields an undefined expression at a point. Does this mean that dy/dx is undefined at that point?" Using the equation $x^2 - 2xy + y^2 = 0$ as a basis for your discussion, write a paragraph that answers the student's question.

✓ QUICK CHECK ANSWERS 2.7

1. $\frac{1}{x+2}$ 2. $\frac{dy}{dx} = \frac{2x-y}{x+3y^2}$ 3. -1 4. $\frac{d^2y}{dx^2} = \sec^2 y \tan y$

2.8 RELATED RATES

In this section we will study related rates problems. In such problems one tries to find the rate at which some quantity is changing by relating the quantity to other quantities whose rates of change are known.

DIFFERENTIATING EQUATIONS TO RELATE RATES

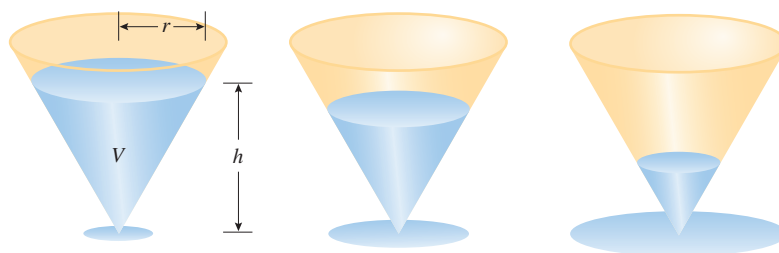
Figure 2.8.1 shows a liquid draining through a conical filter. As the liquid drains, its volume V , height h , and radius r are functions of the elapsed time t , and at each instant these variables are related by the equation

$$V = \frac{\pi}{3} r^2 h$$

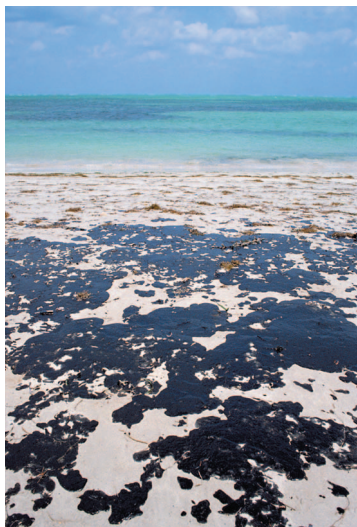
If we were interested in finding the rate of change of the volume V with respect to the time t , we could begin by differentiating both sides of this equation with respect to t to obtain

$$\frac{dV}{dt} = \frac{\pi}{3} \left[r^2 \frac{dh}{dt} + h \left(2r \frac{dr}{dt} \right) \right] = \frac{\pi}{3} \left(r^2 \frac{dh}{dt} + 2rh \frac{dr}{dt} \right)$$

Thus, to find dV/dt at a specific time t from this equation we would need to have values for r , h , dh/dt , and dr/dt at that time. This is called a **related rates problem** because the goal is to find an unknown rate of change by *relating* it to other variables whose values and whose rates of change at time t are known or can be found in some way. Let us begin with a simple example.



▲ Figure 2.8.1



Arni Katz/Phototake

Oil spill from a ruptured tanker.

► **Example 1** Suppose that x and y are differentiable functions of t and are related by the equation $y = x^3$. Find dy/dt at time $t = 1$ if $x = 2$ and $dx/dt = 4$ at time $t = 1$.

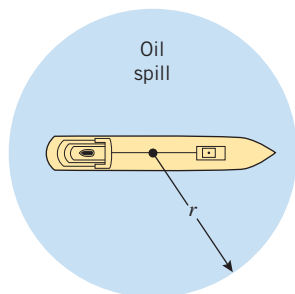
Solution. Using the chain rule to differentiate both sides of the equation $y = x^3$ with respect to t yields

$$\frac{dy}{dt} = \frac{d}{dt}[x^3] = 3x^2 \frac{dx}{dt}$$

Thus, the value of dy/dt at time $t = 1$ is

$$\left. \frac{dy}{dt} \right|_{t=1} = 3(2)^2 \left. \frac{dx}{dt} \right|_{t=1} = 12 \cdot 4 = 48 \quad \blacktriangleleft$$

► **Example 2** Assume that oil spilled from a ruptured tanker spreads in a circular pattern whose radius increases at a constant rate of 2 ft/s. How fast is the area of the spill increasing when the radius of the spill is 60 ft?



▲ Figure 2.8.2

Solution. Let

t = number of seconds elapsed from the time of the spill

r = radius of the spill in feet after t seconds

A = area of the spill in square feet after t seconds

(Figure 2.8.2). We know the rate at which the radius is increasing, and we want to find the rate at which the area is increasing at the instant when $r = 60$; that is, we want to find

$$\left. \frac{dA}{dt} \right|_{r=60} \quad \text{given that} \quad \frac{dr}{dt} = 2 \text{ ft/s}$$

This suggests that we look for an equation relating A and r that we can differentiate with respect to t to produce a relationship between dA/dt and dr/dt . But A is the area of a circle of radius r , so

$$A = \pi r^2 \quad (1)$$

Differentiating both sides of (1) with respect to t yields

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt} \quad (2)$$

Thus, when $r = 60$ the area of the spill is increasing at the rate of

$$\left. \frac{dA}{dt} \right|_{r=60} = 2\pi(60)(2) = 240\pi \text{ ft}^2/\text{s} \approx 754 \text{ ft}^2/\text{s} \quad \blacktriangleleft$$

With some minor variations, the method used in Example 2 can be used to solve a variety of related rates problems. We can break the method down into five steps.

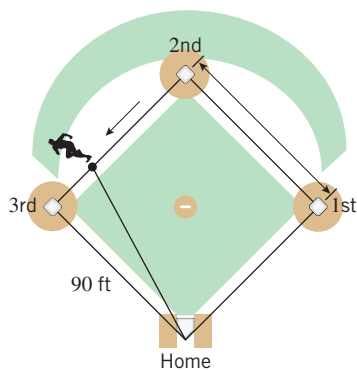
WARNING

We have italicized the word “After” in Step 5 because it is a common error to substitute numerical values before performing the differentiation. For instance, in Example 2 had we substituted the known value of $r = 60$ in (1) before differentiating, we would have obtained $dA/dt = 0$, which is obviously incorrect.

A Strategy for Solving Related Rates Problems

- Step 1.** Assign letters to all quantities that vary with time and any others that seem relevant to the problem. Give a definition for each letter.
- Step 2.** Identify the rates of change that are known and the rate of change that is to be found. Interpret each rate as a derivative.
- Step 3.** Find an equation that relates the variables whose rates of change were identified in Step 2. To do this, it will often be helpful to draw an appropriately labeled figure that illustrates the relationship.
- Step 4.** Differentiate both sides of the equation obtained in Step 3 with respect to time to produce a relationship between the known rates of change and the unknown rate of change.
- Step 5.** *After* completing Step 4, substitute all known values for the rates of change and the variables, and then solve for the unknown rate of change.

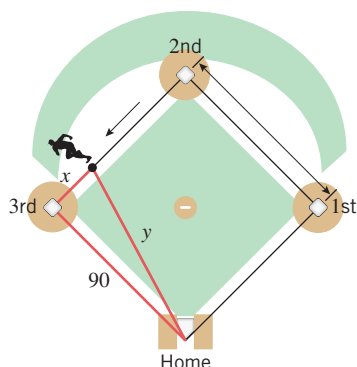
► **Example 3** A baseball diamond is a square whose sides are 90 ft long (Figure 2.8.3). Suppose that a player running from second base to third base has a speed of 30 ft/s at the instant when he is 20 ft from third base. At what rate is the player’s distance from home plate changing at that instant?



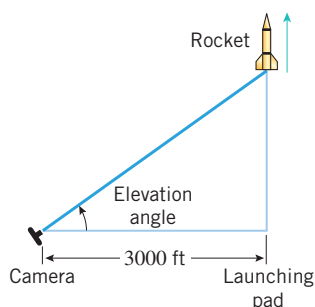
▲ Figure 2.8.3

The quantity

$$\left. \frac{dx}{dt} \right|_{x=20}$$

is negative because x is decreasing with respect to t .

▲ Figure 2.8.4



▲ Figure 2.8.5

Solution. We are given a constant speed with which the player is approaching third base, and we want to find the rate of change of the distance between the player and home plate at a particular instant. Thus, let

t = number of seconds since the player left second base

x = distance in feet from the player to third base

y = distance in feet from the player to home plate

(Figure 2.8.4). Thus, we want to find

$$\left. \frac{dy}{dt} \right|_{x=20} \quad \text{given that} \quad \left. \frac{dx}{dt} \right|_{x=20} = -30 \text{ ft/s}$$

As suggested by Figure 2.8.4, an equation relating the variables x and y can be obtained using the Theorem of Pythagoras:

$$x^2 + 90^2 = y^2 \quad (3)$$

Differentiating both sides of this equation with respect to t yields

$$2x \frac{dx}{dt} = 2y \frac{dy}{dt}$$

from which we obtain

$$\frac{dy}{dt} = \frac{x}{y} \frac{dx}{dt} \quad (4)$$

When $x = 20$, it follows from (3) that

$$y = \sqrt{20^2 + 90^2} = \sqrt{8500} = 10\sqrt{85}$$

so that (4) yields

$$\left. \frac{dy}{dt} \right|_{x=20} = \frac{20}{10\sqrt{85}}(-30) = -\frac{60}{\sqrt{85}} \approx -6.51 \text{ ft/s}$$

The negative sign in the answer tells us that y is decreasing, which makes sense physically from Figure 2.8.4. ◀

► **Example 4** In Figure 2.8.5 we have shown a camera mounted at a point 3000 ft from the base of a rocket launching pad. If the rocket is rising vertically at 880 ft/s when it is 4000 ft above the launching pad, how fast must the camera elevation angle change at that instant to keep the camera aimed at the rocket?

Solution. Let

t = number of seconds elapsed from the time of launch

ϕ = camera elevation angle in radians after t seconds

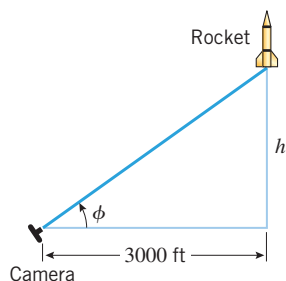
h = height of the rocket in feet after t seconds

(Figure 2.8.6). At each instant the rate at which the camera elevation angle must change is $d\phi/dt$, and the rate at which the rocket is rising is dh/dt . We want to find

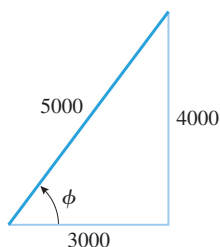
$$\left. \frac{d\phi}{dt} \right|_{h=4000} \quad \text{given that} \quad \left. \frac{dh}{dt} \right|_{h=4000} = 880 \text{ ft/s}$$

From Figure 2.8.6 we see that

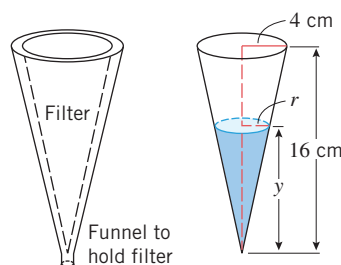
$$\tan \phi = \frac{h}{3000} \quad (5)$$



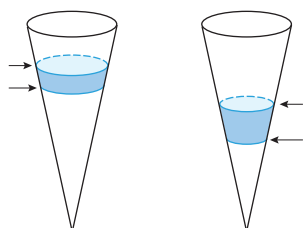
▲ Figure 2.8.6



▲ Figure 2.8.7



▲ Figure 2.8.8



The same volume has drained, but the change in height is greater near the bottom than near the top.

▲ Figure 2.8.9

Differentiating both sides of (5) with respect to t yields

$$(\sec^2 \phi) \frac{d\phi}{dt} = \frac{1}{3000} \frac{dh}{dt} \quad (6)$$

When $h = 4000$, it follows that

$$(\sec \phi)|_{h=4000} = \frac{5000}{3000} = \frac{5}{3}$$

(see Figure 2.8.7), so that from (6)

$$\begin{aligned} \left(\frac{5}{3}\right)^2 \frac{d\phi}{dt} \Big|_{h=4000} &= \frac{1}{3000} \cdot 880 = \frac{22}{75} \\ \frac{d\phi}{dt} \Big|_{h=4000} &= \frac{22}{75} \cdot \frac{9}{25} = \frac{66}{625} \approx 0.11 \text{ rad/s} \approx 6.05 \text{ deg/s} \quad \blacktriangleleft \end{aligned}$$

► **Example 5** Suppose that liquid is to be cleared of sediment by allowing it to drain through a conical filter that is 16 cm high and has a radius of 4 cm at the top (Figure 2.8.8). Suppose also that the liquid is forced out of the cone at a constant rate of $2 \text{ cm}^3/\text{min}$.

- Do you think that the depth of the liquid will decrease at a constant rate? Give a verbal argument that justifies your conclusion.
- Find a formula that expresses the rate at which the depth of the liquid is changing in terms of the depth, and use that formula to determine whether your conclusion in part (a) is correct.
- At what rate is the depth of the liquid changing at the instant when the liquid in the cone is 8 cm deep?

Solution (a). For the volume of liquid to decrease by a *fixed amount*, it requires a greater decrease in depth when the cone is close to empty than when it is almost full (Figure 2.8.9). This suggests that for the volume to decrease at a constant rate, the depth must decrease at an increasing rate.

Solution (b). Let

- t = time elapsed from the initial observation (min)
- V = volume of liquid in the cone at time t (cm^3)
- y = depth of the liquid in the cone at time t (cm)
- r = radius of the liquid surface at time t (cm)

(Figure 2.8.8). At each instant the rate at which the volume of liquid is changing is dV/dt , and the rate at which the depth is changing is dy/dt . We want to express dy/dt in terms of y given that dV/dt has a constant value of $dV/dt = -2$. (We must use a minus sign here because V decreases as t increases.)

From the formula for the volume of a cone, the volume V , the radius r , and the depth y are related by

$$V = \frac{1}{3}\pi r^2 y \quad (7)$$

If we differentiate both sides of (7) with respect to t , the right side will involve the quantity dr/dt . Since we have no direct information about dr/dt , it is desirable to eliminate r from (7) before differentiating. This can be done using similar triangles. From Figure 2.8.8 we see that

$$\frac{r}{y} = \frac{4}{16} \quad \text{or} \quad r = \frac{1}{4}y$$

Substituting this expression in (7) gives

$$V = \frac{\pi}{48}y^3 \quad (8)$$

Differentiating both sides of (8) with respect to t we obtain

$$\frac{dV}{dt} = \frac{\pi}{48} \left(3y^2 \frac{dy}{dt} \right)$$

or

$$\frac{dy}{dt} = \frac{16}{\pi y^2} \frac{dV}{dt} = \frac{16}{\pi y^2} (-2) = -\frac{32}{\pi y^2} \quad (9)$$

which expresses dy/dt in terms of y . The minus sign tells us that y is decreasing with time, and

$$\left| \frac{dy}{dt} \right| = \frac{32}{\pi y^2}$$

tells us how fast y is decreasing. From this formula we see that $|dy/dt|$ increases as y decreases, which confirms our conjecture in part (a) that the depth of the liquid decreases more quickly as the liquid drains through the filter.

Solution (c). The rate at which the depth is changing when the depth is 8 cm can be obtained from (9) with $y = 8$:

$$\left. \frac{dy}{dt} \right|_{y=8} = -\frac{32}{\pi(8^2)} = -\frac{1}{2\pi} \approx -0.16 \text{ cm/min} \quad \blacktriangleleft$$

✓ QUICK CHECK EXERCISES 2.8 (See page 175 for answers.)

1. If $A = x^2$ and $\frac{dx}{dt} = 3$, find $\left. \frac{dA}{dt} \right|_{x=10}$.
2. If $A = x^2$ and $\frac{dA}{dt} = 3$, find $\left. \frac{dx}{dt} \right|_{x=10}$.
3. A 10-foot ladder stands on a horizontal floor and leans against a vertical wall. Use x to denote the distance along the floor from the wall to the foot of the ladder, and use y to denote the distance along the wall from the floor to the

top of the ladder. If the foot of the ladder is dragged away from the wall, find an equation that relates rates of change of x and y with respect to time.

4. Suppose that a block of ice in the shape of a right circular cylinder melts so that it retains its cylindrical shape. Find an equation that relates the rates of change of the volume (V), height (h), and radius (r) of the block of ice.

EXERCISE SET 2.8

1–4 Both x and y denote functions of t that are related by the given equation. Use this equation and the given derivative information to find the specified derivative. ■

1. Equation: $y = 3x + 5$.
 - (a) Given that $dx/dt = 2$, find dy/dt when $x = 1$.
 - (b) Given that $dy/dt = -1$, find dx/dt when $x = 0$.
2. Equation: $x + 4y = 3$.
 - (a) Given that $dx/dt = 1$, find dy/dt when $x = 2$.
 - (b) Given that $dy/dt = 4$, find dx/dt when $x = 3$.
3. Equation: $4x^2 + 9y^2 = 1$.
 - (a) Given that $dx/dt = 3$, find dy/dt when $(x, y) = \left(\frac{1}{2\sqrt{2}}, \frac{1}{3\sqrt{2}} \right)$.

- (b) Given that $dy/dt = 8$, find dx/dt when $(x, y) = \left(\frac{1}{3}, -\frac{\sqrt{5}}{9} \right)$.

4. Equation: $x^2 + y^2 = 2x + 4y$.
 - (a) Given that $dx/dt = -5$, find dy/dt when $(x, y) = (3, 1)$.
 - (b) Given that $dy/dt = 6$, find dx/dt when $(x, y) = (1 + \sqrt{2}, 2 + \sqrt{3})$.

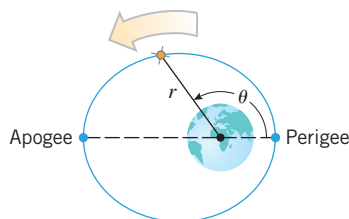
FOCUS ON CONCEPTS

5. Let A be the area of a square whose sides have length x , and assume that x varies with the time t .
 - (a) Draw a picture of the square with the labels A and x placed appropriately.
 - (b) Write an equation that relates A and x .

(cont.)

- (c) Use the equation in part (b) to find an equation that relates dA/dt and dx/dt .
- (d) At a certain instant the sides are 3 ft long and increasing at a rate of 2 ft/min. How fast is the area increasing at that instant?
6. In parts (a)–(d), let A be the area of a circle of radius r , and assume that r increases with the time t .
- (a) Draw a picture of the circle with the labels A and r placed appropriately.
- (b) Write an equation that relates A and r .
- (c) Use the equation in part (b) to find an equation that relates dA/dt and dr/dt .
- (d) At a certain instant the radius is 5 cm and increasing at the rate of 2 cm/s. How fast is the area increasing at that instant?
7. Let V be the volume of a cylinder having height h and radius r , and assume that h and r vary with time.
- (a) How are dV/dt , dh/dt , and dr/dt related?
- (b) At a certain instant, the height is 6 in and increasing at 1 in/s, while the radius is 10 in and decreasing at 1 in/s. How fast is the volume changing at that instant? Is the volume increasing or decreasing at that instant?
8. Let l be the length of a diagonal of a rectangle whose sides have lengths x and y , and assume that x and y vary with time.
- (a) How are dl/dt , dx/dt , and dy/dt related?
- (b) If x increases at a constant rate of $\frac{1}{2}$ ft/s and y decreases at a constant rate of $\frac{1}{4}$ ft/s, how fast is the size of the diagonal changing when $x = 3$ ft and $y = 4$ ft? Is the diagonal increasing or decreasing at that instant?
9. Let θ (in radians) be an acute angle in a right triangle, and let x and y , respectively, be the lengths of the sides adjacent to and opposite θ . Suppose also that x and y vary with time.
- (a) How are $d\theta/dt$, dx/dt , and dy/dt related?
- (b) At a certain instant, $x = 2$ units and is increasing at 1 unit/s, while $y = 2$ units and is decreasing at $\frac{1}{4}$ unit/s. How fast is θ changing at that instant? Is θ increasing or decreasing at that instant?
10. Suppose that $z = x^3y^2$, where both x and y are changing with time. At a certain instant when $x = 1$ and $y = 2$, x is decreasing at the rate of 2 units/s, and y is increasing at the rate of 3 units/s. How fast is z changing at this instant? Is z increasing or decreasing?
11. The minute hand of a certain clock is 4 in long. Starting from the moment when the hand is pointing straight up, how fast is the area of the sector that is swept out by the hand increasing at any instant during the next revolution of the hand?
12. A stone dropped into a still pond sends out a circular ripple whose radius increases at a constant rate of 3 ft/s. How rapidly is the area enclosed by the ripple increasing at the end of 10 s?
13. Oil spilled from a ruptured tanker spreads in a circle whose area increases at a constant rate of 6 mi²/h. How fast is the radius of the spill increasing when the area is 9 mi²?
14. A spherical balloon is inflated so that its volume is increasing at the rate of 3 ft³/min. How fast is the diameter of the balloon increasing when the radius is 1 ft?
15. A spherical balloon is to be deflated so that its radius decreases at a constant rate of 15 cm/min. At what rate must air be removed when the radius is 9 cm?
16. A 17 ft ladder is leaning against a wall. If the bottom of the ladder is pulled along the ground away from the wall at a constant rate of 5 ft/s, how fast will the top of the ladder be moving down the wall when it is 8 ft above the ground?
17. A 13 ft ladder is leaning against a wall. If the top of the ladder slips down the wall at a rate of 2 ft/s, how fast will the foot be moving away from the wall when the top is 5 ft above the ground?
18. A 10 ft plank is leaning against a wall. If at a certain instant the bottom of the plank is 2 ft from the wall and is being pushed toward the wall at the rate of 6 in/s, how fast is the acute angle that the plank makes with the ground increasing?
19. A softball diamond is a square whose sides are 60 ft long. Suppose that a player running from first to second base has a speed of 25 ft/s at the instant when she is 10 ft from second base. At what rate is the player's distance from home plate changing at that instant?
20. A rocket, rising vertically, is tracked by a radar station that is on the ground 5 mi from the launchpad. How fast is the rocket rising when it is 4 mi high and its distance from the radar station is increasing at a rate of 2000 mi/h?
21. For the camera and rocket shown in Figure 2.8.5, at what rate is the camera-to-rocket distance changing when the rocket is 4000 ft up and rising vertically at 880 ft/s?
22. For the camera and rocket shown in Figure 2.8.5, at what rate is the rocket rising when the elevation angle is $\pi/4$ radians and increasing at a rate of 0.2 rad/s?
23. A satellite is in an elliptical orbit around the Earth. Its distance r (in miles) from the center of the Earth is given by
- $$r = \frac{4995}{1 + 0.12 \cos \theta}$$
- where θ is the angle measured from the point on the orbit nearest the Earth's surface (see the accompanying figure on the next page).
- (a) Find the altitude of the satellite at *perigee* (the point nearest the surface of the Earth) and at *apogee* (the point farthest from the surface of the Earth). Use 3960 mi as the radius of the Earth.
- (b) At the instant when θ is 120° , the angle θ is increasing at the rate of $2.7^\circ/\text{min}$. Find the altitude of the

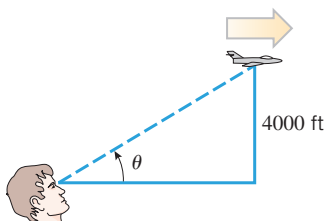
satellite and the rate at which the altitude is changing at this instant. Express the rate in units of mi/min.



◀ Figure Ex-23

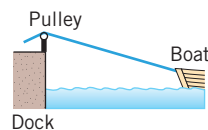
24. An aircraft is flying horizontally at a constant height of 4000 ft above a fixed observation point (see the accompanying figure). At a certain instant the angle of elevation θ is 30° and decreasing, and the speed of the aircraft is 300 mi/h.

- How fast is θ decreasing at this instant? Express the result in units of deg/s.
- How fast is the distance between the aircraft and the observation point changing at this instant? Express the result in units of ft/s. Use 1 mi = 5280 ft.



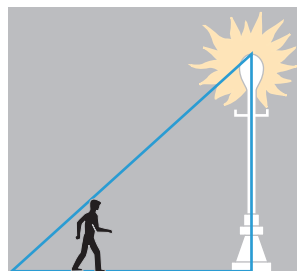
◀ Figure Ex-24

- A conical water tank with vertex down has a radius of 10 ft at the top and is 24 ft high. If water flows into the tank at a rate of $20 \text{ ft}^3/\text{min}$, how fast is the depth of the water increasing when the water is 16 ft deep?
- Grain pouring from a chute at the rate of $8 \text{ ft}^3/\text{min}$ forms a conical pile whose height is always twice its radius. How fast is the height of the pile increasing at the instant when the pile is 6 ft high?
- Sand pouring from a chute forms a conical pile whose height is always equal to the diameter. If the height increases at a constant rate of 5 ft/min, at what rate is sand pouring from the chute when the pile is 10 ft high?
- Wheat is poured through a chute at the rate of $10 \text{ ft}^3/\text{min}$ and falls in a conical pile whose bottom radius is always half the altitude. How fast will the circumference of the base be increasing when the pile is 8 ft high?
- An aircraft is climbing at a 30° angle to the horizontal. How fast is the aircraft gaining altitude if its speed is 500 mi/h?
- A boat is pulled into a dock by means of a rope attached to a pulley on the dock (see the accompanying figure). The rope is attached to the bow of the boat at a point 10 ft below the pulley. If the rope is pulled through the pulley at a rate of 20 ft/min, at what rate will the boat be approaching the dock when 125 ft of rope is out?



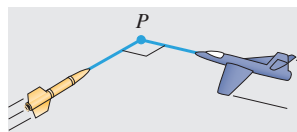
◀ Figure Ex-30

- For the boat in Exercise 30, how fast must the rope be pulled if we want the boat to approach the dock at a rate of 12 ft/min at the instant when 125 ft of rope is out?
- A man 6 ft tall is walking at the rate of 3 ft/s toward a streetlight 18 ft high (see the accompanying figure).
 - At what rate is his shadow length changing?
 - How fast is the tip of his shadow moving?



◀ Figure Ex-32

- A beacon that makes one revolution every 10 s is located on a ship anchored 4 kilometers from a straight shoreline. How fast is the beam moving along the shoreline when it makes an angle of 45° with the shore?
- An aircraft is flying at a constant altitude with a constant speed of 600 mi/h. An anti-aircraft missile is fired on a straight line perpendicular to the flight path of the aircraft so that it will hit the aircraft at a point P (see the accompanying figure). At the instant the aircraft is 2 mi from the impact point P the missile is 4 mi from P and flying at 1200 mi/h. At that instant, how rapidly is the distance between missile and aircraft decreasing?



◀ Figure Ex-34

- Solve Exercise 34 under the assumption that the angle between the flight paths is 120° instead of the assumption that the paths are perpendicular. [Hint: Use the law of cosines.]
- A police helicopter is flying due north at 100 mi/h and at a constant altitude of $\frac{1}{2}$ mi. Below, a car is traveling west on a highway at 75 mi/h. At the moment the helicopter crosses over the highway the car is 2 mi east of the helicopter.
 - How fast is the distance between the car and helicopter changing at the moment the helicopter crosses the highway?
 - Is the distance between the car and helicopter increasing or decreasing at that moment?

37. A particle is moving along the curve whose equation is

$$\frac{xy^3}{1+y^2} = \frac{8}{5}$$

Assume that the x -coordinate is increasing at the rate of 6 units/s when the particle is at the point $(1, 2)$.

- (a) At what rate is the y -coordinate of the point changing at that instant?
 (b) Is the particle rising or falling at that instant?
38. A point P is moving along the curve whose equation is $y = \sqrt{x^3 + 17}$. When P is at $(2, 5)$, y is increasing at the rate of 2 units/s. How fast is x changing?
39. A point P is moving along the line whose equation is $y = 2x$. How fast is the distance between P and the point $(3, 0)$ changing at the instant when P is at $(3, 6)$ if x is decreasing at the rate of 2 units/s at that instant?
40. A point P is moving along the curve whose equation is $y = \sqrt{x}$. Suppose that x is increasing at the rate of 4 units/s when $x = 3$.
 (a) How fast is the distance between P and the point $(2, 0)$ changing at this instant?
 (b) How fast is the angle of inclination of the line segment from P to $(2, 0)$ changing at this instant?
41. A particle is moving along the curve $y = x/(x^2 + 1)$. Find all values of x at which the rate of change of x with respect to time is three times that of y . [Assume that dx/dt is never zero.]
42. A particle is moving along the curve $16x^2 + 9y^2 = 144$. Find all points (x, y) at which the rates of change of x and y with respect to time are equal. [Assume that dx/dt and dy/dt are never both zero at the same point.]
43. The **thin lens equation** in physics is

$$\frac{1}{s} + \frac{1}{S} = \frac{1}{f}$$

where s is the object distance from the lens, S is the image distance from the lens, and f is the focal length of the lens. Suppose that a certain lens has a focal length of 6 cm and that an object is moving toward the lens at the rate of 2 cm/s. How fast is the image distance changing at the instant when the object is 10 cm from the lens? Is the image moving away from the lens or toward the lens?

44. Water is stored in a cone-shaped reservoir (vertex down). Assuming the water evaporates at a rate proportional to the surface area exposed to the air, show that the depth of the water will decrease at a constant rate that does not depend on the dimensions of the reservoir.
45. A meteor enters the Earth's atmosphere and burns up at a rate that, at each instant, is proportional to its surface area. Assuming that the meteor is always spherical, show that the radius decreases at a constant rate.
46. On a certain clock the minute hand is 4 in long and the hour hand is 3 in long. How fast is the distance between the tips of the hands changing at 9 o'clock?
47. Coffee is poured at a uniform rate of $20 \text{ cm}^3/\text{s}$ into a cup whose inside is shaped like a truncated cone (see the accompanying figure). If the upper and lower radii of the cup are 4 cm and 2 cm and the height of the cup is 6 cm, how fast will the coffee level be rising when the coffee is halfway up? [Hint: Extend the cup downward to form a cone.]



◀ Figure Ex-47

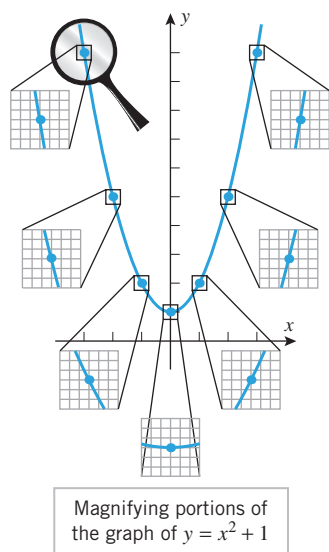
✓ QUICK CHECK ANSWERS 2.8

1. 60 2. $\frac{3}{20}$ 3. $x \frac{dx}{dt} + y \frac{dy}{dt} = 0$ 4. $\frac{dV}{dt} = 2\pi r h \frac{dr}{dt} + \pi r^2 \frac{dh}{dt}$

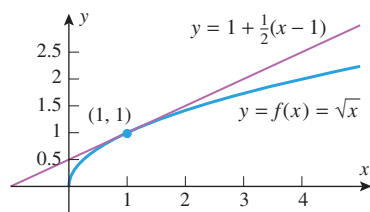
2.9 LOCAL LINEAR APPROXIMATION; DIFFERENTIALS

In this section we will show how derivatives can be used to approximate nonlinear functions by linear functions. Also, up to now we have been interpreting dy/dx as a single entity representing the derivative. In this section we will define the quantities dx and dy themselves, thereby allowing us to interpret dy/dx as an actual ratio.

Recall from Section 2.2 that if a function f is differentiable at x_0 , then a sufficiently magnified portion of the graph of f centered at the point $P(x_0, f(x_0))$ takes on the appearance of a straight line segment. Figure 2.9.1 illustrates this at several points on the graph of $y = x^2 + 1$. For this reason, a function that is differentiable at x_0 is sometimes said to be **locally linear** at x_0 .



▲ Figure 2.9.1



▲ Figure 2.9.2

Examples 1 and 2 illustrate important ideas and are not meant to suggest that you should use local linear approximations for computations that your calculating utility can perform. The main application of local linear approximation is in modeling problems where it is useful to replace complicated functions by simpler ones.

The line that best approximates the graph of f in the vicinity of $P(x_0, f(x_0))$ is the tangent line to the graph of f at x_0 , given by the equation

$$y = f(x_0) + f'(x_0)(x - x_0)$$

[see Formula (3) of Section 2.2]. Thus, for values of x near x_0 we can approximate values of $f(x)$ by

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) \quad (1)$$

This is called the **local linear approximation** of f at x_0 . This formula can also be expressed in terms of the increment $\Delta x = x - x_0$ as

$$f(x_0 + \Delta x) \approx f(x_0) + f'(x_0)\Delta x \quad (2)$$

► Example 1

- Find the local linear approximation of $f(x) = \sqrt{x}$ at $x_0 = 1$.
- Use the local linear approximation obtained in part (a) to approximate $\sqrt{1.1}$, and compare your approximation to the result produced directly by a calculating utility.

Solution (a). Since $f'(x) = 1/(2\sqrt{x})$, it follows from (1) that the local linear approximation of $\sqrt{x}$ at a point x_0 is

$$\sqrt{x} \approx \sqrt{x_0} + \frac{1}{2\sqrt{x_0}}(x - x_0)$$

Thus, the local linear approximation at $x_0 = 1$ is

$$\sqrt{x} \approx 1 + \frac{1}{2}(x - 1) \quad (3)$$

The graphs of $y = \sqrt{x}$ and the local linear approximation $y = 1 + \frac{1}{2}(x - 1)$ are shown in Figure 2.9.2.

Solution (b). Applying (3) with $x = 1.1$ yields

$$\sqrt{1.1} \approx 1 + \frac{1}{2}(1.1 - 1) = 1.05$$

Since the tangent line $y = 1 + \frac{1}{2}(x - 1)$ in Figure 2.9.2 lies above the graph of $f(x) = \sqrt{x}$, we would expect this approximation to be slightly too large. This expectation is confirmed by the calculator approximation $\sqrt{1.1} \approx 1.04881$. ◀

► Example 2

- Find the local linear approximation of $f(x) = \sin x$ at $x_0 = 0$.
- Use the local linear approximation obtained in part (a) to approximate $\sin 2^\circ$, and compare your approximation to the result produced directly by your calculating device.

Solution (a). Since $f'(x) = \cos x$, it follows from (1) that the local linear approximation of $\sin x$ at a point x_0 is

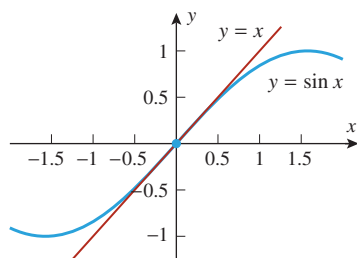
$$\sin x \approx \sin x_0 + (\cos x_0)(x - x_0)$$

Thus, the local linear approximation at $x_0 = 0$ is

$$\sin x \approx \sin 0 + (\cos 0)(x - 0)$$

which simplifies to

$$\sin x \approx x \quad (4)$$



▲ Figure 2.9.3

Solution (b). The variable x in (4) is in radian measure, so we must first convert 2° to radians before we can apply this approximation. Since

$$2^\circ = 2 \left(\frac{\pi}{180} \right) = \frac{\pi}{90} \approx 0.0349066 \text{ radian}$$

it follows from (4) that $\sin 2^\circ \approx 0.0349066$. Comparing the two graphs in Figure 2.9.3, we would expect this approximation to be slightly larger than the exact value. The calculator approximation $\sin 2^\circ \approx 0.0348995$ shows that this is indeed the case. ◀

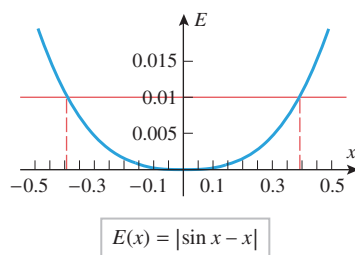
■ ERROR IN LOCAL LINEAR APPROXIMATIONS

As a general rule, the accuracy of the local linear approximation to $f(x)$ at x_0 will deteriorate as x gets progressively farther from x_0 . To illustrate this for the approximation $\sin x \approx x$ in Example 2, let us graph the function

$$E(x) = |\sin x - x|$$

which is the absolute value of the error in the approximation (Figure 2.9.4).

In Figure 2.9.4, the graph shows how the absolute error in the local linear approximation of $\sin x$ increases as x moves progressively farther from 0 in either the positive or negative direction. The graph also tells us that for values of x between the two vertical lines, the absolute error does not exceed 0.01. Thus, for example, we could use the local linear approximation $\sin x \approx x$ for all values of x in the interval $-0.35 < x < 0.35$ (radians) with confidence that the approximation is within ± 0.01 of the exact value.



▲ Figure 2.9.4

■ DIFFERENTIALS

Newton and Leibniz each used a different notation when they published their discoveries of calculus, thereby creating a notational divide between Britain and the European continent that lasted for more than 50 years. The **Leibniz notation** dy/dx eventually prevailed because it suggests correct formulas in a natural way, the chain rule

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

being a good example.

Up to now we have interpreted dy/dx as a single entity representing the derivative of y with respect to x ; the symbols “ dy ” and “ dx ,” which are called **differentials**, have had no meanings attached to them. Our next goal is to define these symbols in such a way that dy/dx can be treated as an actual ratio. To do this, assume that f is differentiable at a point x , *define* dx to be an independent variable that can have any real value, and *define* dy by the formula

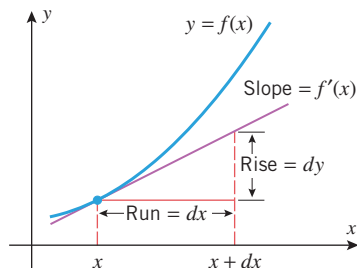
$$dy = f'(x) dx \quad (5)$$

If $dx \neq 0$, then we can divide both sides of (5) by dx to obtain

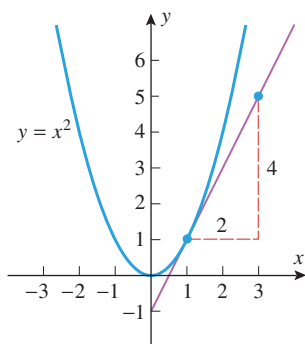
$$\frac{dy}{dx} = f'(x) \quad (6)$$

Thus, we have achieved our goal of defining dy and dx so their ratio is $f'(x)$. Formula (5) is said to express (6) in **differential form**.

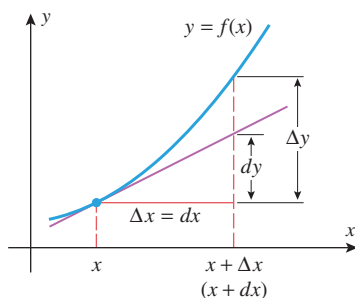
To interpret (5) geometrically, note that $f'(x)$ is the slope of the tangent line to the graph of f at x . The differentials dy and dx can be viewed as a corresponding rise and run of this tangent line (Figure 2.9.5).



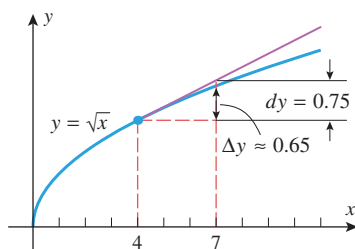
▲ Figure 2.9.5



▲ Figure 2.9.6



▲ Figure 2.9.7



▲ Figure 2.9.8

► **Example 3** Express the derivative with respect to x of $y = x^2$ in differential form, and discuss the relationship between dy and dx at $x = 1$.

Solution. The derivative of y with respect to x is $dy/dx = 2x$, which can be expressed in differential form as

$$dy = 2x \, dx$$

When $x = 1$ this becomes

$$dy = 2 \, dx$$

This tells us that if we travel along the tangent line to the curve $y = x^2$ at $x = 1$, then a change of dx units in x produces a change of $2 \, dx$ units in y . Thus, for example, a run of $dx = 2$ units produces a rise of $dy = 4$ units along the tangent line (Figure 2.9.6). ◀

It is important to understand the distinction between the increment Δy and the differential dy . To see the difference, let us assign the independent variables dx and Δx the same value, so $dx = \Delta x$. Then Δy represents the change in y that occurs when we start at x and travel *along the curve* $y = f(x)$ until we have moved $\Delta x (= dx)$ units in the x -direction, while dy represents the change in y that occurs if we start at x and travel *along the tangent line* until we have moved $dx (= \Delta x)$ units in the x -direction (Figure 2.9.7).

► **Example 4** Let $y = \sqrt{x}$.

(a) Find formulas for Δy and dy .

(b) Evaluate Δy and dy at $x = 4$ with $dx = \Delta x = 3$. Then make a sketch of $y = \sqrt{x}$ showing the values of Δy and dy in the picture.

Solution (a). With $y = f(x) = \sqrt{x}$ we obtain

$$\Delta y = f(x + \Delta x) - f(x) = \sqrt{x + \Delta x} - \sqrt{x}$$

and

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}, \quad \text{so} \quad dy = \frac{1}{2\sqrt{x}} \, dx$$

Solution (b). At $x = 4$ with $dx = \Delta x = 3$,

$$\Delta y = \sqrt{7} - \sqrt{4} \approx 0.65$$

and

$$dy = \frac{1}{2\sqrt{4}}(3) = \frac{3}{4} = 0.75$$

Figure 2.9.8 shows the curve $y = \sqrt{x}$ together with Δy and dy . ◀

LOCAL LINEAR APPROXIMATION FROM THE DIFFERENTIAL POINT OF VIEW

Although Δy and dy are generally different, the differential dy will nonetheless be a good approximation of Δy provided $dx = \Delta x$ is close to 0. To see this, recall from Section 2.2 that

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

It follows that if Δx is close to 0, then we will have $f'(x) \approx \Delta y / \Delta x$ or, equivalently,

$$\Delta y \approx f'(x) \Delta x$$

If we agree to let $dx = \Delta x$, then we can rewrite this as

$$\Delta y \approx f'(x) \, dx = dy \tag{7}$$

In words, this states that for values of dx near zero the differential dy closely approximates the increment Δy (Figure 2.9.7). But this is to be expected since the graph of the tangent line at x is the local linear approximation of the graph of f .

■ ERROR PROPAGATION



© Michael Newman/PhotoEdit
Real-world measurements inevitably have small errors.

Note that measurement error is positive if the measured value is greater than the exact value and is negative if it is less than the exact value. The sign of the propagated error conveys similar information.

Explain why an error estimate of at most $\pm \frac{1}{32}$ inch is reasonable for a ruler that is calibrated in sixteenths of an inch.

In real-world applications, small errors in measured quantities will invariably occur. These measurement errors are of importance in scientific research—all scientific measurements come with measurement errors included. For example, your height might be measured as 170 ± 0.5 cm, meaning that your exact height lies somewhere between 169.5 and 170.5 cm. Researchers often must use these inexactly measured quantities to compute other quantities, thereby *propagating* the errors from the measured quantities to the computed quantities. This phenomenon is called **error propagation**. Researchers must be able to estimate errors in the computed quantities. Our goal is to show how to estimate these errors using local linear approximation and differentials. For this purpose, suppose

x_0 is the exact value of the quantity being measured

$y_0 = f(x_0)$ is the exact value of the quantity being computed

x is the measured value of x_0

$y = f(x)$ is the computed value of y

We define

$dx (= \Delta x) = x - x_0$ to be the **measurement error** of x

$\Delta y = f(x) - f(x_0)$ to be the **propagated error** of y

It follows from (7) with x_0 replacing x that the propagated error Δy can be approximated by

$$\Delta y \approx dy = f'(x_0) dx \quad (8)$$

Unfortunately, there is a practical difficulty in applying this formula since the value of x_0 is unknown. (Keep in mind that only the measured value x is known to the researcher.) This being the case, it is standard practice in research to use the measured value x in place of x_0 in (8) and use the approximation

$$\Delta y \approx dy = f'(x) dx \quad (9)$$

for the propagated error.

► **Example 5** Suppose that the side of a square is measured with a ruler to be 10 inches with a measurement error of at most $\pm \frac{1}{32}$ in. Estimate the error in the computed area of the square.

Solution. Let x denote the exact length of a side and y the exact area so that $y = x^2$. It follows from (9) with $f(x) = x^2$ that if dx is the measurement error, then the propagated error Δy can be approximated as

$$\Delta y \approx dy = 2x dx$$

Substituting the measured value $x = 10$ into this equation yields

$$dy = 20 dx \quad (10)$$

But to say that the measurement error is at most $\pm \frac{1}{32}$ means that

$$-\frac{1}{32} \leq dx \leq \frac{1}{32}$$

Multiplying these inequalities through by 20 and applying (10) yields

$$20 \left(-\frac{1}{32}\right) \leq dy \leq 20 \left(\frac{1}{32}\right) \quad \text{or equivalently} \quad -\frac{5}{8} \leq dy \leq \frac{5}{8}$$

Thus, the propagated error in the area is estimated to be within $\pm \frac{5}{8}$ in². ◀

If the true value of a quantity is q and a measurement or calculation produces an error Δq , then $\Delta q/q$ is called the **relative error** in the measurement or calculation; when expressed

as a percentage, $\Delta q/q$ is called the **percentage error**. As a practical matter, the true value q is usually unknown, so that the measured or calculated value of q is used instead; and the relative error is approximated by dq/q .

► **Example 6** The diameter of a polyurethane sphere is measured with percentage error within $\pm 0.4\%$. Estimate the percentage error in the calculated volume of the sphere.

Solution. A sphere of diameter x has radius $r = x/2$ and volume

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi \left(\frac{x}{2}\right)^3 = \frac{1}{6}\pi x^3$$

Therefore

$$\frac{dV}{dx} = \frac{1}{2}\pi x^2 \quad \text{and} \quad dV = \frac{1}{2}\pi x^2 dx$$

The relative error in V is then approximately

$$\frac{dV}{V} = \frac{\frac{1}{2}\pi x^2 dx}{\frac{1}{6}\pi x^3} = 3 \frac{dx}{x} \quad (11)$$

We are given that the percentage error in the measured value of x is within $\pm 0.4\%$, which means that

$$-0.004 \leq \frac{dx}{x} \leq 0.004$$

Multiplying these inequalities through by 3 and applying (11) yields

$$-0.012 \leq \frac{dV}{V} \leq 0.012$$

Thus, we estimate the percentage error in the calculated value of V to be within $\pm 1.2\%$.



© Sergiy Serdyuk/iStockphoto

Polyurethane spheres are used in the maintenance of oil and gas pipelines.

■ MORE NOTATION; DIFFERENTIAL FORMULAS

The symbol df is another common notation for the differential of a function $y = f(x)$. For example, if $f(x) = \sin x$, then we can write $df = \cos x dx$. We can also view the symbol “ d ” as an *operator* that acts on a function to produce the corresponding differential. For example, $d[x^2] = 2x dx$, $d[\sin x] = \cos x dx$, and so on. All of the general rules of differentiation then have corresponding differential versions:

DERIVATIVE FORMULA	DIFFERENTIAL FORMULA
$\frac{d}{dx}[c] = 0$	$d[c] = 0$
$\frac{d}{dx}[cf] = c \frac{df}{dx}$	$d[cf] = c df$
$\frac{d}{dx}[f+g] = \frac{df}{dx} + \frac{dg}{dx}$	$d[f+g] = df + dg$
$\frac{d}{dx}[fg] = f \frac{dg}{dx} + g \frac{df}{dx}$	$d[fg] = f dg + g df$
$\frac{d}{dx}\left[\frac{f}{g}\right] = \frac{g \frac{df}{dx} - f \frac{dg}{dx}}{g^2}$	$d\left[\frac{f}{g}\right] = \frac{g df - f dg}{g^2}$

For example,

$$\begin{aligned} d[x^2 \sin x] &= (x^2 \cos x + 2x \sin x) dx \\ &= x^2(\cos x dx) + (2x dx) \sin x \\ &= x^2 d[\sin x] + (\sin x) d[x^2] \end{aligned}$$

illustrates the differential version of the product rule.

QUICK CHECK EXERCISES 2.9 (See page 183 for answers.)

- The local linear approximation of f at x_0 uses the _____ line to the graph of $y = f(x)$ at $x = x_0$ to approximate values of _____ for values of x near _____.
- Find an equation for the local linear approximation to $y = 5 - x^2$ at $x_0 = 2$.
- Let $y = 5 - x^2$. Find dy and Δy at $x = 2$ with $dx = \Delta x = 0.1$.
- The intensity of light from a light source is a function $I = f(x)$ of the distance x from the light source. Suppose that a small gemstone is measured to be 10 m from a light source, $f(10) = 0.2 \text{ W/m}^2$, and $f'(10) = -0.04 \text{ W/m}^3$. If the distance $x = 10$ m was obtained with a measurement error within ± 0.05 m, estimate the percentage error in the calculated intensity of the light on the gemstone.

EXERCISE SET 2.9 Graphing Utility

- Use Formula (1) to obtain the local linear approximation of x^3 at $x_0 = 1$.
 - Use Formula (2) to rewrite the approximation obtained in part (a) in terms of Δx .
 - Use the result obtained in part (a) to approximate $(1.02)^3$, and confirm that the formula obtained in part (b) produces the same result.
- Use Formula (1) to obtain the local linear approximation of $1/x$ at $x_0 = 2$.
 - Use Formula (2) to rewrite the approximation obtained in part (a) in terms of Δx .
 - Use the result obtained in part (a) to approximate $1/2.05$, and confirm that the formula obtained in part (b) produces the same result.
- Find the local linear approximation of the function $f(x) = \sqrt{1+x}$ at $x_0 = 0$, and use it to approximate $\sqrt{0.9}$ and $\sqrt{1.1}$.
 - Graph f and its tangent line at x_0 together, and use the graphs to illustrate the relationship between the exact values and the approximations of $\sqrt{0.9}$ and $\sqrt{1.1}$.
- A student claims that whenever a local linear approximation is used to approximate the square root of a number, the approximation is too large.
 - Write a few sentences that make the student's claim precise, and justify this claim geometrically.
 - Verify the student's claim algebraically using approximation (1).
- $f(x) = \sqrt{x}$; $\sqrt{1+\Delta x} \approx 1 + \frac{1}{2}\Delta x$
- $f(x) = \frac{1}{2+x}$; $\frac{1}{3+\Delta x} \approx \frac{1}{3} - \frac{1}{9}\Delta x$
- $f(x) = (4+x)^3$; $(5+\Delta x)^3 \approx 125 + 75\Delta x$
-  **13–16** Confirm that the formula is the local linear approximation at $x_0 = 0$, and use a graphing utility to estimate an interval of x -values on which the error is at most ± 0.1 . ■
- $\sqrt{x+3} \approx \sqrt{3} + \frac{1}{2\sqrt{3}}x$
- $\frac{1}{\sqrt{9-x}} \approx \frac{1}{3} + \frac{1}{54}x$
- $\tan 2x \approx 2x$
- $\frac{1}{(1+2x)^5} \approx 1 - 10x$
- Use the local linear approximation of $\sin x$ at $x_0 = 0$ obtained in Example 2 to approximate $\sin 1^\circ$, and compare the approximation to the result produced directly by your calculating device.
 - How would you choose x_0 to approximate $\sin 44^\circ$?
 - Approximate $\sin 44^\circ$; compare the approximation to the result produced directly by your calculating device.
- Use the local linear approximation of $\tan x$ at $x_0 = 0$ to approximate $\tan 2^\circ$, and compare the approximation to the result produced directly by your calculating device.
 - How would you choose x_0 to approximate $\tan 61^\circ$?
 - Approximate $\tan 61^\circ$; compare the approximation to the result produced directly by your calculating device.

FOCUS ON CONCEPTS

5–10 Confirm that the stated formula is the local linear approximation at $x_0 = 0$. ■

- $(1+x)^{15} \approx 1 + 15x$
- $\frac{1}{\sqrt{1-x}} \approx 1 + \frac{1}{2}x$
- $\tan x \approx x$
- $\frac{1}{1+x} \approx 1 - x$

9–12 Confirm that the stated formula is the local linear approximation of f at $x_0 = 1$, where $\Delta x = x - 1$. ■

- $f(x) = x^4$; $(1+\Delta x)^4 \approx 1 + 4\Delta x$

19–27 Use an appropriate local linear approximation to estimate the value of the given quantity. ■

- $(3.02)^4$
- $(1.97)^3$
- $\sqrt{65}$
- $\sqrt{24}$
- $\sqrt{80.9}$
- $\sqrt{36.03}$
- $\sin 0.1$
- $\tan 0.2$
- $\cos 31^\circ$

FOCUS ON CONCEPTS

- The approximation $(1+x)^k \approx 1 + kx$ is commonly used by engineers for quick calculations.
 - Derive this result, and use it to make a rough estimate of $(1.001)^{37}$.
 - Compare your estimate to that produced directly by your calculating device.

(cont.)

(c) If k is a positive integer, how is the approximation $(1+x)^k \approx 1+kx$ related to the expansion of $(1+x)^k$ using the binomial theorem?

29. Use the approximation $(1+x)^k \approx 1+kx$, along with some mental arithmetic to show that $\sqrt[3]{8.24} \approx 2.02$ and $4.08^{3/2} \approx 8.24$.
30. Referring to the accompanying figure, suppose that the angle of elevation of the top of the building, as measured from a point 500 ft from its base, is found to be $\theta = 6^\circ$. Use an appropriate local linear approximation, along with some mental arithmetic to show that the building is about 52 ft high.

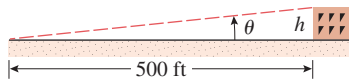


Figure Ex-30

31. (a) Let $y = 1/x$. Find dy and Δy at $x = 1$ with $dx = \Delta x = -0.5$.
 (b) Sketch the graph of $y = 1/x$, showing dy and Δy in the picture.
32. (a) Let $y = \sqrt{x}$. Find dy and Δy at $x = 9$ with $dx = \Delta x = -1$.
 (b) Sketch the graph of $y = \sqrt{x}$, showing dy and Δy in the picture.

33–36 Find formulas for dy and Δy . ■

33. $y = x^3$ 34. $y = 8x - 4$
 35. $y = x^2 - 2x + 1$ 36. $y = \sin x$

37–40 Find the differential dy . ■

37. (a) $y = 4x^3 - 7x^2$ (b) $y = x \cos x$
 38. (a) $y = 1/x$ (b) $y = 5 \tan x$
 39. (a) $y = x\sqrt{1-x}$ (b) $y = (1+x)^{-17}$
 40. (a) $y = \frac{1}{x^3 - 1}$ (b) $y = \frac{1-x^3}{2-x}$

41–44 True–False Determine whether the statement is true or false. Explain your answer. ■

41. A differential dy is defined to be a very small change in y .
 42. The error in approximation (2) is the same as the error in approximation (7).
 43. A local linear approximation to a function can never be identically equal to the function.
 44. A local linear approximation to a nonconstant function can never be constant.

45–48 Use the differential dy to approximate Δy when x changes as indicated. ■

45. $y = \sqrt{3x-2}$; from $x = 2$ to $x = 2.03$
 46. $y = \sqrt{x^2+8}$; from $x = 1$ to $x = 0.97$

47. $y = \frac{x}{x^2+1}$; from $x = 2$ to $x = 1.96$

48. $y = x\sqrt{8x+1}$; from $x = 3$ to $x = 3.05$

49. The side of a square is measured to be 10 ft, with a possible error of ± 0.1 ft.

- (a) Use differentials to estimate the error in the calculated area.
 (b) Estimate the percentage errors in the side and the area.

50. The side of a cube is measured to be 25 cm, with a possible error of ± 1 cm.

- (a) Use differentials to estimate the error in the calculated volume.
 (b) Estimate the percentage errors in the side and volume.

51. The hypotenuse of a right triangle is known to be 10 in exactly, and one of the acute angles is measured to be 30° , with a possible error of $\pm 1^\circ$.

- (a) Use differentials to estimate the errors in the sides opposite and adjacent to the measured angle.
 (b) Estimate the percentage errors in the sides.

52. One side of a right triangle is known to be 25 cm exactly. The angle opposite to this side is measured to be 60° , with a possible error of $\pm 0.5^\circ$.

- (a) Use differentials to estimate the errors in the adjacent side and the hypotenuse.
 (b) Estimate the percentage errors in the adjacent side and hypotenuse.

53. The electrical resistance R of a certain wire is given by $R = k/r^2$, where k is a constant and r is the radius of the wire. Assuming that the radius r has a possible error of $\pm 5\%$, use differentials to estimate the percentage error in R . (Assume k is exact.)

54. A 12-foot ladder leaning against a wall makes an angle θ with the floor. If the top of the ladder is h feet up the wall, express h in terms of θ and then use dh to estimate the change in h if θ changes from 60° to 59° .

55. The area of a right triangle with a hypotenuse of H is calculated using the formula $A = \frac{1}{4}H^2 \sin 2\theta$, where θ is one of the acute angles. Use differentials to approximate the error in calculating A if $H = 4$ cm (exactly) and θ is measured to be 30° , with a possible error of $\pm 15'$.

56. The side of a square is measured with a possible percentage error of $\pm 1\%$. Use differentials to estimate the percentage error in the area.

57. The side of a cube is measured with a possible percentage error of $\pm 2\%$. Use differentials to estimate the percentage error in the volume.

58. The volume of a sphere is to be computed from a measured value of its radius. Estimate the maximum permissible percentage error in the measurement if the percentage error in the volume must be kept within $\pm 3\%$. ($V = \frac{4}{3}\pi r^3$ is the volume of a sphere of radius r .)

59. The area of a circle is to be computed from a measured value of its diameter. Estimate the maximum permissible

percentage error in the measurement if the percentage error in the area must be kept within $\pm 1\%$.

60. A steel cube with 1-inch sides is coated with 0.01 inch of copper. Use differentials to estimate the volume of copper in the coating. [Hint: Let ΔV be the change in the volume of the cube.]
61. A metal rod 15 cm long and 5 cm in diameter is to be covered (except for the ends) with insulation that is 0.1 cm thick. Use differentials to estimate the volume of insulation. [Hint: Let ΔV be the change in volume of the rod.]
62. The time required for one complete oscillation of a pendulum is called its **period**. If L is the length of the pendulum and the oscillation is small, then the period is given by $P = 2\pi\sqrt{L/g}$, where g is the constant acceleration due to gravity. Use differentials to show that the percentage error in P is approximately half the percentage error in L .
63. If the temperature T of a metal rod of length L is changed by an amount ΔT , then the length will change by the amount $\Delta L = \alpha L \Delta T$, where α is called the **coefficient of linear expansion**. For moderate changes in temperature α is taken as constant.
 - (a) Suppose that a rod 40 cm long at 20°C is found to be 40.006 cm long when the temperature is raised to 30°C . Find α .

- (b) If an aluminum pole is 180 cm long at 15°C , how long is the pole if the temperature is raised to 40°C ? [Take $\alpha = 2.3 \times 10^{-5}/^\circ\text{C}$.]

64. If the temperature T of a solid or liquid of volume V is changed by an amount ΔT , then the volume will change by the amount $\Delta V = \beta V \Delta T$, where β is called the **coefficient of volume expansion**. For moderate changes in temperature β is taken as constant. Suppose that a tank truck loads 4000 gallons of ethyl alcohol at a temperature of 35°C and delivers its load sometime later at a temperature of 15°C . Using $\beta = 7.5 \times 10^{-4}/^\circ\text{C}$ for ethyl alcohol, find the number of gallons delivered.
65. **Writing** Explain why the local linear approximation of a function value is equivalent to the use of a differential to approximate a change in the function.
66. **Writing** The local linear approximation

$$\sin x \approx x$$

is known as the *small angle approximation* and has both practical and theoretical applications. Do some research on some of these applications, and write a short report on the results of your investigations.

✓ QUICK CHECK ANSWERS 2.9

1. tangent; $f(x)$; x_0
2. $y = 1 + (-4)(x - 2)$ or $y = -4x + 9$
3. $dy = -0.4$, $\Delta y = -0.41$
4. within $\pm 1\%$

CHAPTER 2 REVIEW EXERCISES



Graphing Utility



CAS

1. Explain the difference between average and instantaneous rates of change, and discuss how they are calculated.
2. In parts (a)–(d), use the function $y = \frac{1}{2}x^2$.
 - (a) Find the average rate of change of y with respect to x over the interval $[3, 4]$.
 - (b) Find the instantaneous rate of change of y with respect to x at $x = 3$.
 - (c) Find the instantaneous rate of change of y with respect to x at a general x -value.
 - (d) Sketch the graph of $y = \frac{1}{2}x^2$ together with the secant line whose slope is given by the result in part (a), and indicate graphically the slope of the tangent line that corresponds to the result in part (b).
3. Complete each part for the function $f(x) = x^2 + 1$.
 - (a) Find the slope of the tangent line to the graph of f at a general x -value.
 - (b) Find the slope of the tangent line to the graph of f at $x = 2$.
4. A car is traveling on a straight road that is 120 mi long. For the first 100 mi the car travels at an average velocity of 50

mi/h. Show that no matter how fast the car travels for the final 20 mi it cannot bring the average velocity up to 60 mi/h for the entire trip.

5. At time $t = 0$ a car moves into the passing lane to pass a slow-moving truck. The average velocity of the car from $t = 1$ to $t = 1 + h$ is

$$v_{\text{ave}} = \frac{3(h+1)^{2.5} + 580h - 3}{10h}$$

Estimate the instantaneous velocity of the car at $t = 1$, where time is in seconds and distance is in feet.



6. A skydiver jumps from an airplane. Suppose that the distance she falls during the first t seconds before her parachute opens is $s(t) = 976((0.835)^t - 1) + 176t$, where s is in feet. Graph s versus t for $0 \leq t \leq 20$, and use your graph to estimate the instantaneous velocity at $t = 15$.
7. A particle moves on a line away from its initial position so that after t hours it is $s = 3t^2 + t$ miles from its initial position.

(cont.)

- (a) Find the average velocity of the particle over the interval $[1, 3]$.
 (b) Find the instantaneous velocity at $t = 1$.
8. State the definition of a derivative, and give two interpretations of it.
9. Use the definition of a derivative to find dy/dx , and check your answer by calculating the derivative using appropriate derivative formulas.
- (a) $y = \sqrt{9 - 4x}$ (b) $y = \frac{x}{x + 1}$

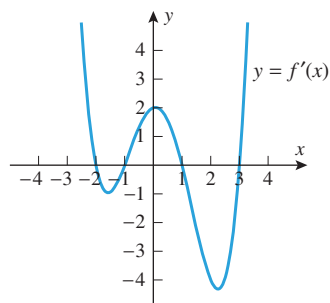
10. Suppose that $f(x) = \begin{cases} x^2 - 1, & x \leq 1 \\ k(x - 1), & x > 1. \end{cases}$

For what values of k is f

- (a) continuous? (b) differentiable?

11. The accompanying figure shows the graph of $y = f'(x)$ for an unspecified function f .

- (a) For what values of x does the curve $y = f(x)$ have a horizontal tangent line?
 (b) Over what intervals does the curve $y = f(x)$ have tangent lines with positive slope?
 (c) Over what intervals does the curve $y = f(x)$ have tangent lines with negative slope?
 (d) Given that $g(x) = f(x) \sin x$, find $g''(0)$.

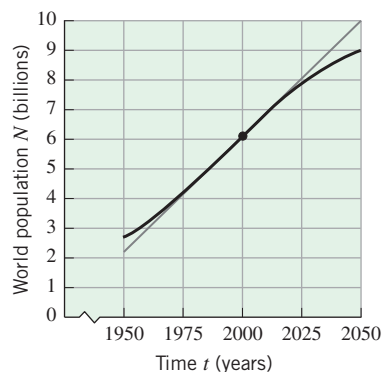


◀ Figure Ex-11

12. Sketch the graph of a function f for which $f(0) = 1$, $f'(0) = 0$, $f'(x) > 0$ if $x < 0$, and $f'(x) < 0$ if $x > 0$.
13. According to the U.S. Bureau of the Census, the estimated and projected midyear world population, N , in billions for the years 1950, 1975, 2000, 2025, and 2050 was 2.555, 4.088, 6.080, 7.841, and 9.104, respectively. Although the increase in population is not a continuous function of the time t , we can apply the ideas in this section if we are willing to approximate the graph of N versus t by a continuous curve, as shown in the accompanying figure.
- (a) Use the tangent line at $t = 2000$ shown in the figure to approximate the value of dN/dt there. Interpret your result as a rate of change.
- (b) The instantaneous **growth rate** is defined as

$$\frac{dN/dt}{N}$$

Use your answer to part (a) to approximate the instantaneous growth rate at the start of the year 2000. Express the result as a percentage and include the proper units.



◀ Figure Ex-13

14. Use a graphing utility to graph the function

$$f(x) = |x^4 - x - 1| - x$$

and estimate the values of x where the derivative of this function does not exist.

- 15–18 (a) Use a CAS to find $f'(x)$ via Definition 2.2.1; (b) check the result by finding the derivative by hand; (c) use the CAS to find $f''(x)$. ■

15. $f(x) = x^2 \sin x$

16. $f(x) = \sqrt{x} + \cos^2 x$

17. $f(x) = \frac{2x^2 - x + 5}{3x + 2}$

18. $f(x) = \frac{\tan x}{1 + x^2}$

19. The amount of water in a tank t minutes after it has started to drain is given by $W = 100(t - 15)^2$ gal.

- (a) At what rate is the water running out at the end of 5 min?
 (b) What is the average rate at which the water flows out during the first 5 min?

20. Use the formula $V = l^3$ for the volume of a cube of side l to find

- (a) the average rate at which the volume of a cube changes with l as l increases from $l = 2$ to $l = 4$
 (b) the instantaneous rate at which the volume of a cube changes with l when $l = 5$.

- 21–22 Zoom in on the graph of f on an interval containing $x = x_0$ until the graph looks like a straight line. Estimate the slope of this line and then check your answer by finding the exact value of $f'(x_0)$. ■

21. (a) $f(x) = x^2 - 1$, $x_0 = 1.8$

(b) $f(x) = \frac{x^2}{x - 2}$, $x_0 = 3.5$

22. (a) $f(x) = x^3 - x^2 + 1$, $x_0 = 2.3$

(b) $f(x) = \frac{x}{x^2 + 1}$, $x_0 = -0.5$

23. Suppose that a function f is differentiable at $x = 1$ and

$$\lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = 5$$

Find $f(1)$ and $f'(1)$.

24. Suppose that a function f is differentiable at $x = 2$ and

$$\lim_{x \rightarrow 2} \frac{x^3 f(x) - 24}{x - 2} = 28$$

Find $f(2)$ and $f'(2)$.

25. Find the equations of all lines through the origin that are tangent to the curve $y = x^3 - 9x^2 - 16x$.
26. Find all values of x for which the tangent line to the curve $y = 2x^3 - x^2$ is perpendicular to the line $x + 4y = 10$.
27. Let $f(x) = x^2$. Show that for any distinct values of a and b , the slope of the tangent line to $y = f(x)$ at $x = \frac{1}{2}(a + b)$ is equal to the slope of the secant line through the points (a, a^2) and (b, b^2) . Draw a picture to illustrate this result.
28. In each part, evaluate the expression given that $f(1) = 1$, $g(1) = -2$, $f'(1) = 3$, and $g'(1) = -1$.

(a) $\left. \frac{d}{dx} [f(x)g(x)] \right|_{x=1}$ (b) $\left. \frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] \right|_{x=1}$

(c) $\left. \frac{d}{dx} [\sqrt{f(x)}] \right|_{x=1}$ (d) $\left. \frac{d}{dx} [f(1)g'(1)] \right|_{x=1}$

29–32 Find $f'(x)$. ■

29. (a) $f(x) = x^8 - 3\sqrt{x} + 5x^{-3}$
 (b) $f(x) = (2x + 1)^{101}(5x^2 - 7)$
30. (a) $f(x) = \sin x + 2 \cos^3 x$
 (b) $f(x) = (1 + \sec x)(x^2 - \tan x)$
31. (a) $f(x) = \sqrt{3x + 1}(x - 1)^2$
 (b) $f(x) = \left(\frac{3x + 1}{x^2} \right)^3$
32. (a) $f(x) = \cot \left(\frac{\csc 2x}{x^3 + 5} \right)$ (b) $f(x) = \frac{1}{2x + \sin^3 x}$

33–34 Find the values of x at which the curve $y = f(x)$ has a horizontal tangent line. ■

33. $f(x) = (2x + 7)^6(x - 2)^5$ 34. $f(x) = \frac{(x - 3)^4}{x^2 + 2x}$

35. Find all lines that are simultaneously tangent to the graph of $y = x^2 + 1$ and to the graph of $y = -x^2 - 1$.
36. (a) Let n denote an even positive integer. Generalize the result of Exercise 35 by finding all lines that are simultaneously tangent to the graph of $y = x^n + n - 1$ and to the graph of $y = -x^n - n + 1$.
 (b) Let n denote an odd positive integer. Are there any lines that are simultaneously tangent to the graph of $y = x^n + n - 1$ and to the graph of $y = -x^n - n + 1$? Explain.

37. Find all values of x for which the line that is tangent to $y = 3x - \tan x$ is parallel to the line $y - x = 2$.

38. Approximate the values of x at which the tangent line to the graph of $y = x^3 - \sin x$ is horizontal.

39. Suppose that $f(x) = M \sin x + N \cos x$ for some constants M and N . If $f(\pi/4) = 3$ and $f'(\pi/4) = 1$, find an equation for the tangent line to $y = f(x)$ at $x = 3\pi/4$.

40. Suppose that $f(x) = M \tan x + N \sec x$ for some constants M and N . If $f(\pi/4) = 2$ and $f'(\pi/4) = 0$, find an equation for the tangent line to $y = f(x)$ at $x = 0$.

41. Suppose that $f'(x) = 2x \cdot f(x)$ and $f(2) = 5$.
 (a) Find $g'(\pi/3)$ if $g(x) = f(\sec x)$.
 (b) Find $h'(2)$ if $h(x) = [f(x)/(x - 1)]^4$.

42–44 Find dy/dx . ■

42. $y = \sqrt[4]{6x - 5}$ 43. $y = \sqrt[3]{x^2 + x}$

44. $y = \frac{(3 - 2x)^{4/3}}{x^2}$

- 45–46** (a) Find dy/dx by differentiating implicitly. (b) Solve the equation for y as a function of x , and find dy/dx from that equation. (c) Confirm that the two results are consistent by expressing the derivative in part (a) as a function of x alone. ■

45. $x^3 + xy - 2x = 1$ 46. $xy = x - y$

47–50 Find dy/dx by implicit differentiation. ■

47. $\frac{1}{y} + \frac{1}{x} = 1$ 48. $x^3 - y^3 = 6xy$

49. $\sec(xy) = y$ 50. $x^2 = \frac{\cot y}{1 + \csc y}$

51–52 Find d^2y/dx^2 by implicit differentiation. ■

51. $3x^2 - 4y^2 = 7$ 52. $2xy - y^2 = 3$

53. Use implicit differentiation to find the slope of the tangent line to the curve $y = x \tan(\pi y/2)$, $x > 0$, $y > 0$ (*the quadratrix of Hippias*) at the point $(\frac{1}{2}, \frac{1}{2})$.

54. At what point(s) is the tangent line to the curve $y^2 = 2x^3$ perpendicular to the line $4x - 3y + 1 = 0$?

55. Prove that if P and Q are two distinct points on the rotated ellipse $x^2 + xy + y^2 = 4$ such that P , Q , and the origin are collinear, then the tangent lines to the ellipse at P and Q are parallel.

56. Find the coordinates of the point in the first quadrant at which the tangent line to the curve $x^3 - xy + y^3 = 0$ is parallel to the x -axis.

57. Find the coordinates of the point in the first quadrant at which the tangent line to the curve $x^3 - xy + y^3 = 0$ is parallel to the y -axis.

58. Use implicit differentiation to show that the equation of the tangent line to the curve $y^2 = kx$ at (x_0, y_0) is

$$y_0 y = \frac{1}{2} k(x + x_0)$$

59. An oil slick on a lake is surrounded by a floating circular containment boom. As the boom is pulled in, the circular containment area shrinks. If the boom is pulled in at the rate of 5 m/min, at what rate is the containment area shrinking when the containment area has a diameter of 100 m?

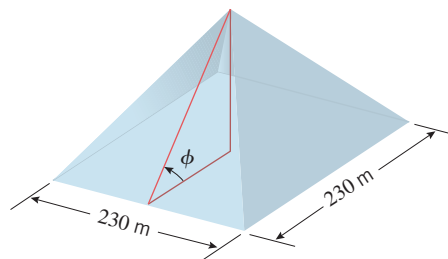
60. The hypotenuse of a right triangle is growing at a constant rate of a centimeters per second and one leg is decreasing at a constant rate of b centimeters per second. How fast is

the acute angle between the hypotenuse and the other leg changing at the instant when both legs are 1 cm?

61. In each part, use the given information to find Δx , Δy , and dy .
 - (a) $y = 1/(x - 1)$; x decreases from 2 to 1.5.
 - (b) $y = \tan x$; x increases from $-\pi/4$ to 0.
 - (c) $y = \sqrt{25 - x^2}$; x increases from 0 to 3.
62. Use an appropriate local linear approximation to estimate the value of $\cot 46^\circ$, and compare your answer to the value obtained with a calculating device.
63. The base of the Great Pyramid at Giza is a square that is 230 m on each side.
 - (a) As illustrated in the accompanying figure, suppose that an archaeologist standing at the center of a side measures the angle of elevation of the apex to be $\phi = 51^\circ$

with an error of $\pm 0.5^\circ$. What can the archaeologist reasonably say about the height of the pyramid?

- (b) Use differentials to estimate the allowable error in the elevation angle that will ensure that the error in calculating the height is at most ± 5 m.



▲ Figure Ex-63

CHAPTER 2 MAKING CONNECTIONS

1. Suppose that f is a function with the properties (i) f is differentiable everywhere, (ii) $f(x + y) = f(x)f(y)$ for all values of x and y , (iii) $f(0) \neq 0$, and (iv) $f'(0) = 1$.
 - (a) Show that $f(0) = 1$. [Hint: Consider $f(0 + 0)$.]
 - (b) Show that $f(x) > 0$ for all values of x . [Hint: First show that $f(x) \neq 0$ for any x by considering $f(x - x)$.]
 - (c) Use the definition of derivative (Definition 2.2.1) to show that $f'(x) = f(x)$ for all values of x .
2. Suppose that f and g are functions each of which has the properties (i)–(iv) in Exercise 1.
 - (a) Show that $y = f(2x)$ satisfies the equation $y' = 2y$ in two ways: using property (ii), and by directly applying the chain rule (Theorem 2.6.1).
 - (b) If k is any constant, show that $y = f(kx)$ satisfies the equation $y' = ky$.
 - (c) Find a value of k such that $y = f(x)g(x)$ satisfies the equation $y' = ky$.
 - (d) If $h = f/g$, find $h'(x)$. Make a conjecture about the relationship between f and g .
3. (a) Apply the product rule (Theorem 2.4.1) twice to show that if f , g , and h are differentiable functions, then $f \cdot g \cdot h$ is differentiable and

$$(f \cdot g \cdot h)' = f' \cdot g \cdot h + f \cdot g' \cdot h + f \cdot g \cdot h'$$
 - (b) Suppose that f , g , h , and k are differentiable functions. Derive a formula for $(f \cdot g \cdot h \cdot k)'$.

- (c) Based on the result in part (a), make a conjecture about a formula differentiating a product of n functions. Prove your formula using induction.

4. (a) Apply the quotient rule (Theorem 2.4.2) twice to show that if f , g , and h are differentiable functions, then $(f/g)/h$ is differentiable where it is defined and

$$[(f/g)/h]' = \frac{f' \cdot g \cdot h - f \cdot g' \cdot h - f \cdot g \cdot h'}{g^2 h^2}$$

- (b) Derive the derivative formula of part (a) by first simplifying $(f/g)/h$ and then applying the quotient and product rules.
- (c) Apply the quotient rule (Theorem 2.4.2) twice to derive a formula for $[f/(g/h)]'$.
- (d) Derive the derivative formula of part (c) by first simplifying $f/(g/h)$ and then applying the quotient and product rules.
5. Assume that $h(x) = f(x)/g(x)$ is differentiable. Derive the quotient rule formula for $h'(x)$ (Theorem 2.4.2) in two ways:
 - (a) Write $h(x) = f(x) \cdot [g(x)]^{-1}$ and use the product and chain rules (Theorems 2.4.1 and 2.6.1) to differentiate h .
 - (b) Write $f(x) = h(x) \cdot g(x)$ and use the product rule to derive a formula for $h'(x)$.



EXPANDING THE CALCULUS HORIZON

To learn how derivatives can be used in the field of robotics, see the module entitled **Robotics** at:

www.wiley.com/college/anton